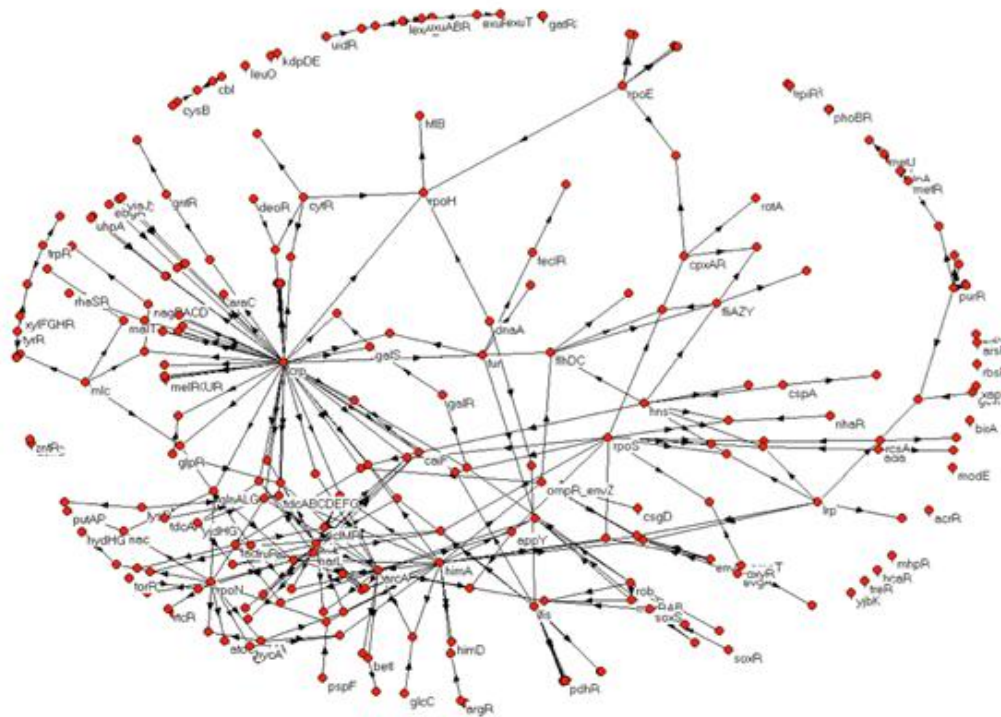


Computational Biology

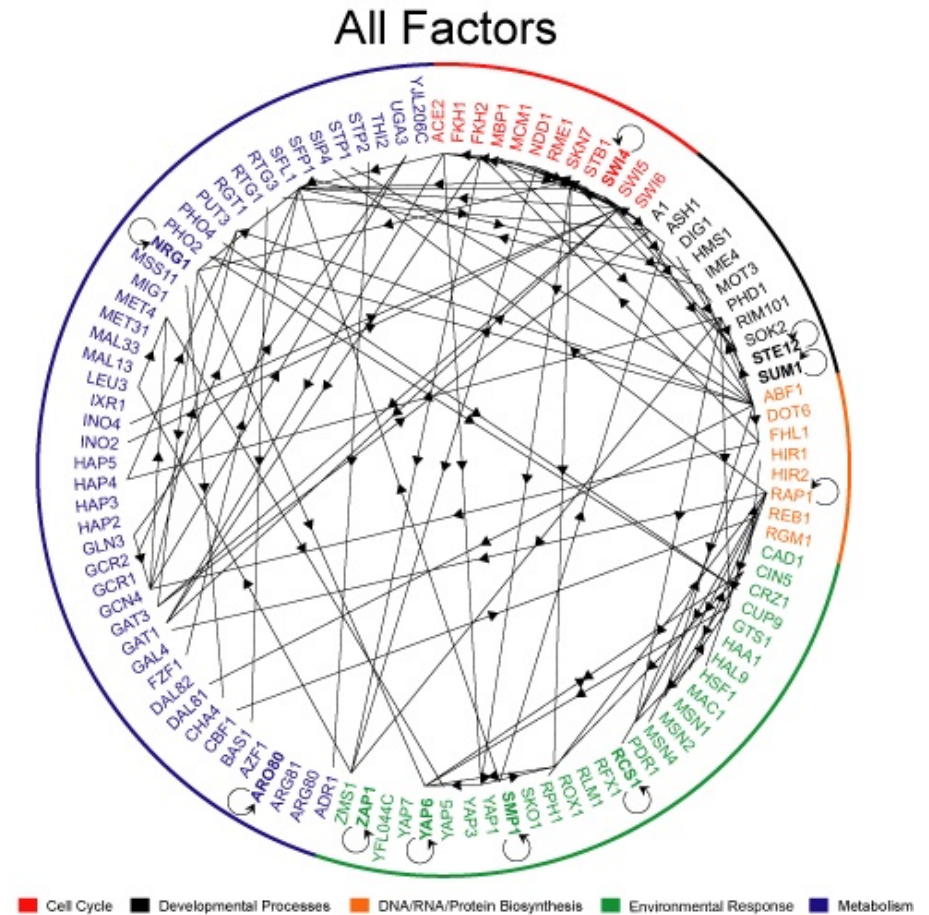
Lecture 3: *Gene Regulation Networks*



Gene regulation, chemical equations, motifs

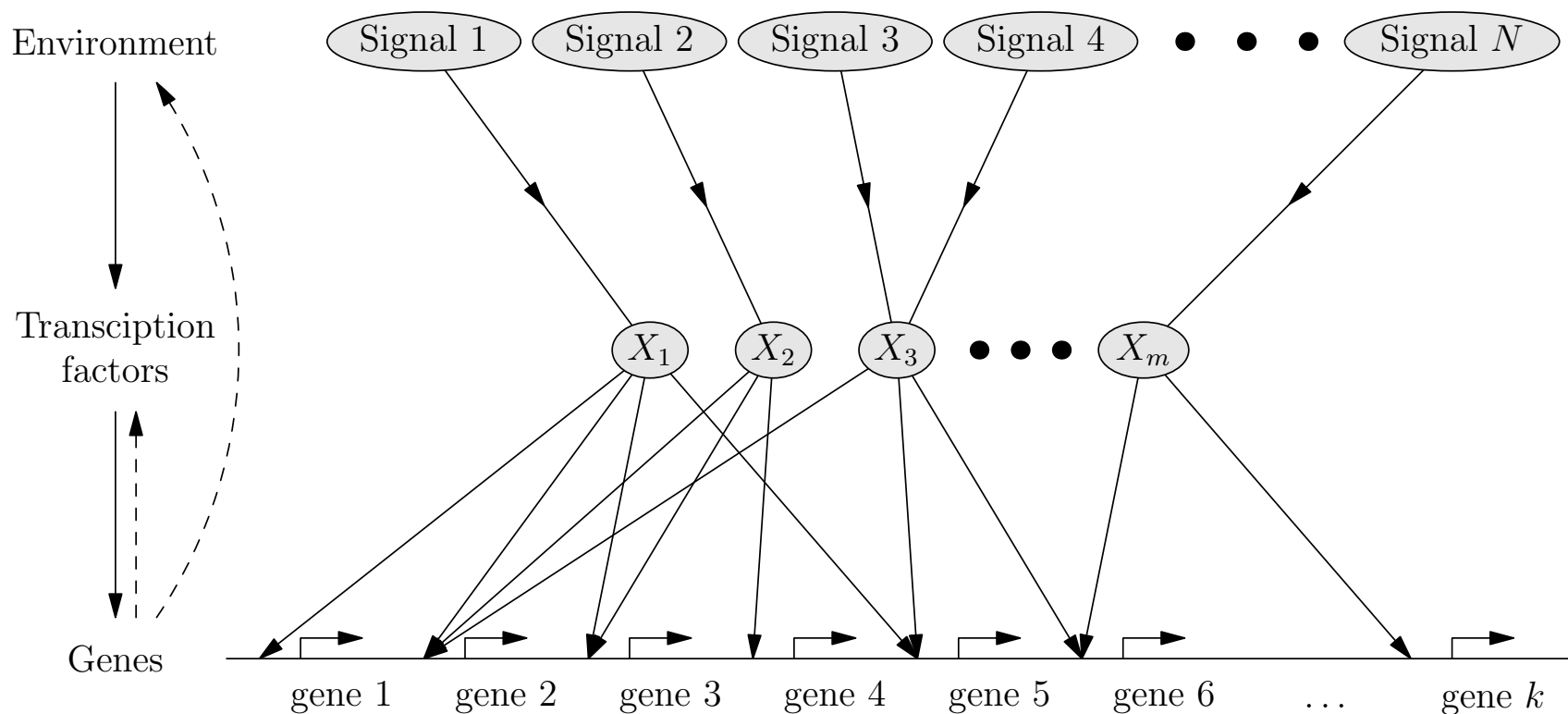
Outline

1. **Gene Regulation**
2. Chemical Equations
3. Motifs
4. Quiz

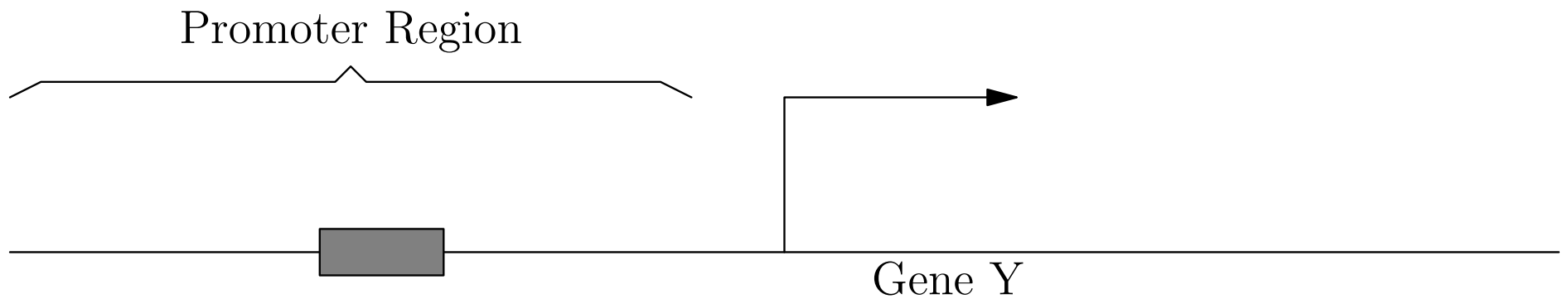


Transcription Networks

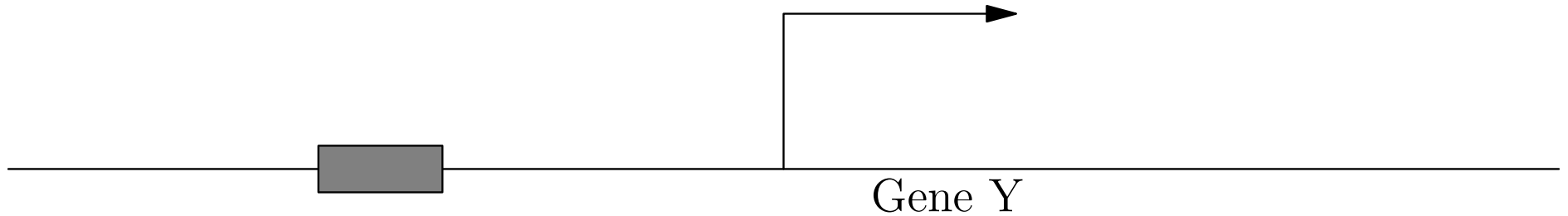
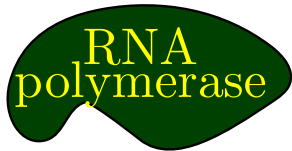
- Signalling molecules combine with transcription factors (proteins) which excite or inhibit the transcription of other proteins



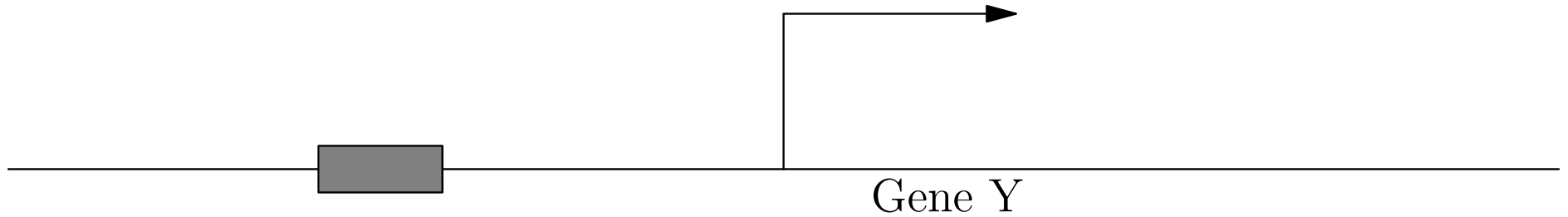
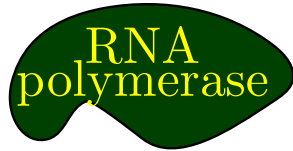
Gene Transcription Regulation



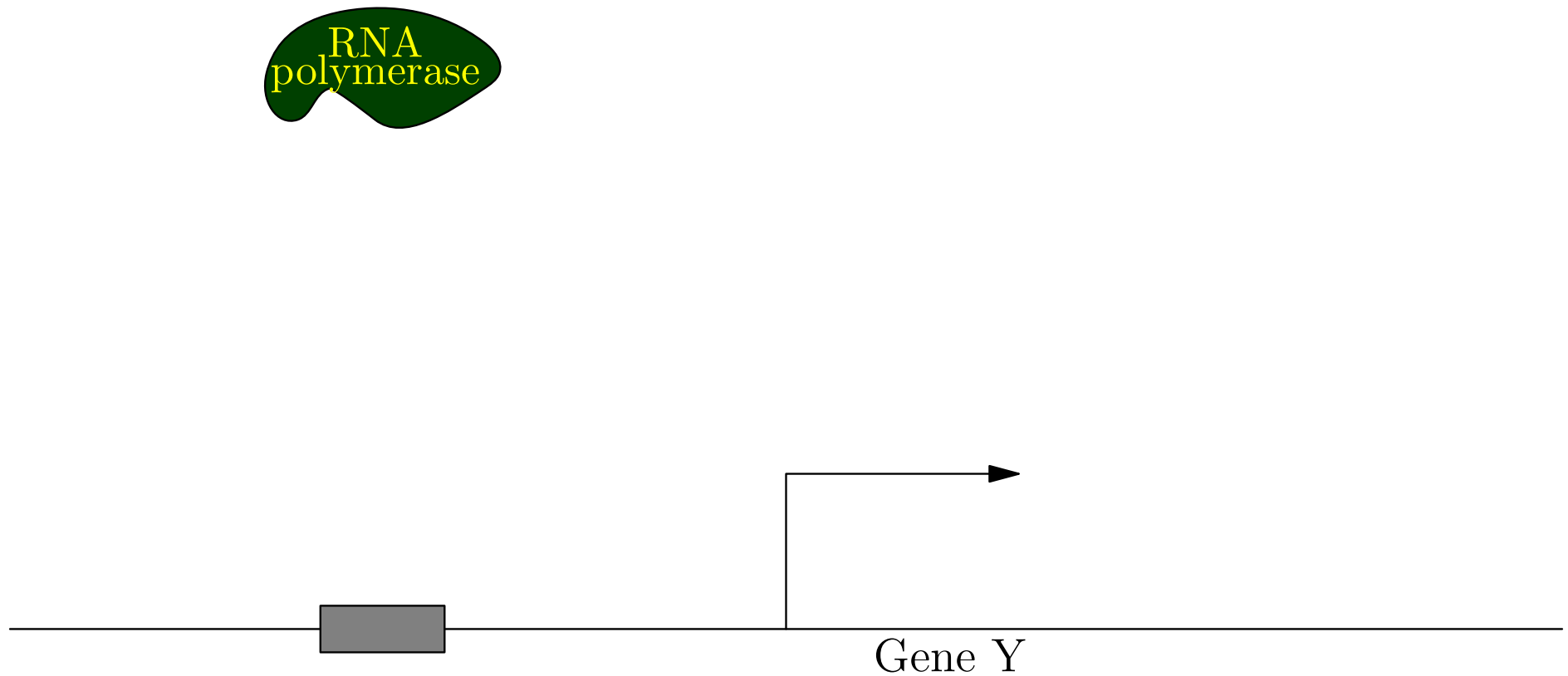
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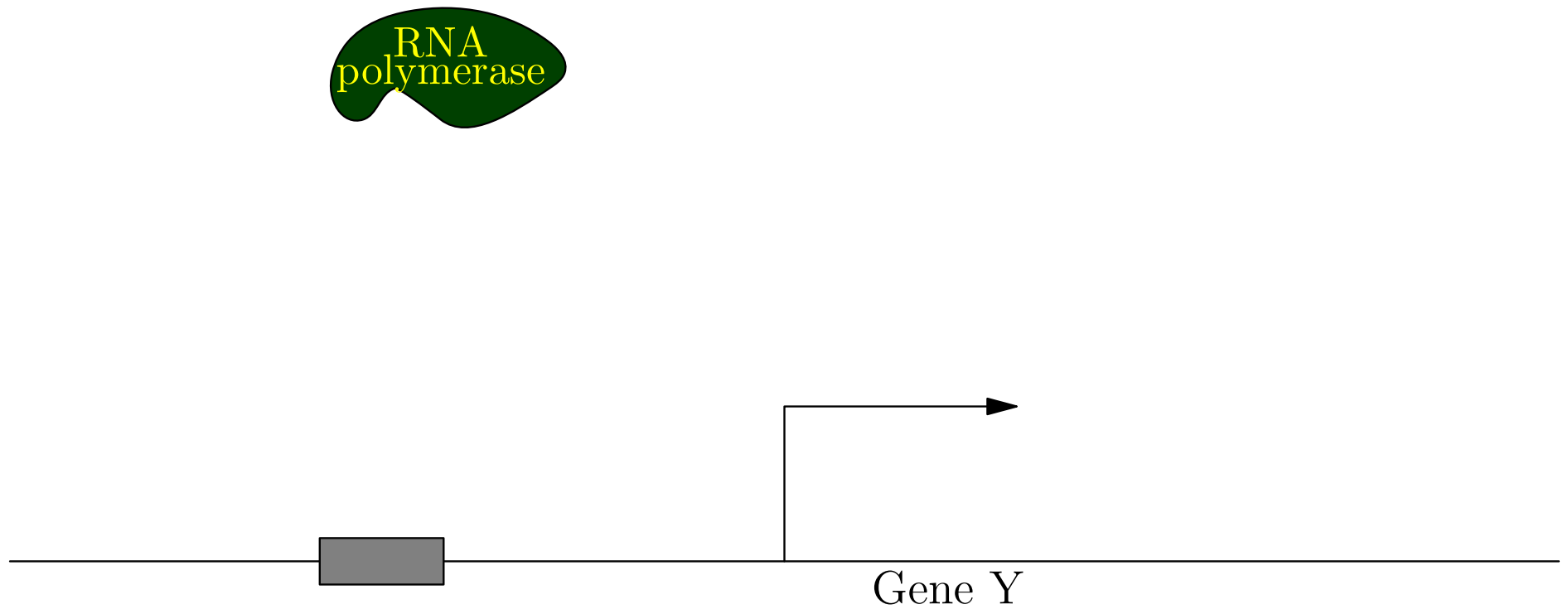
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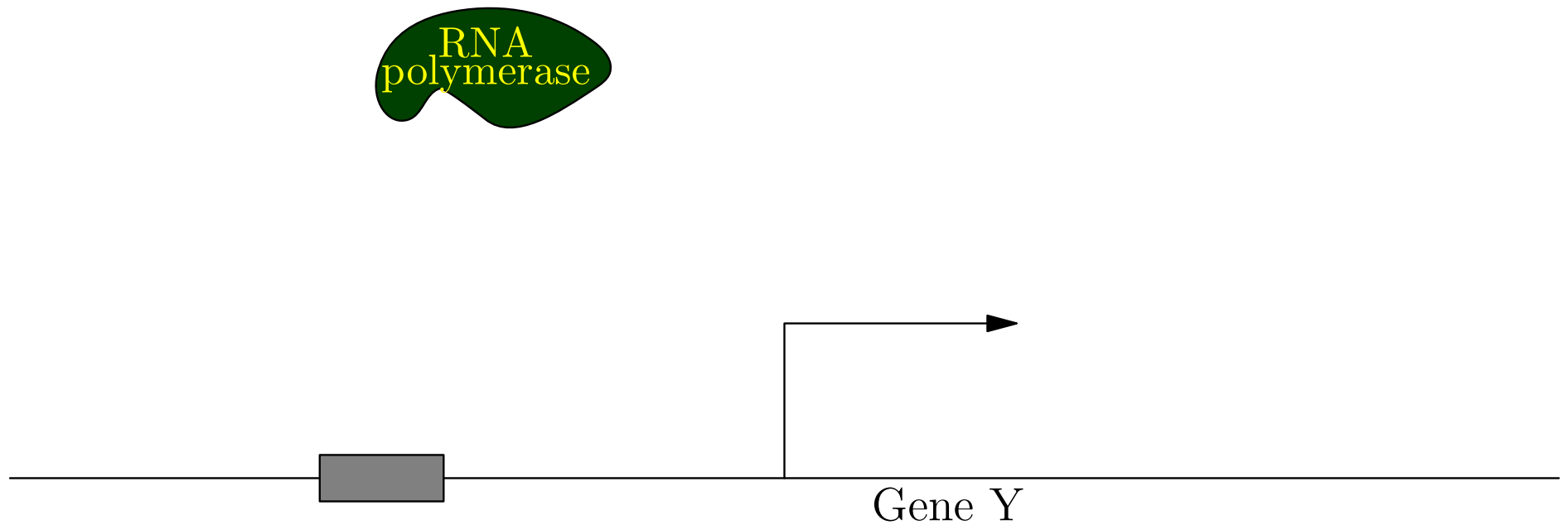
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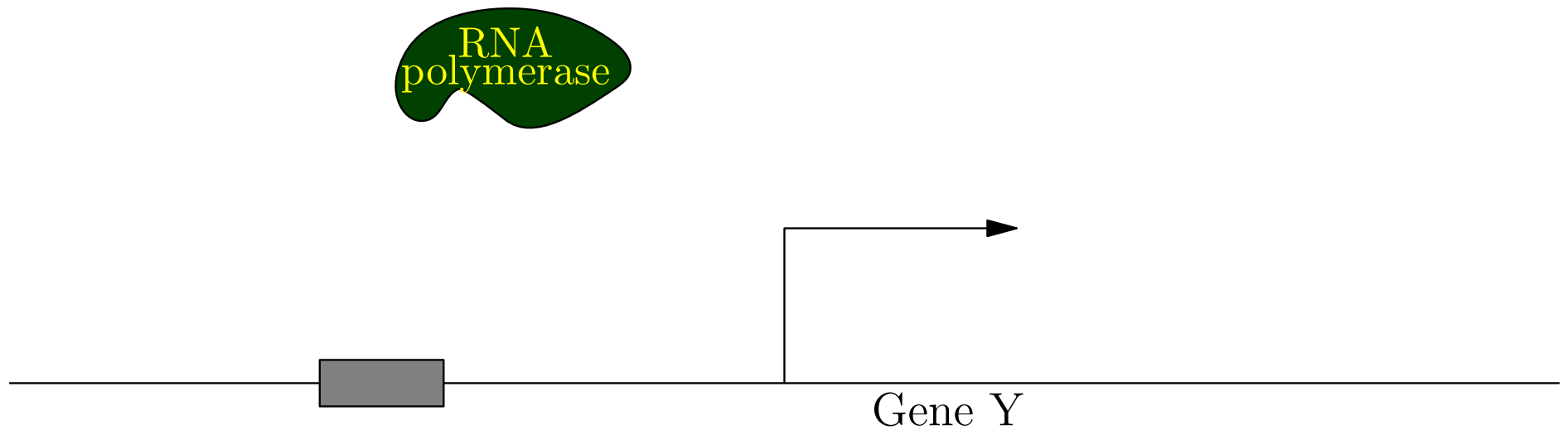
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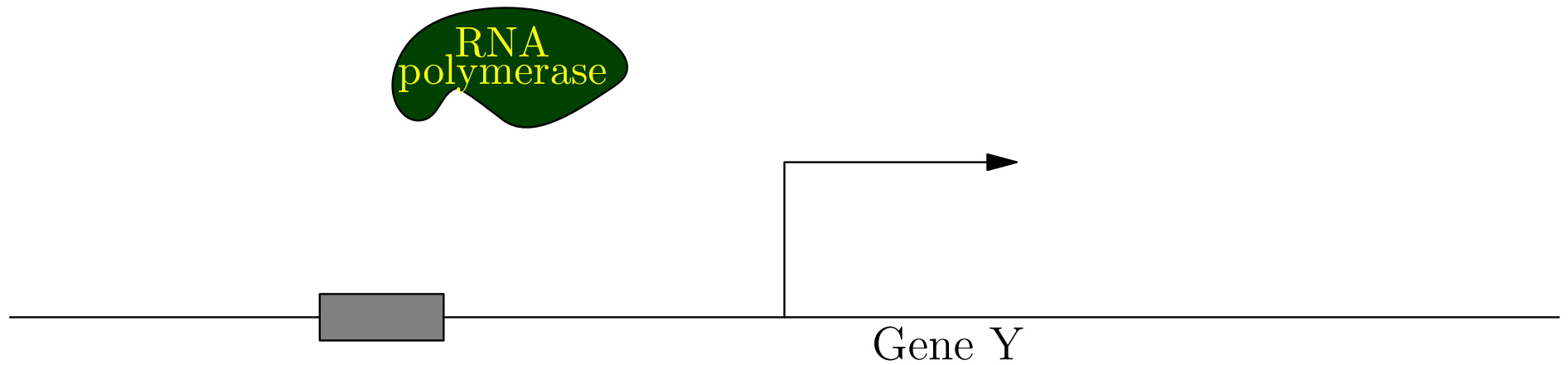
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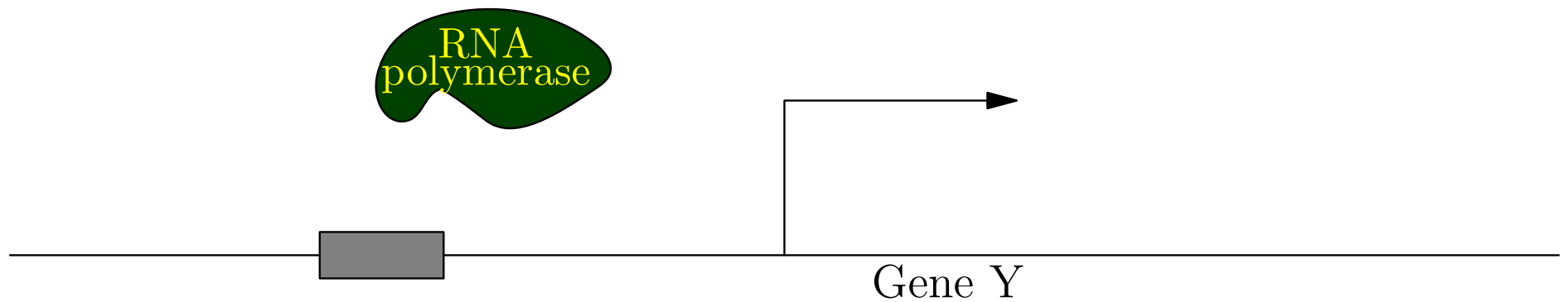
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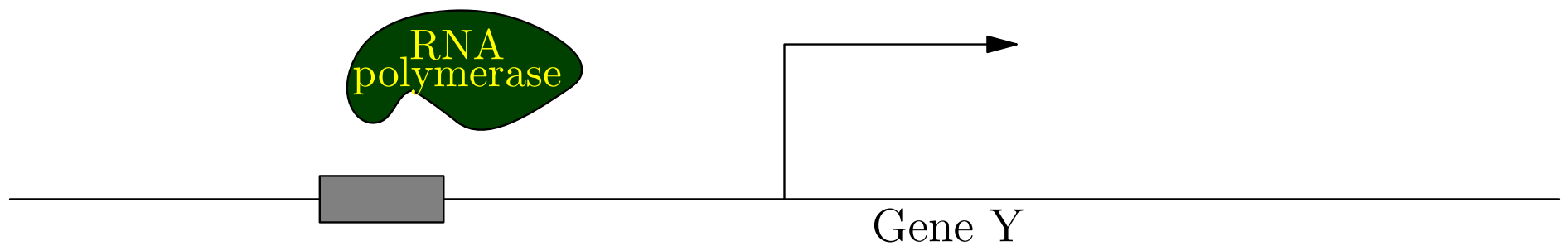
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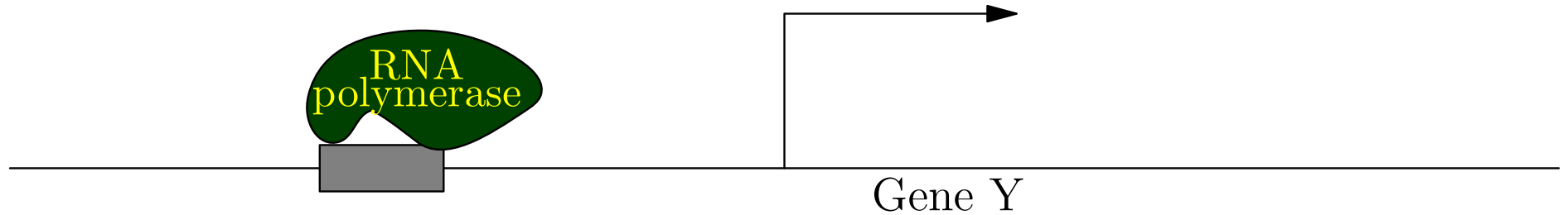
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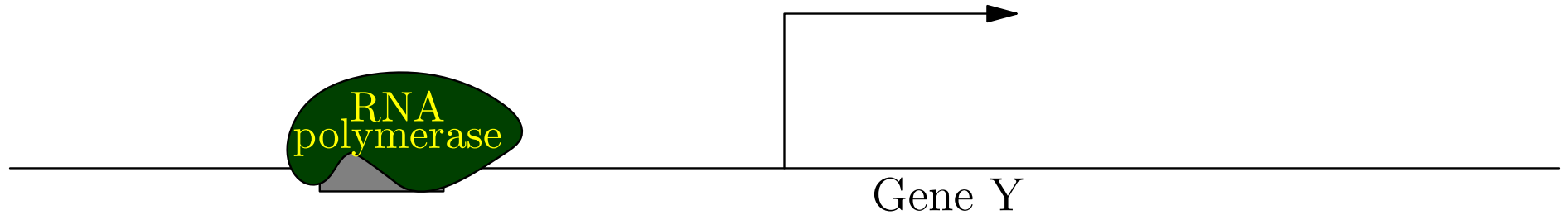
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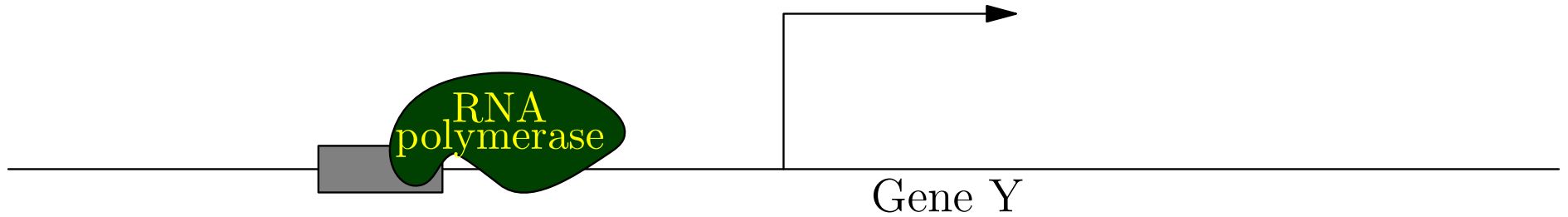
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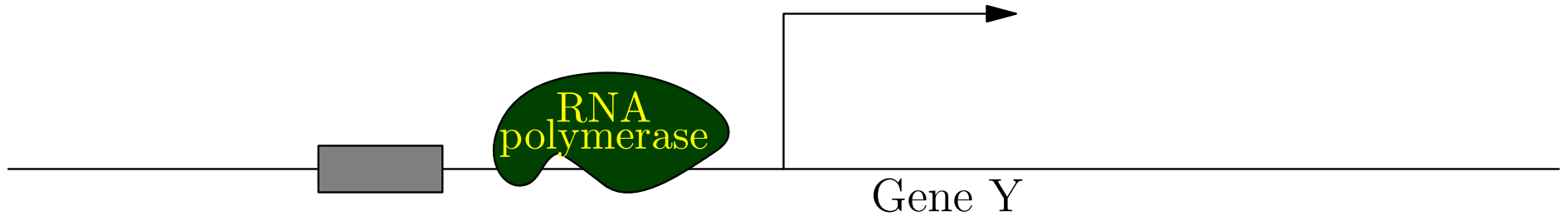
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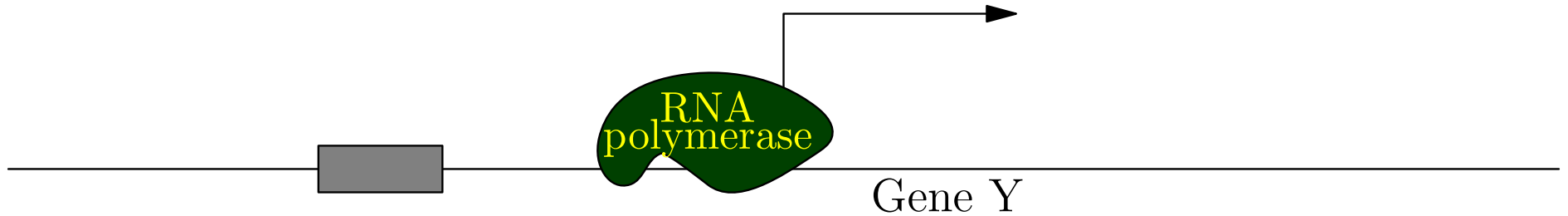
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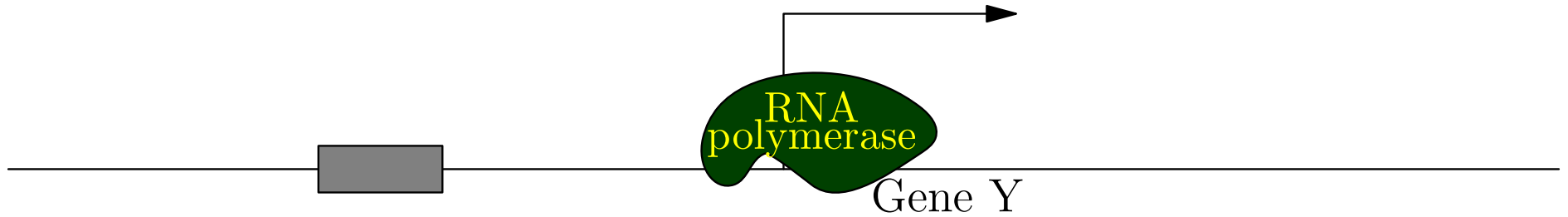
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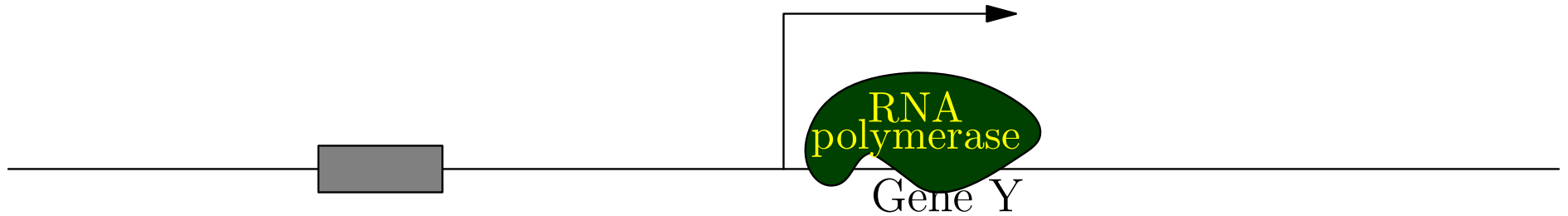
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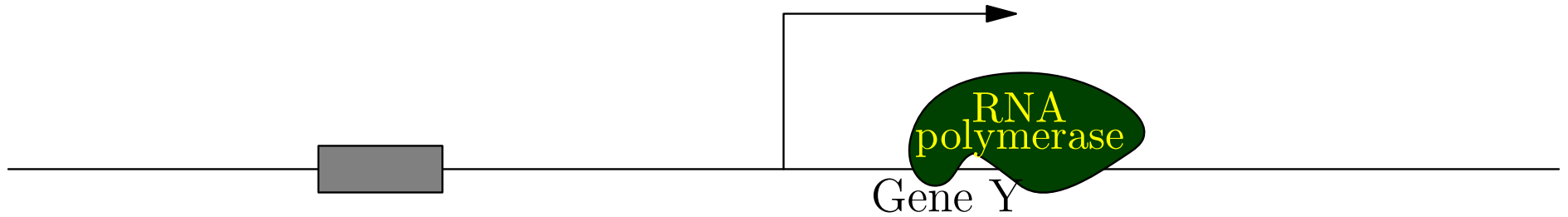
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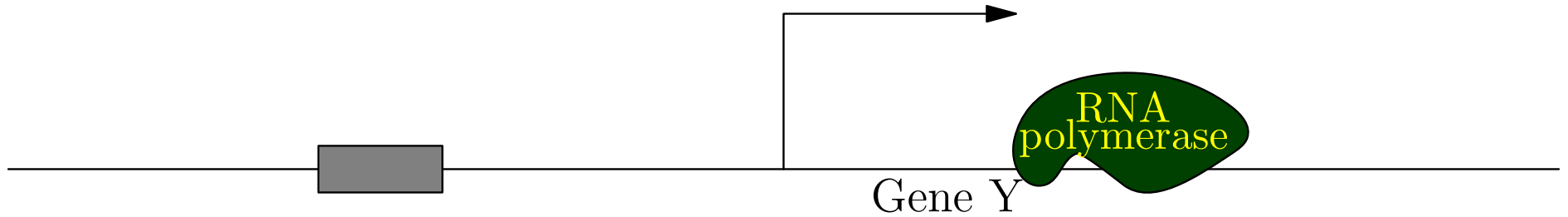
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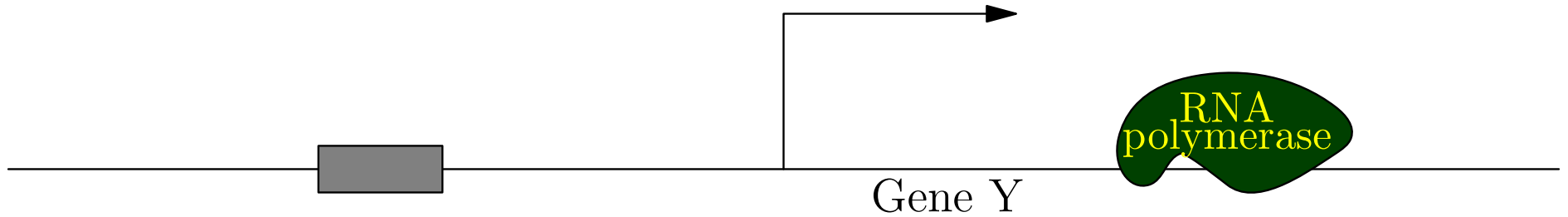
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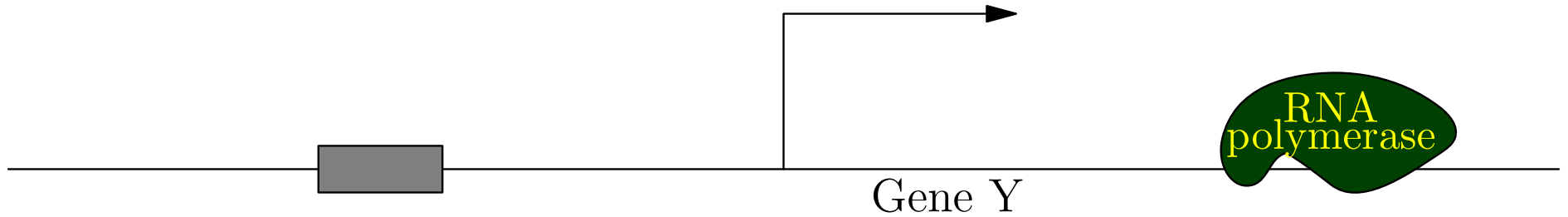
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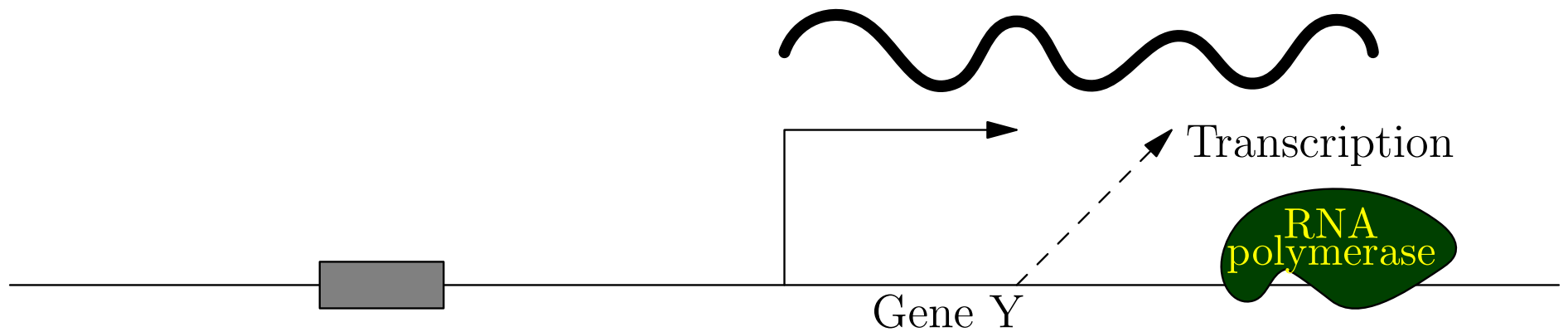
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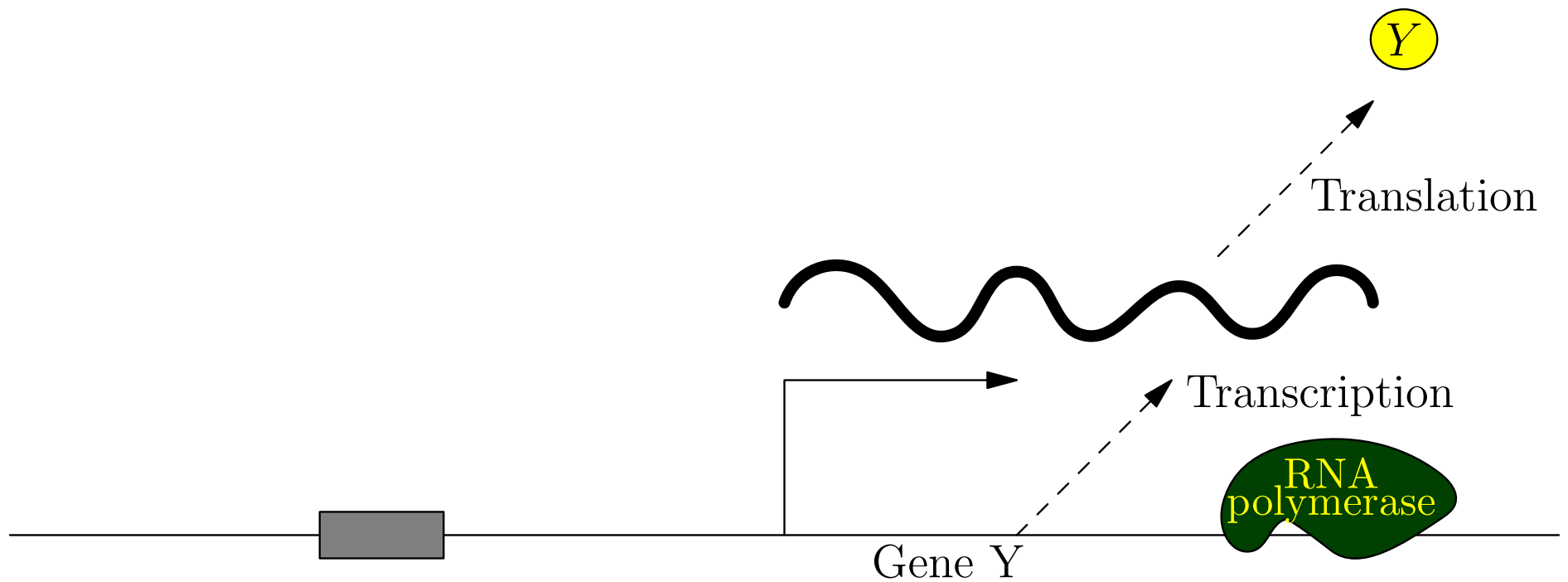
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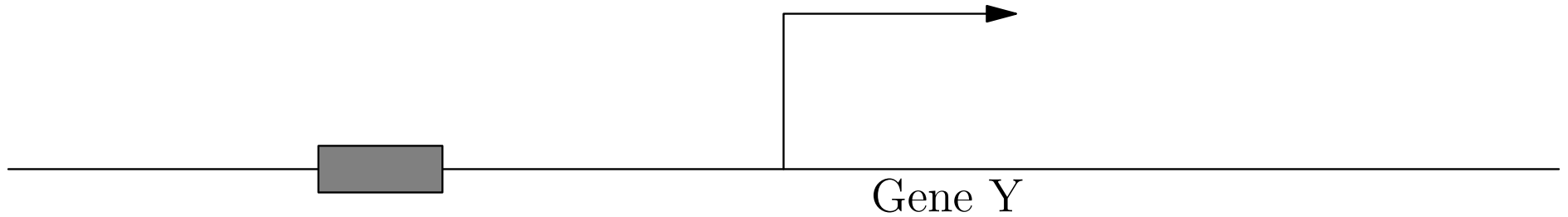
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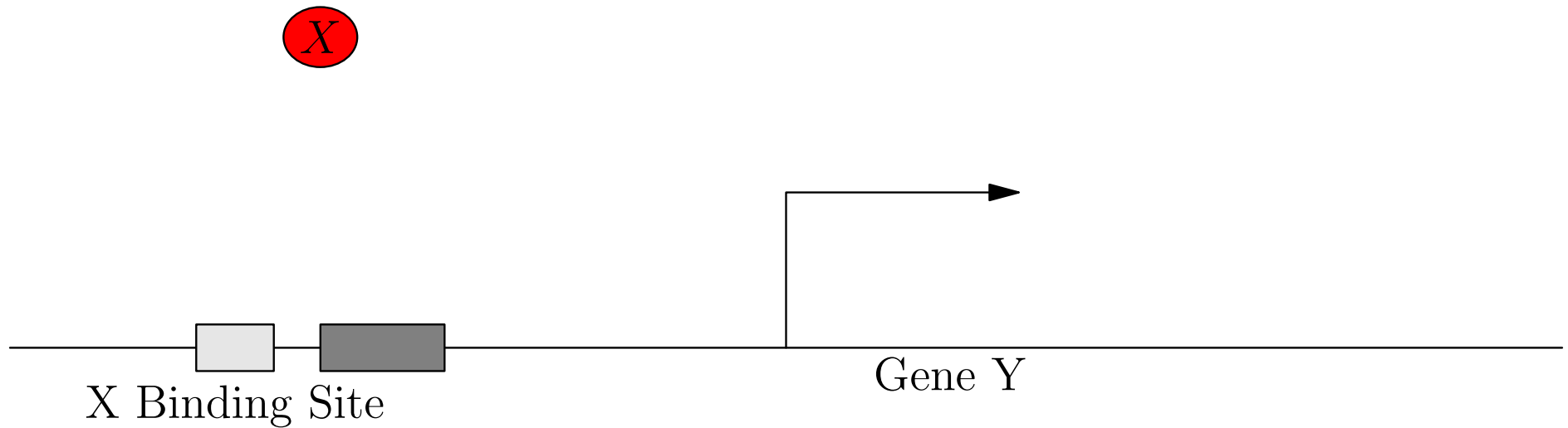
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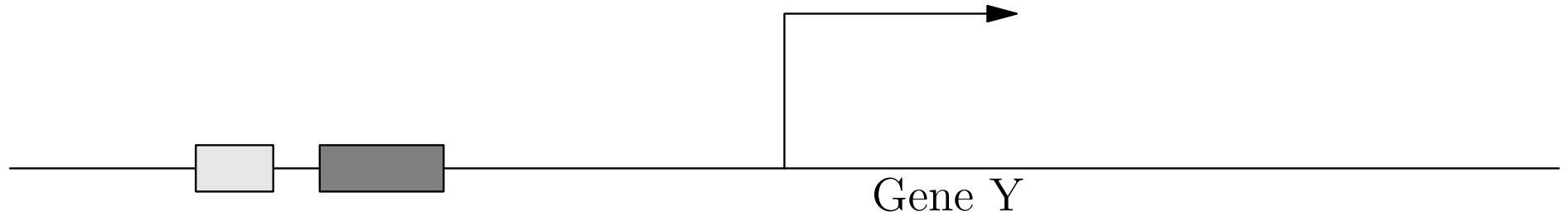
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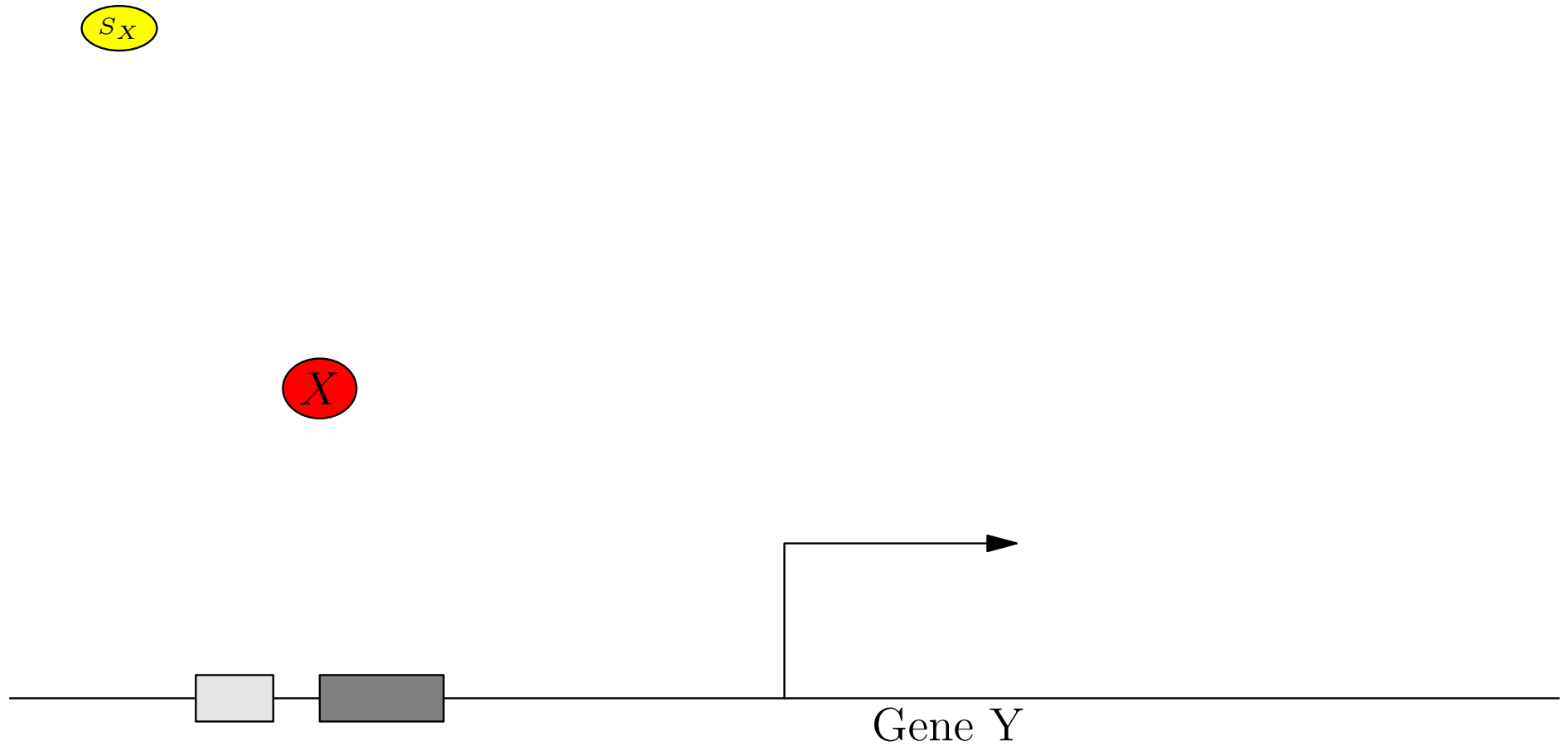
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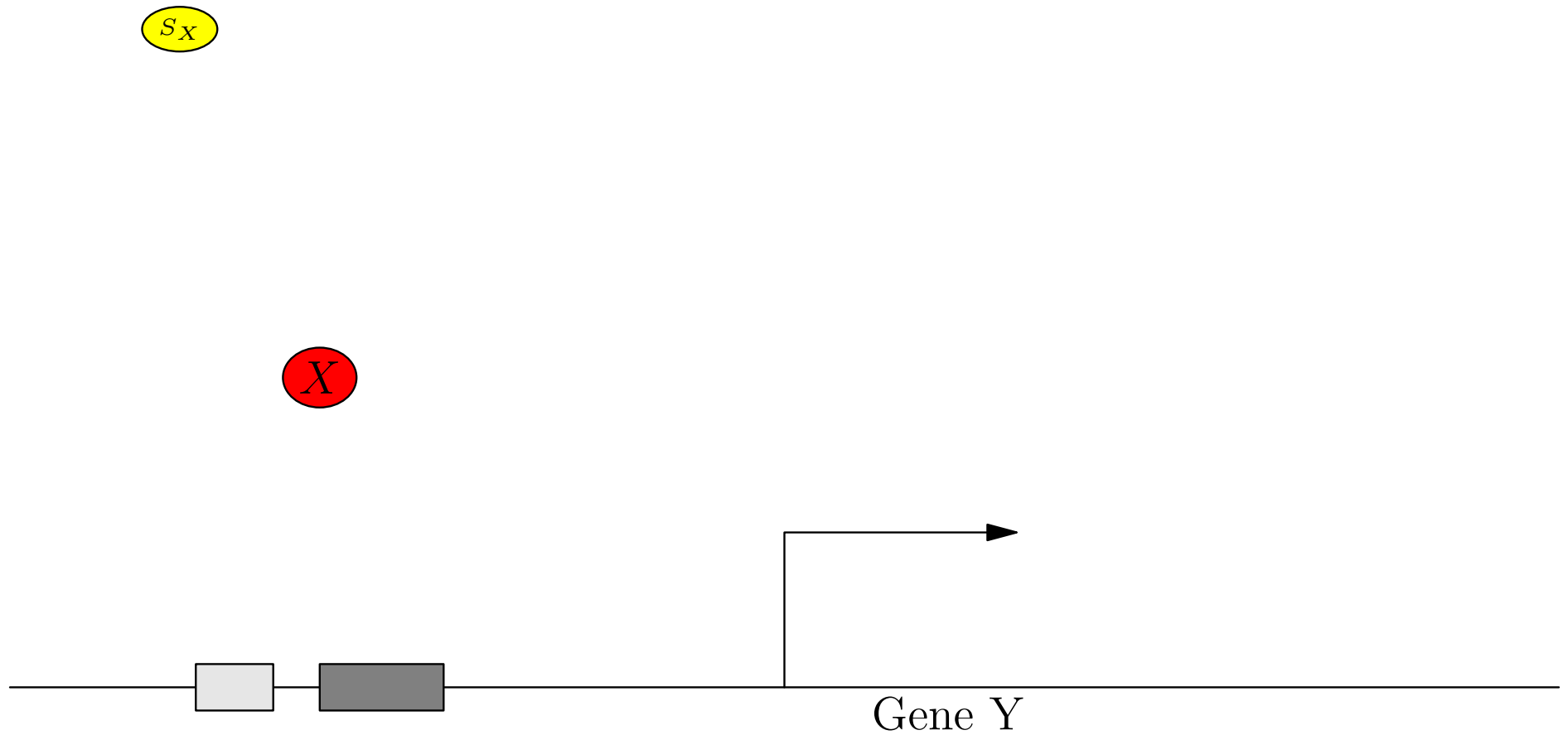
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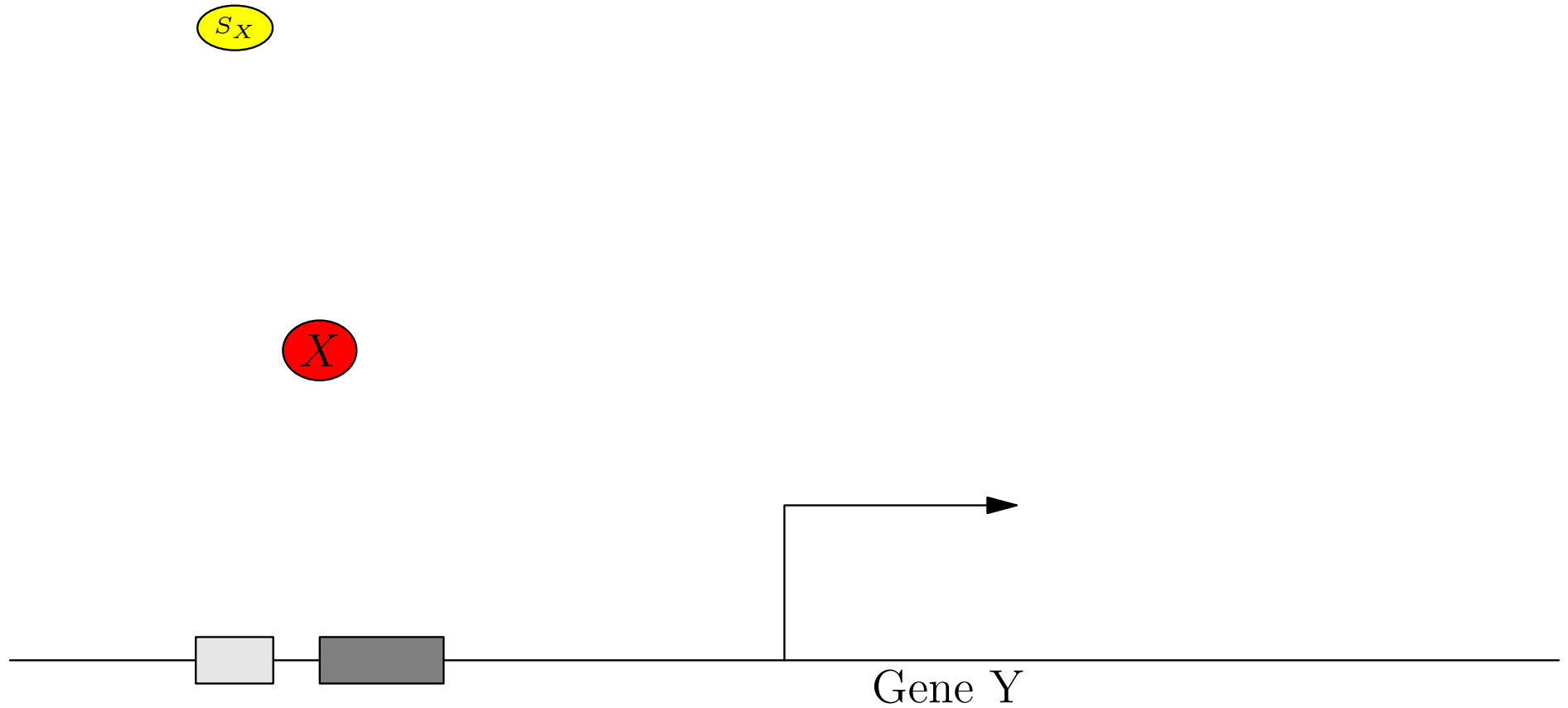
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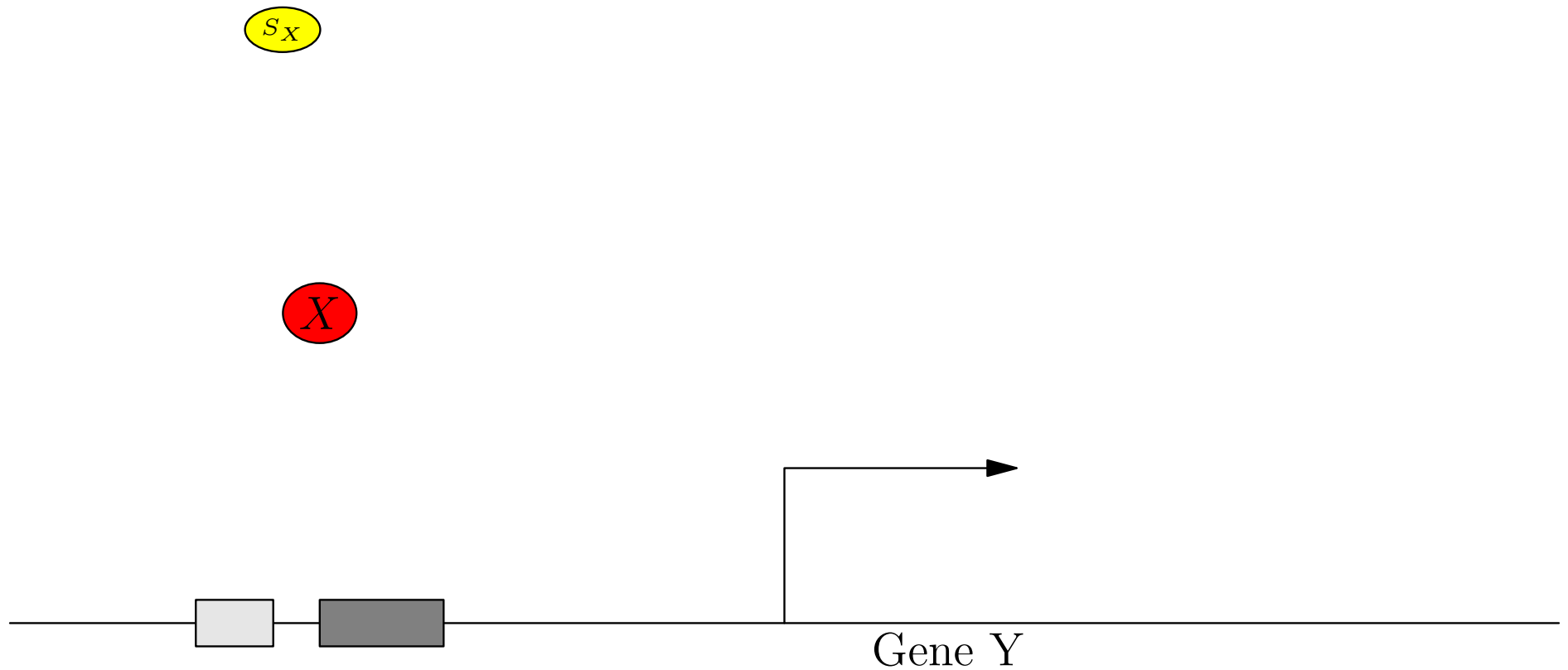
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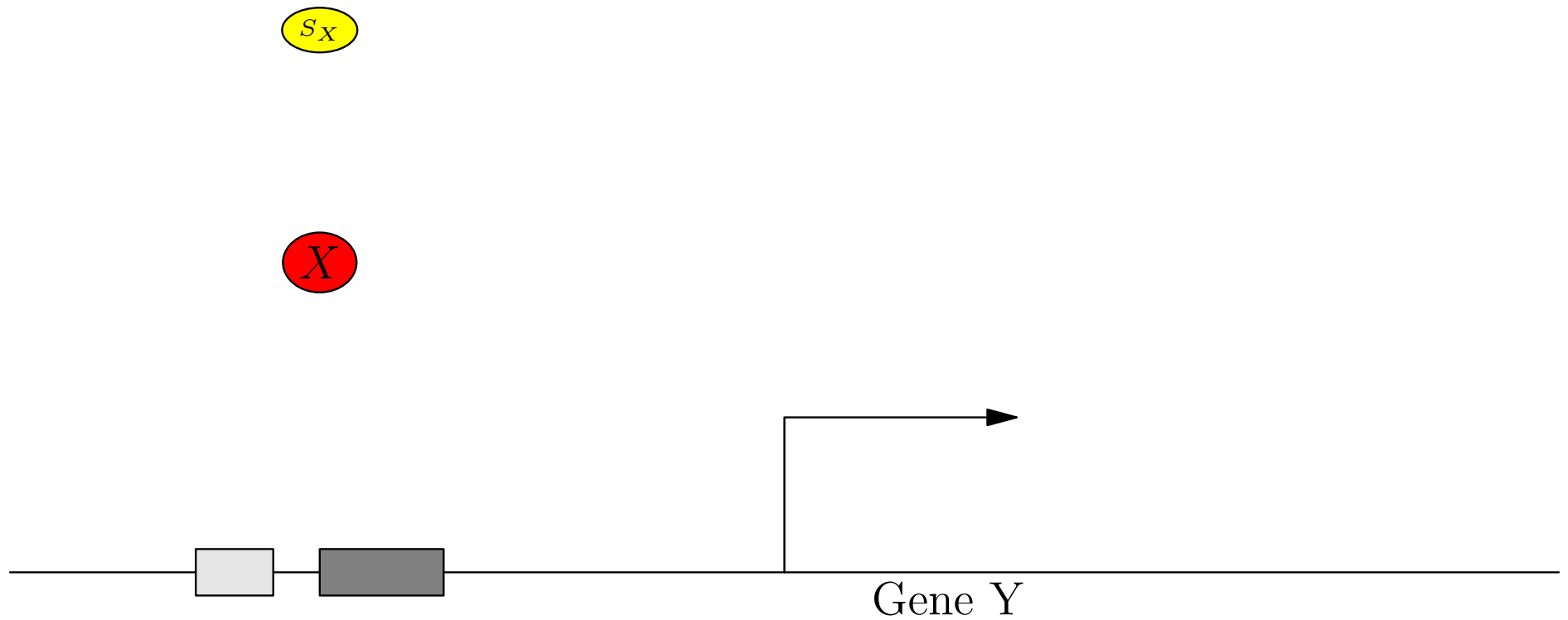
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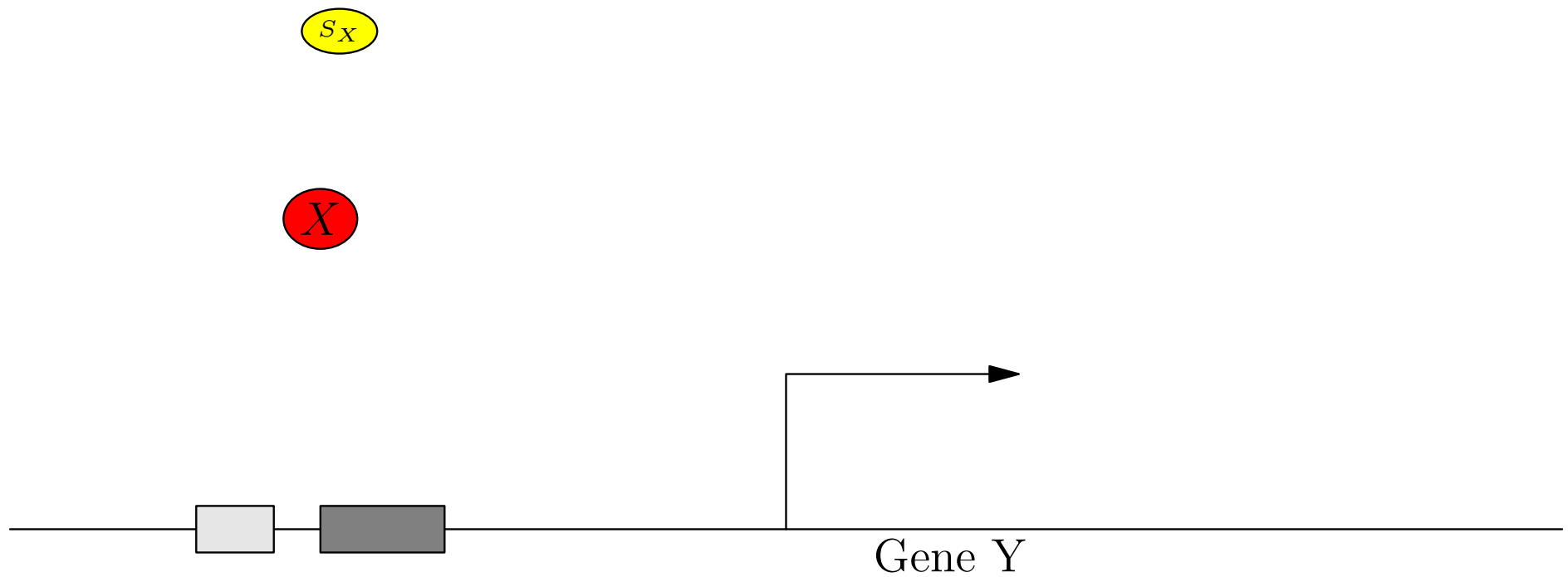
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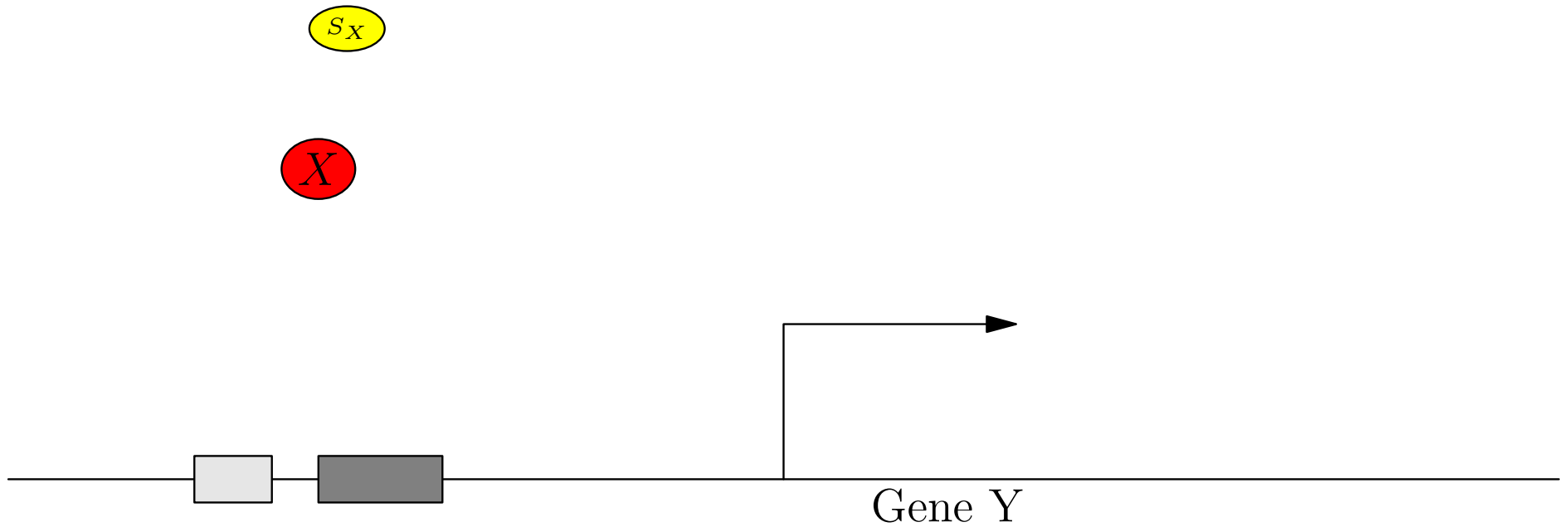
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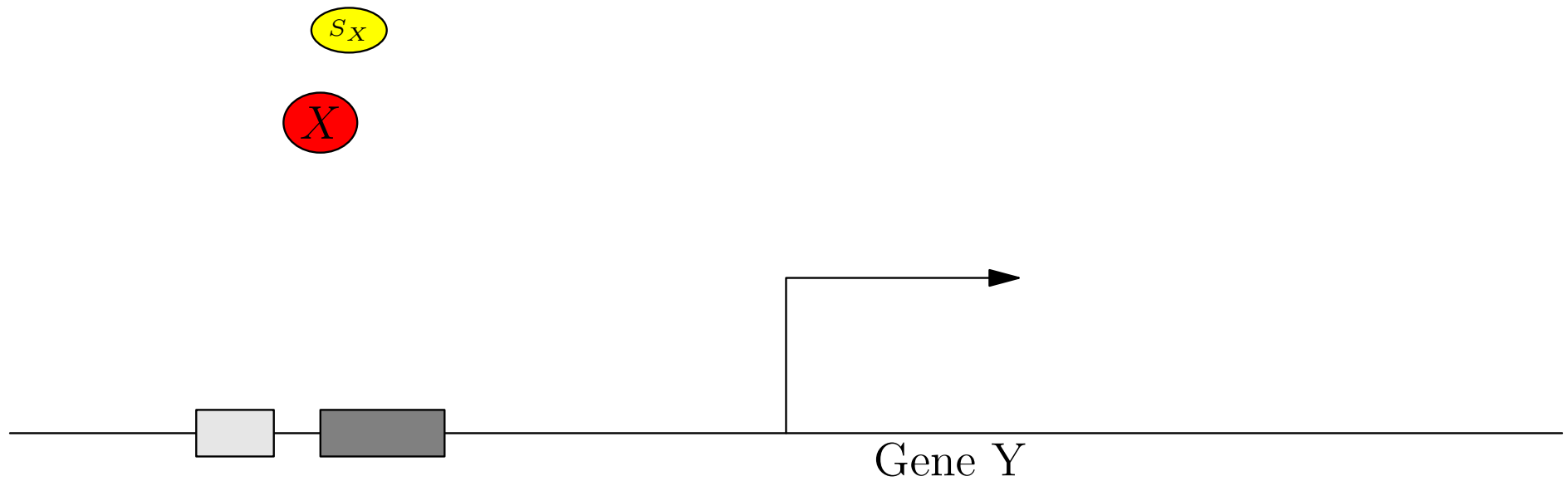
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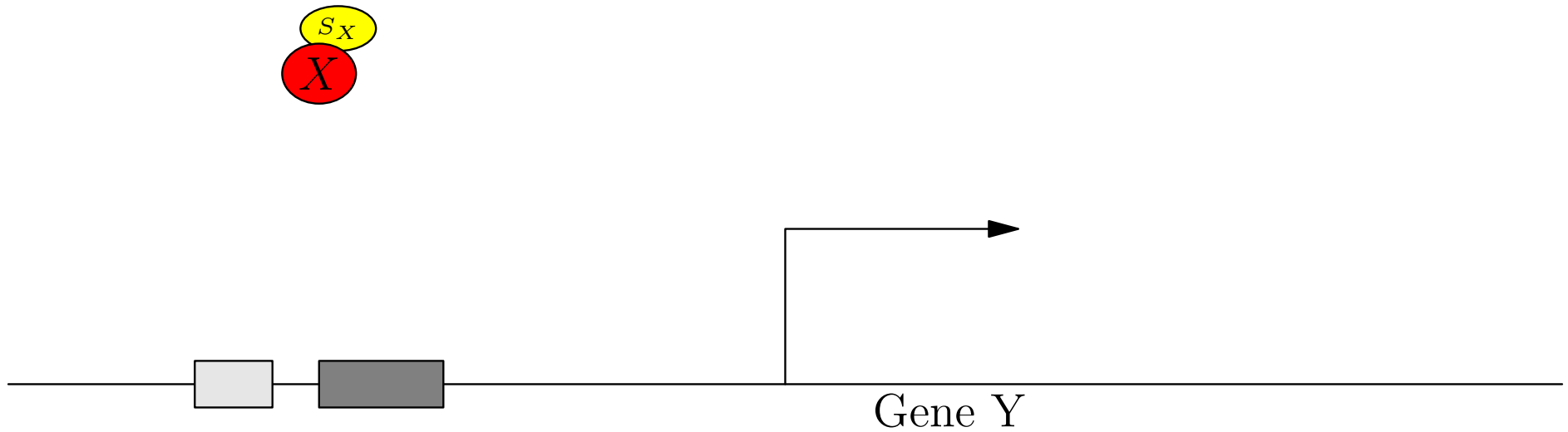
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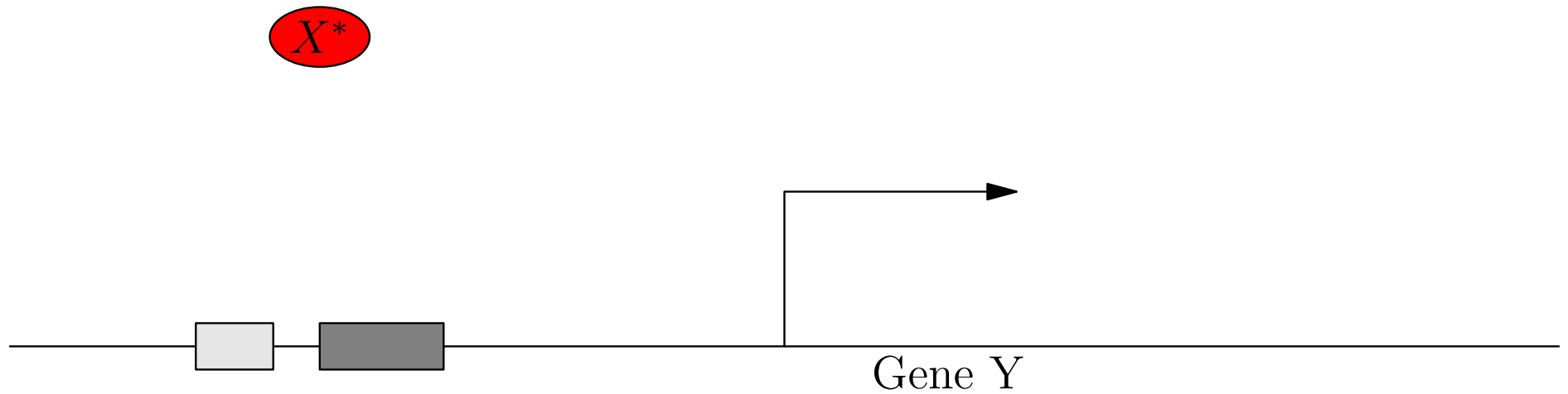
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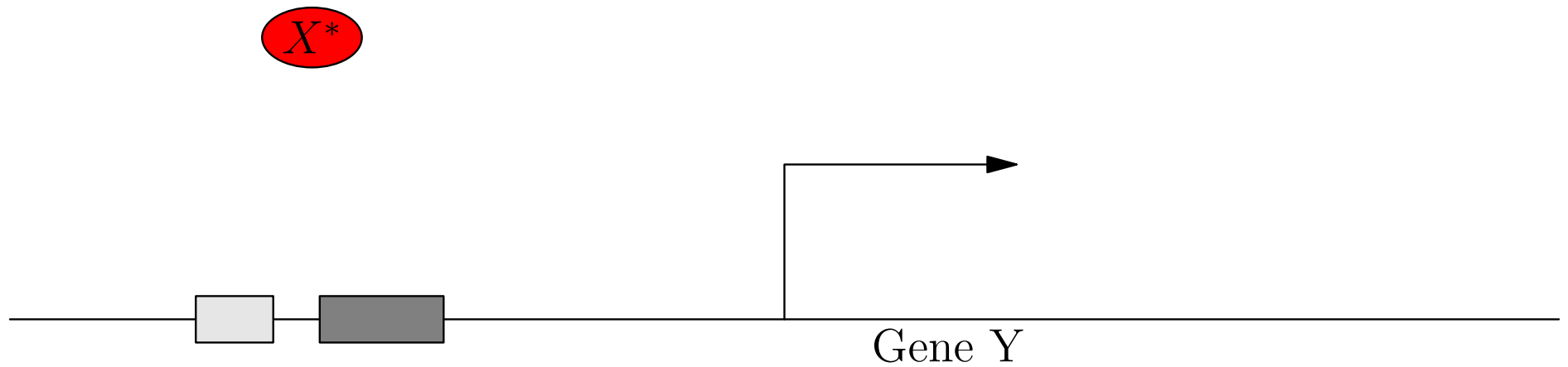
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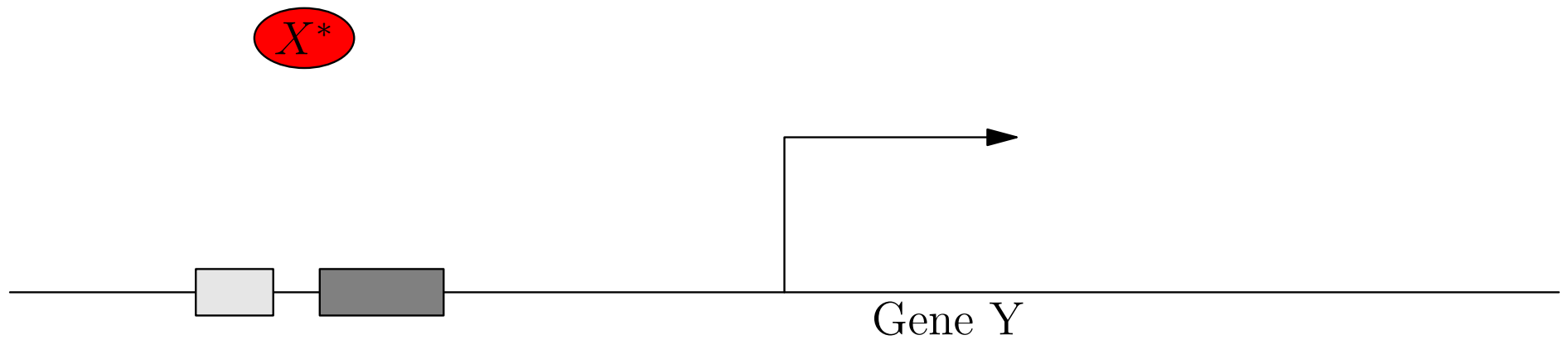
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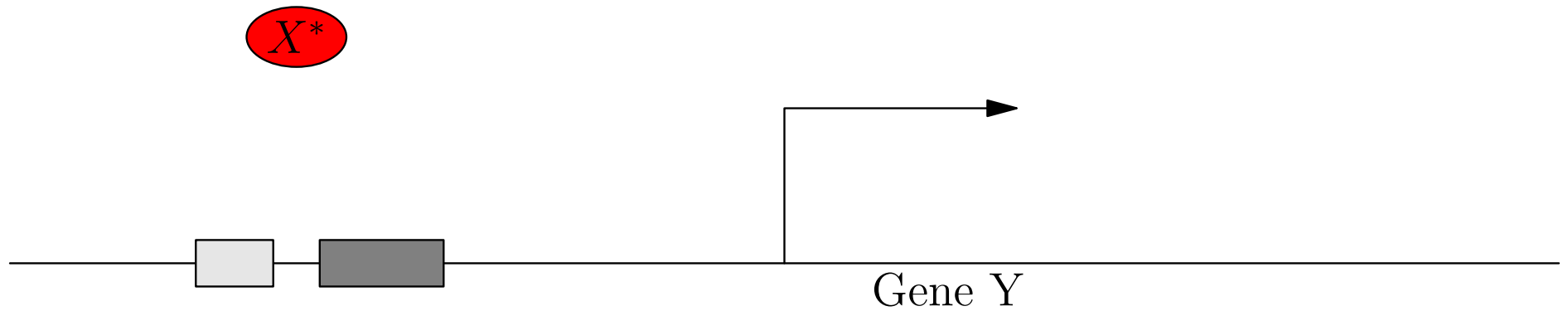
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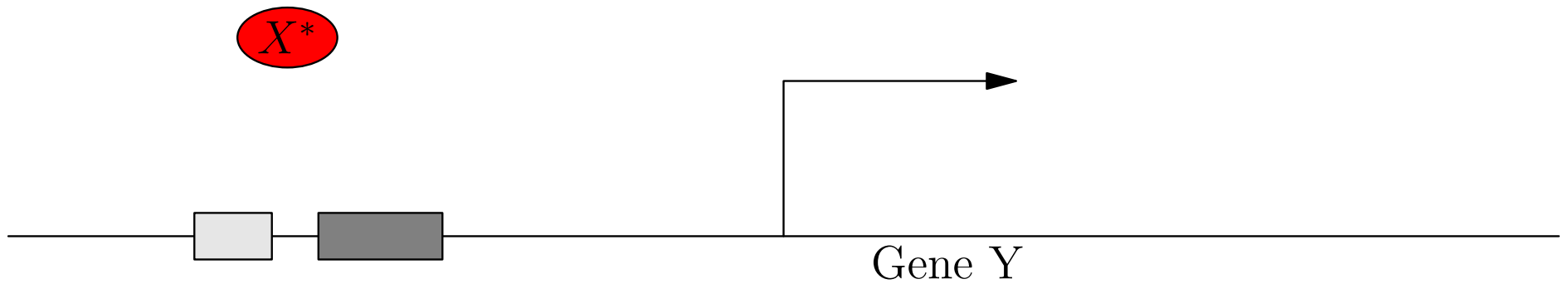
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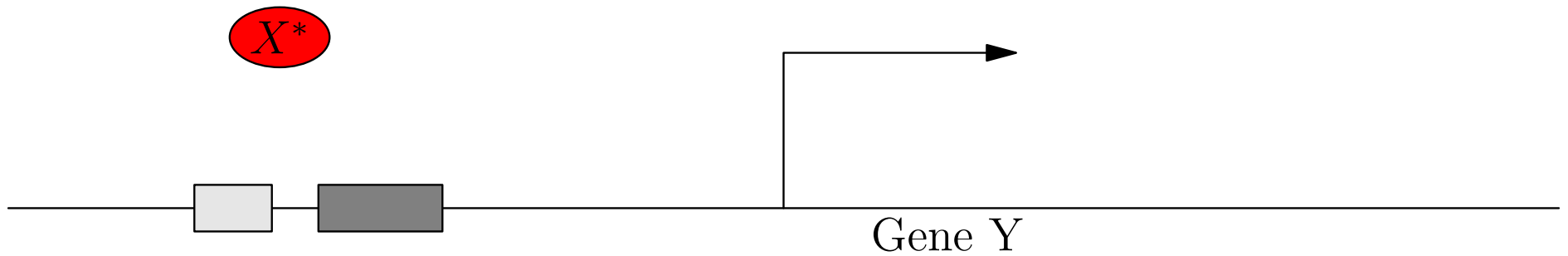
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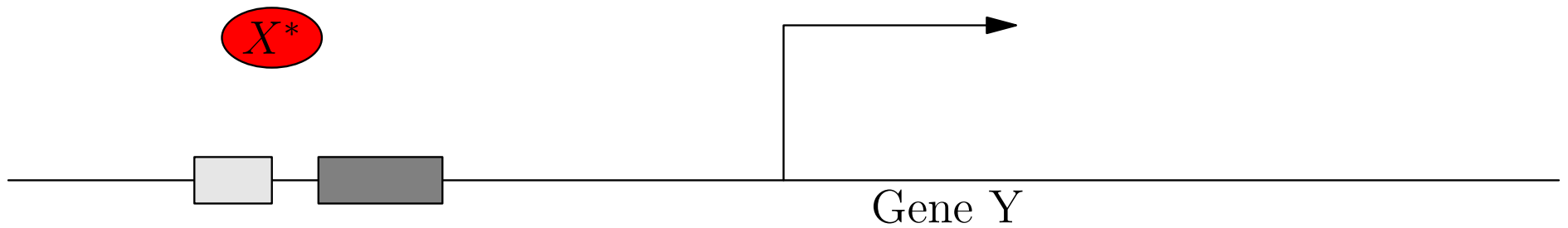
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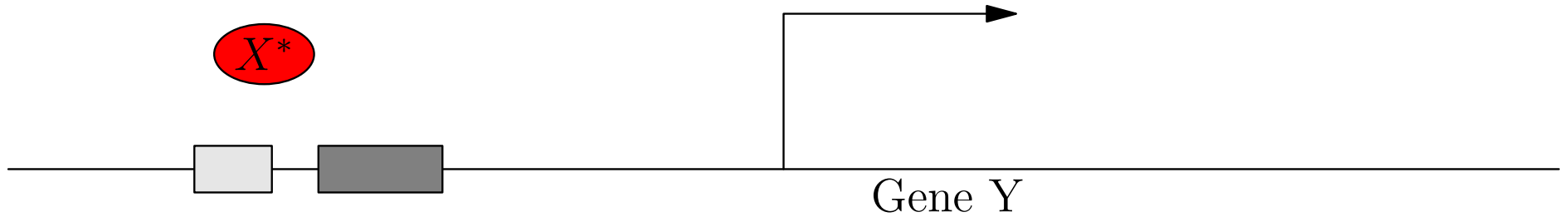
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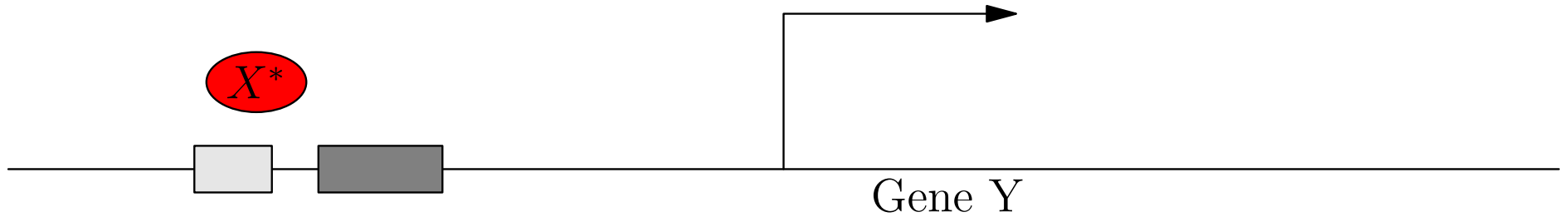
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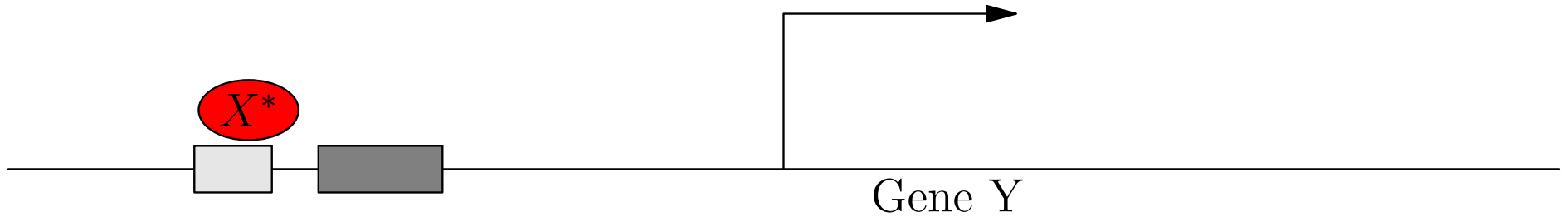
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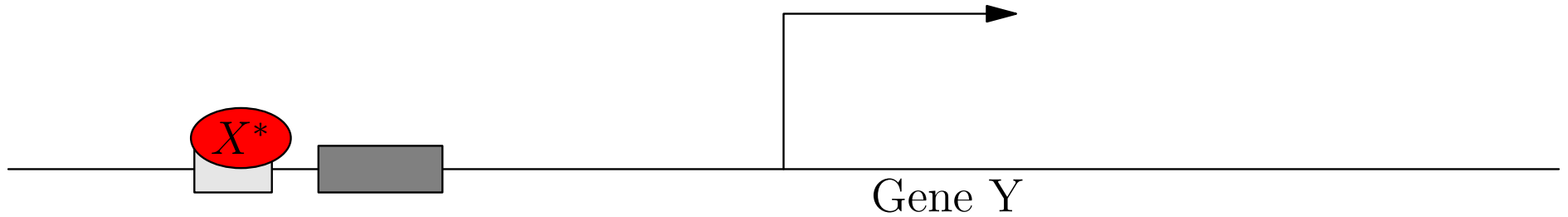
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Gene Transcription Regulation

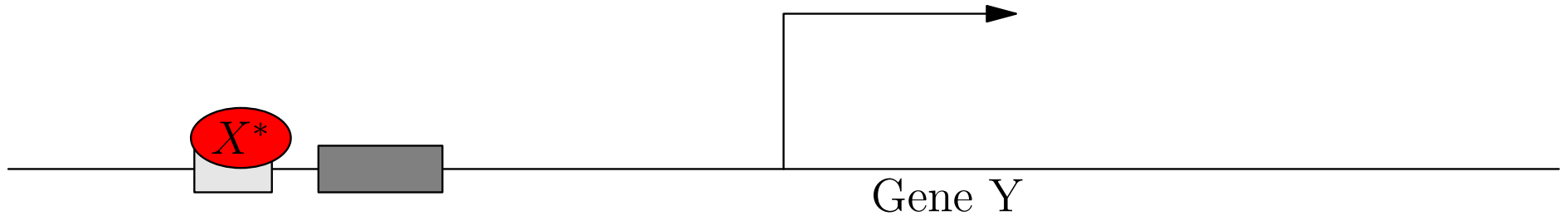
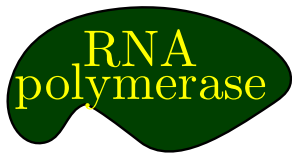


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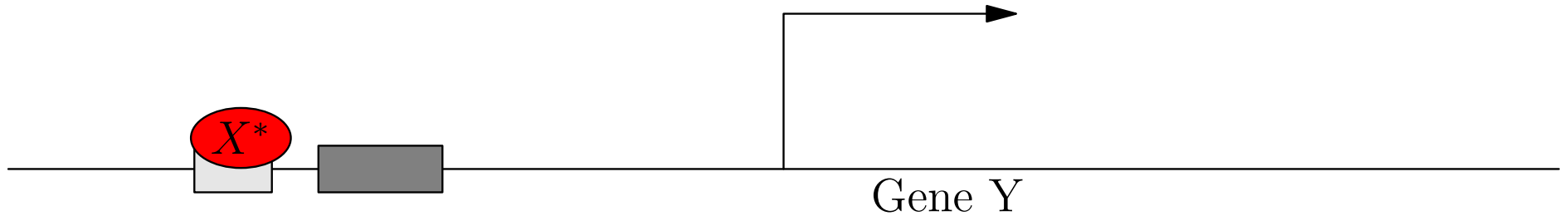
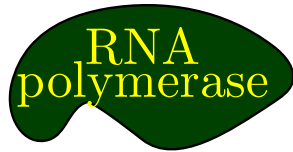
Gene Transcription Regulation

Excititory Transcription Factor



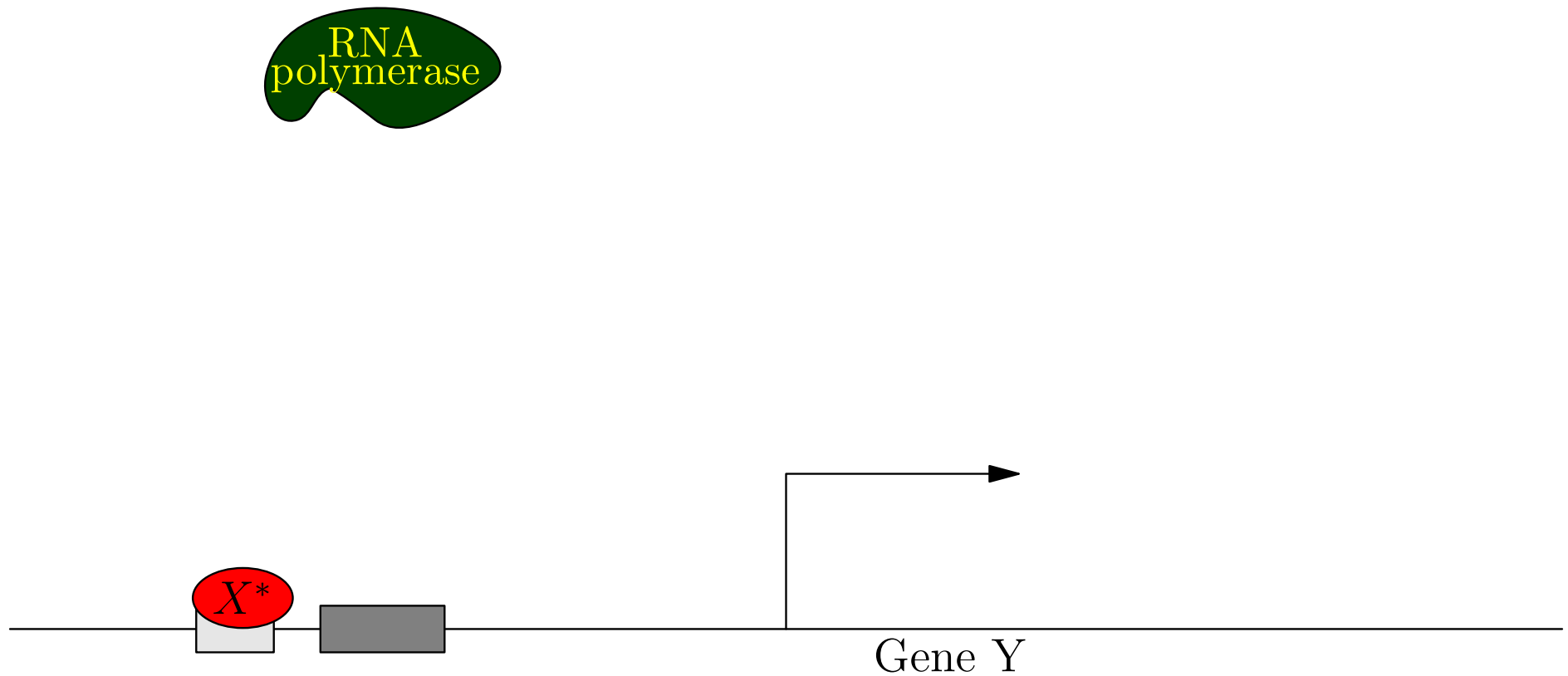
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Excititory Transcription Factor



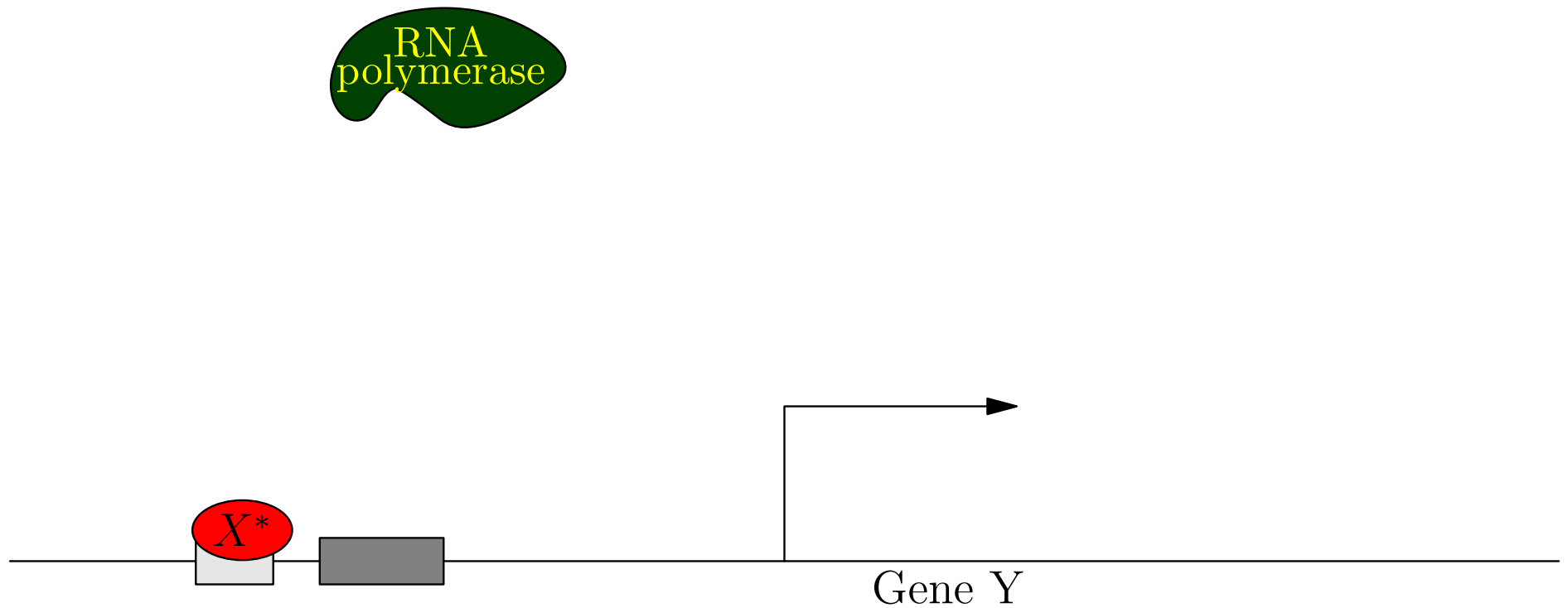
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Excititory Transcription Factor



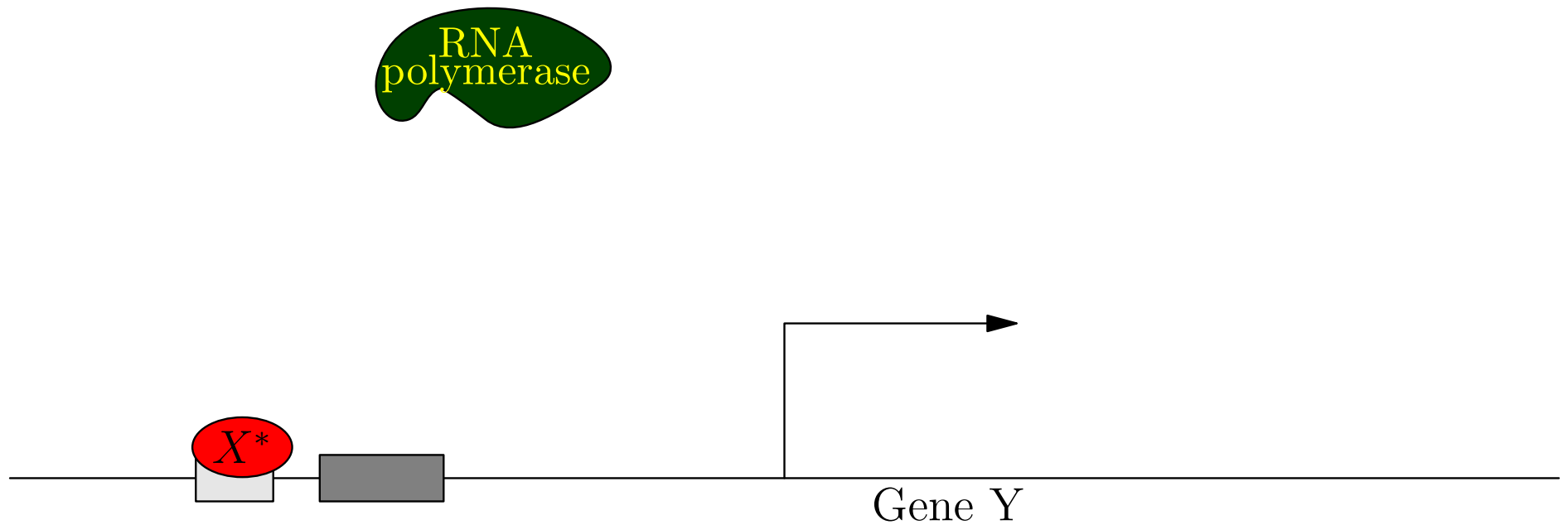
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Excititory Transcription Factor



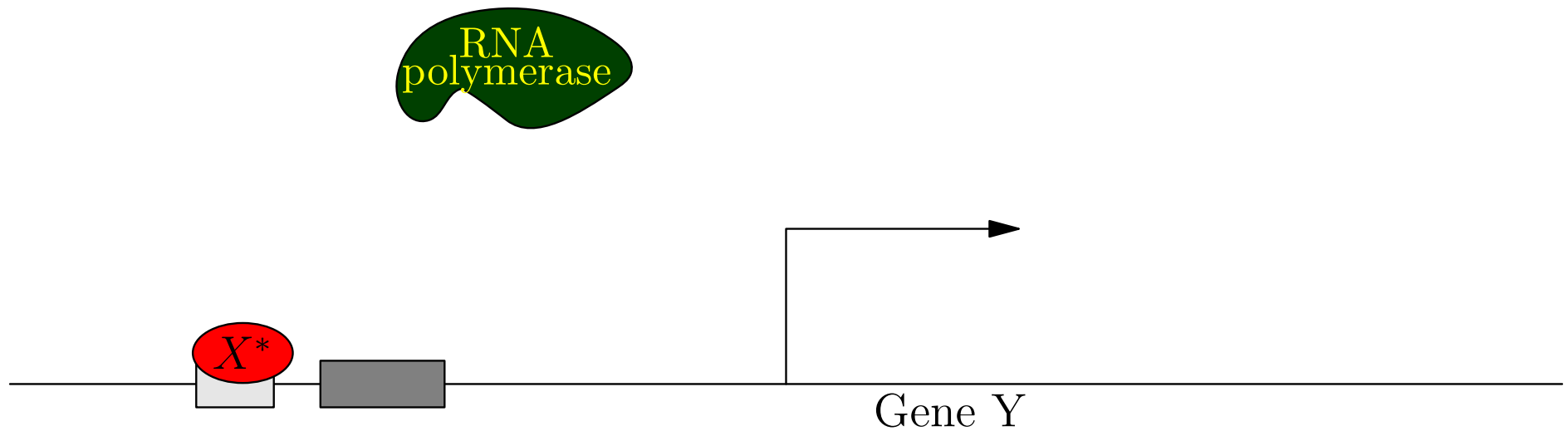
Gene Transcription Regulation

Excititory Transcription Factor



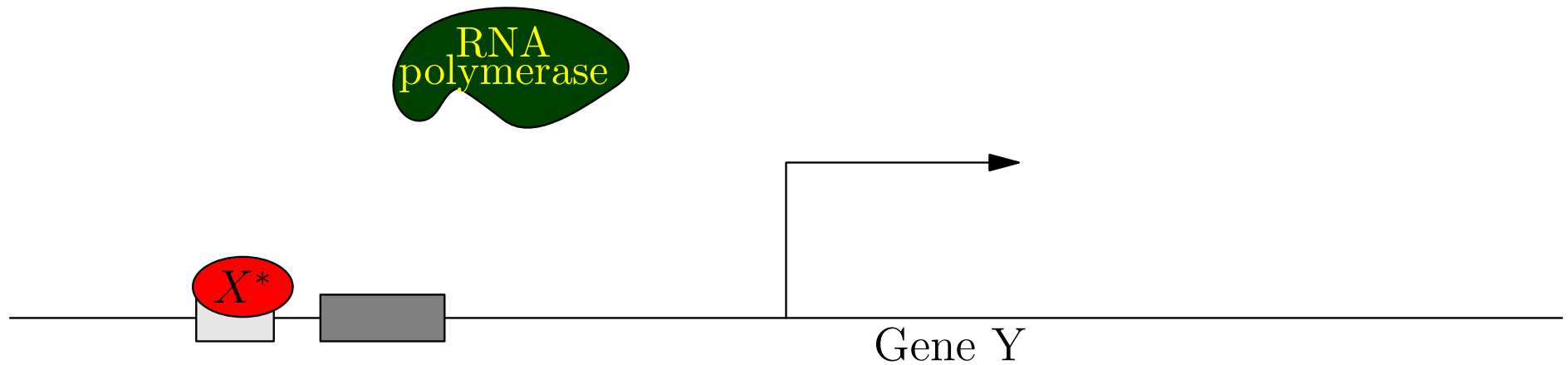
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Excititory Transcription Factor



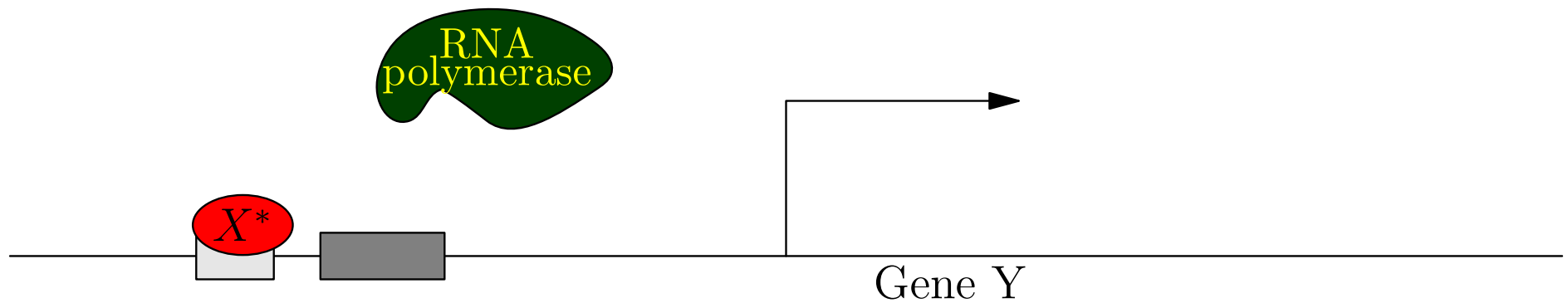
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Excititory Transcription Factor



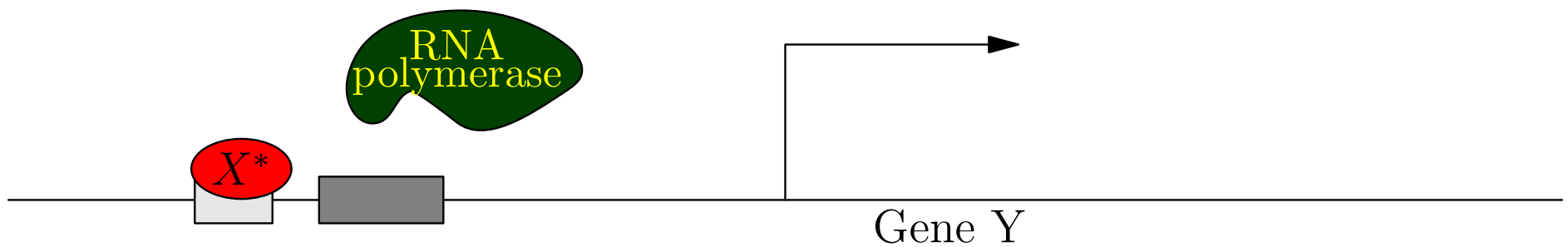
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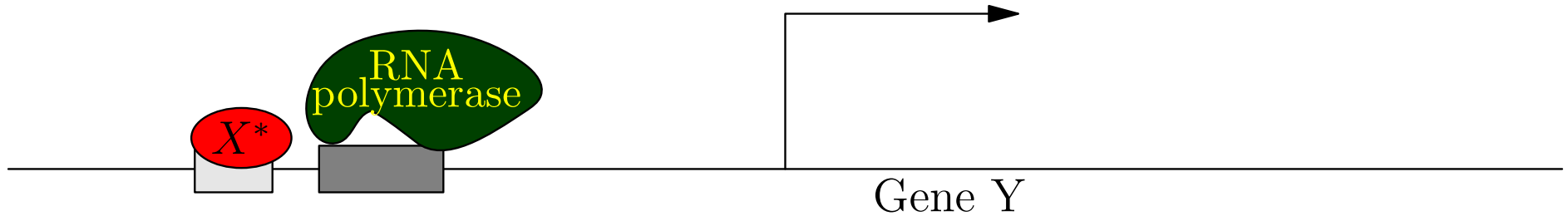
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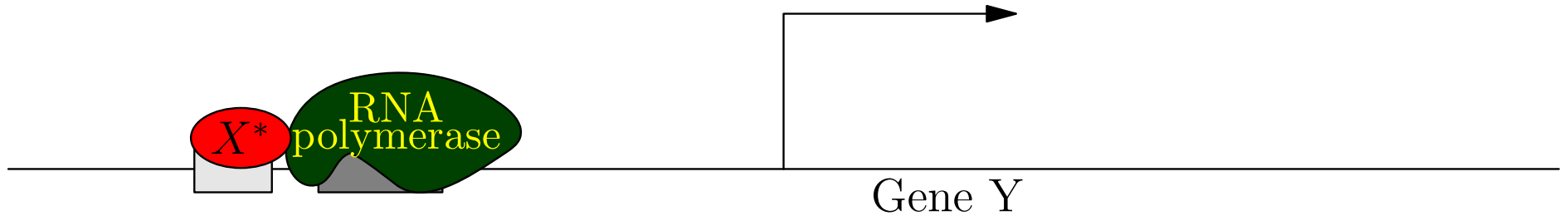
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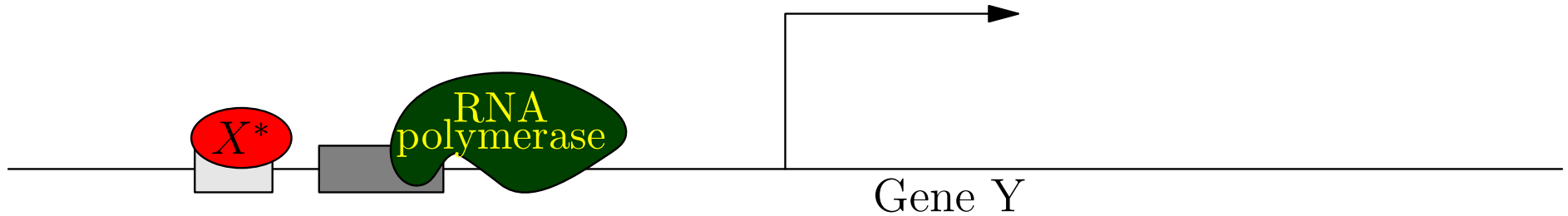
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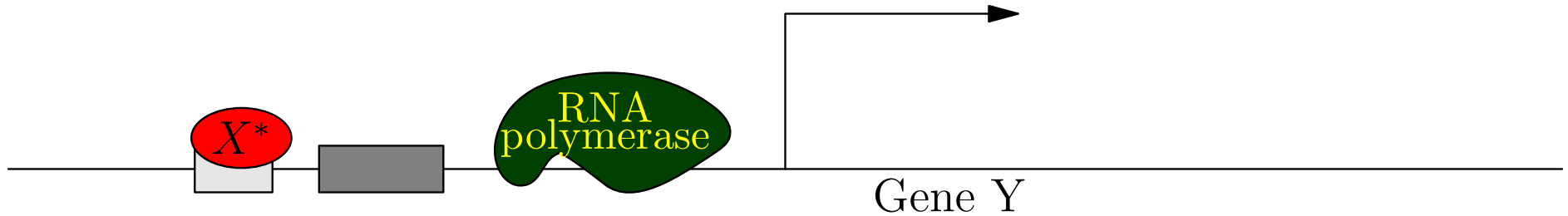
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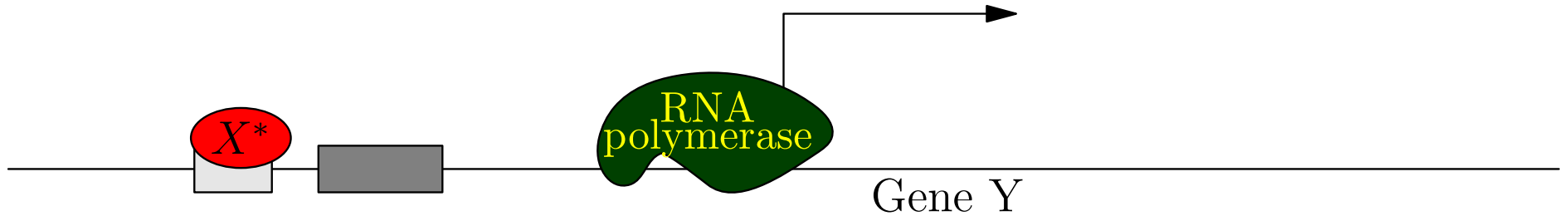
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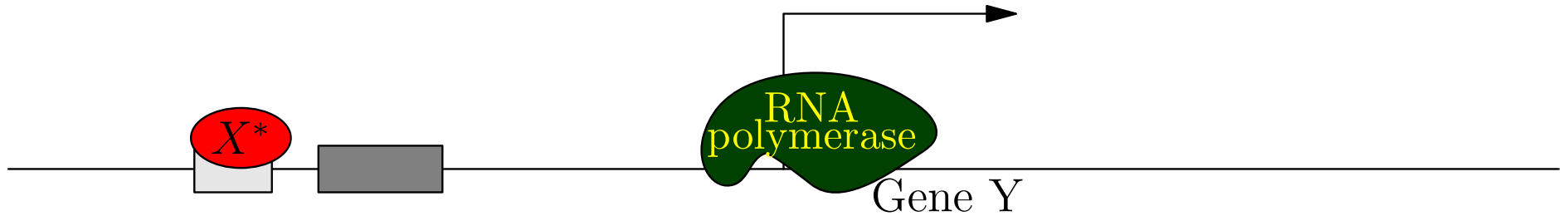
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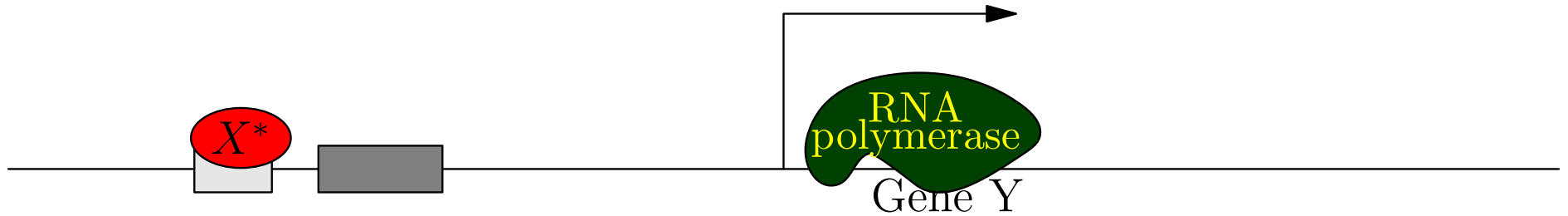
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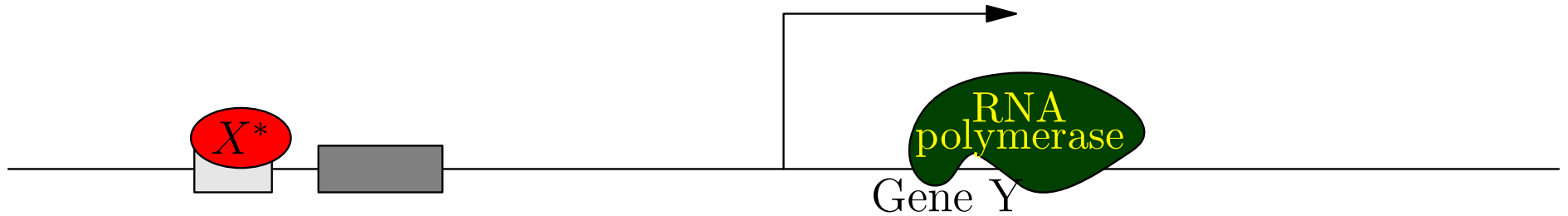
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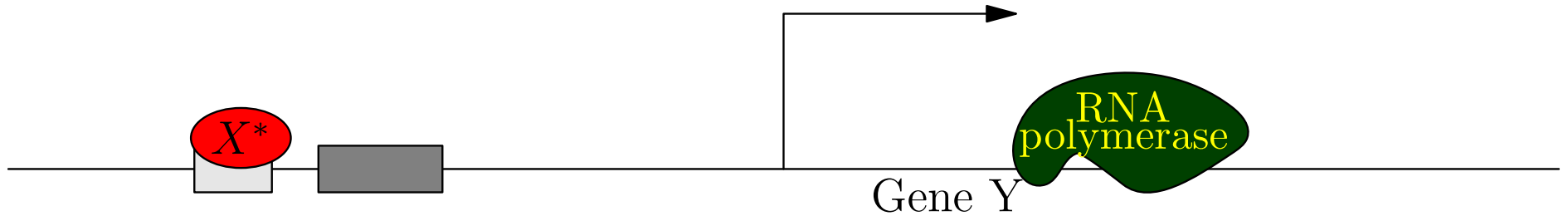
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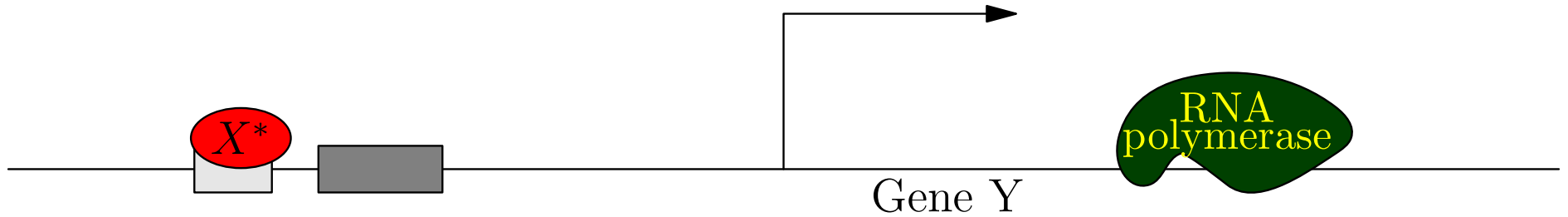
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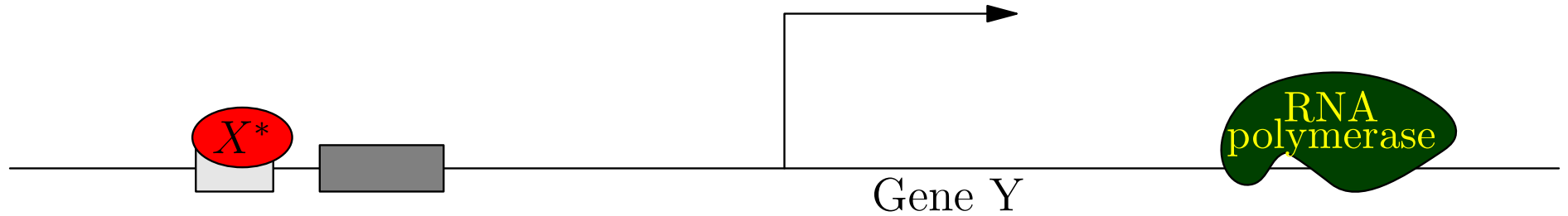
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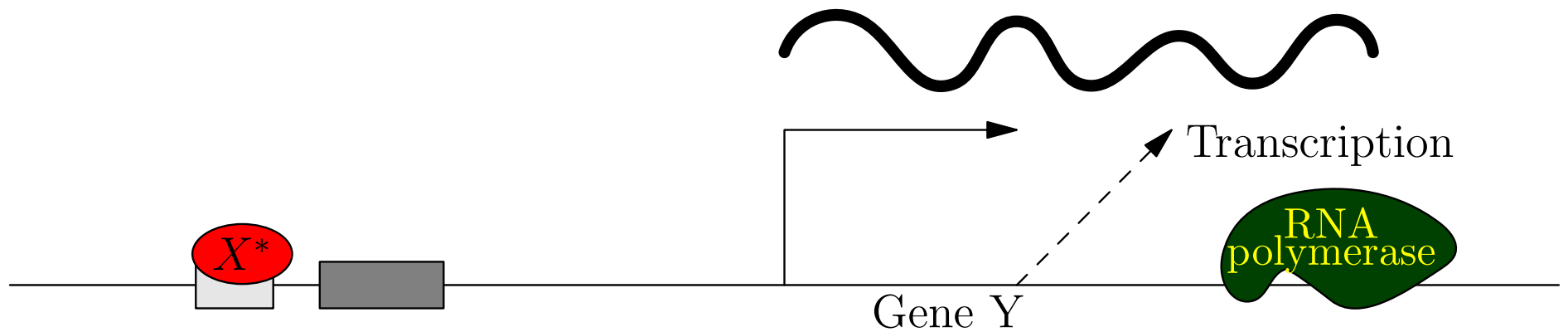
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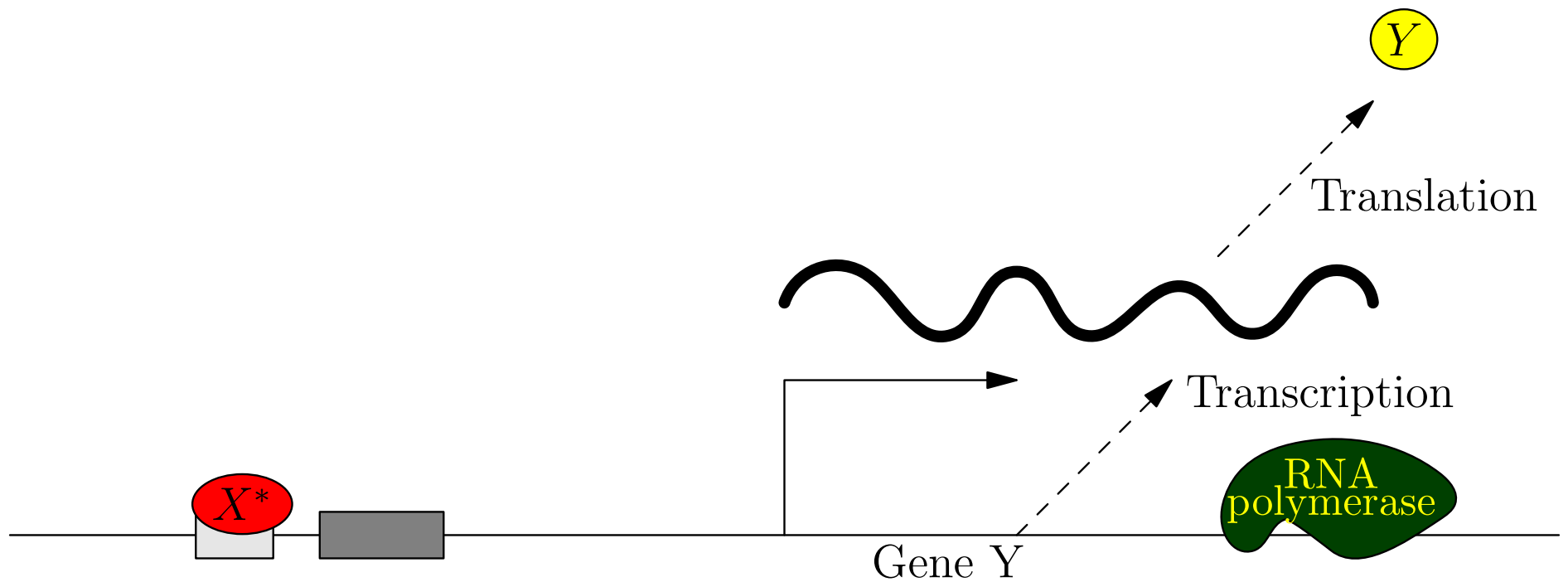
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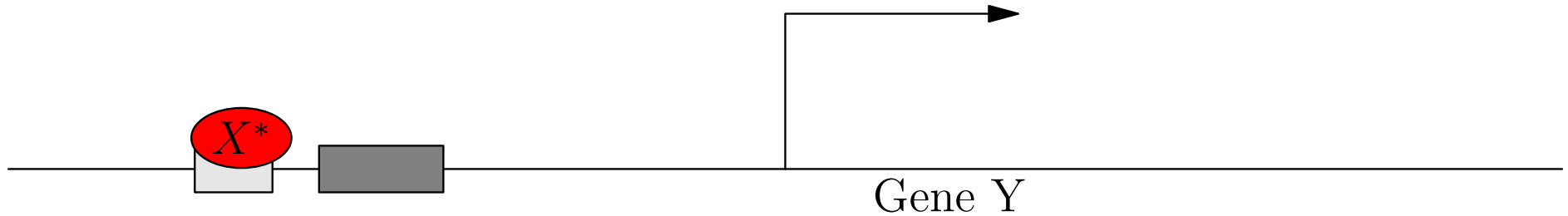
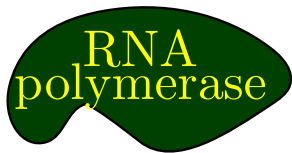
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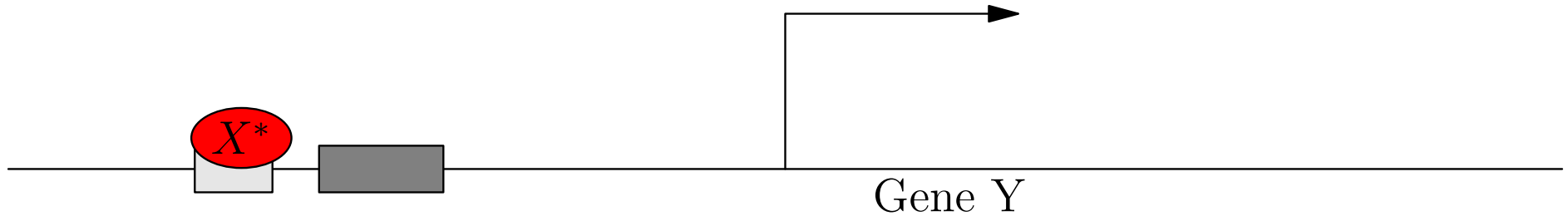
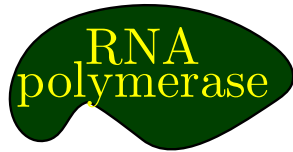
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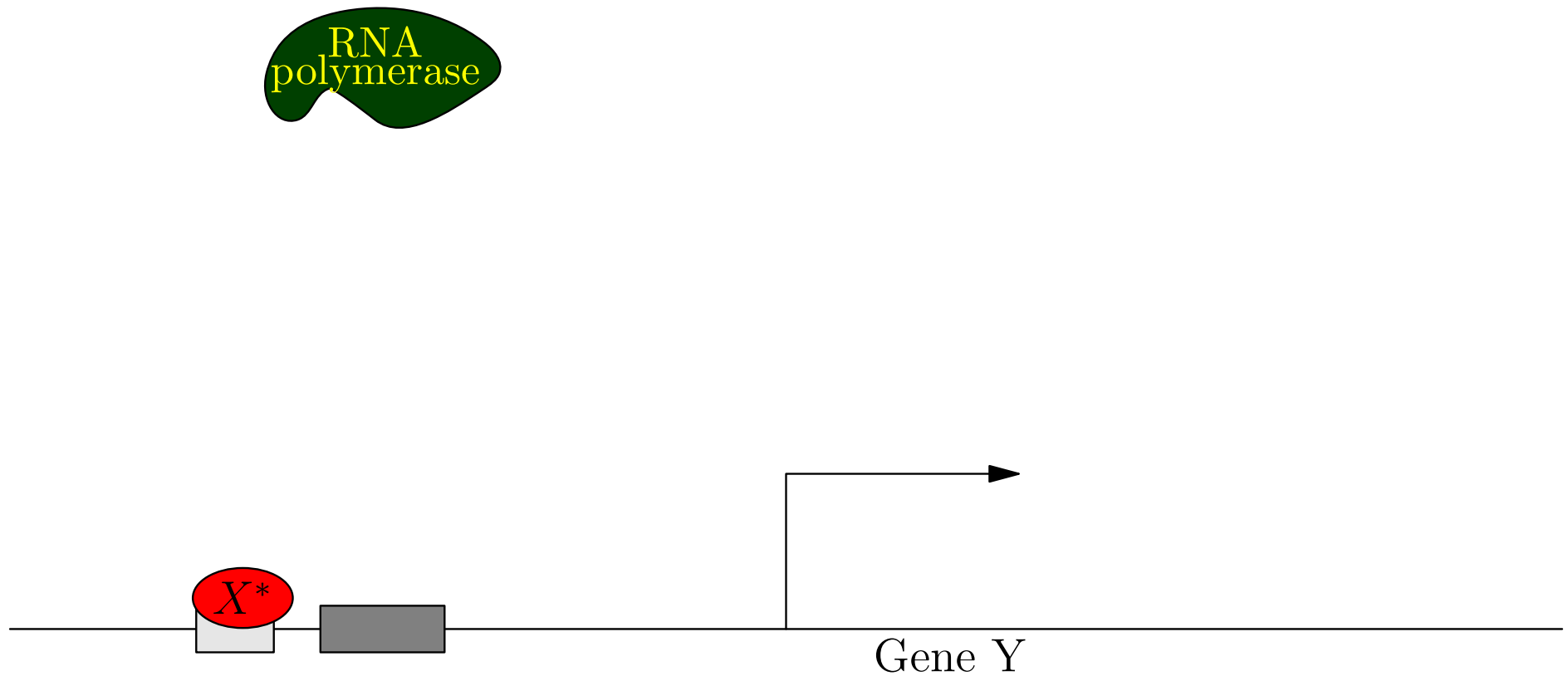
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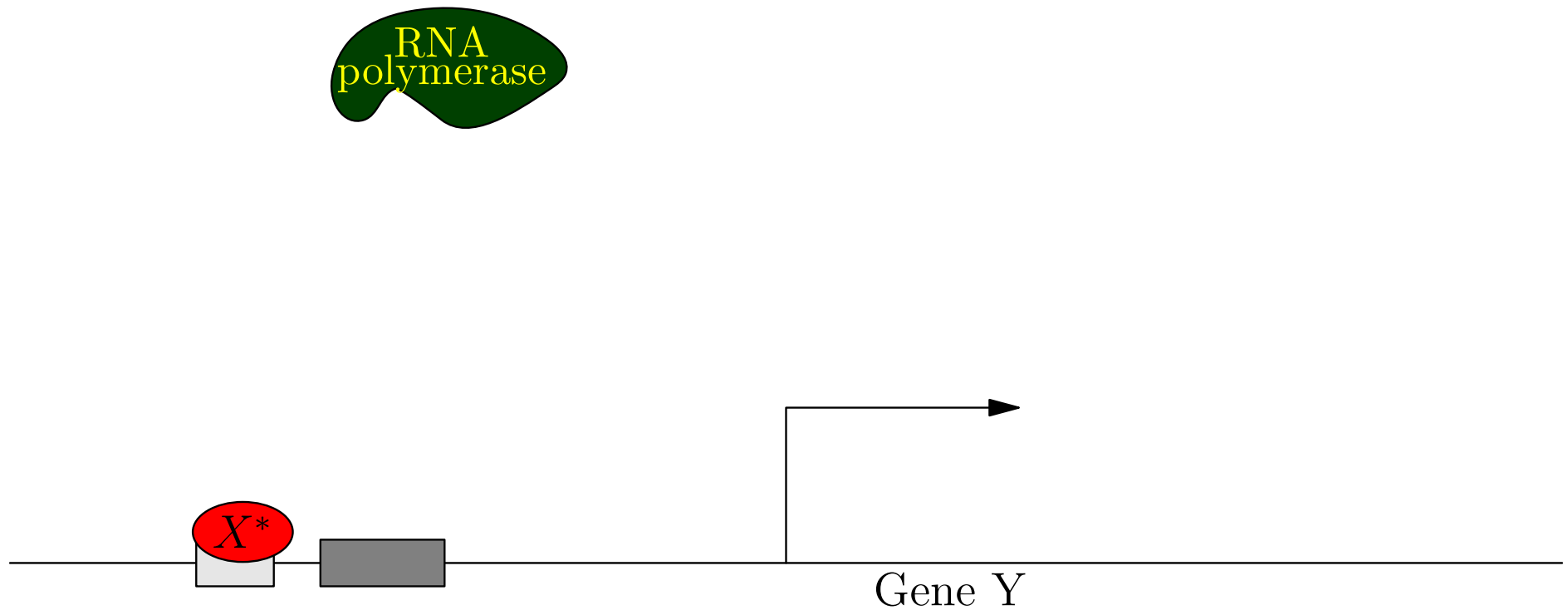
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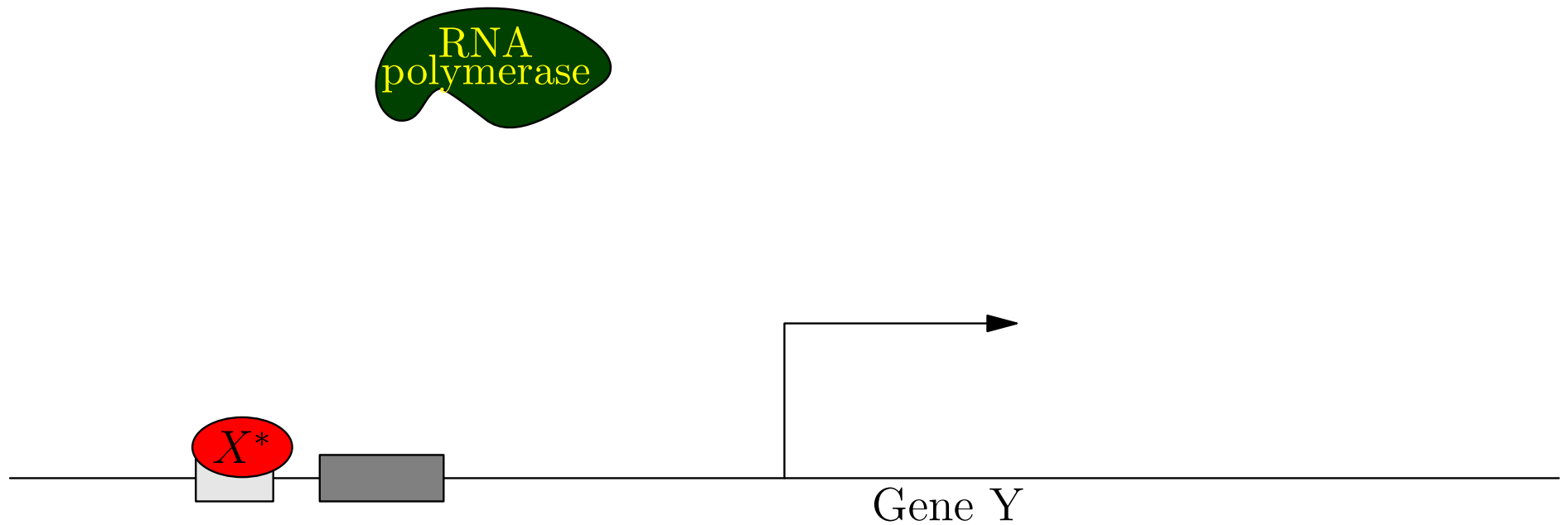
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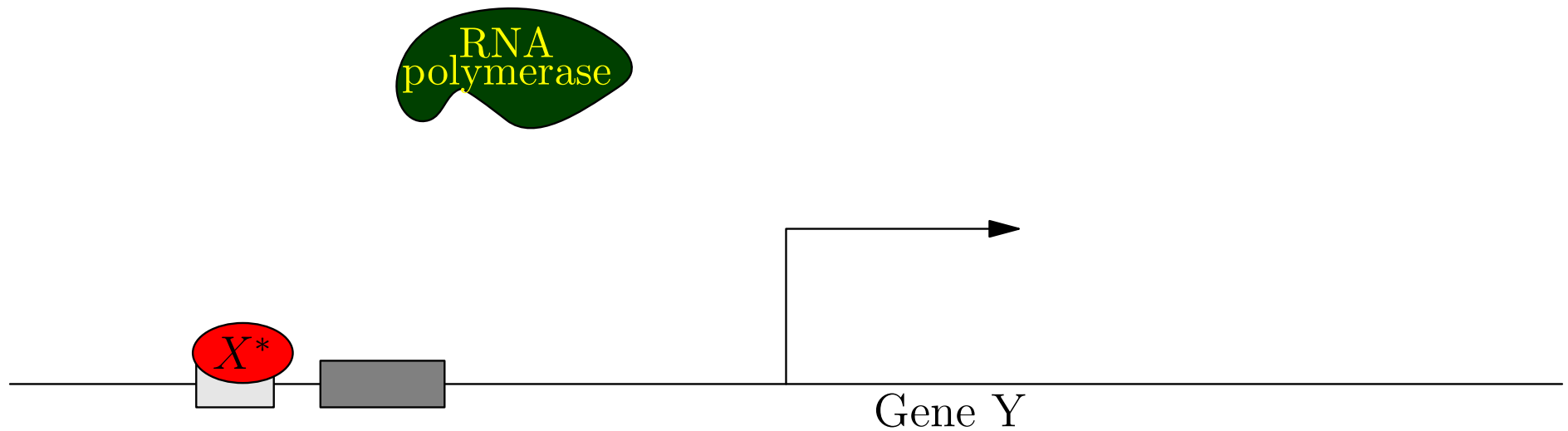
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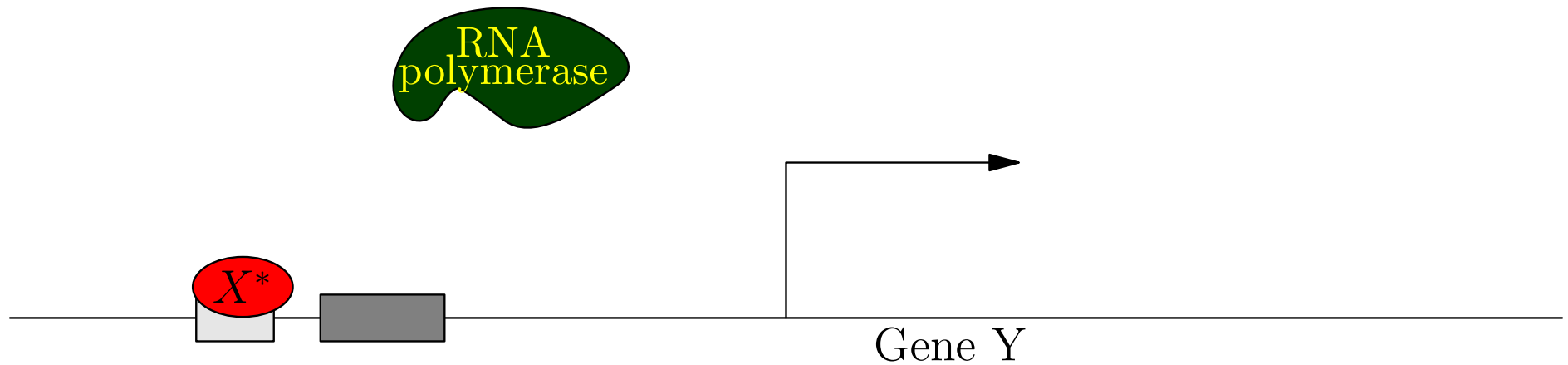
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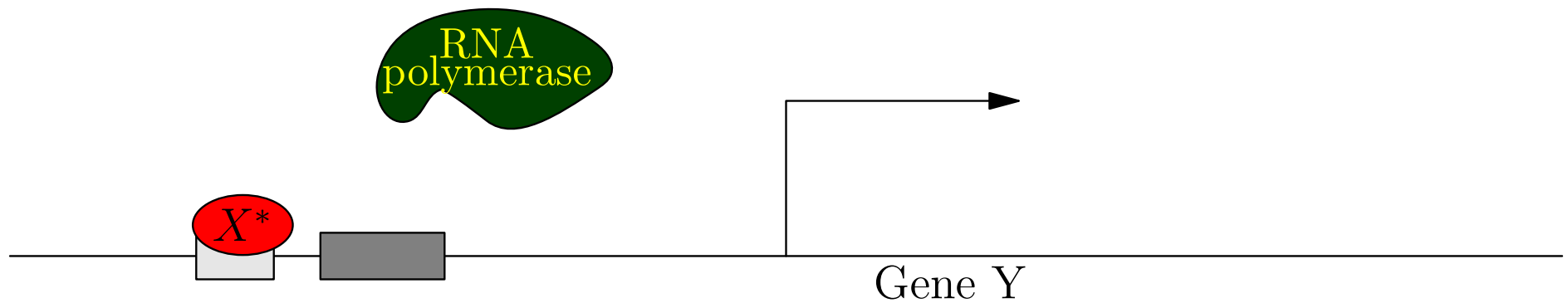
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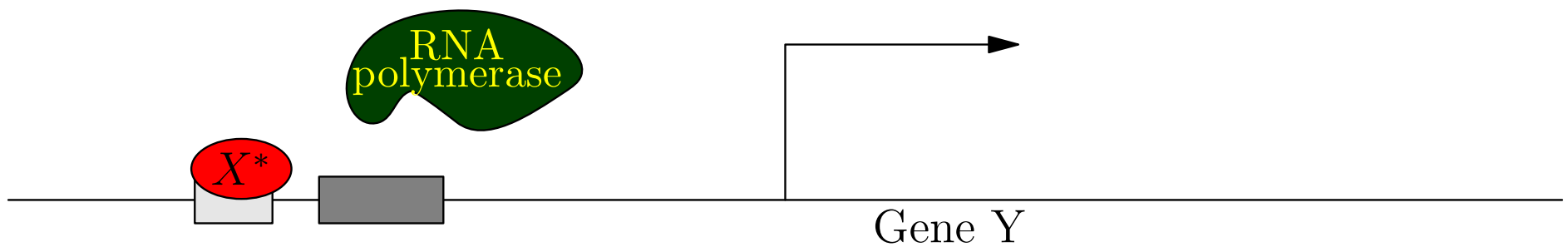
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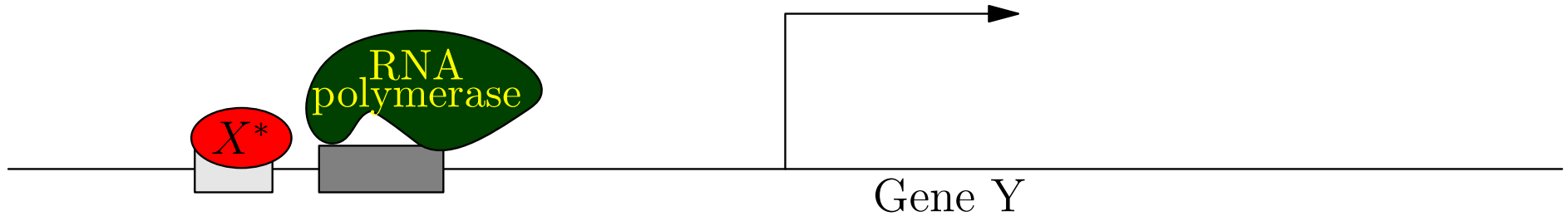
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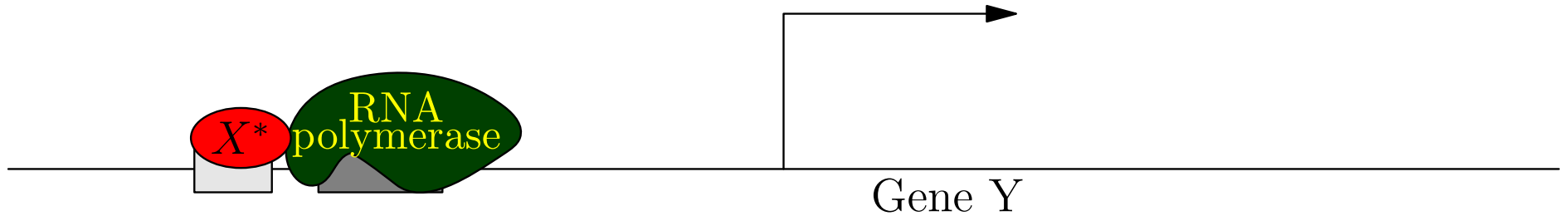
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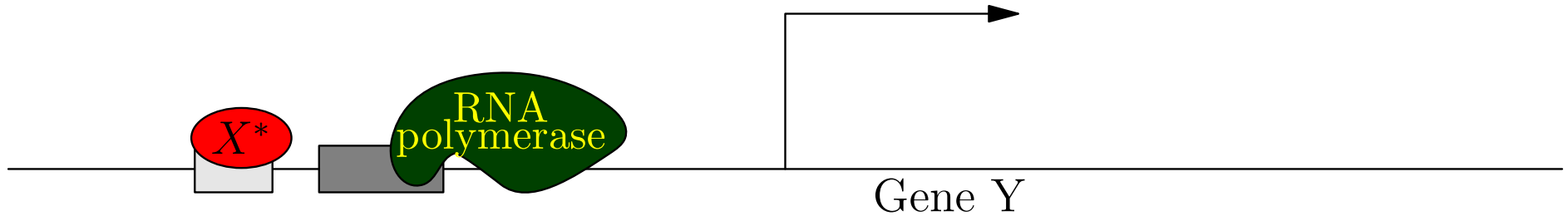
Gene Transcription Regulation

Excititory Transcription Factor



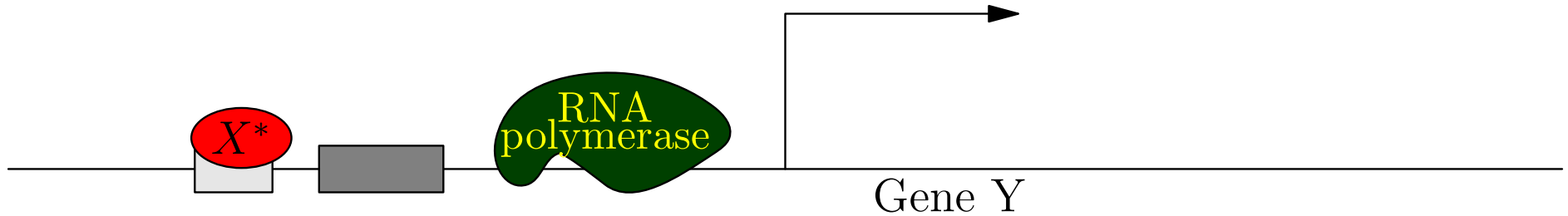
Gene Transcription Regulation

Excititory Transcription Factor



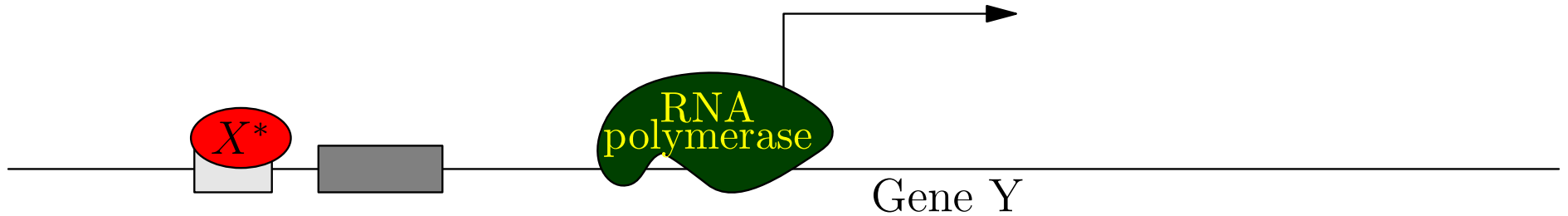
Gene Transcription Regulation

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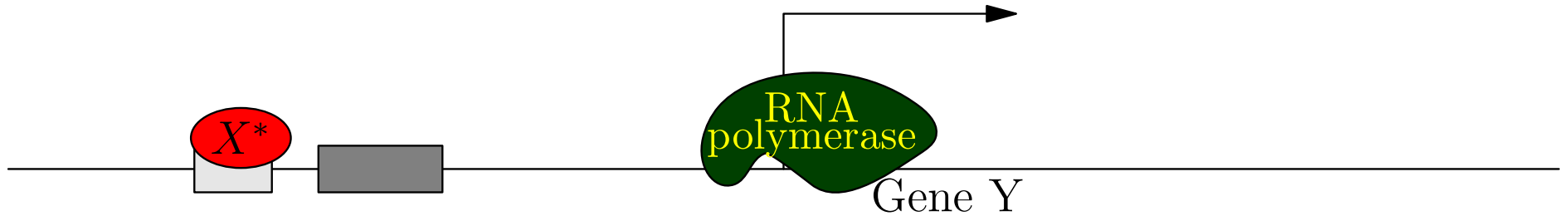
Gene Transcription Regulation

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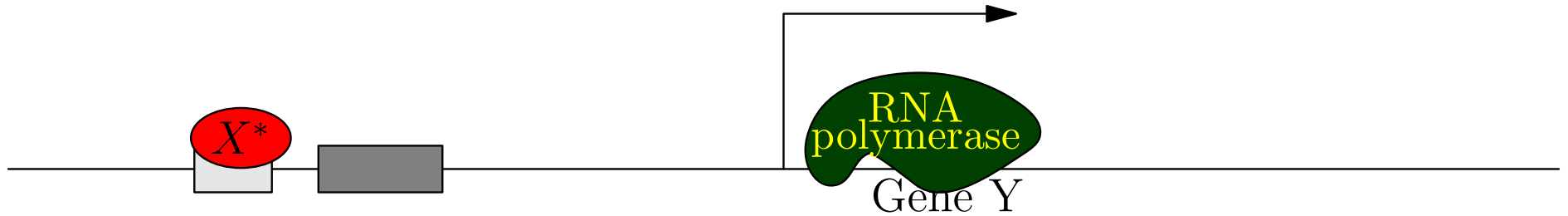
Gene Transcription Regulation

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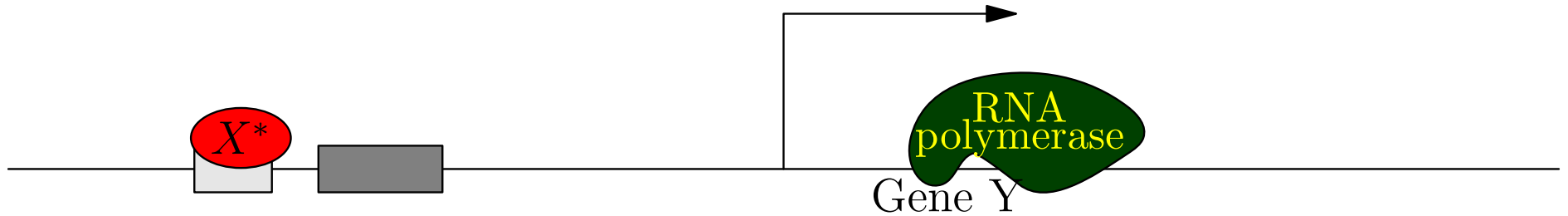
Gene Transcription Regulation

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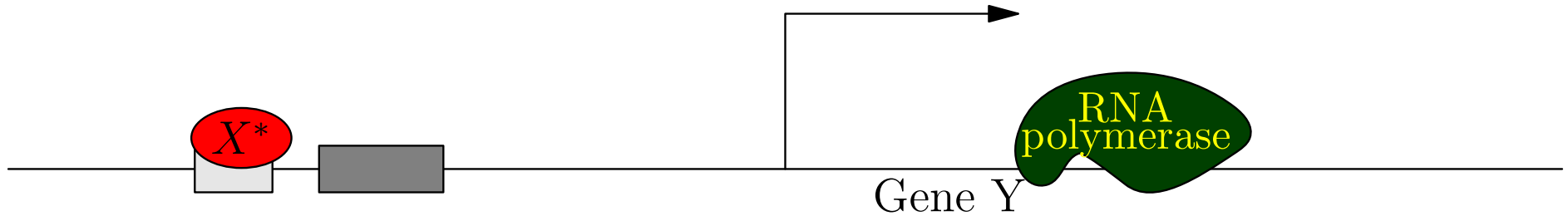
Gene Transcription Regulation

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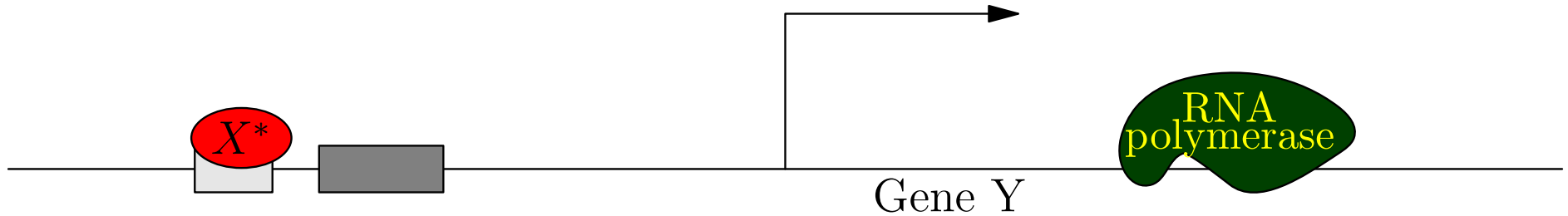
Gene Transcription Regulation

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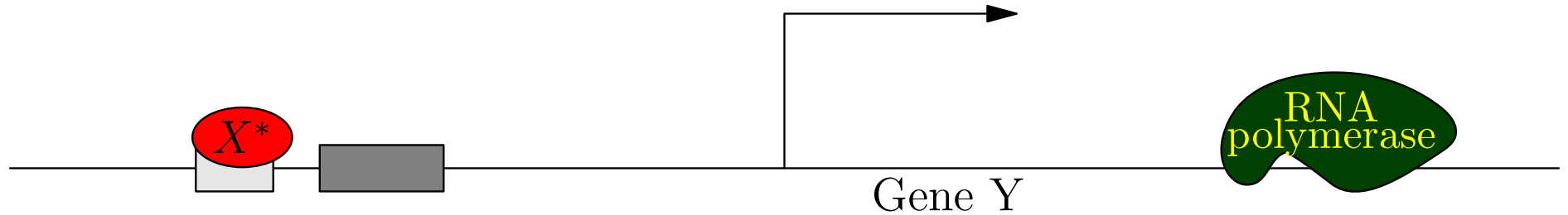
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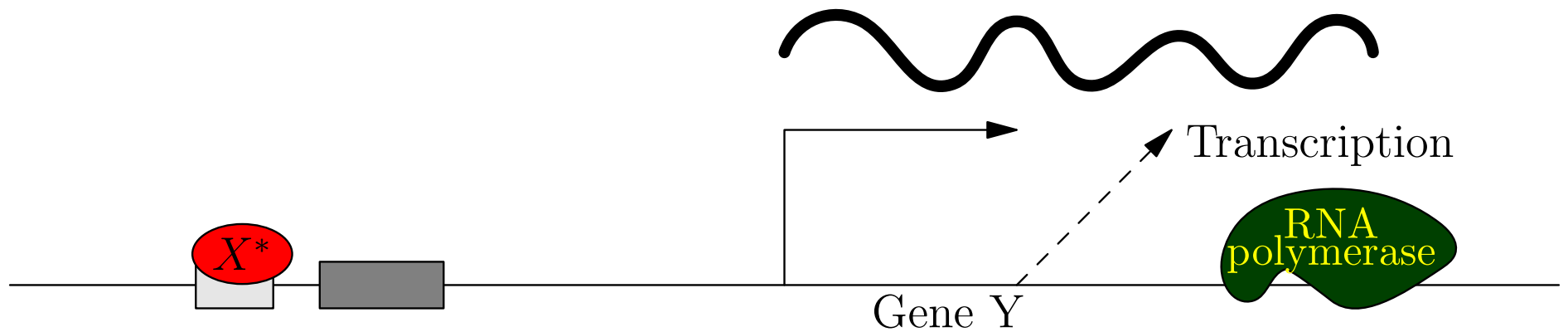
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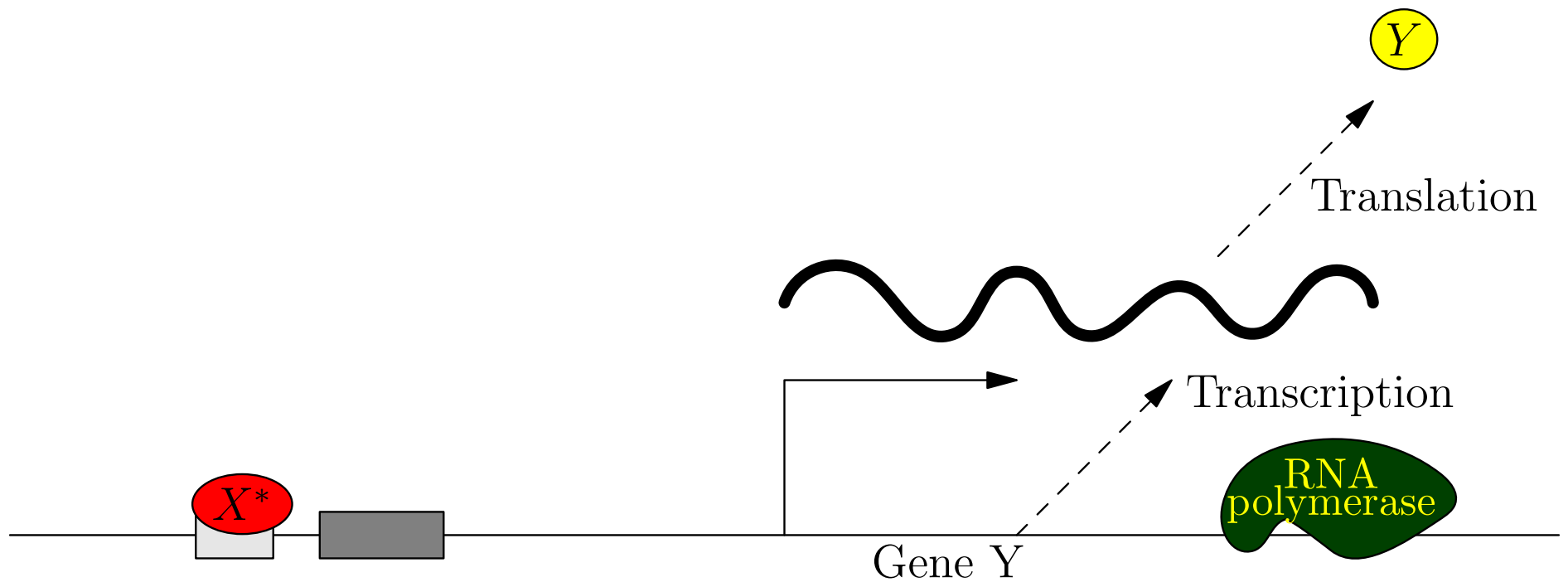
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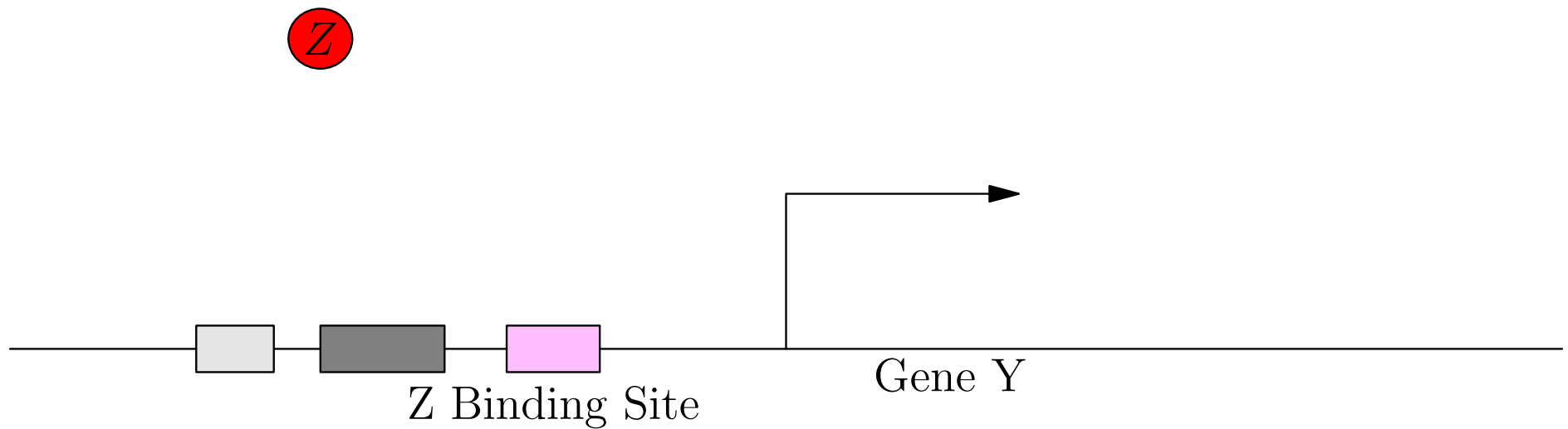


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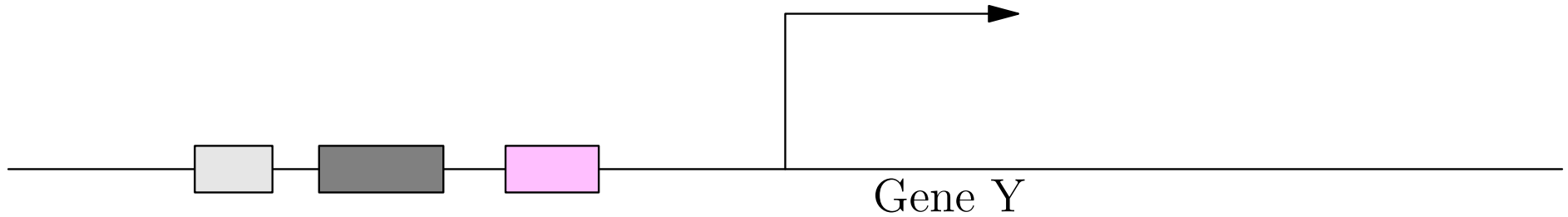
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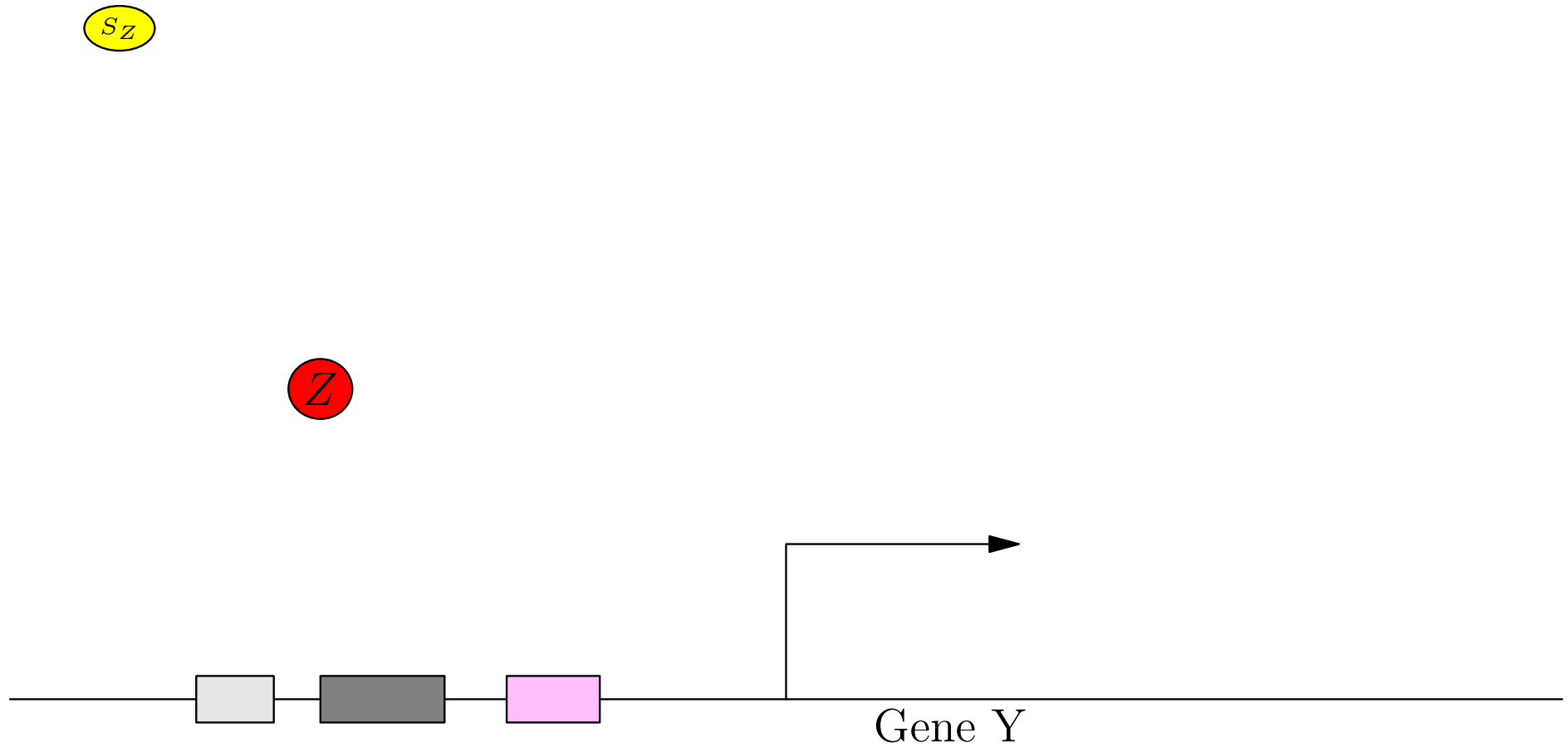
Gene Transcription Regulation

S_Z

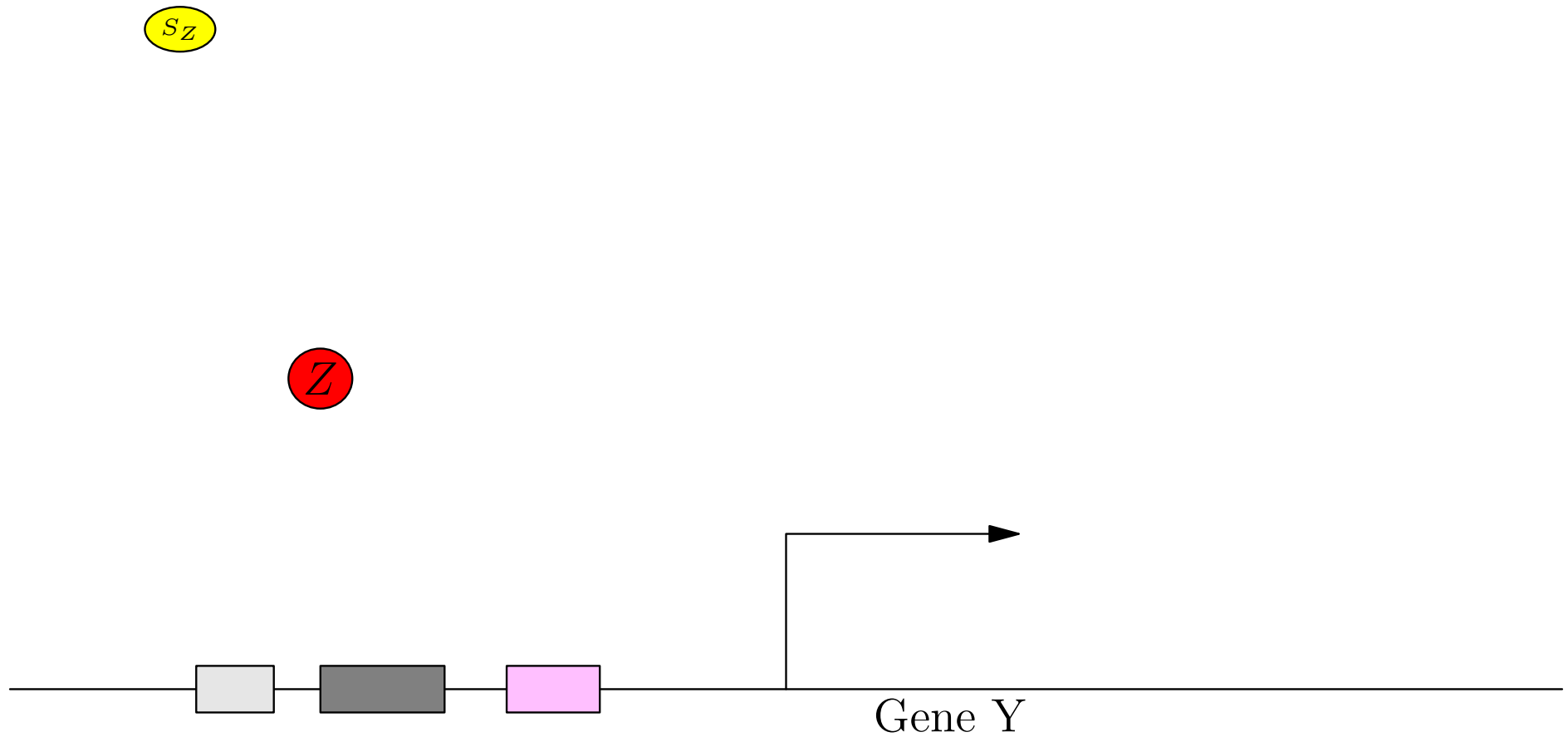
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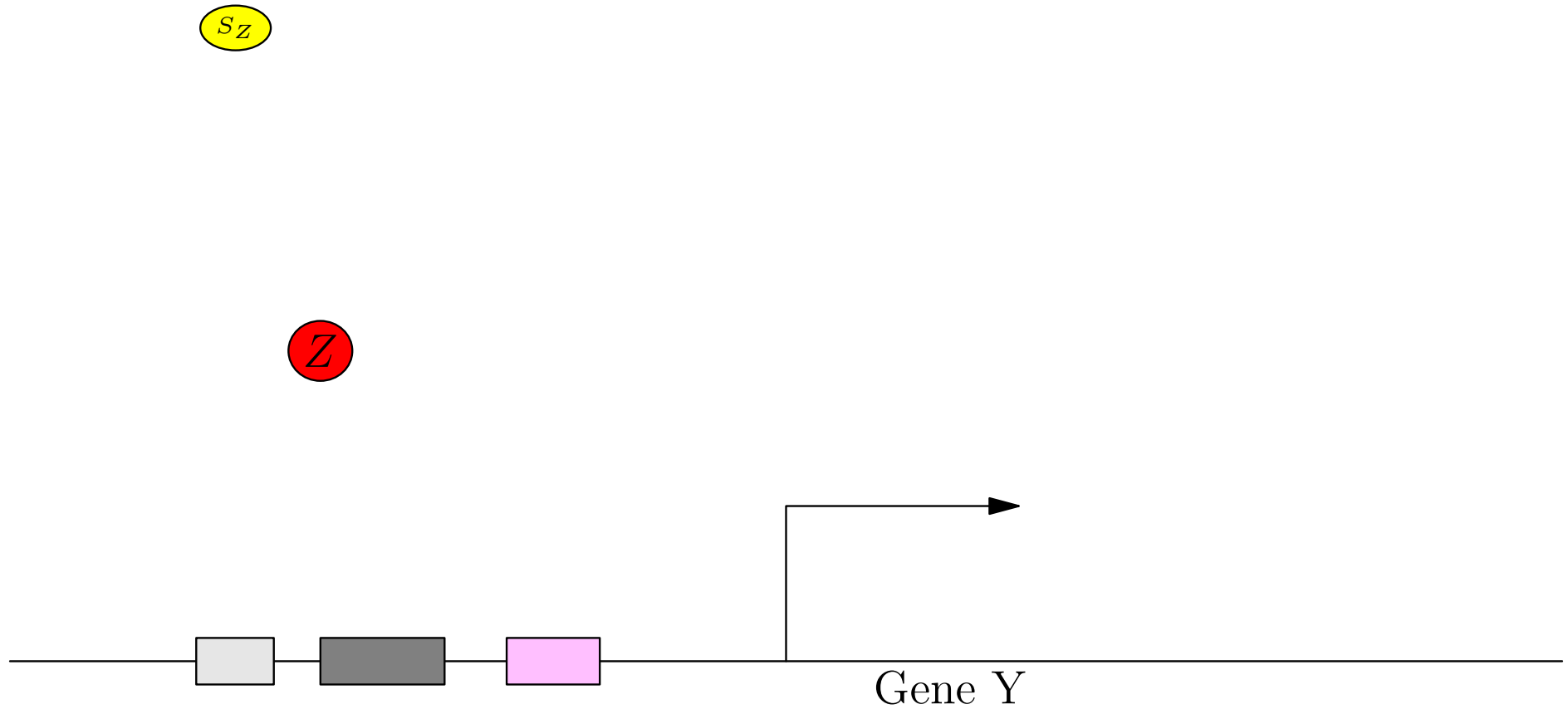
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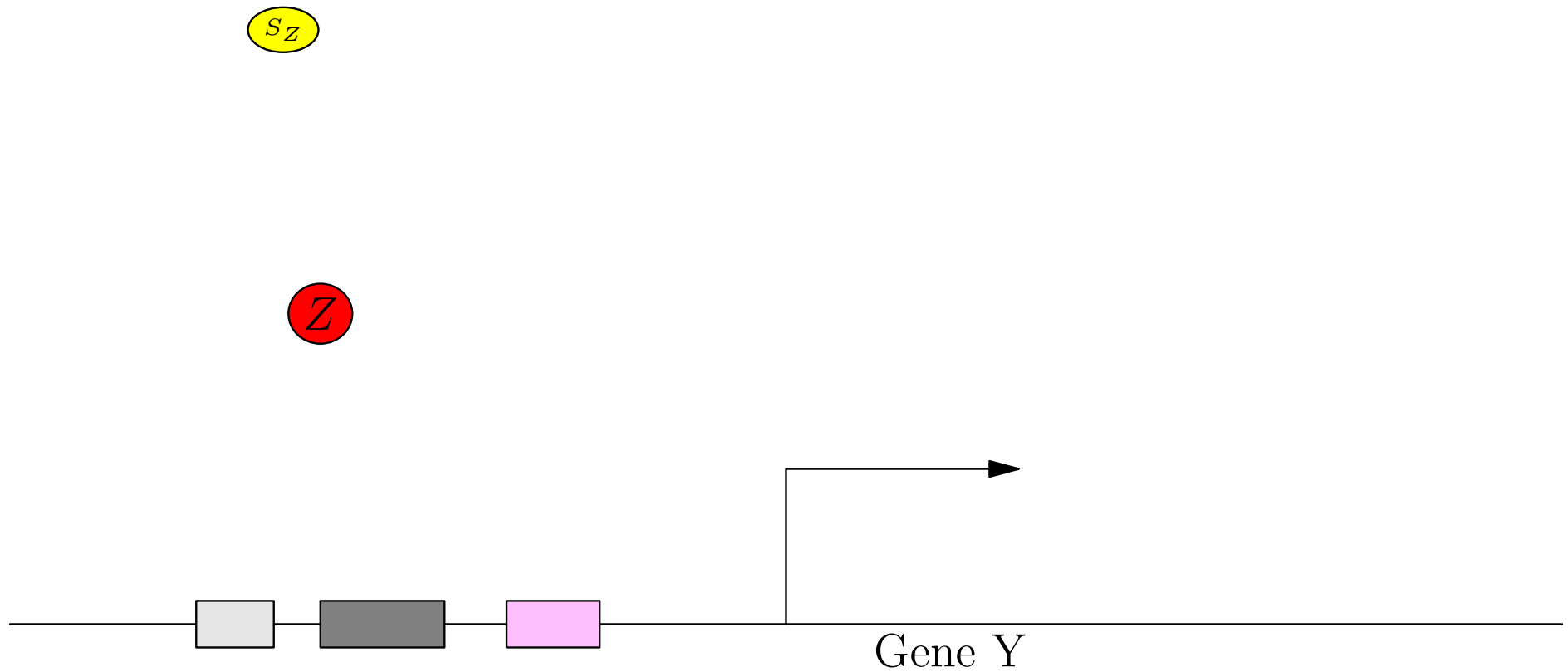
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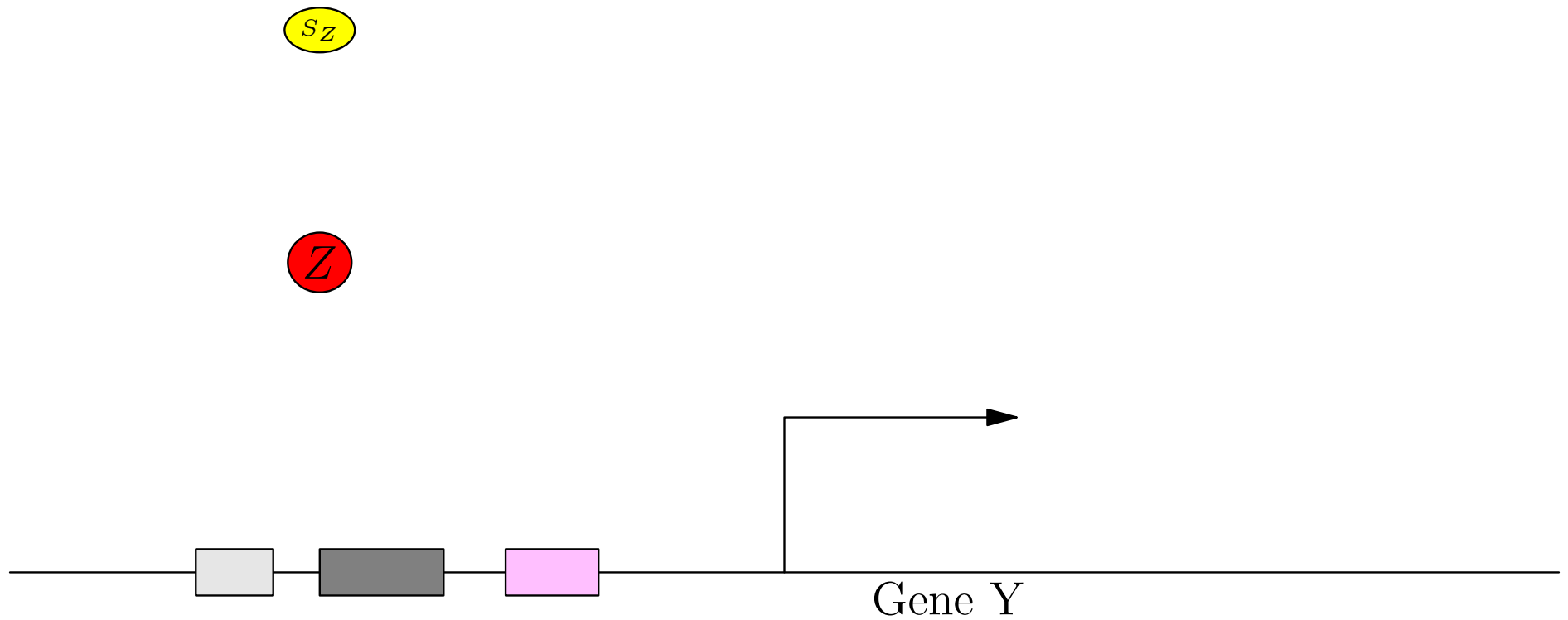
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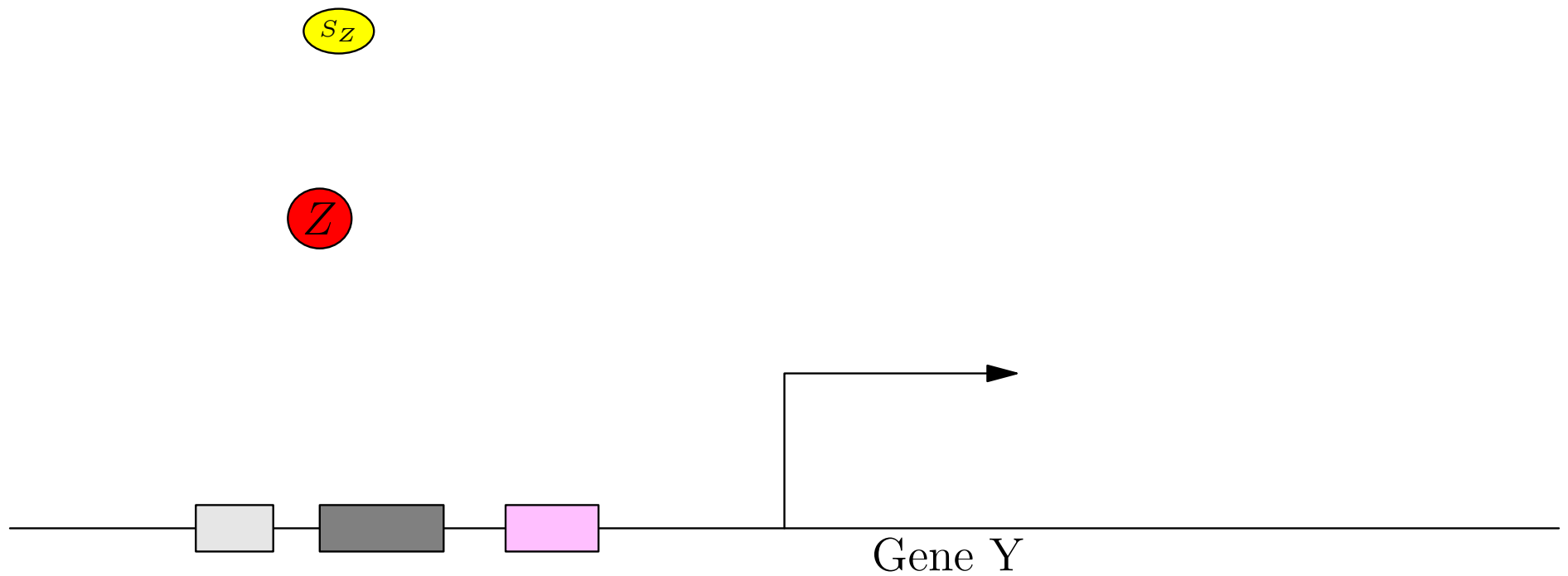
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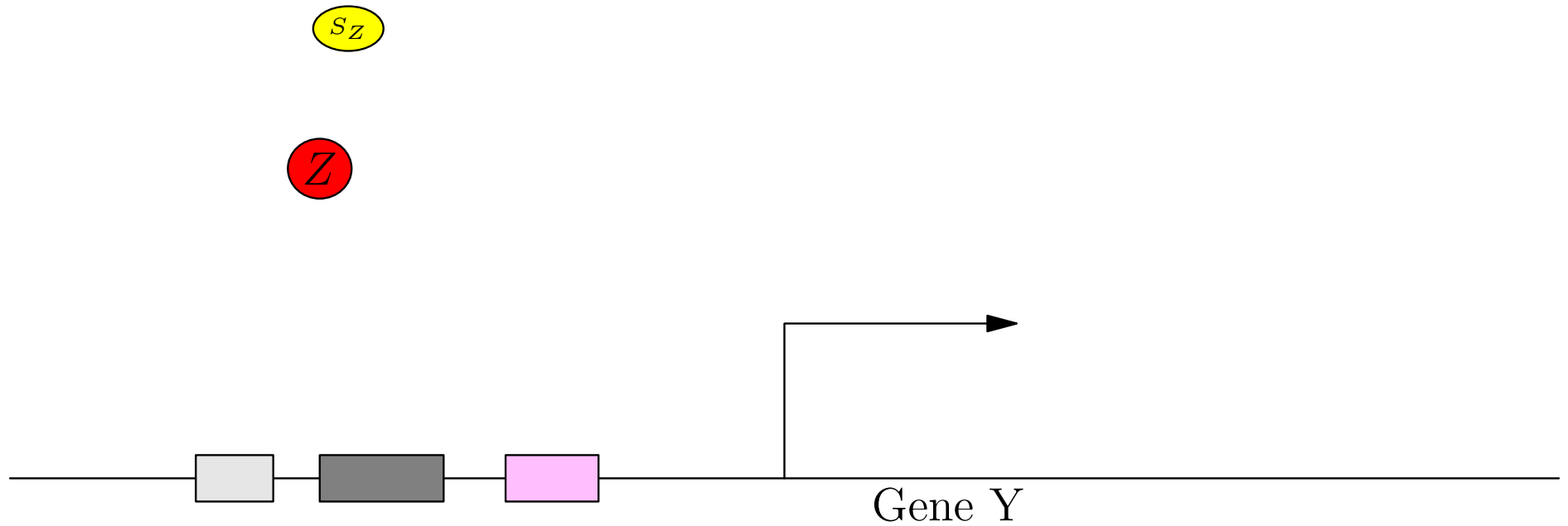
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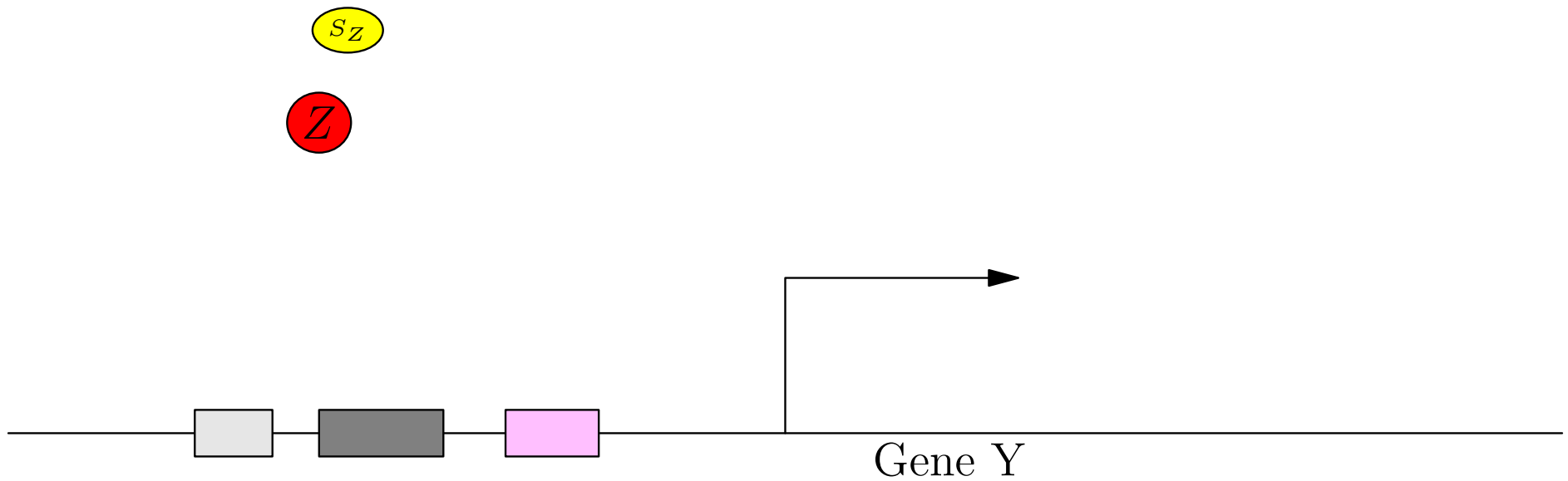
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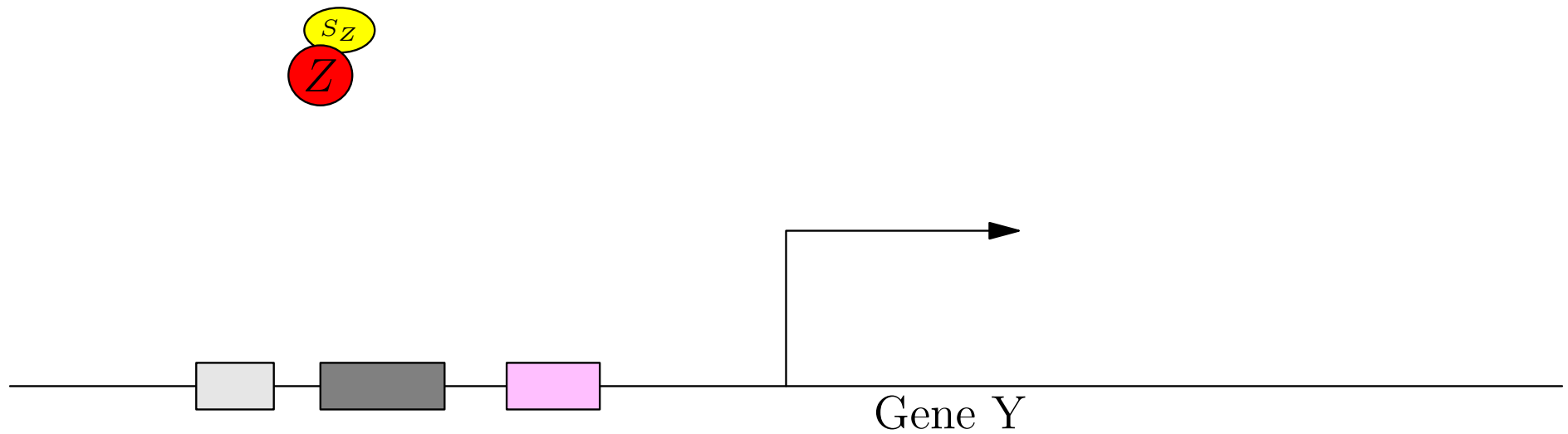
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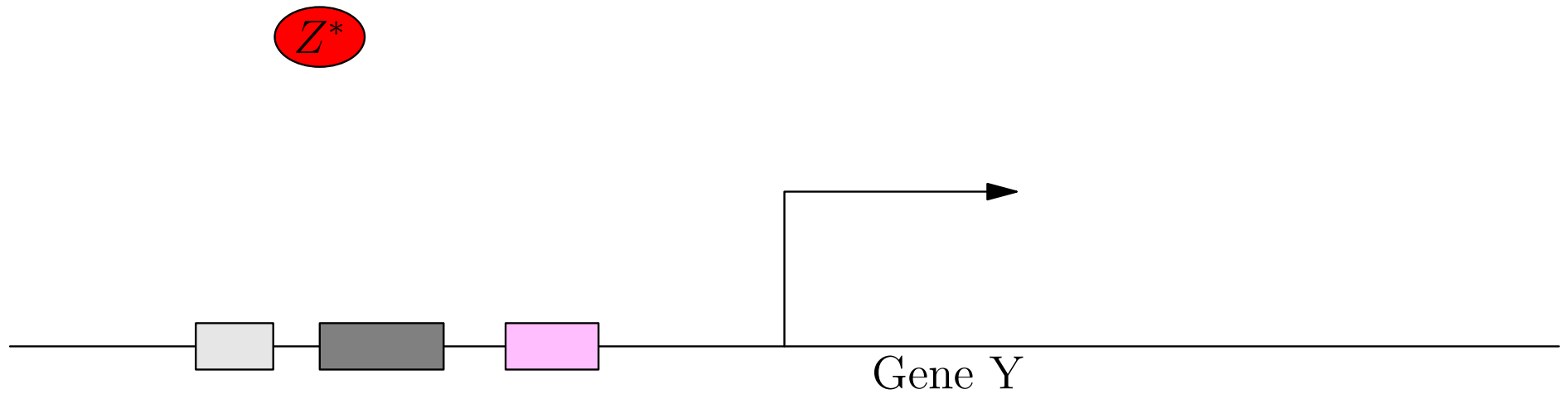
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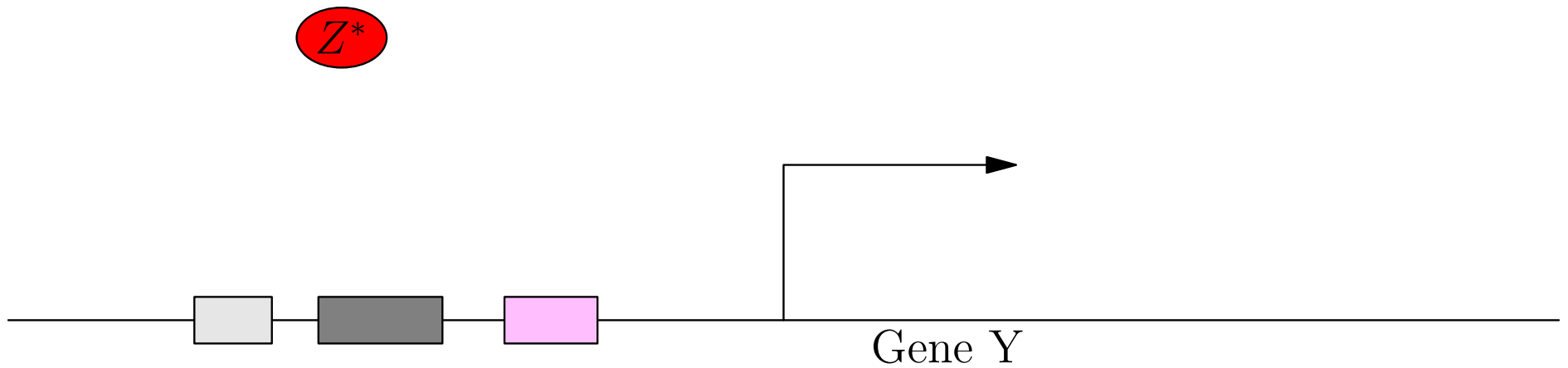
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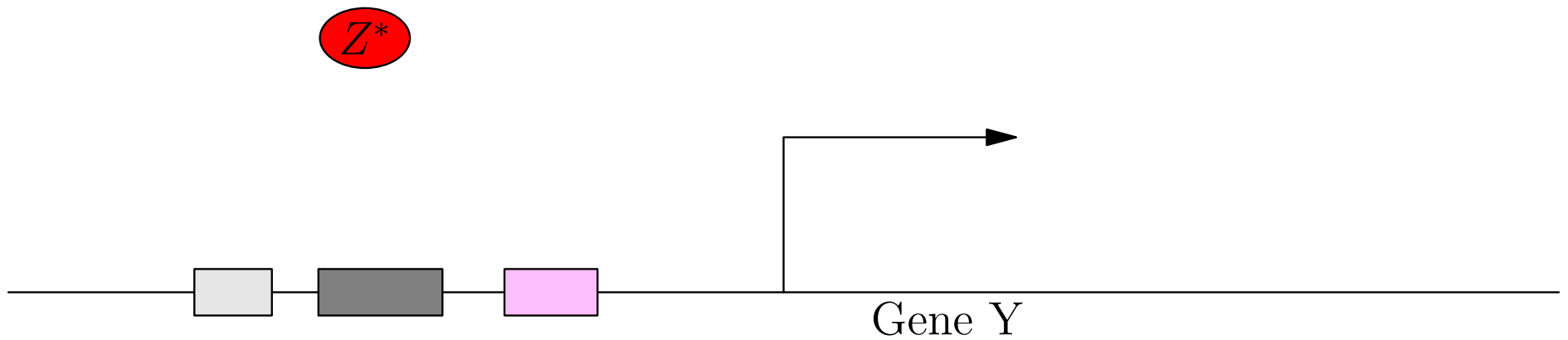
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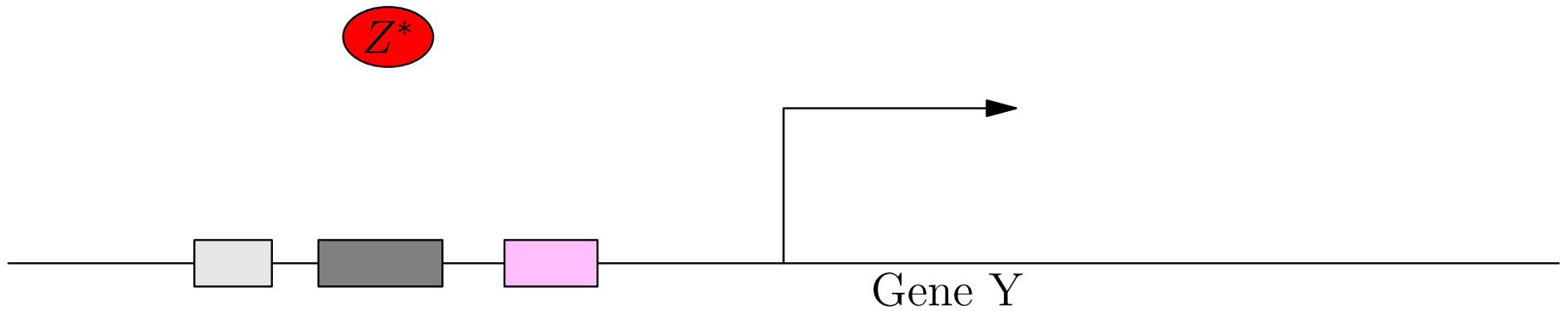
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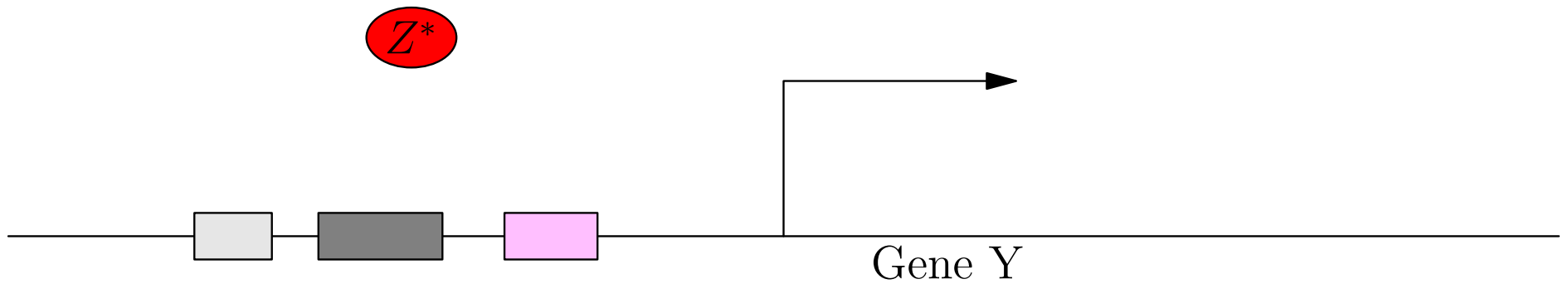
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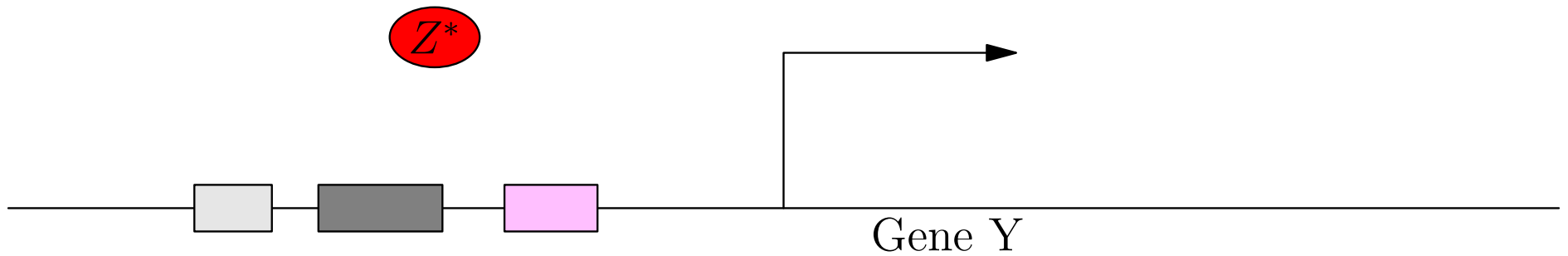
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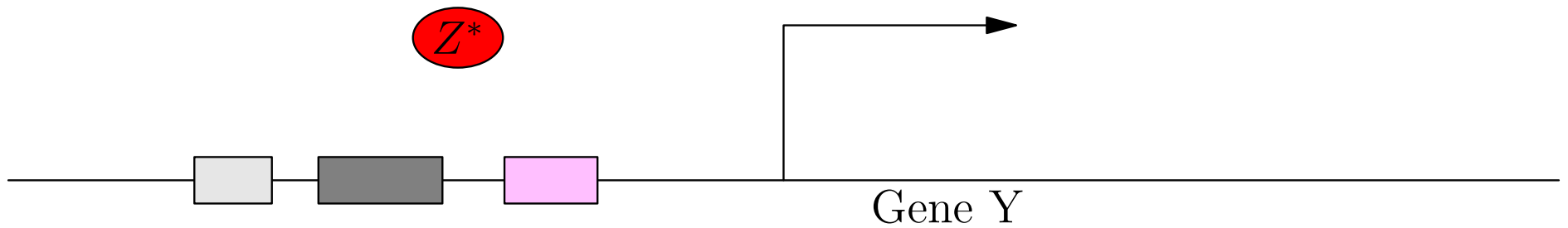
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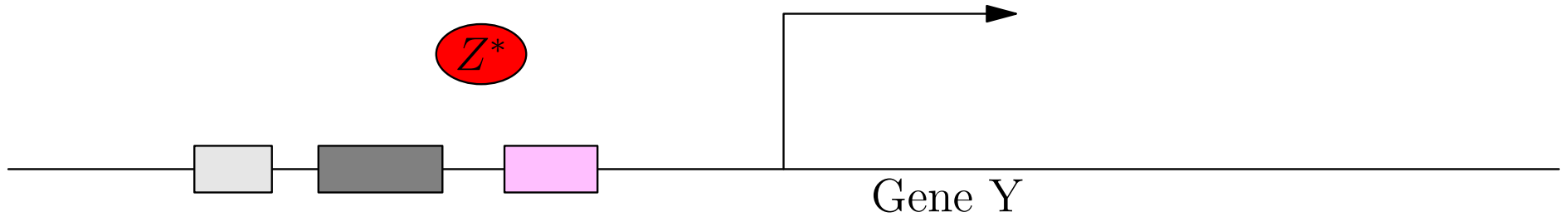
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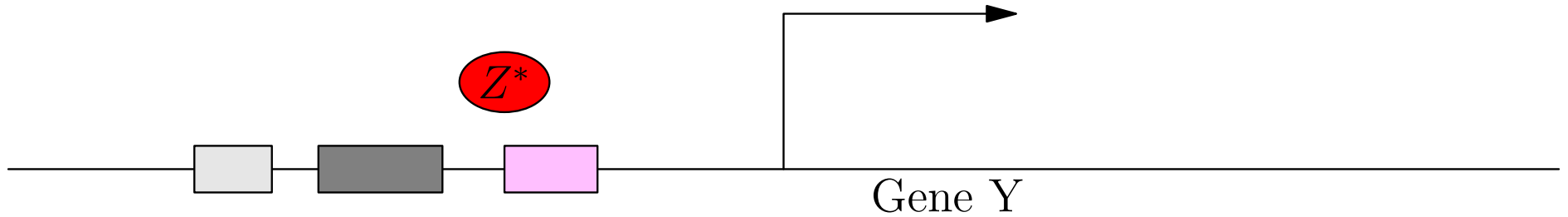
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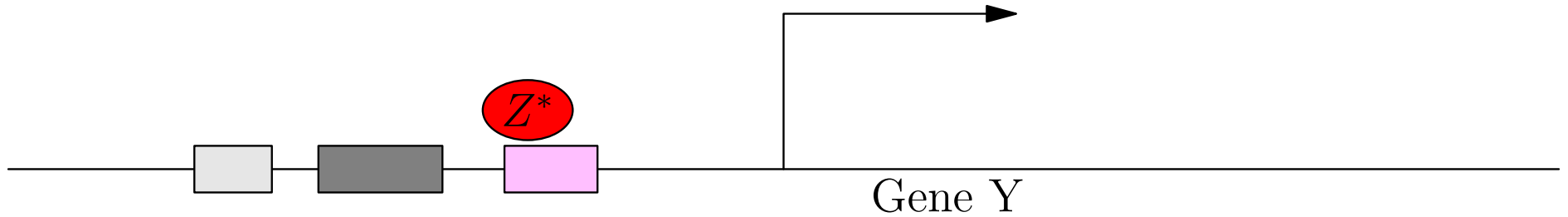
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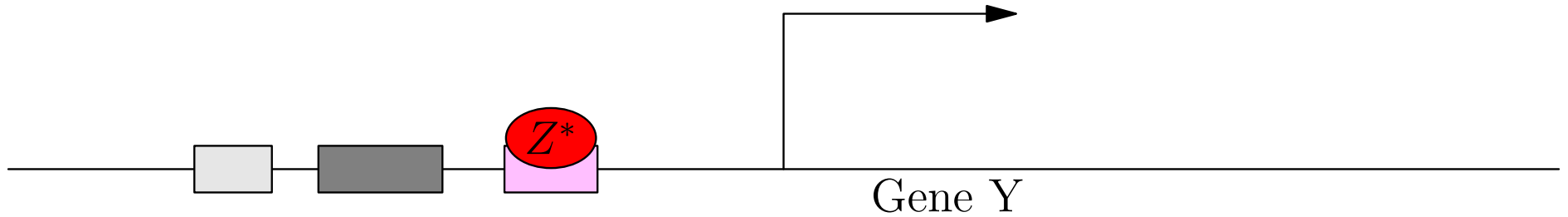
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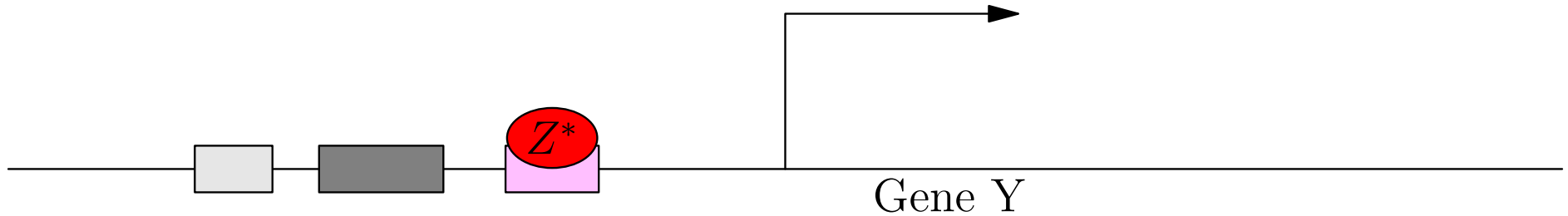
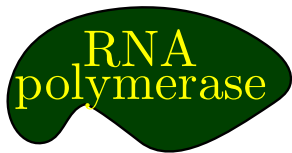


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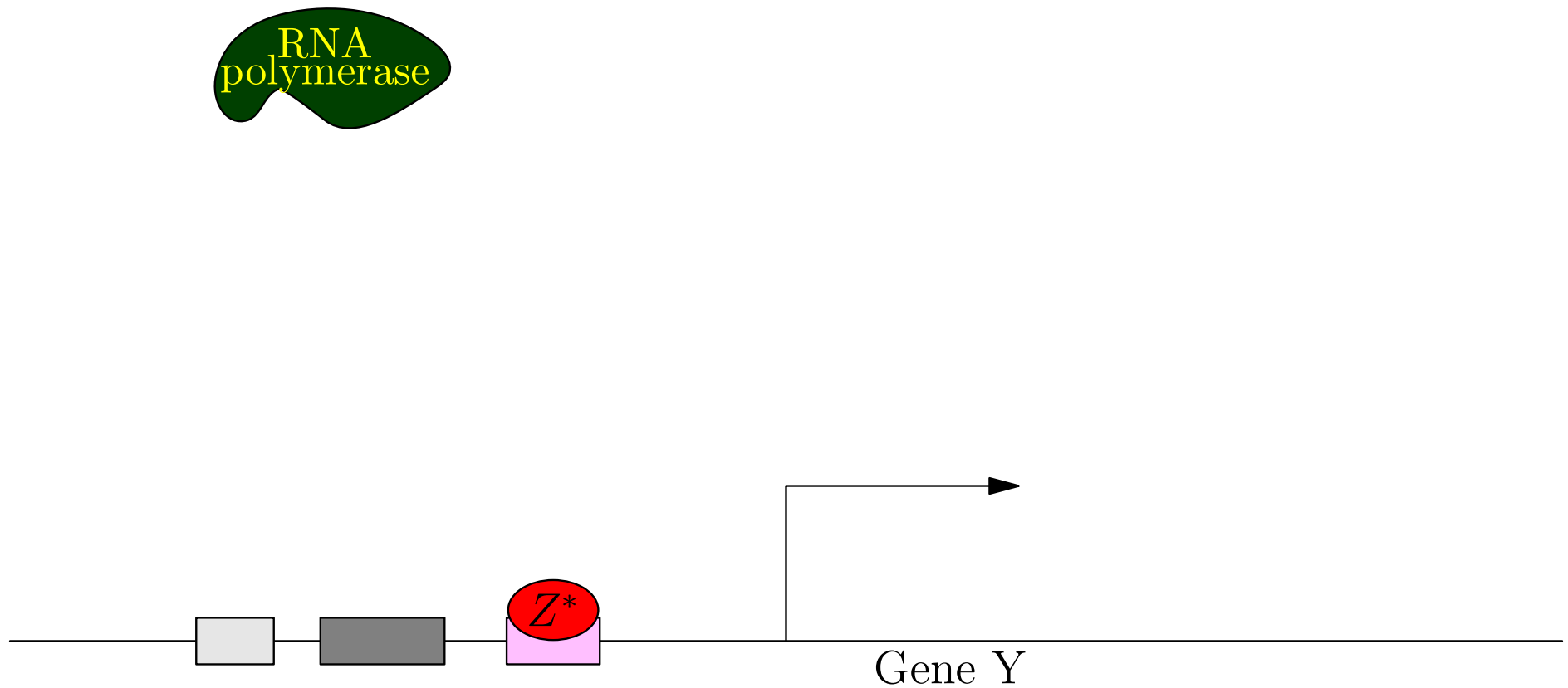
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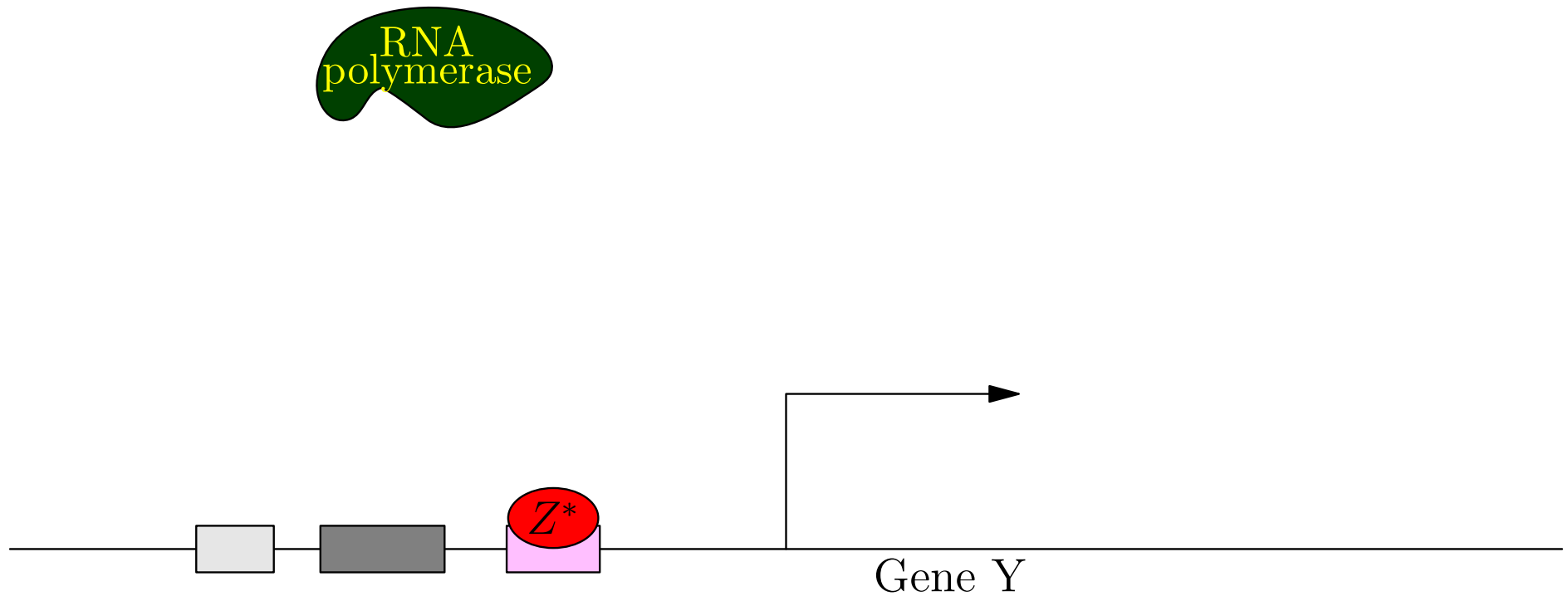
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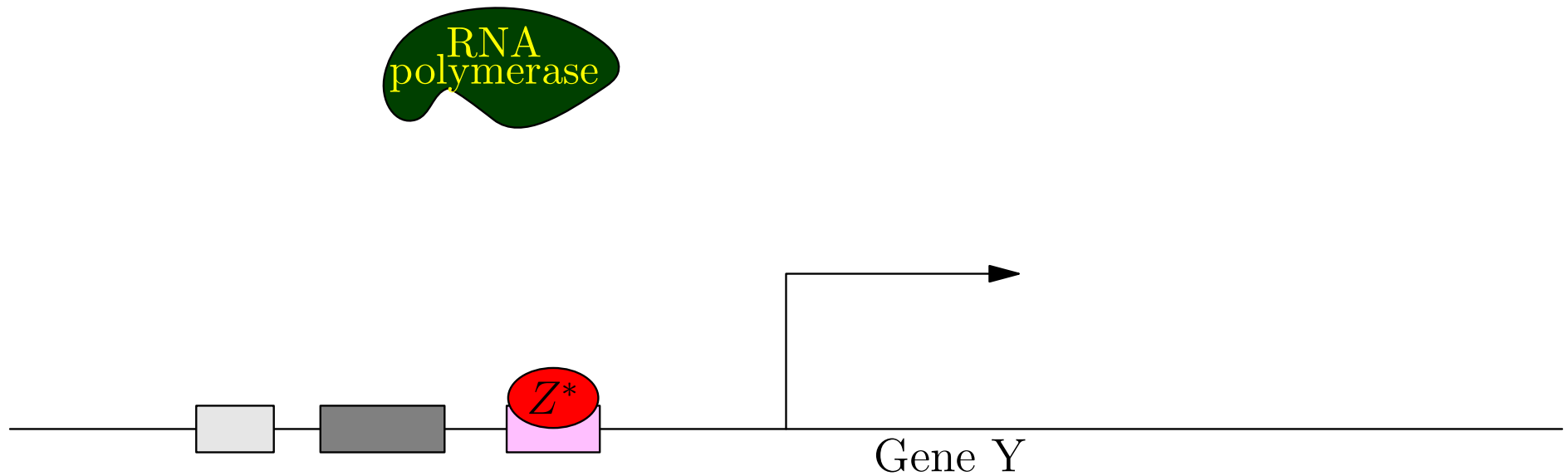
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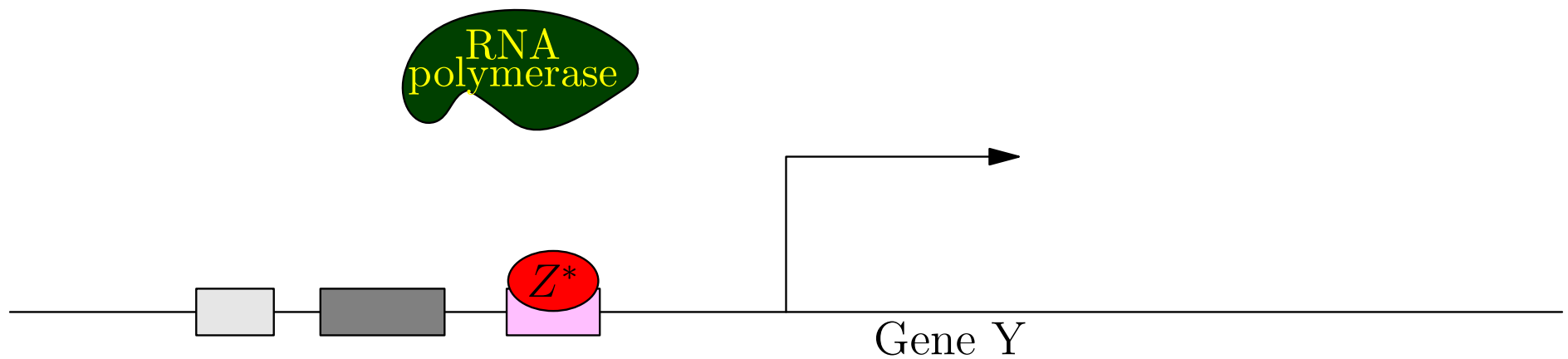
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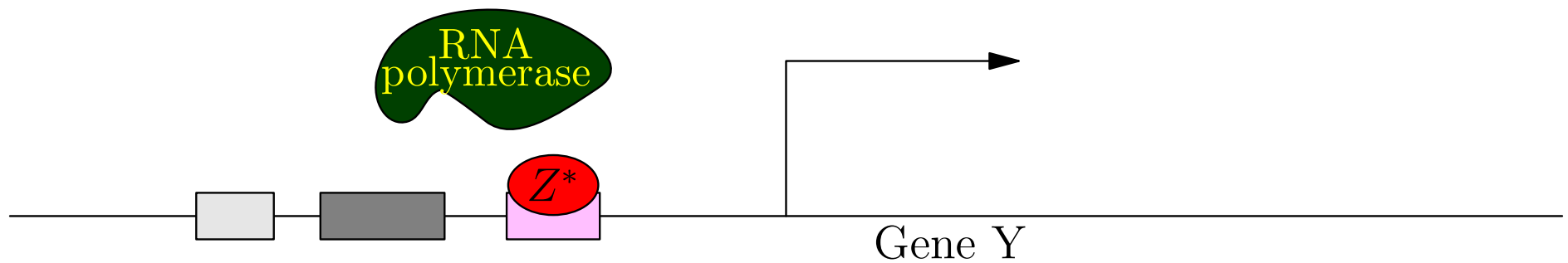
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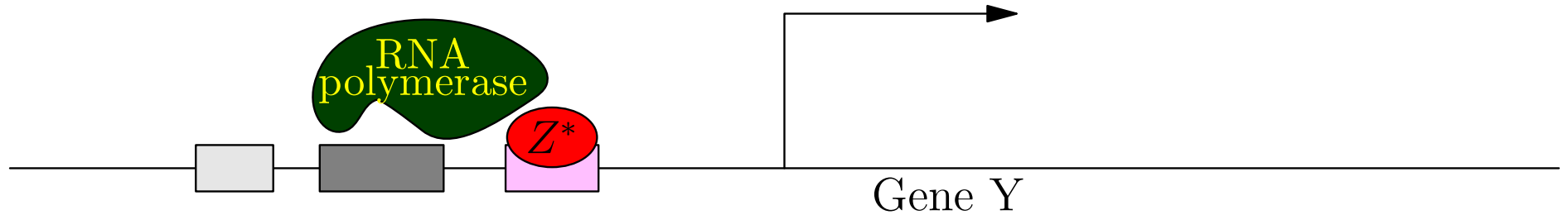
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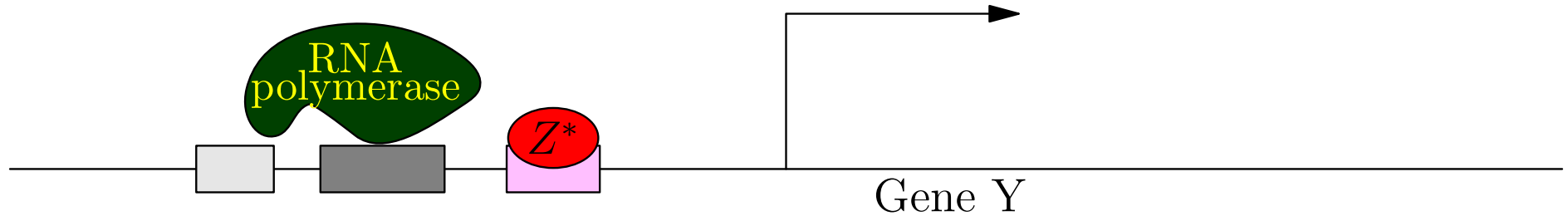
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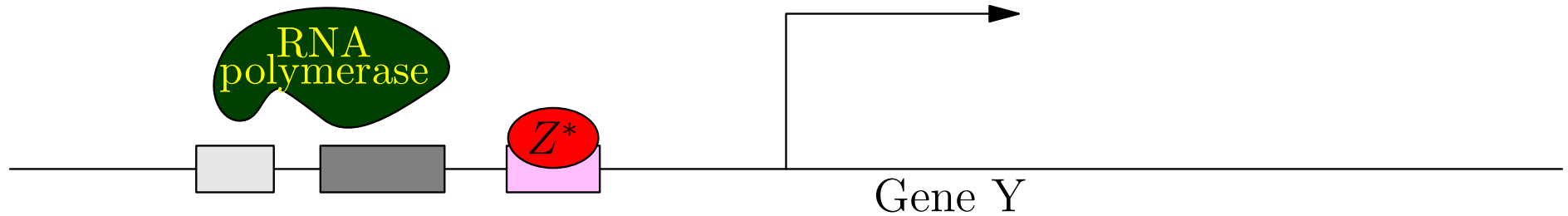
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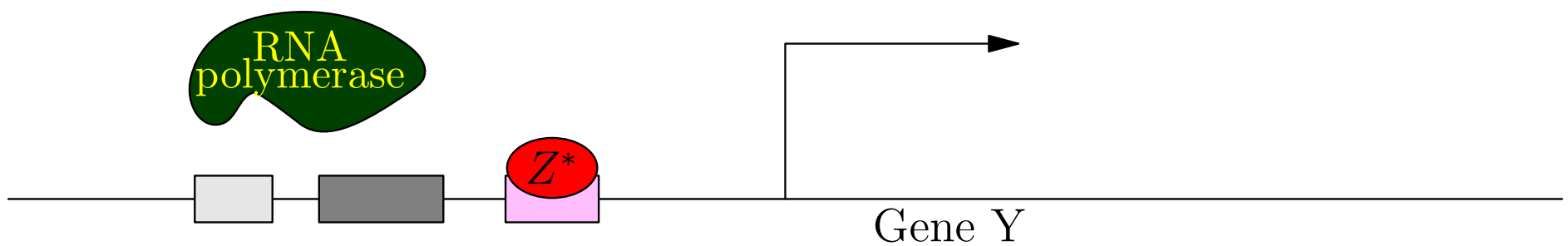
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Time Scales

- Reactions happen on different time scales
 - ★ Binding of small (signalling) molecules: ~ 1 msec
 - ★ Binding of active transcription factor: ~ 1 sec
 - ★ Transcription + translation of gene: ~ 5 min
 - ★ 50% change in concentration of protein: ~ 1 hour
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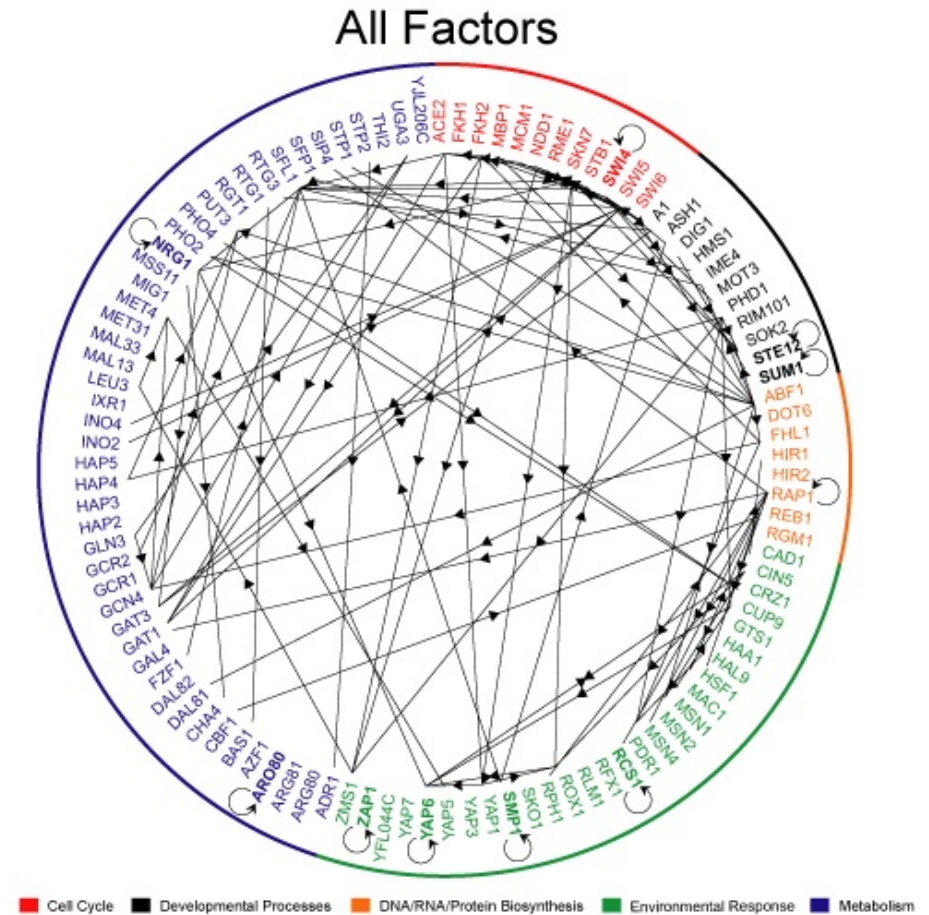
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Outline

1. Gene Regulation
2. Chemical Equations
3. Motifs
4. Quiz



Chemical Equations

- We can write down the mechanism that underlie gene regulation in terms of chemical equations



- These are massively simplified (but we can ignore reactions that happen quickly)
- The rate of reaction depends on the number/concentration of the molecules on the left-hand side and the reaction rate k_i

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Changes on Average

- Chemical equations happen at random, so the rate equations describe the average changes



- Let $[X]$ denote the (mean) concentration of X then

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- $\Omega = VA$, $[X] = X/\Omega$ and $[Y] = Y/\Omega$

Fluctuations

- In chemistry we are used to there being so many molecules that the average rate equations provide a very good approximation of what really happens
- This is often true in biology, but not always
- Cells are small and some proteins can occur in very small numbers
- We return to the problem of modelling systems with small number of molecules in the last lecture
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- Our molecular equation depend on the product of concentrations of the components on the left-hand side
- This can lead to strong non-linearities
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- We showed earlier that proteins are often *activated* by signalling molecules combining with the protein to form a complex
- Proteins can have multiple sub-units that each bind with a signalling molecule
- A complex might only be stable when the protein has bound some number of signalling molecules

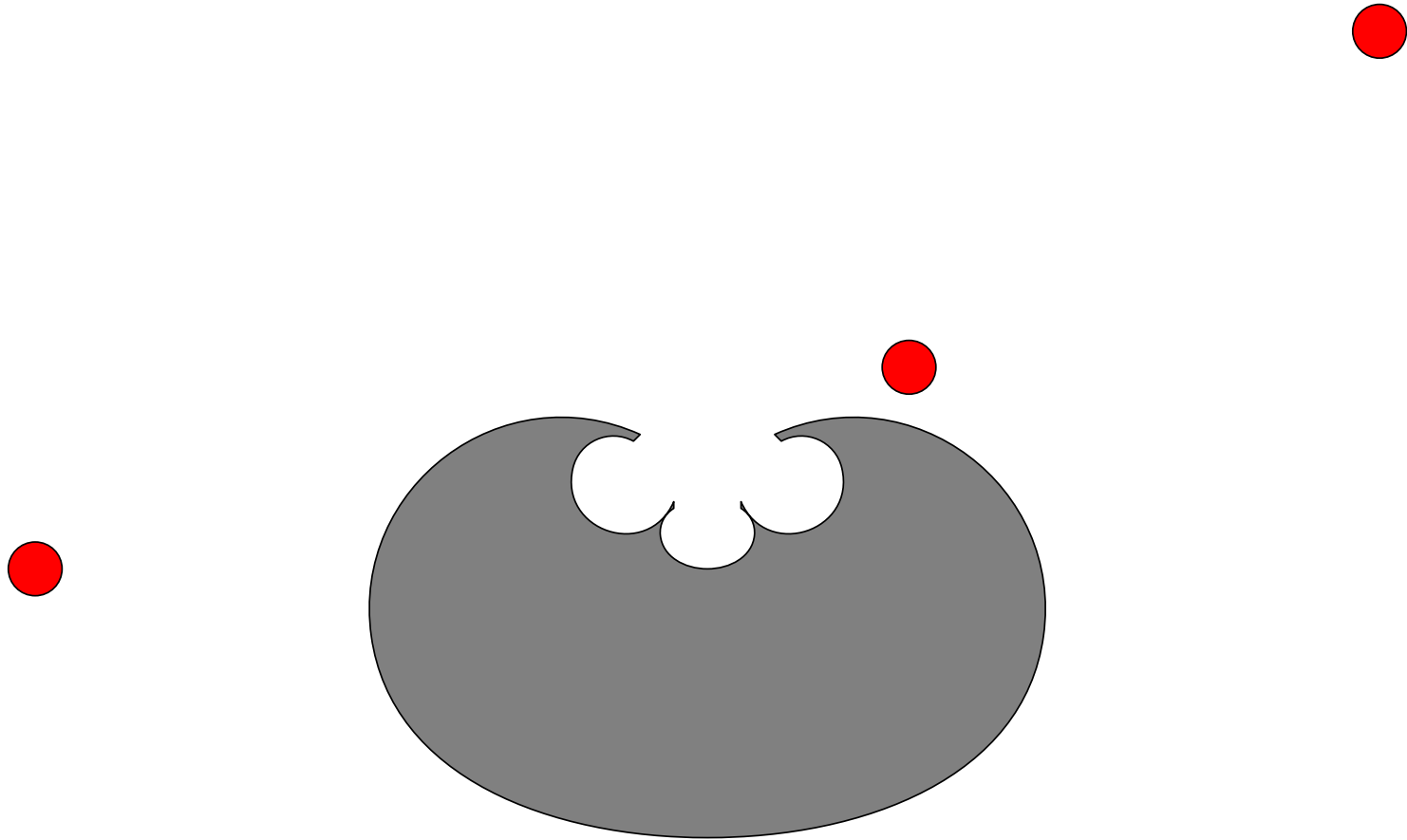
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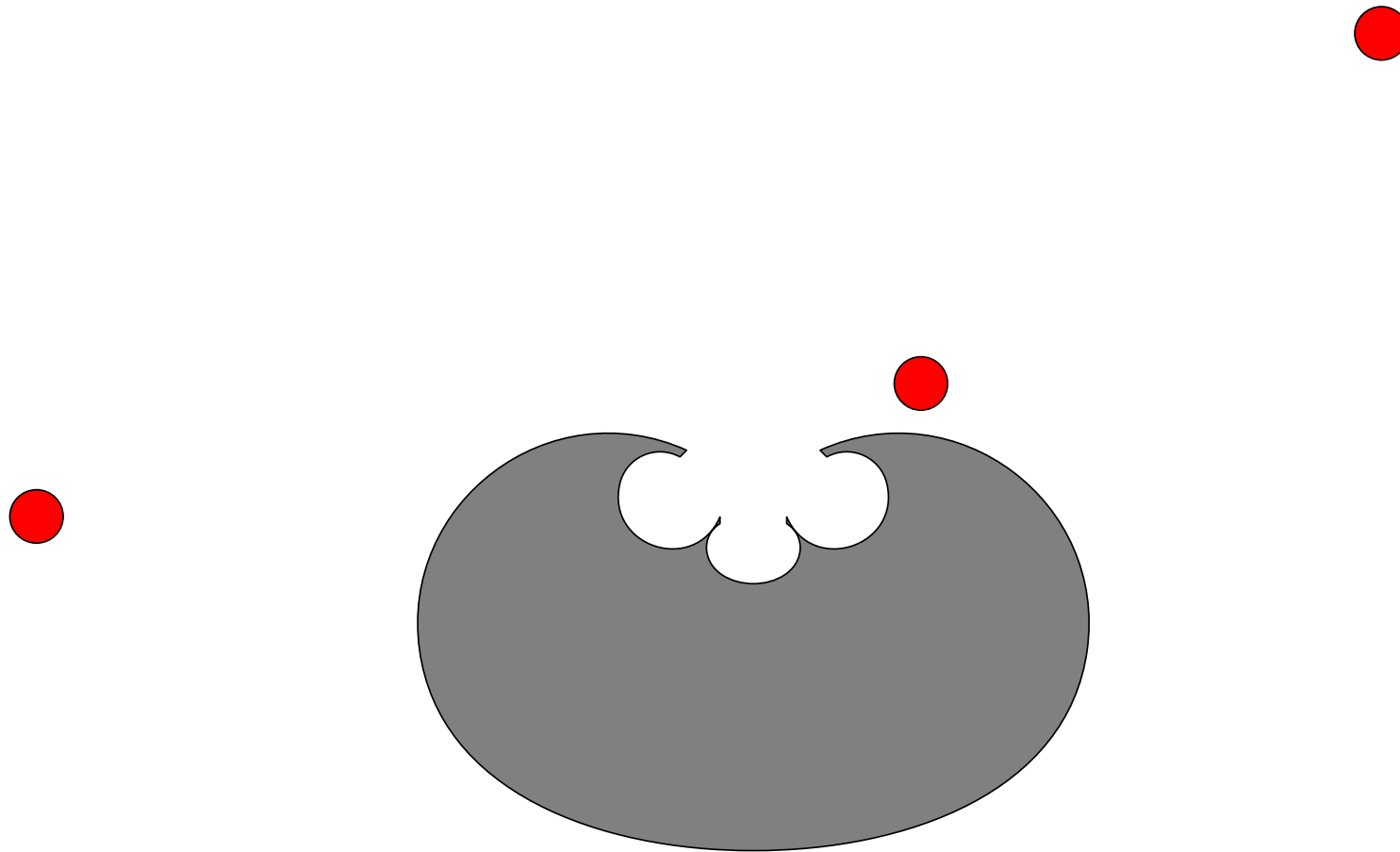
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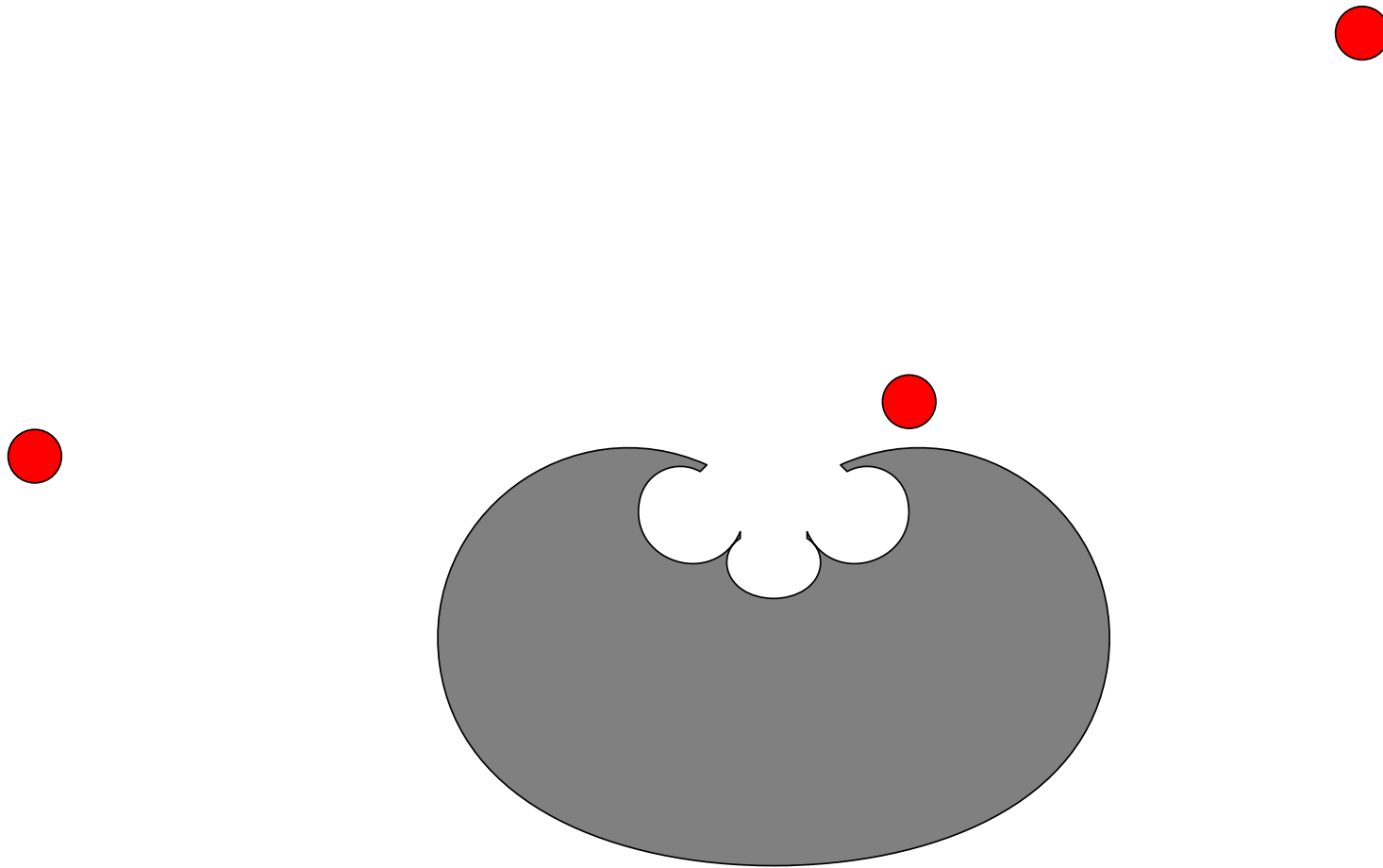
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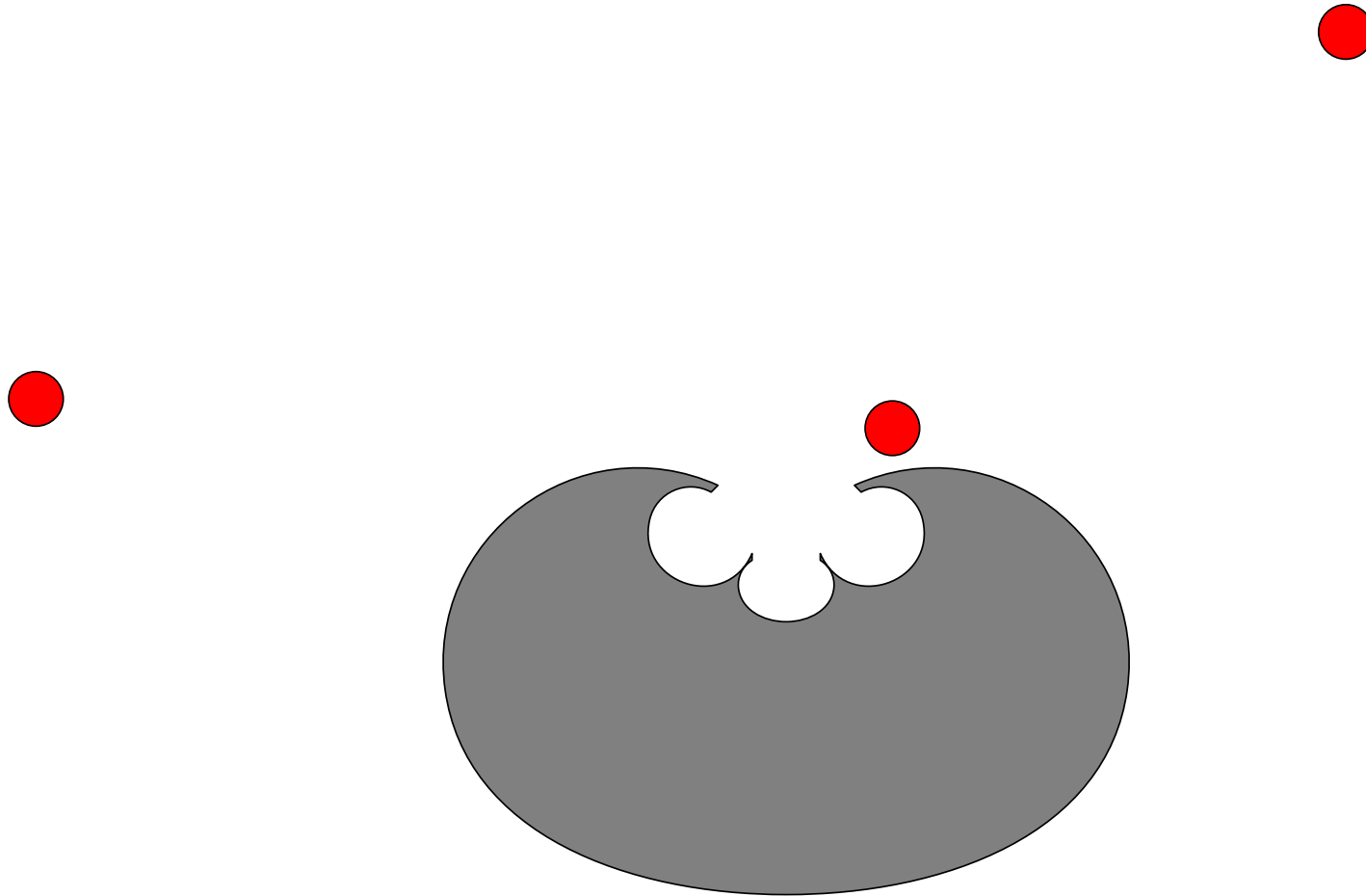
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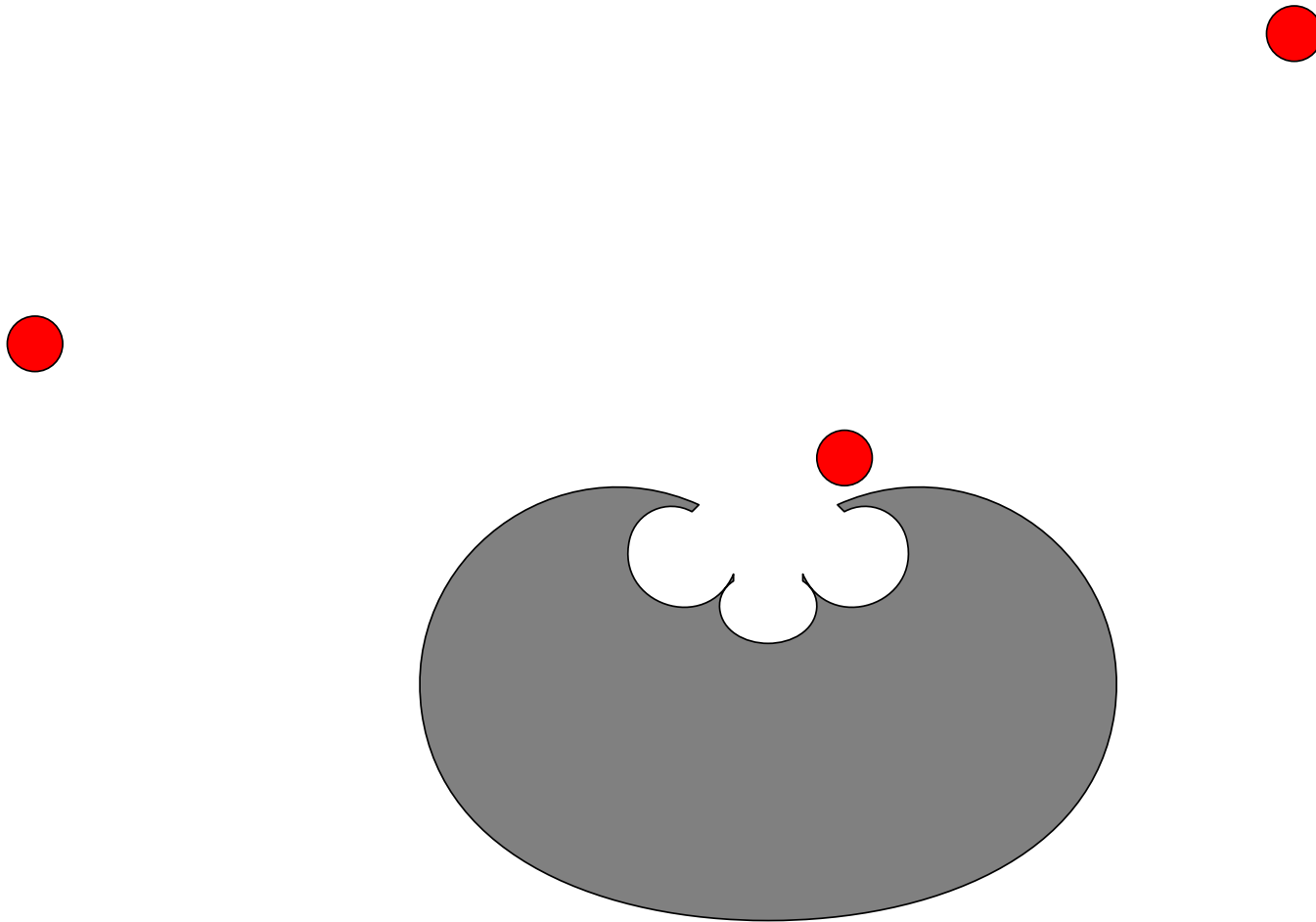
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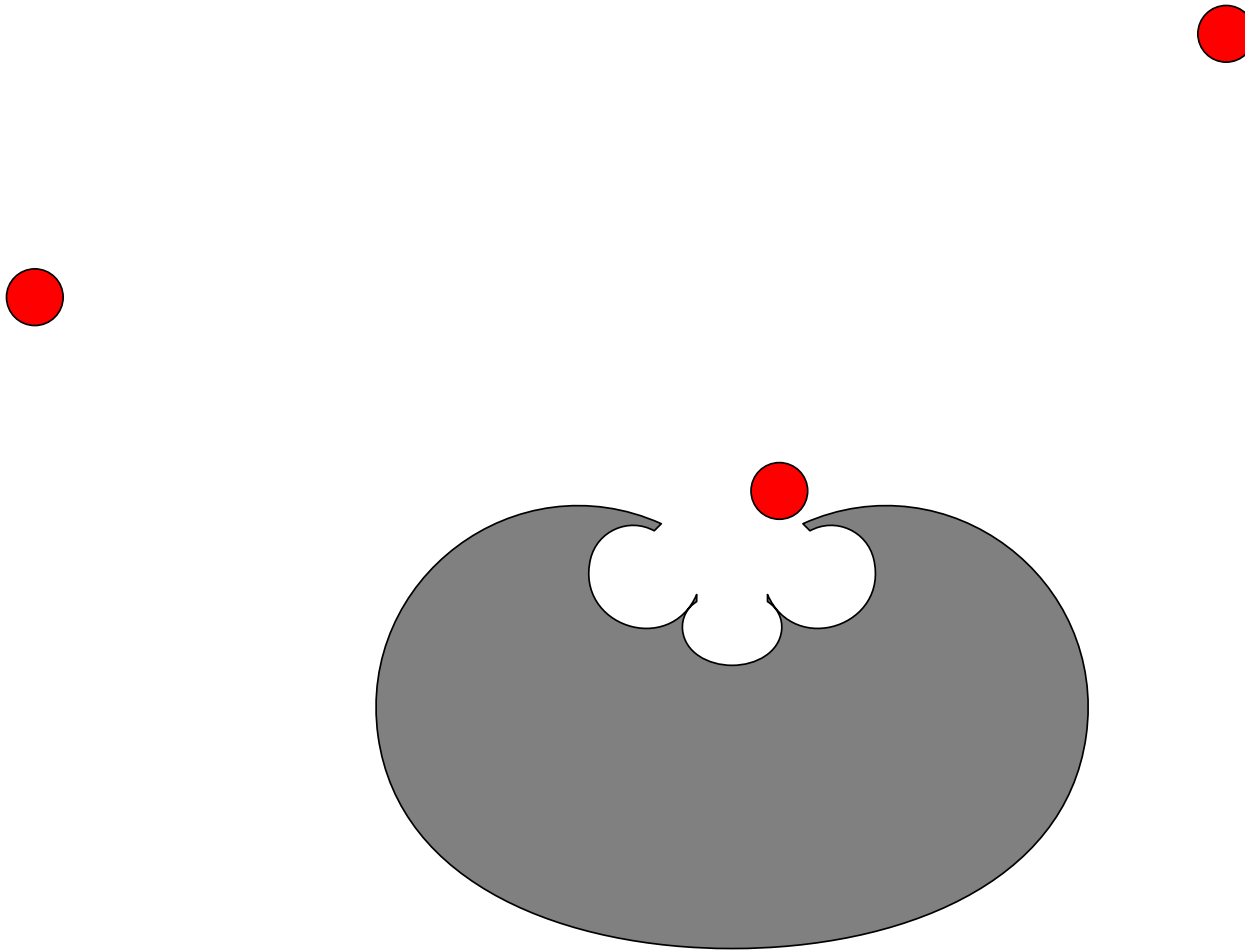
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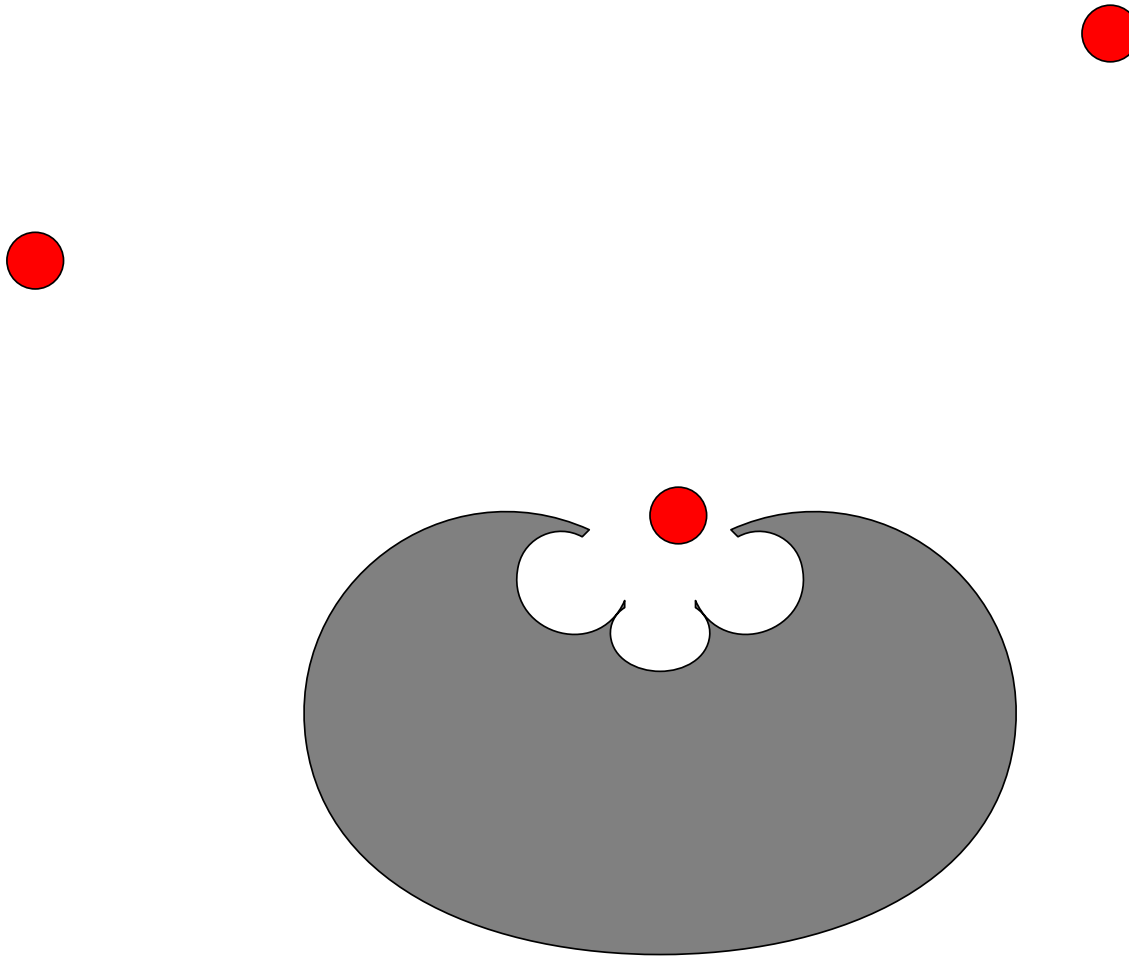
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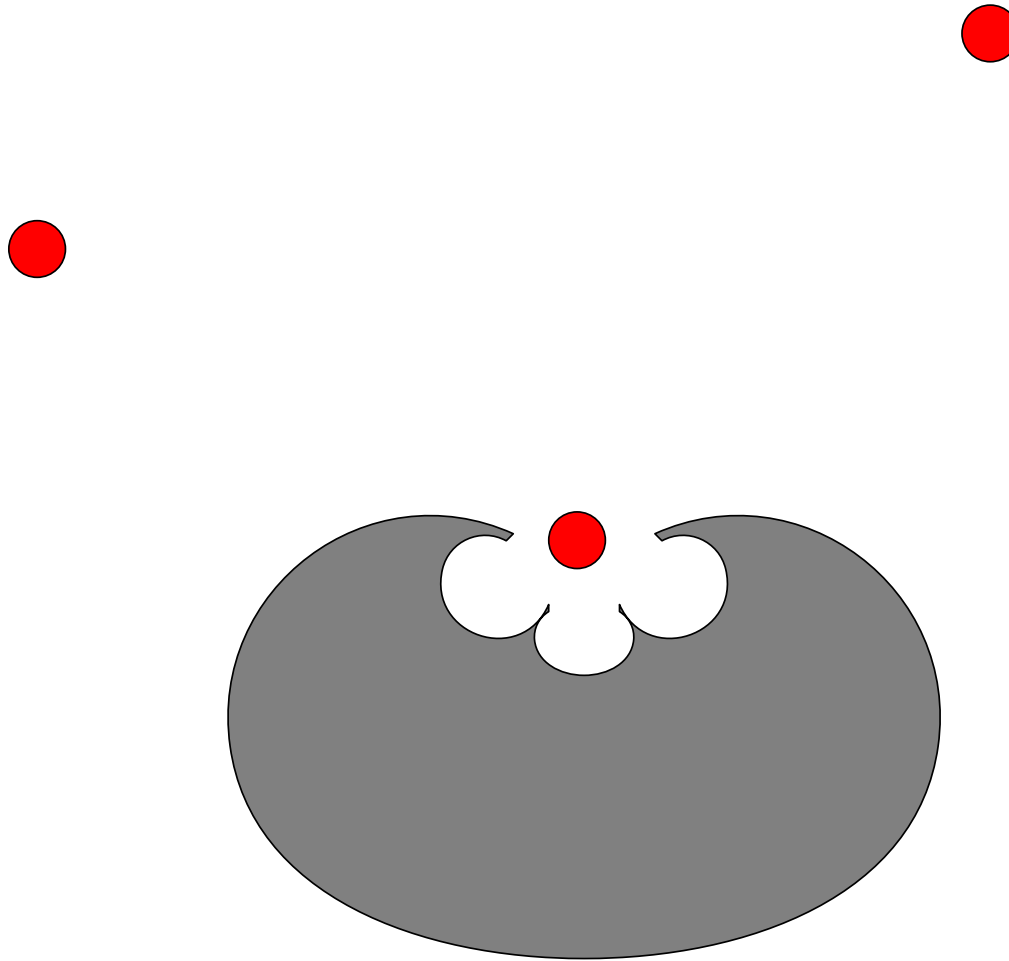
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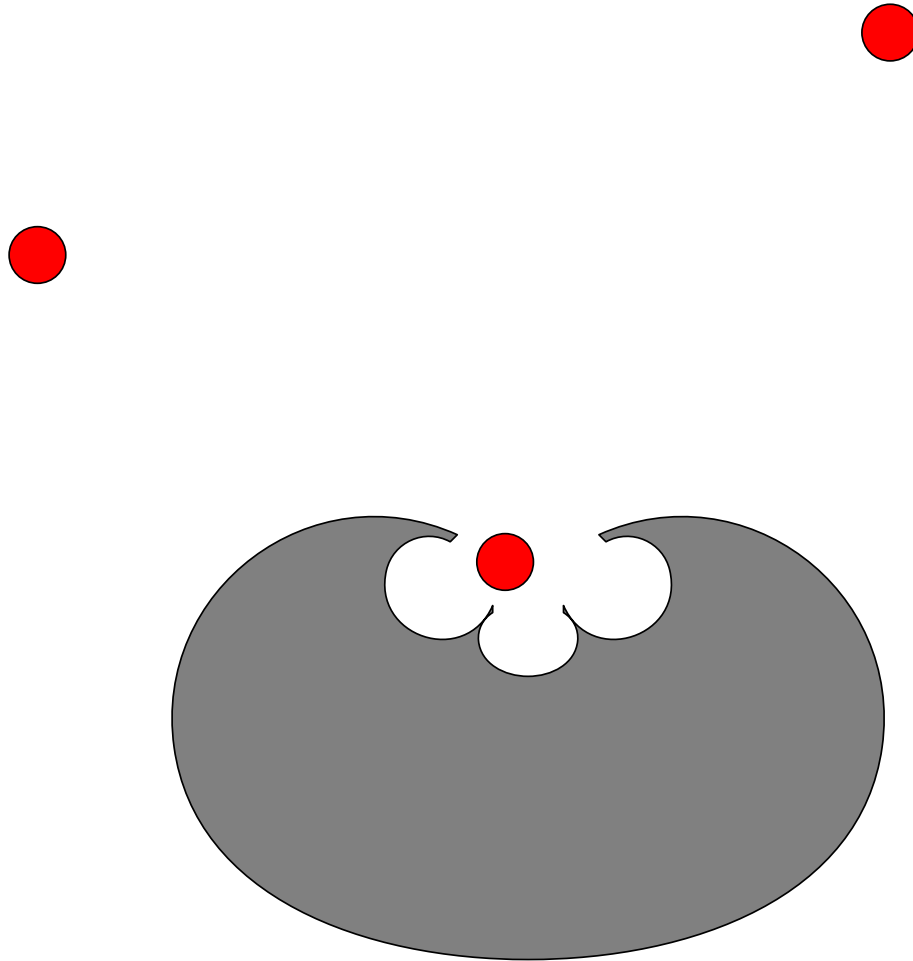
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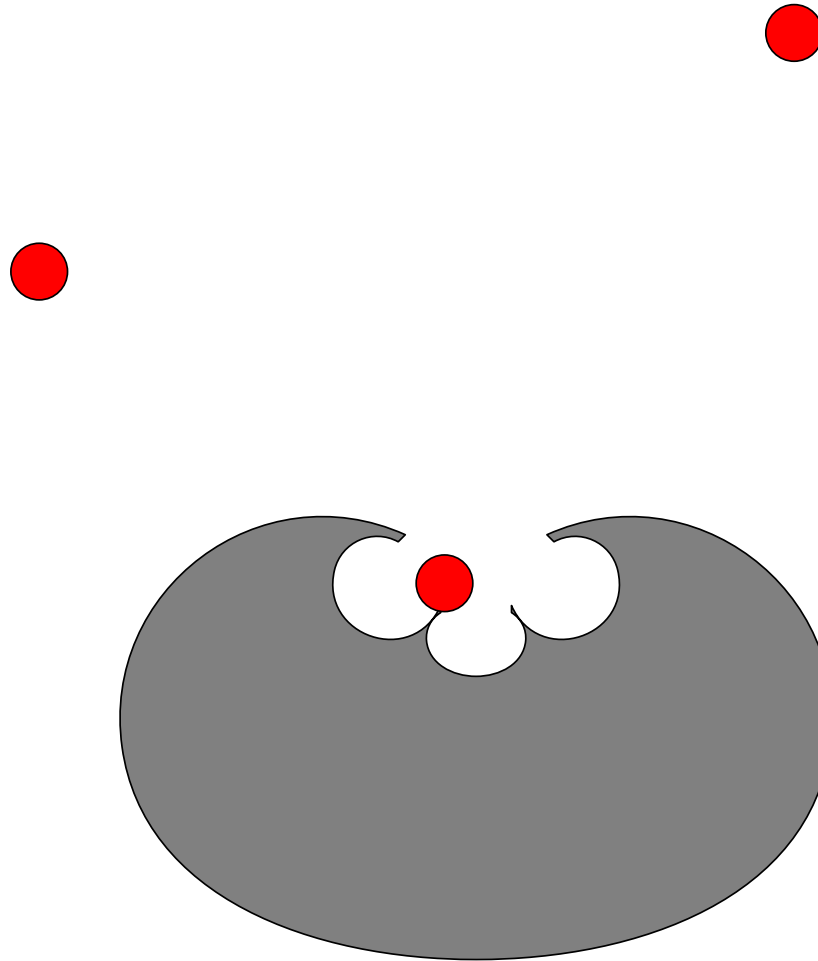
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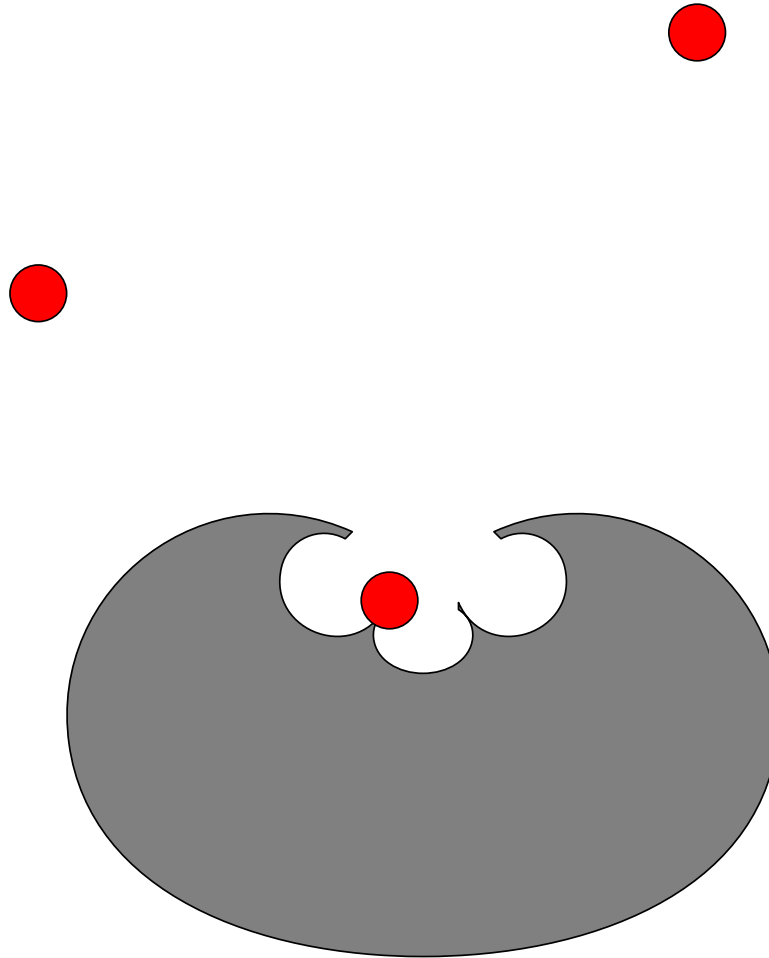
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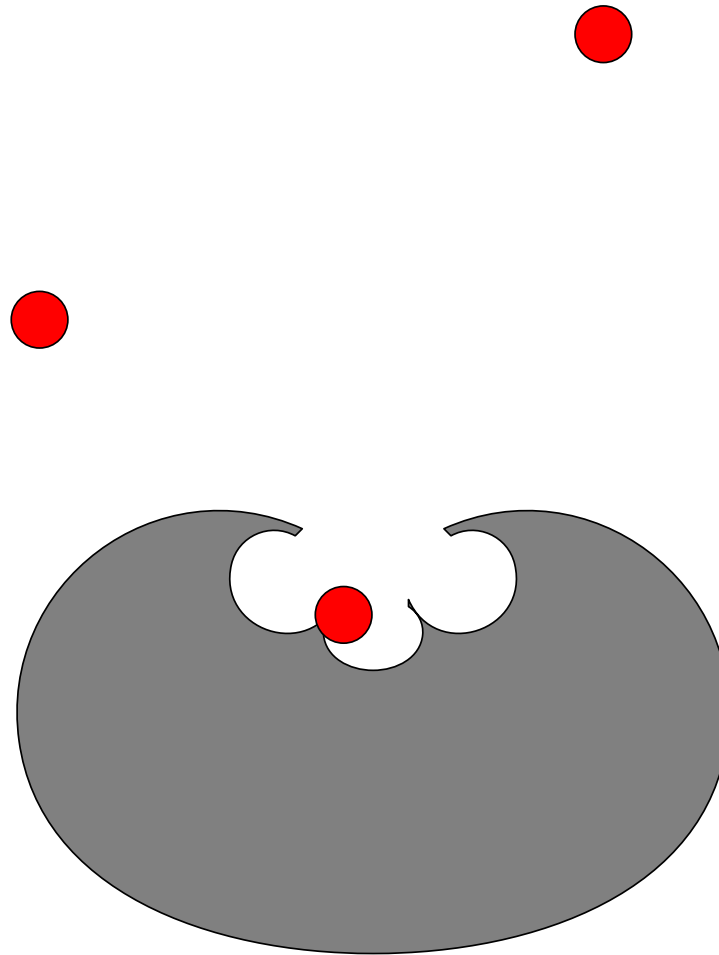
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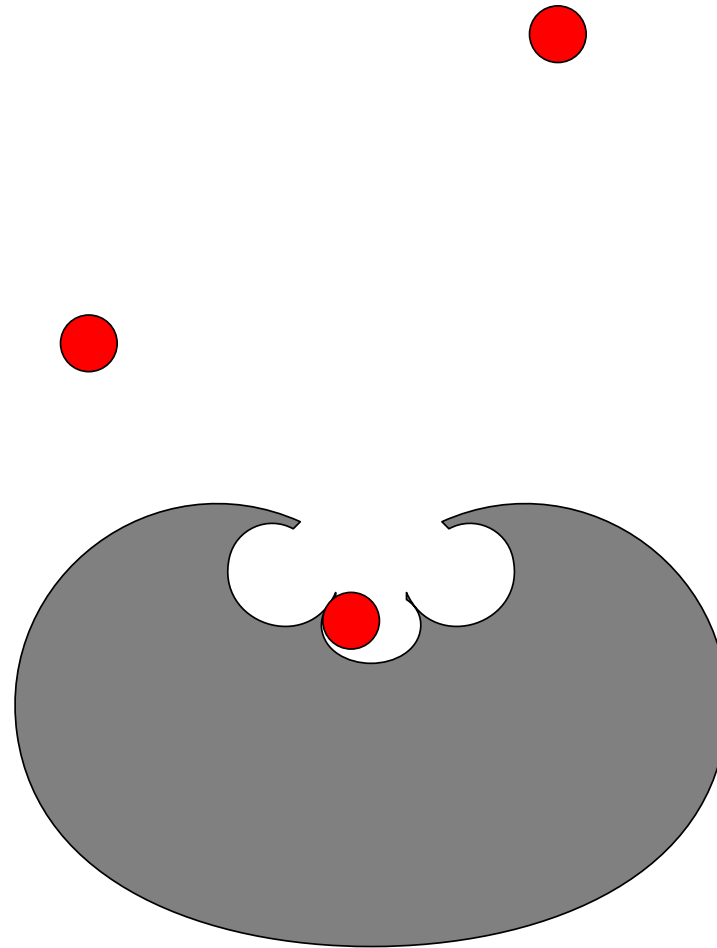
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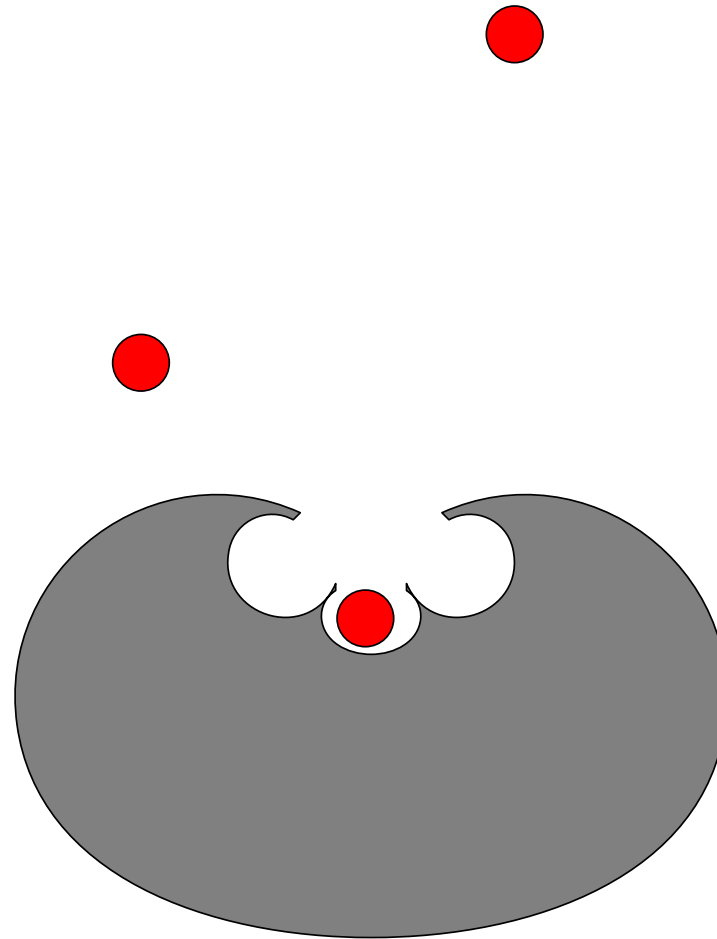
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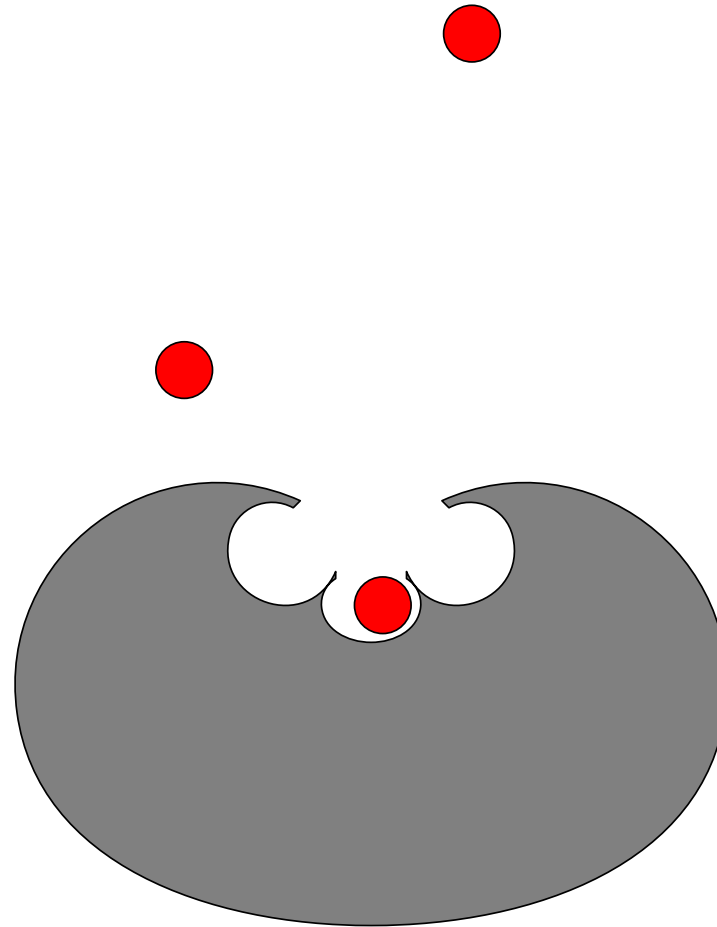
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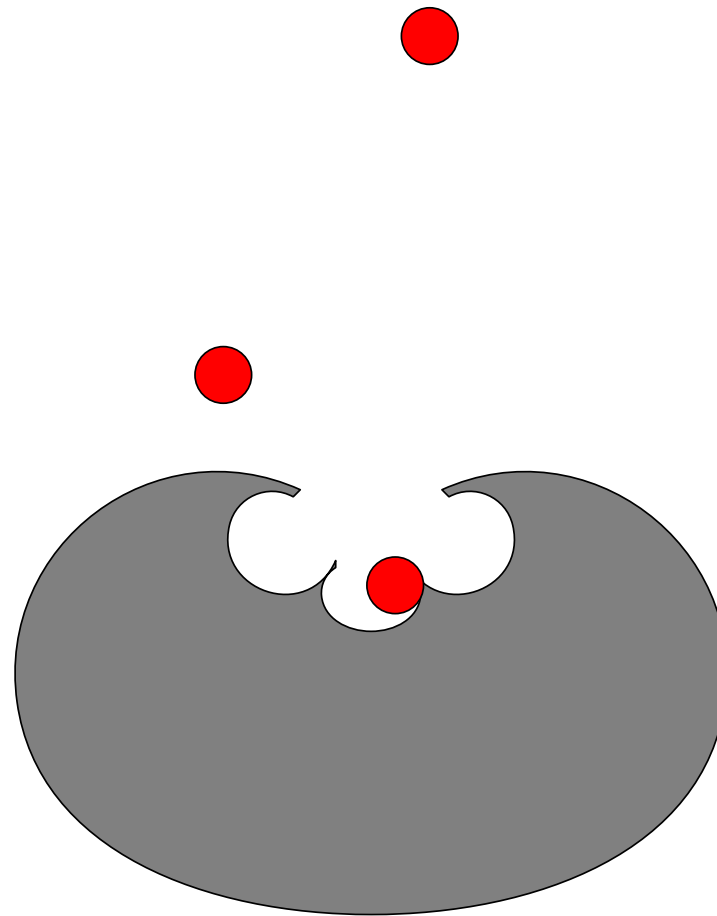
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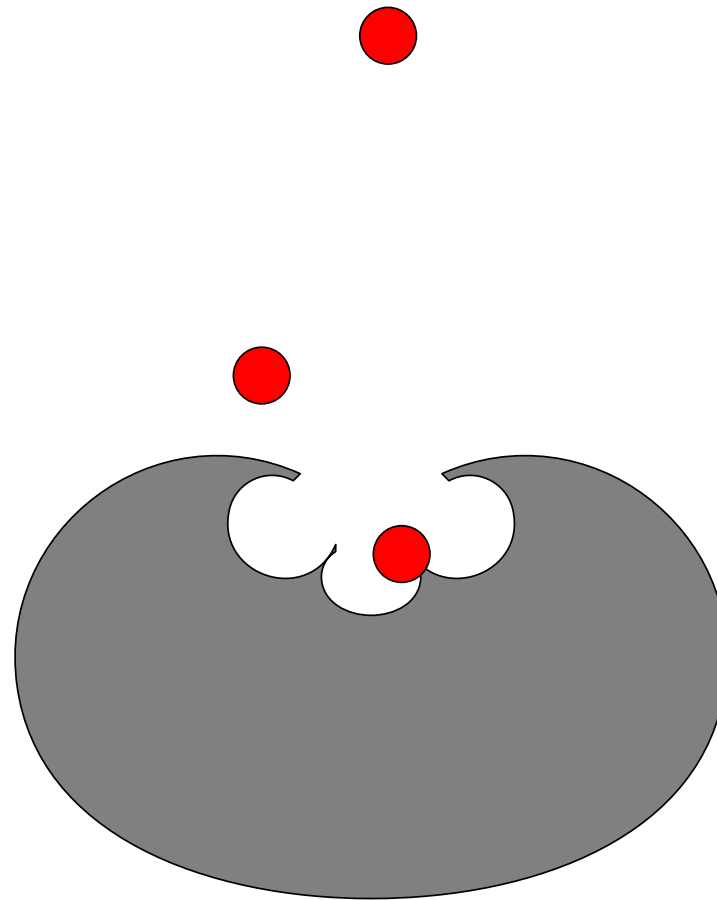
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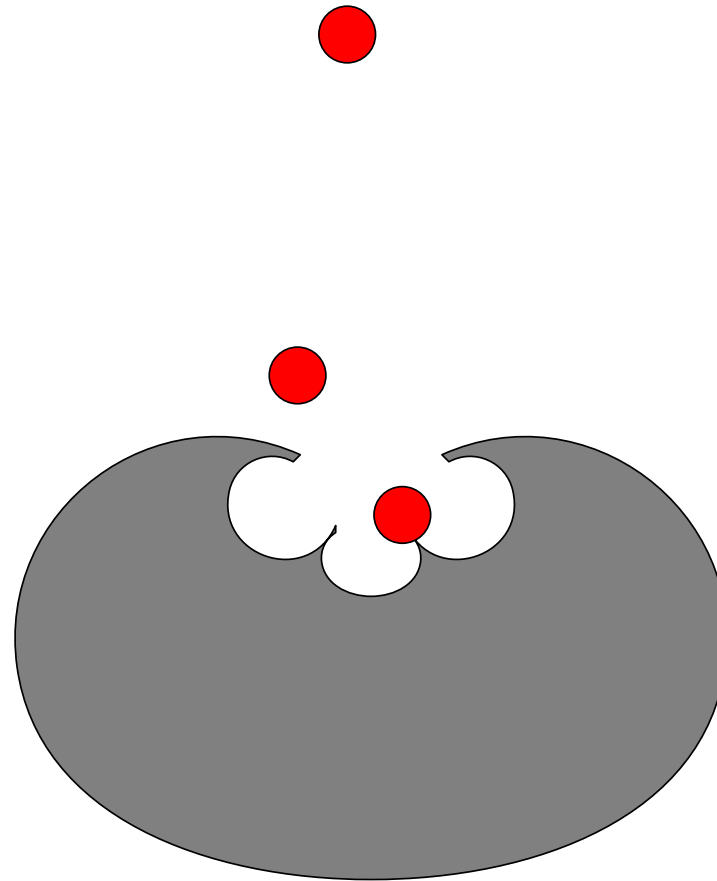
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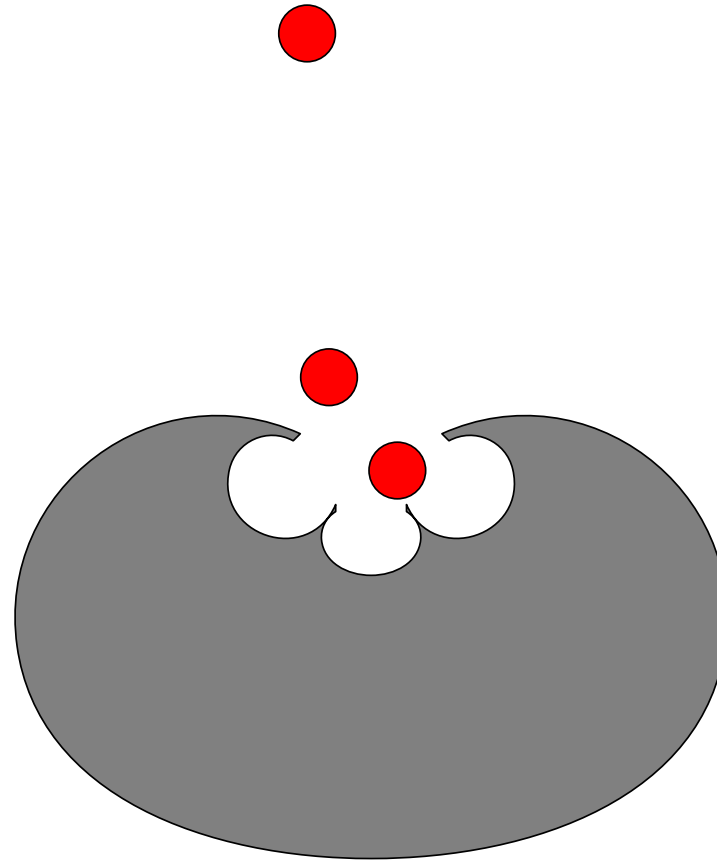
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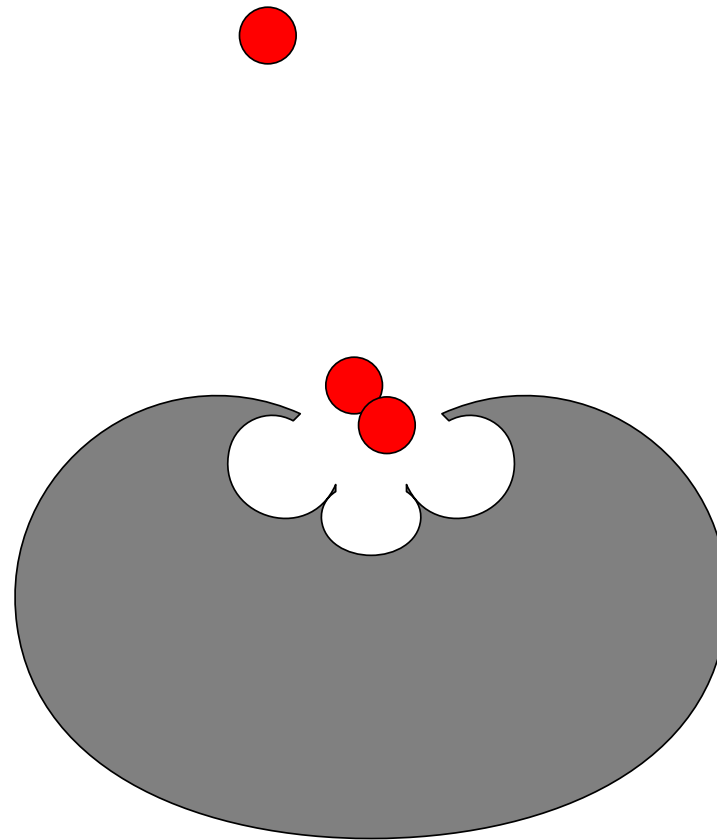
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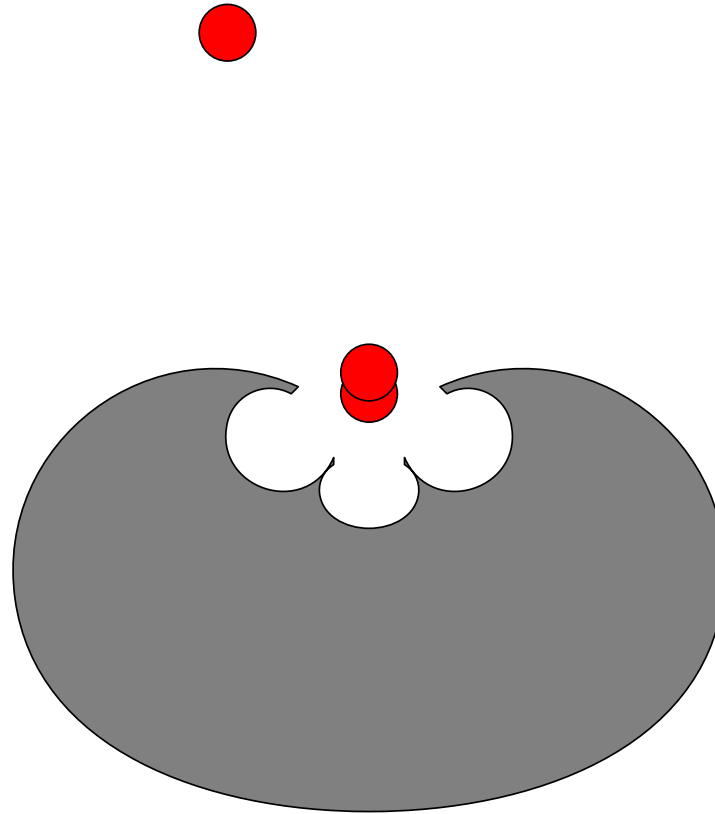
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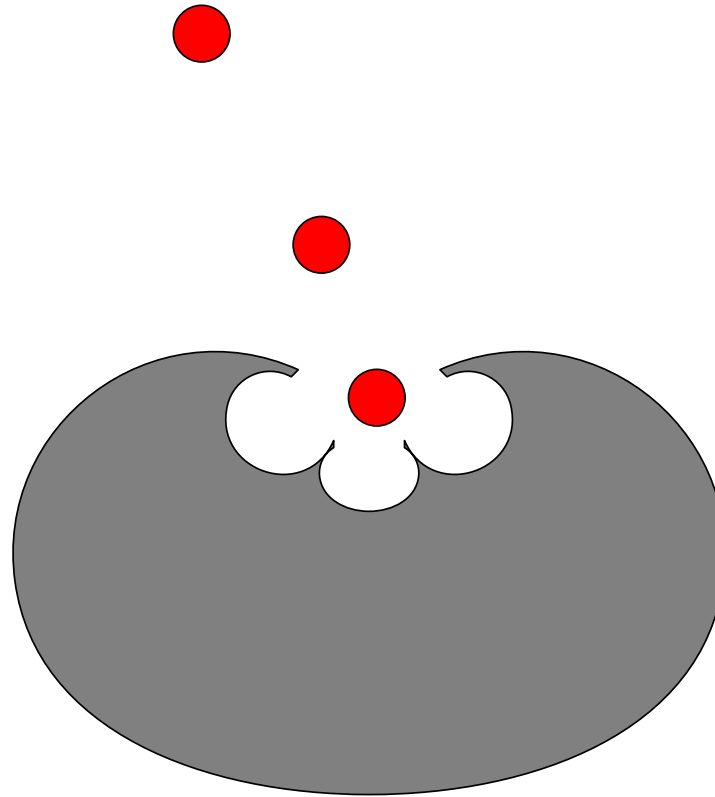
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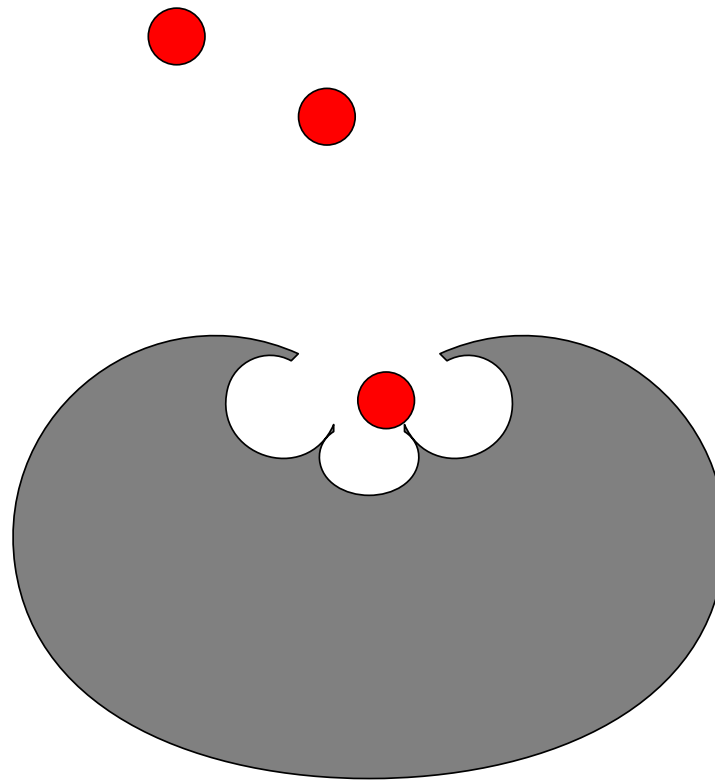
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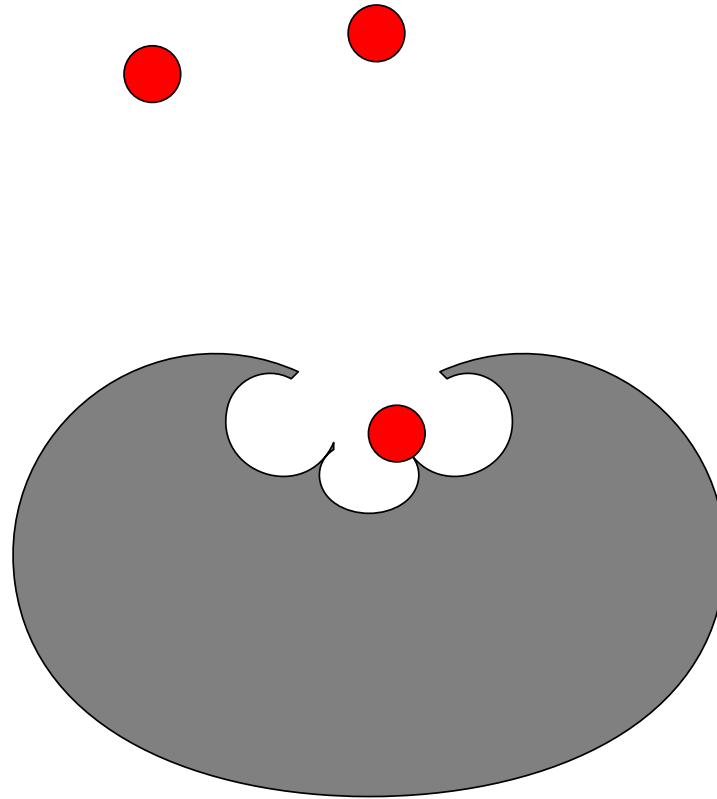
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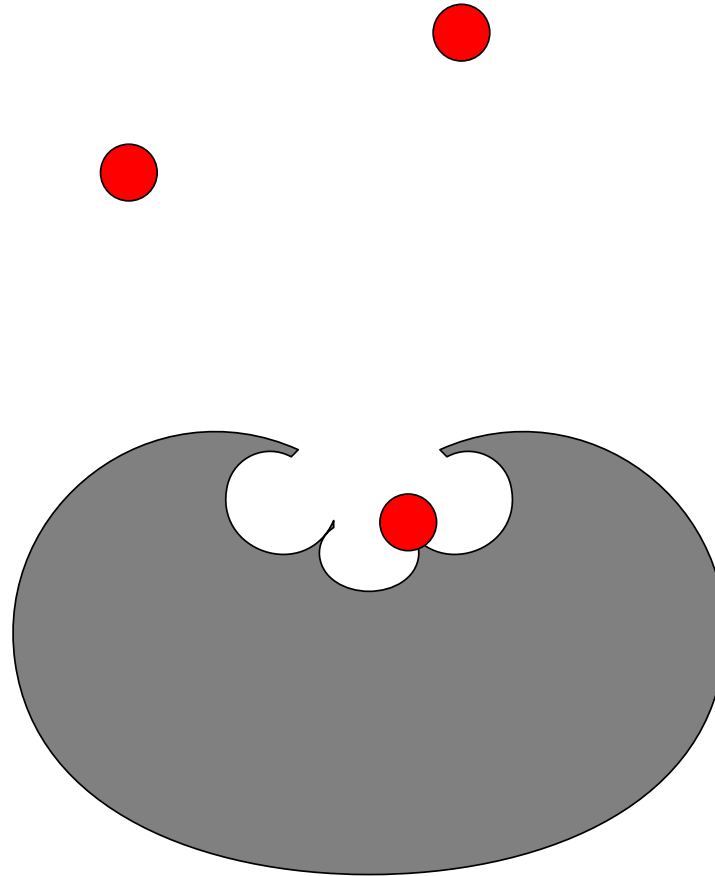
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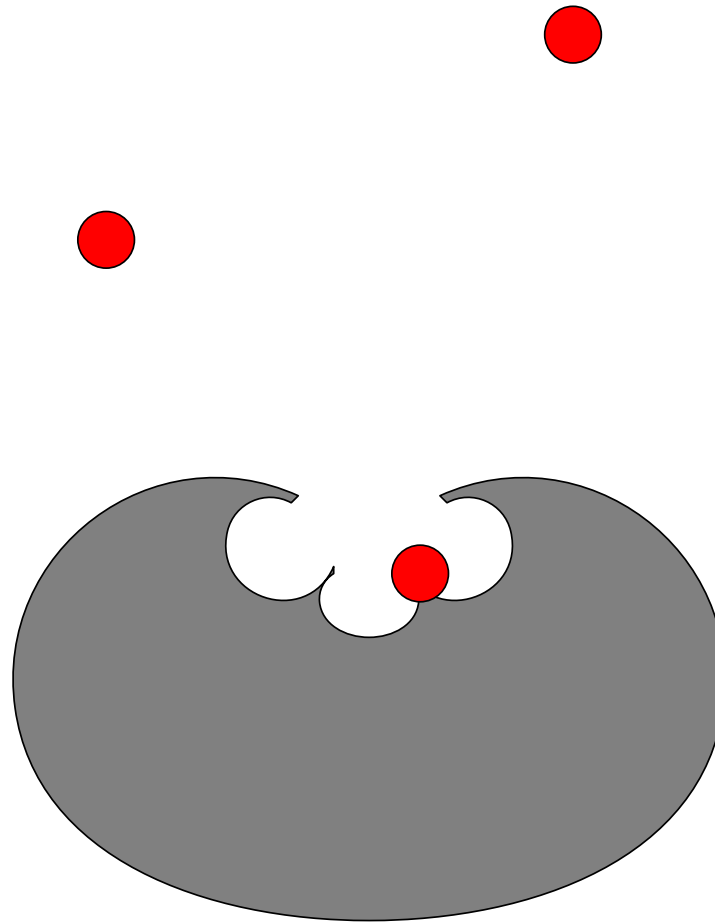
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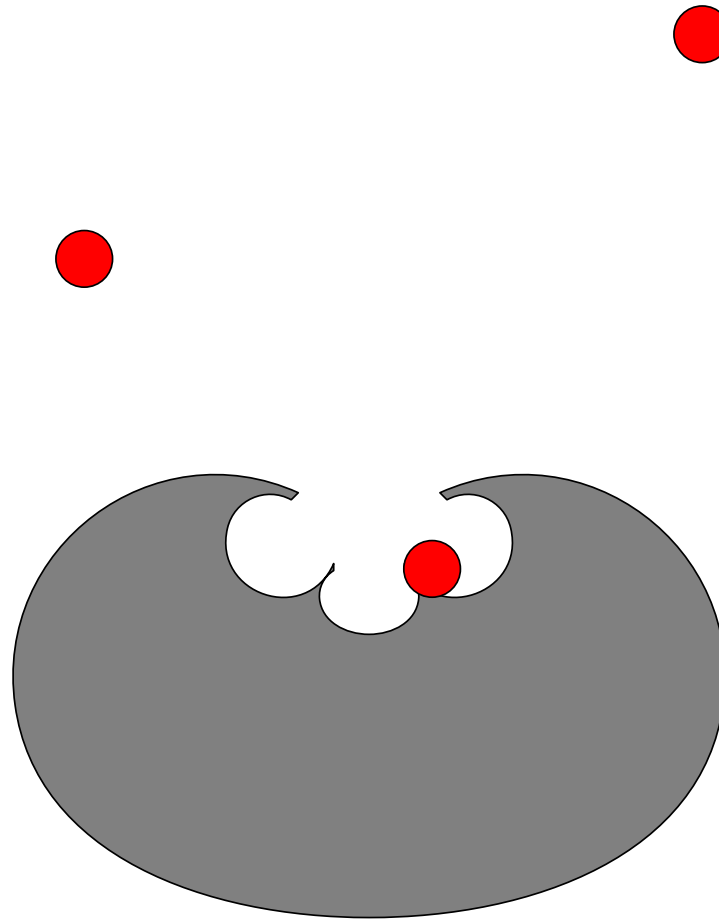
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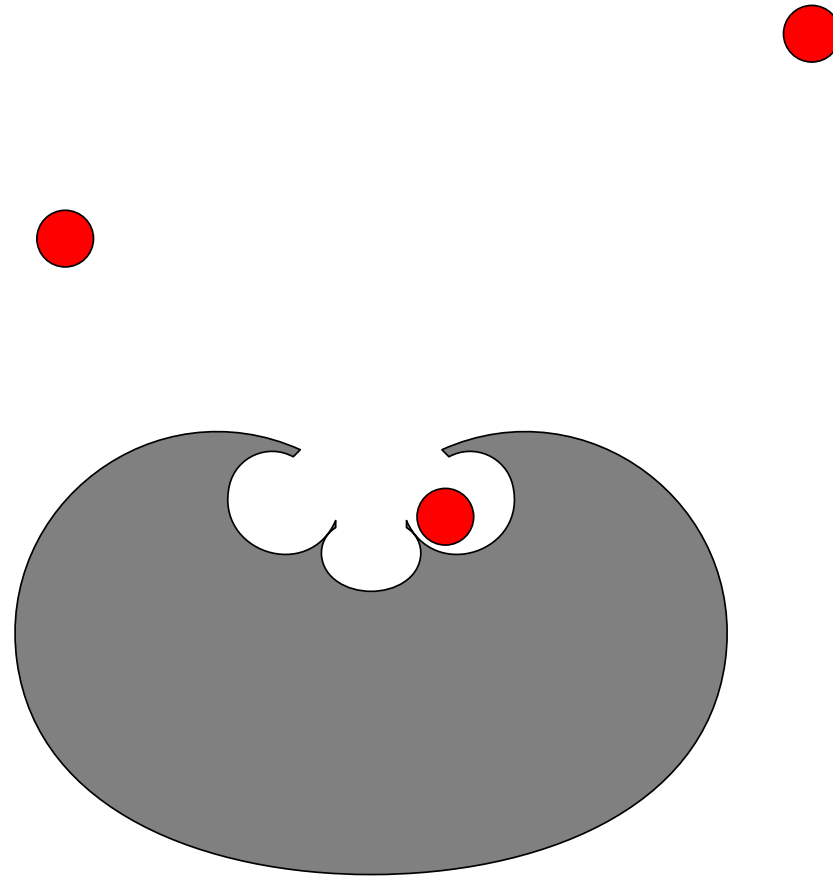
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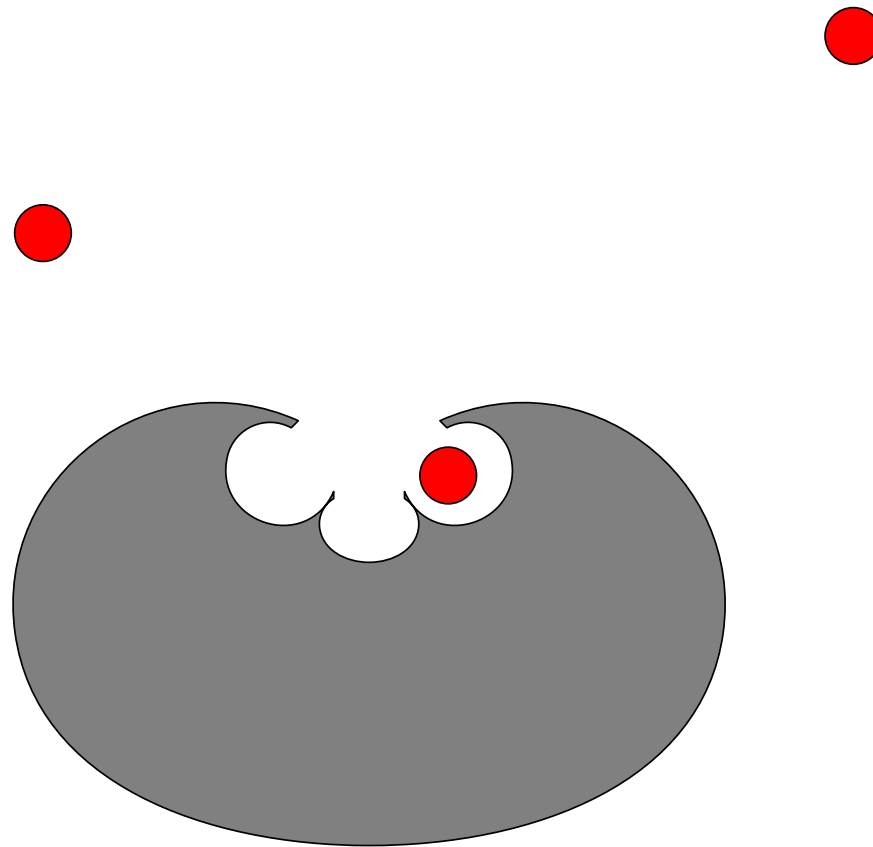
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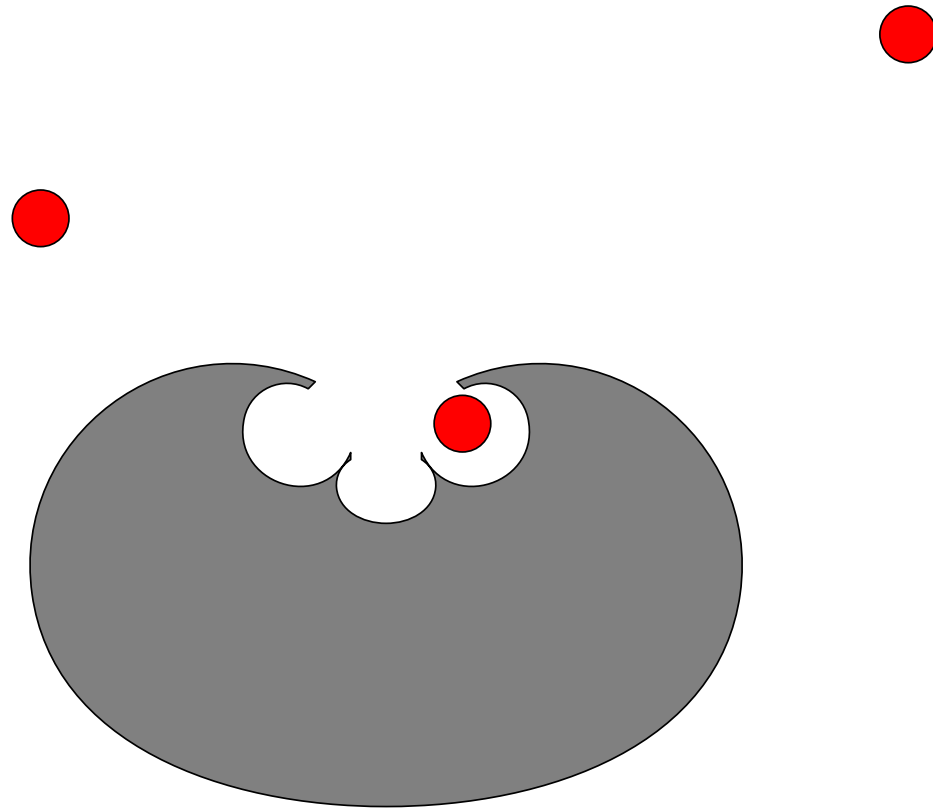
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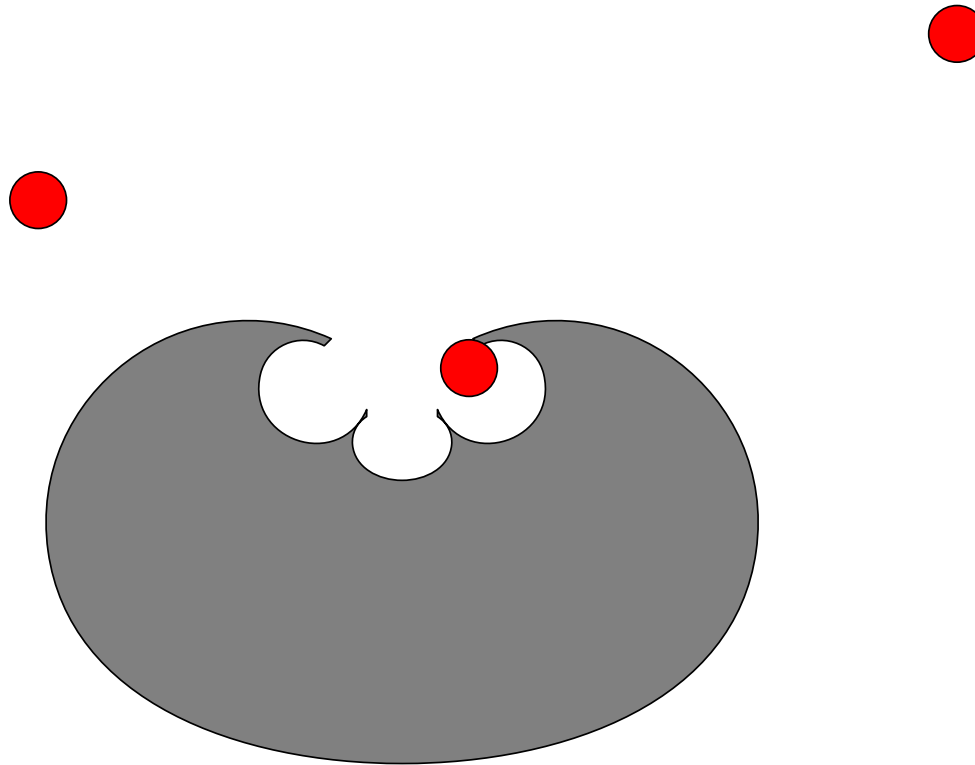
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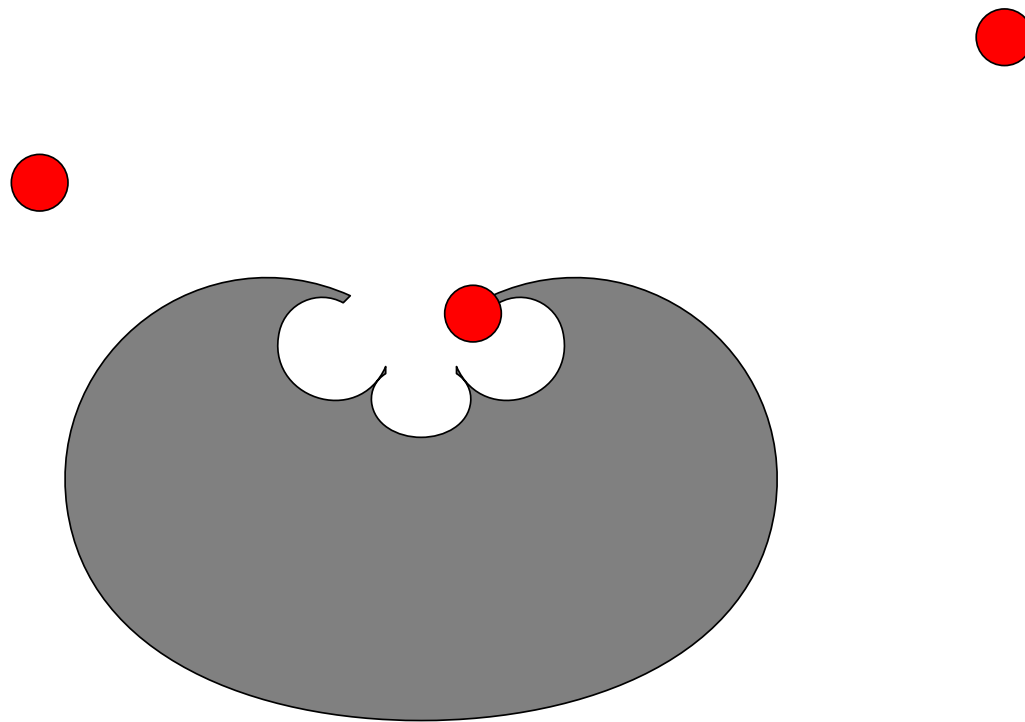
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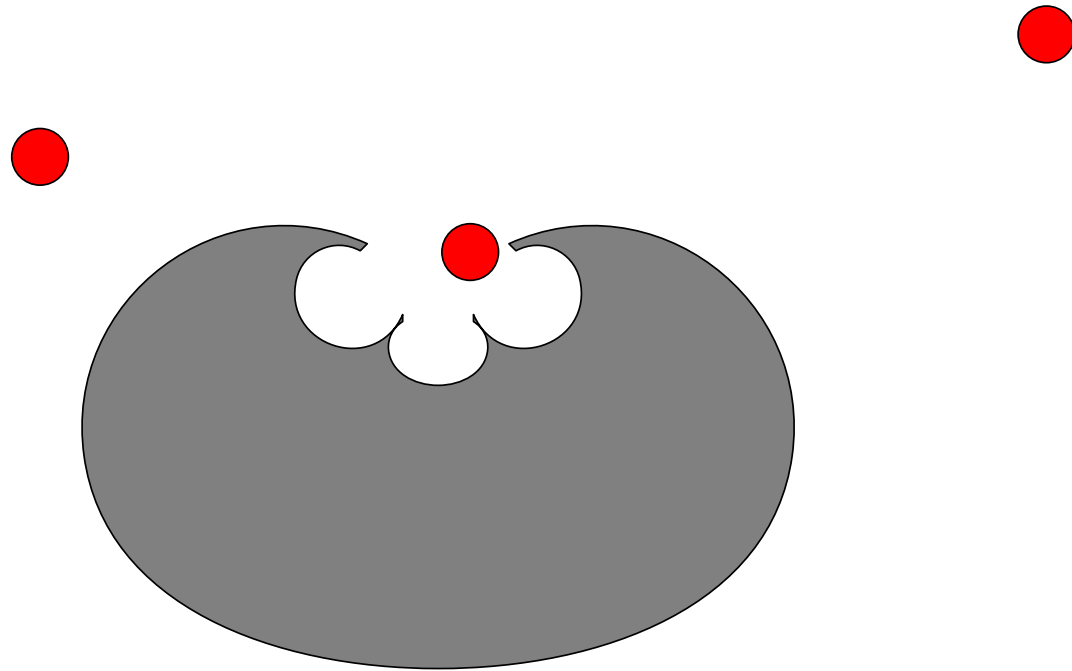
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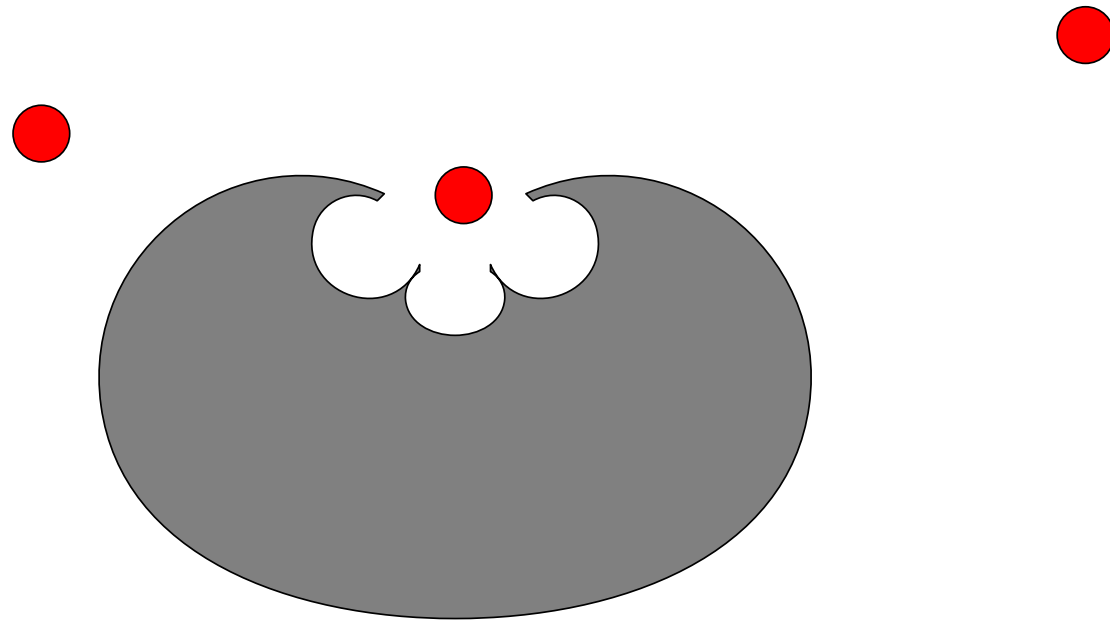
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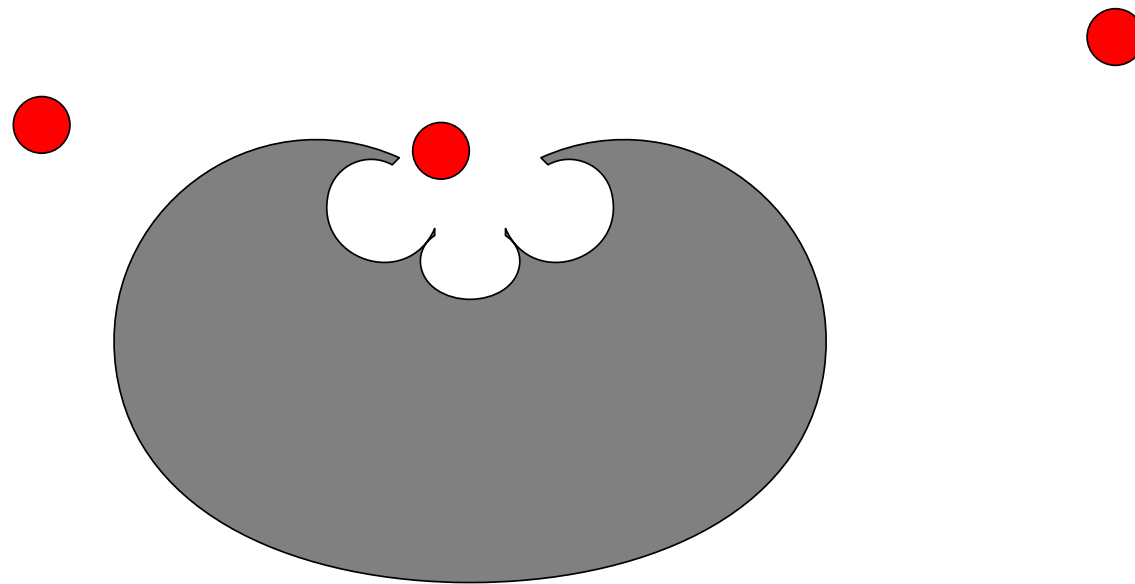
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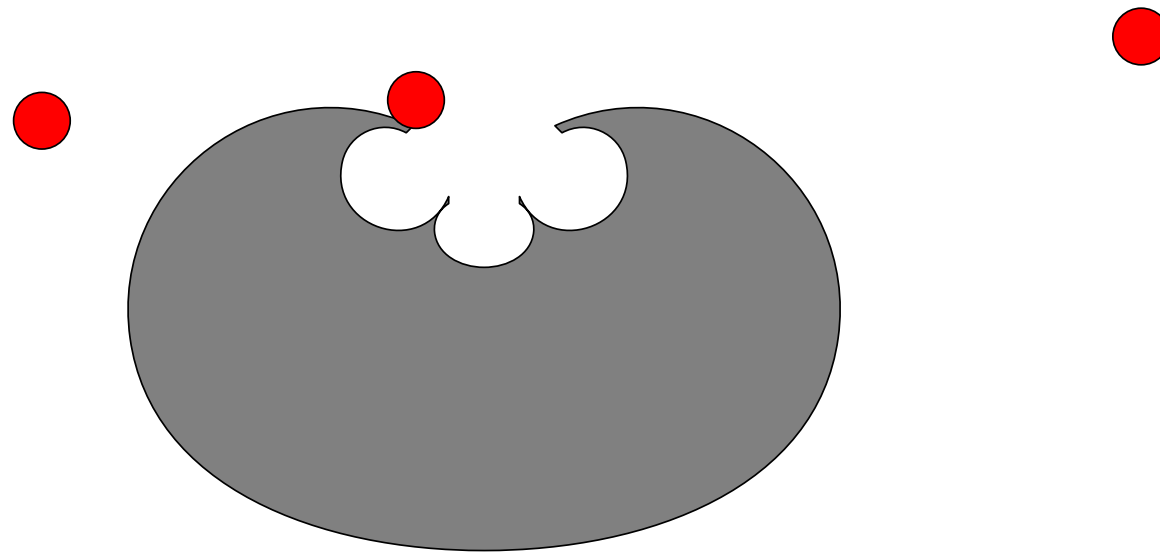
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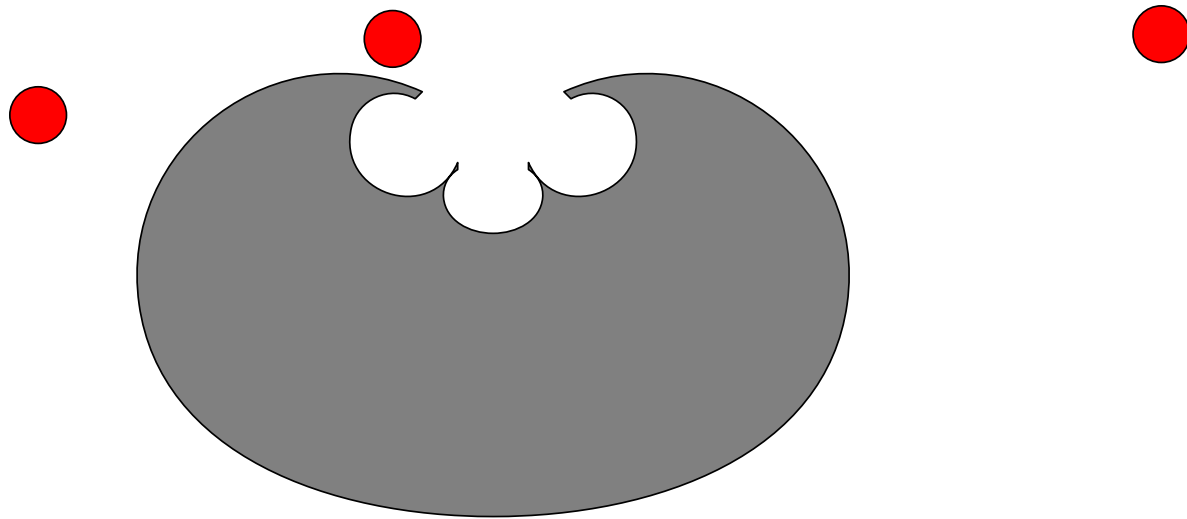
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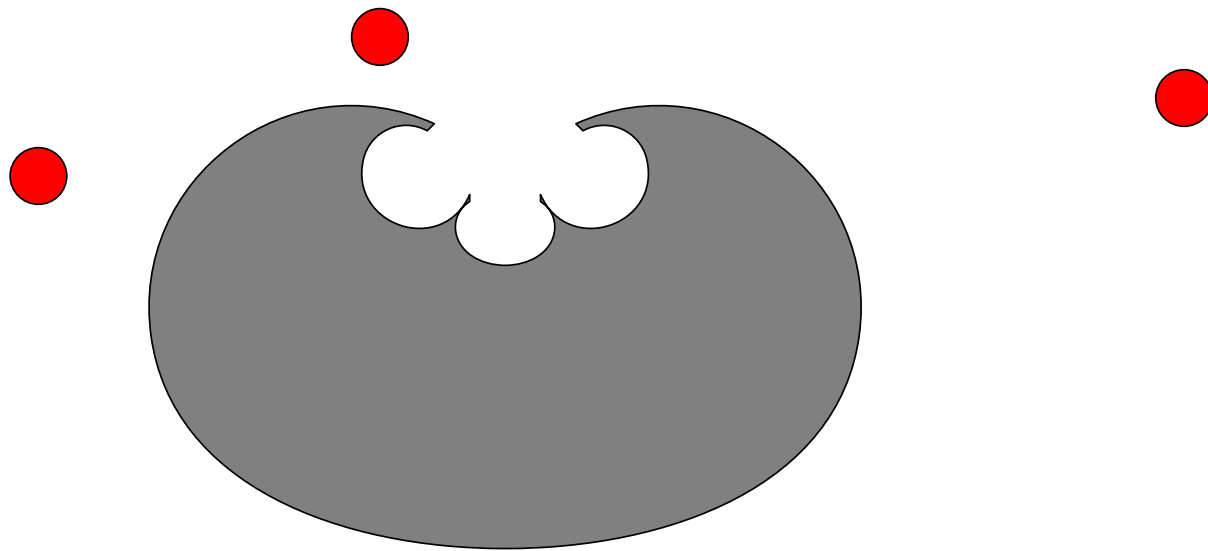
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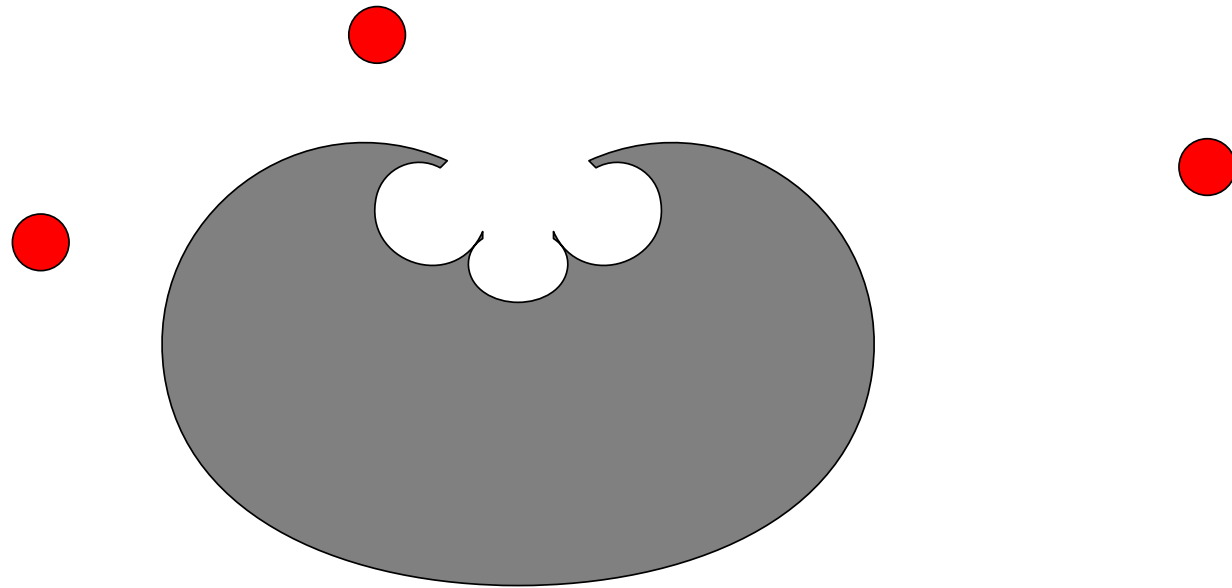
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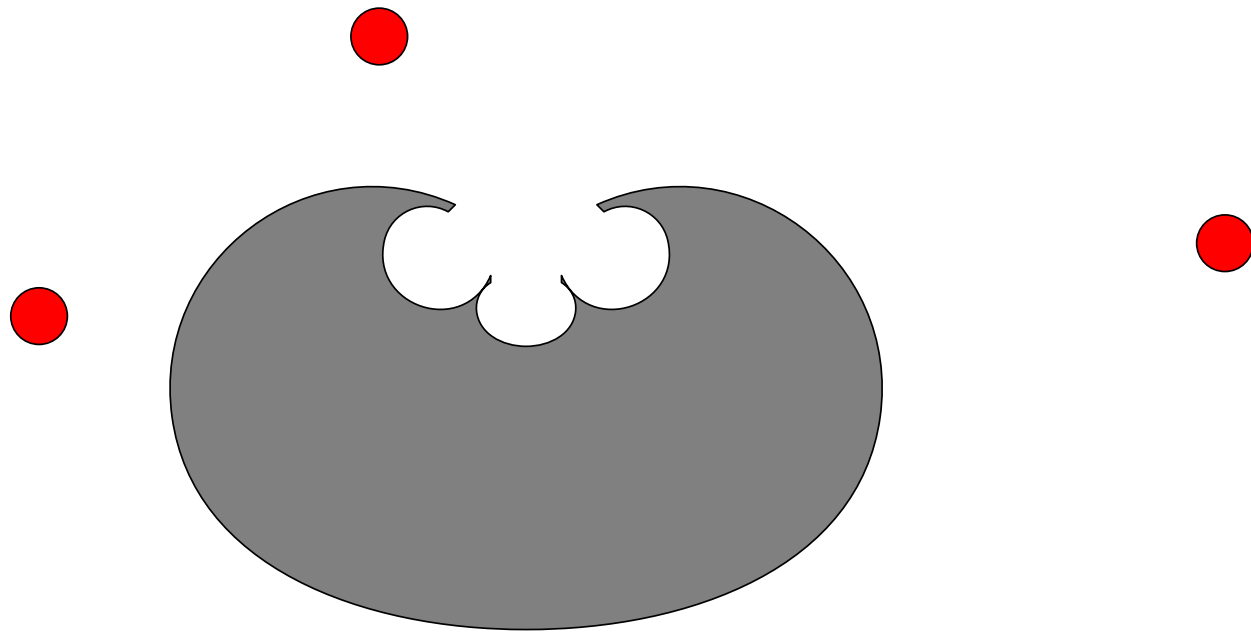
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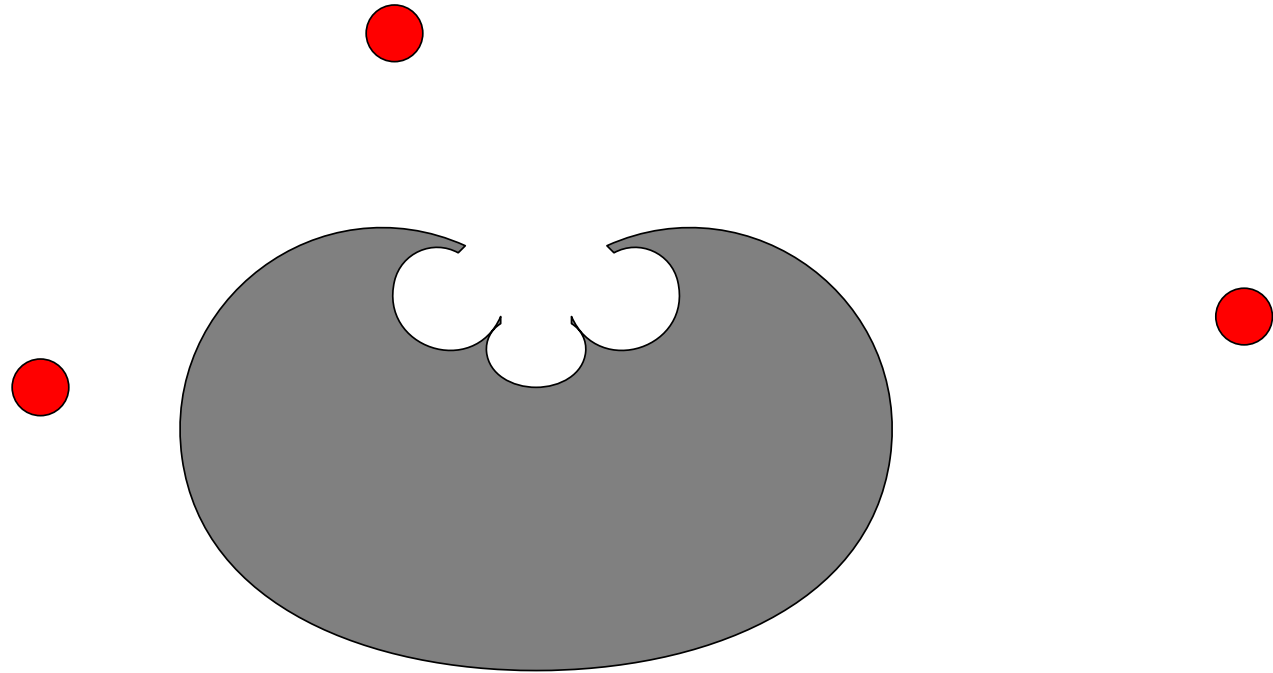
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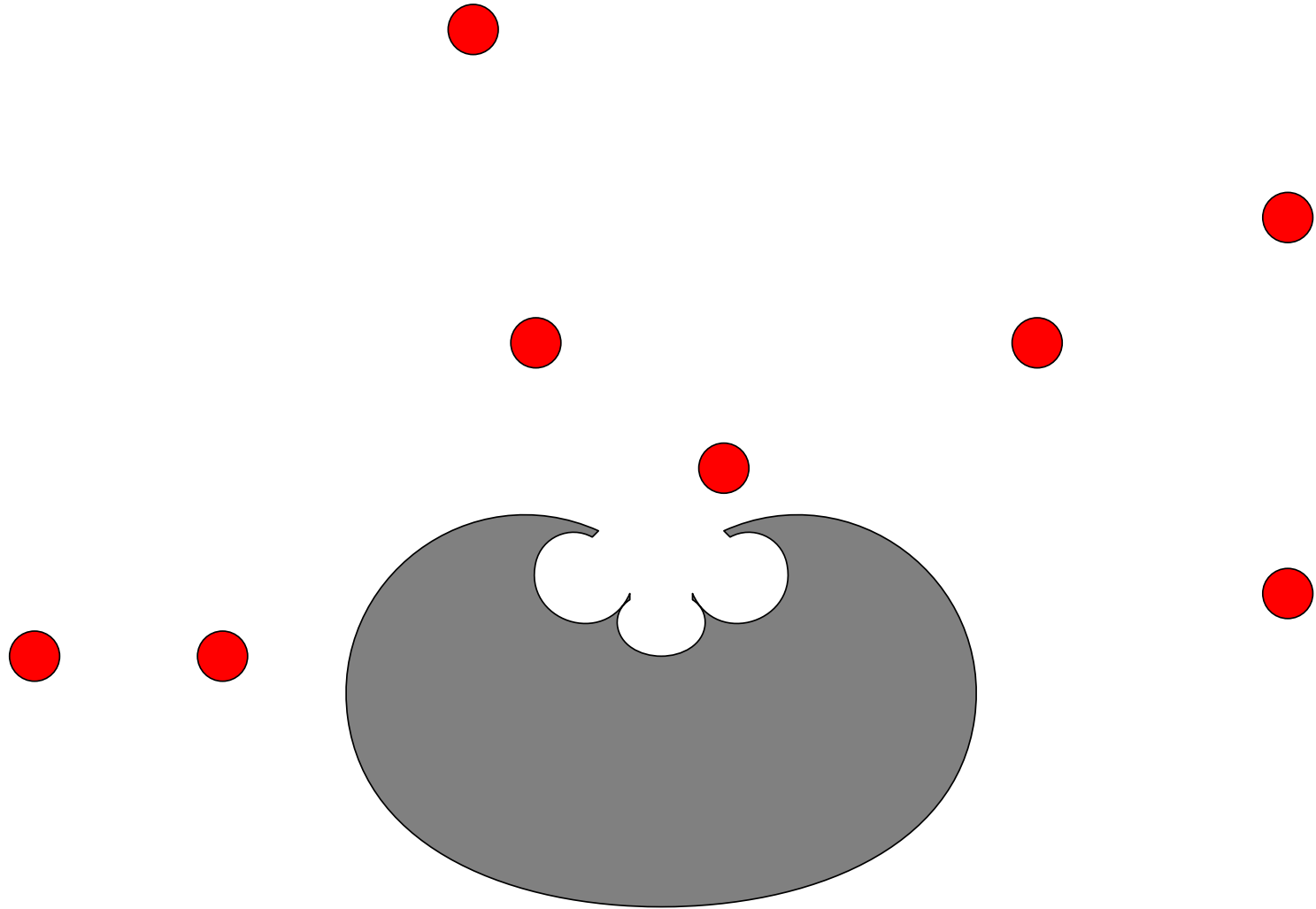
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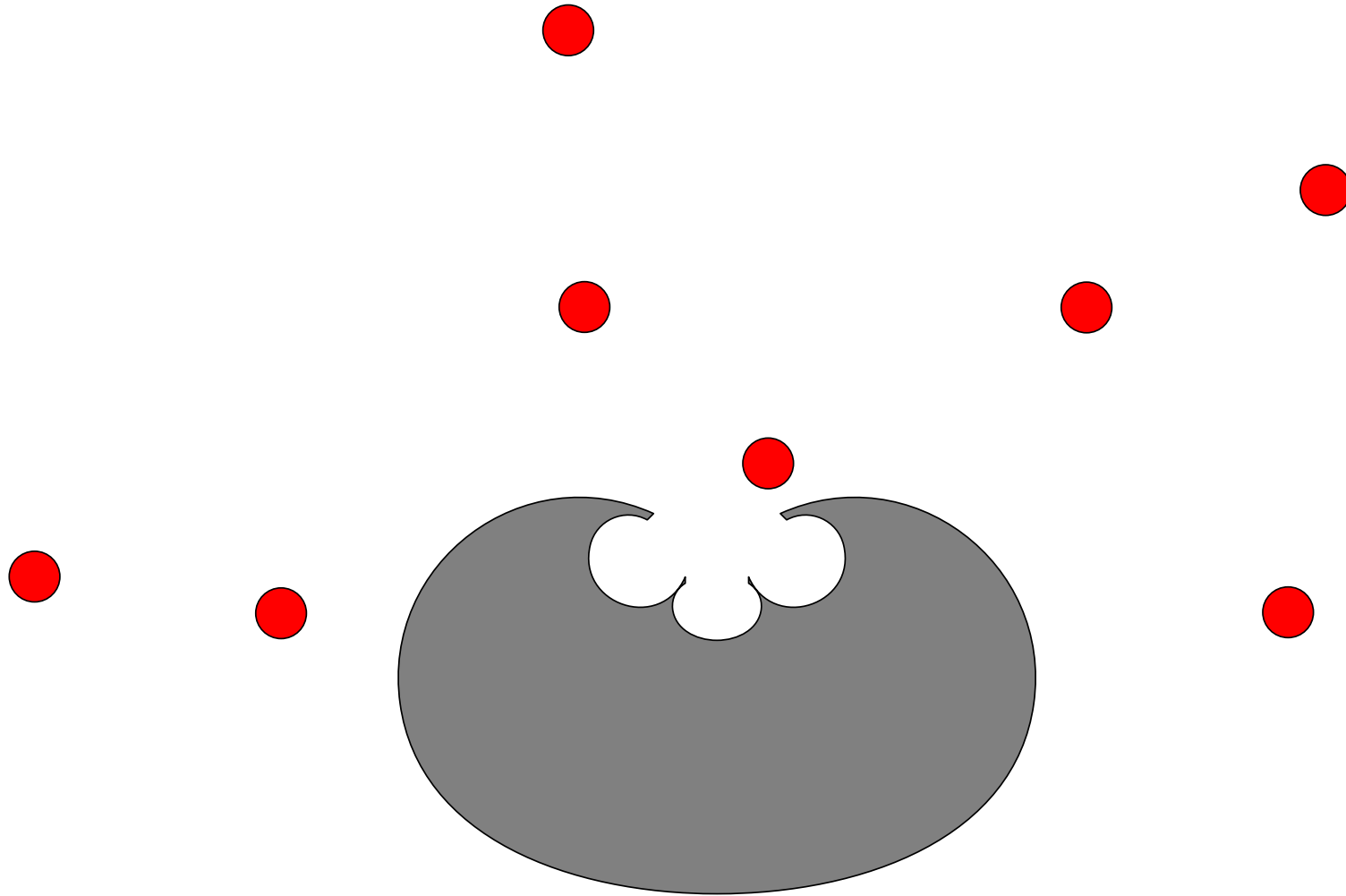
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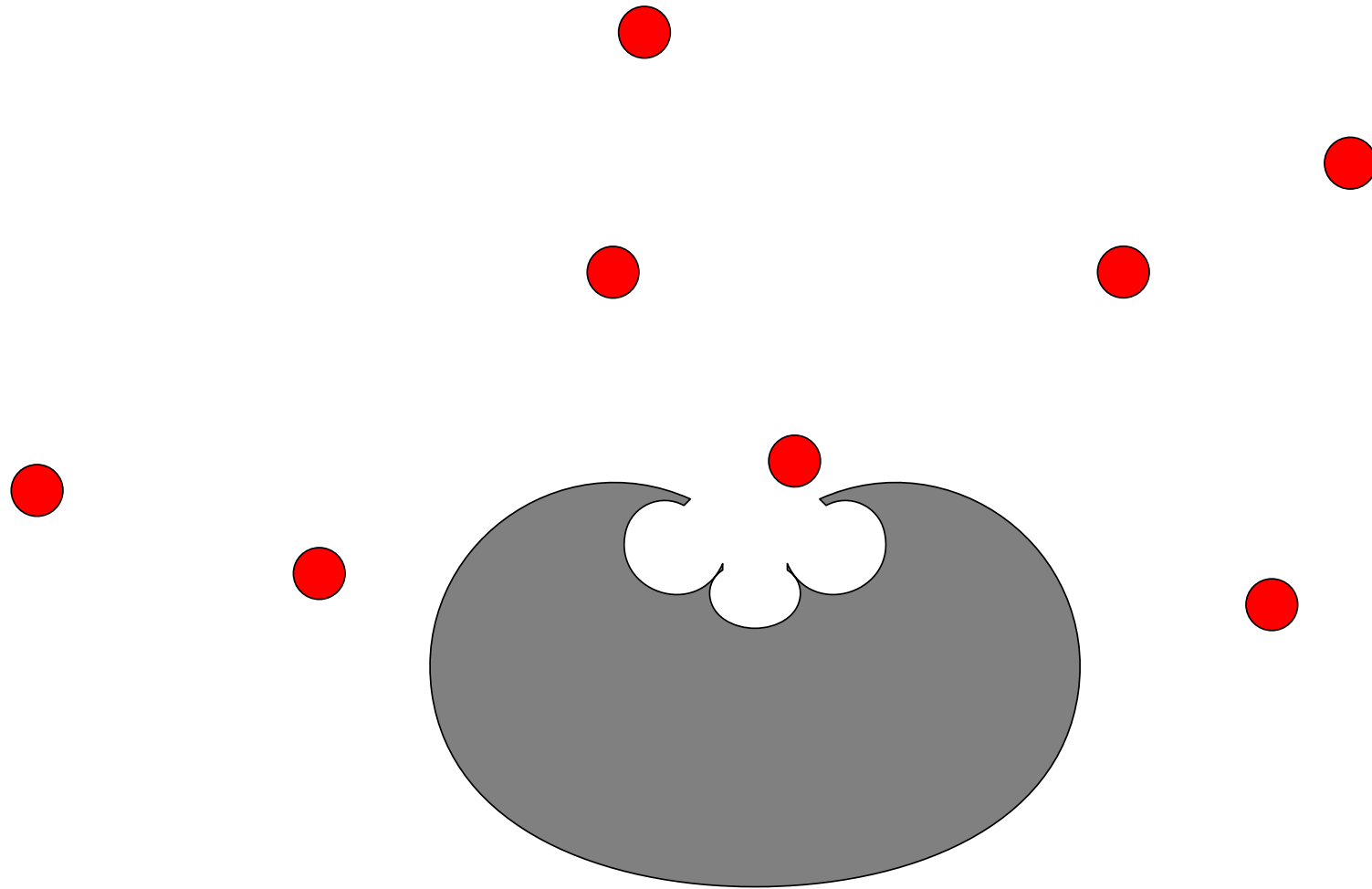
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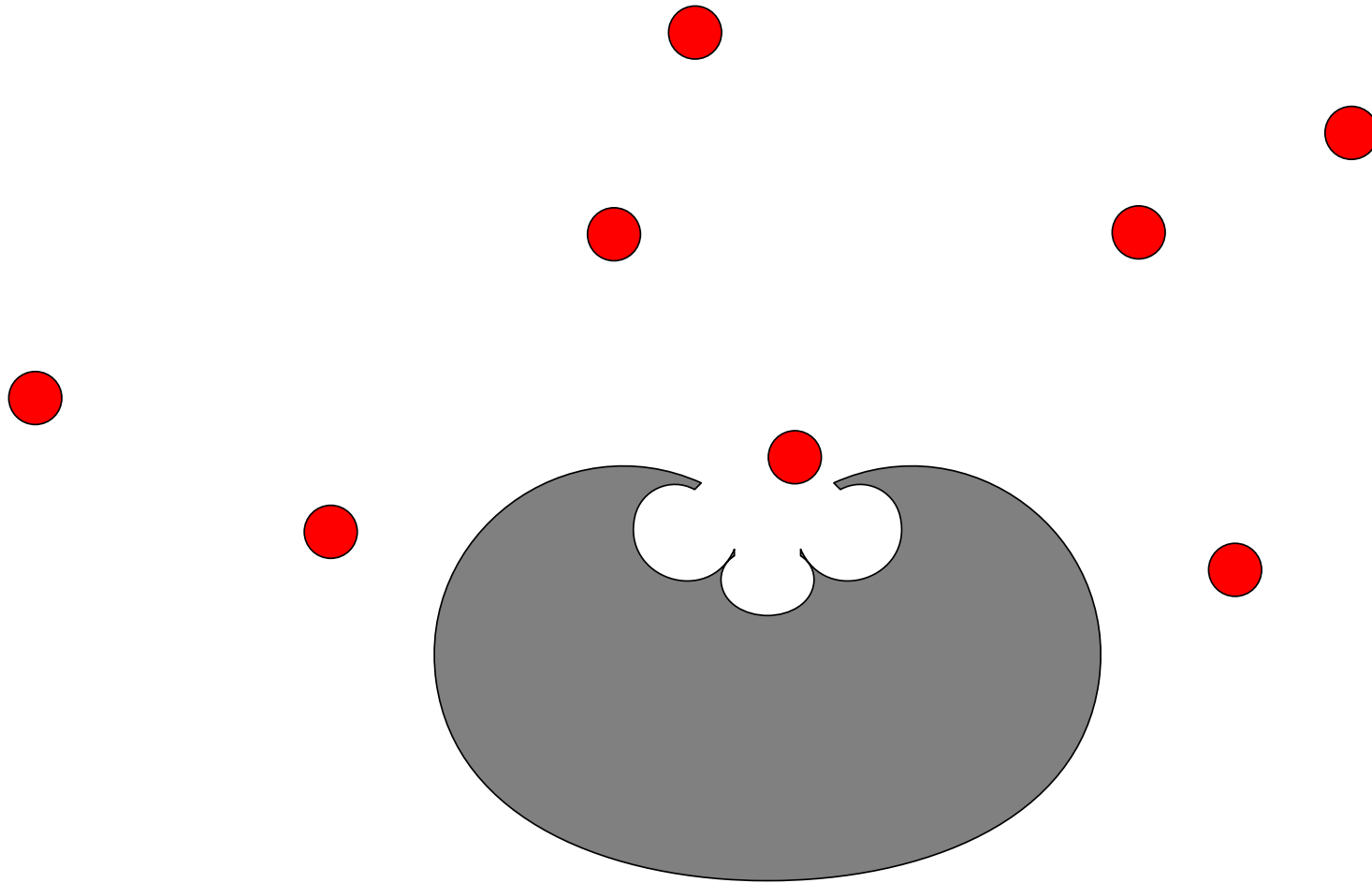
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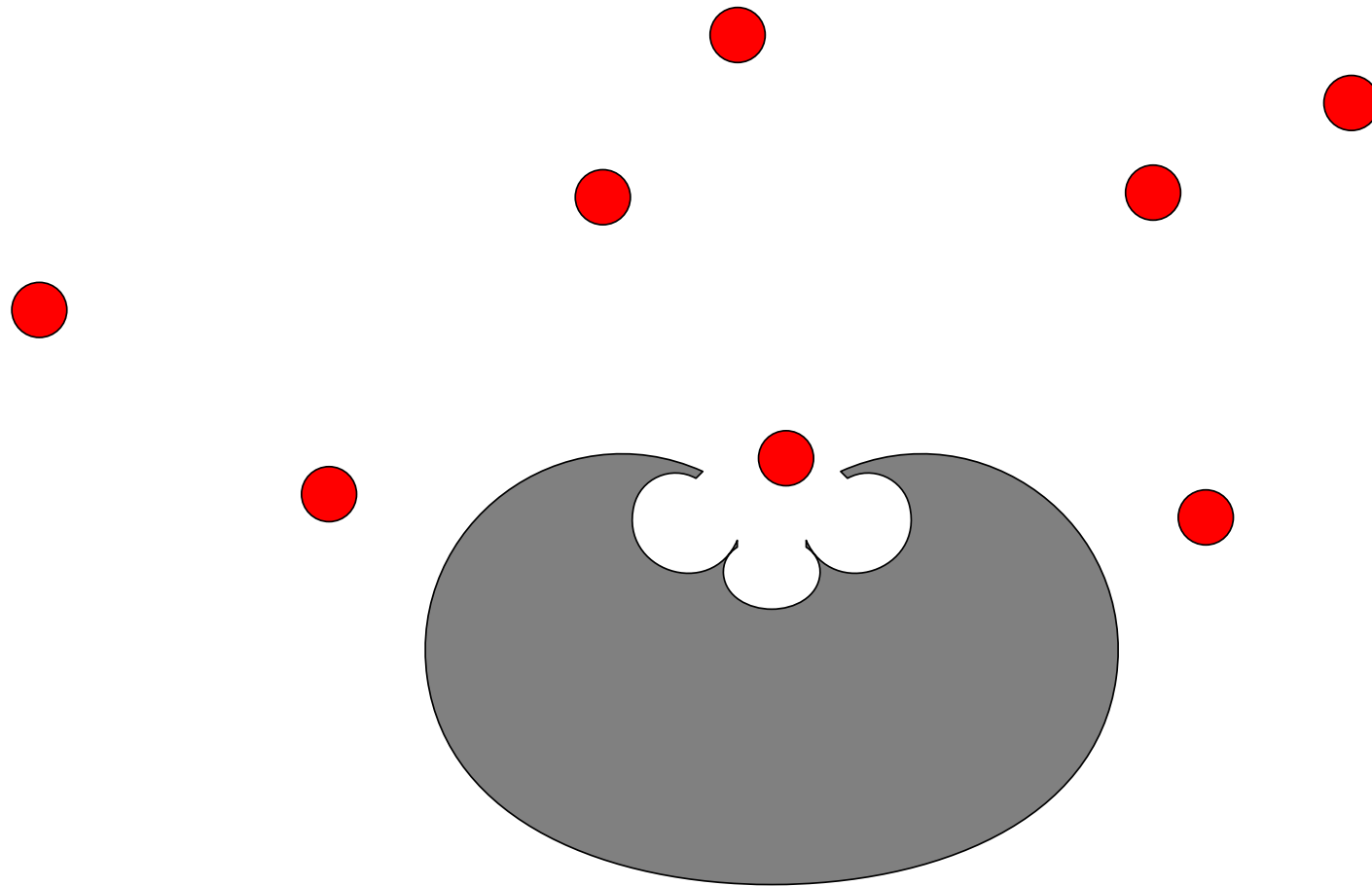
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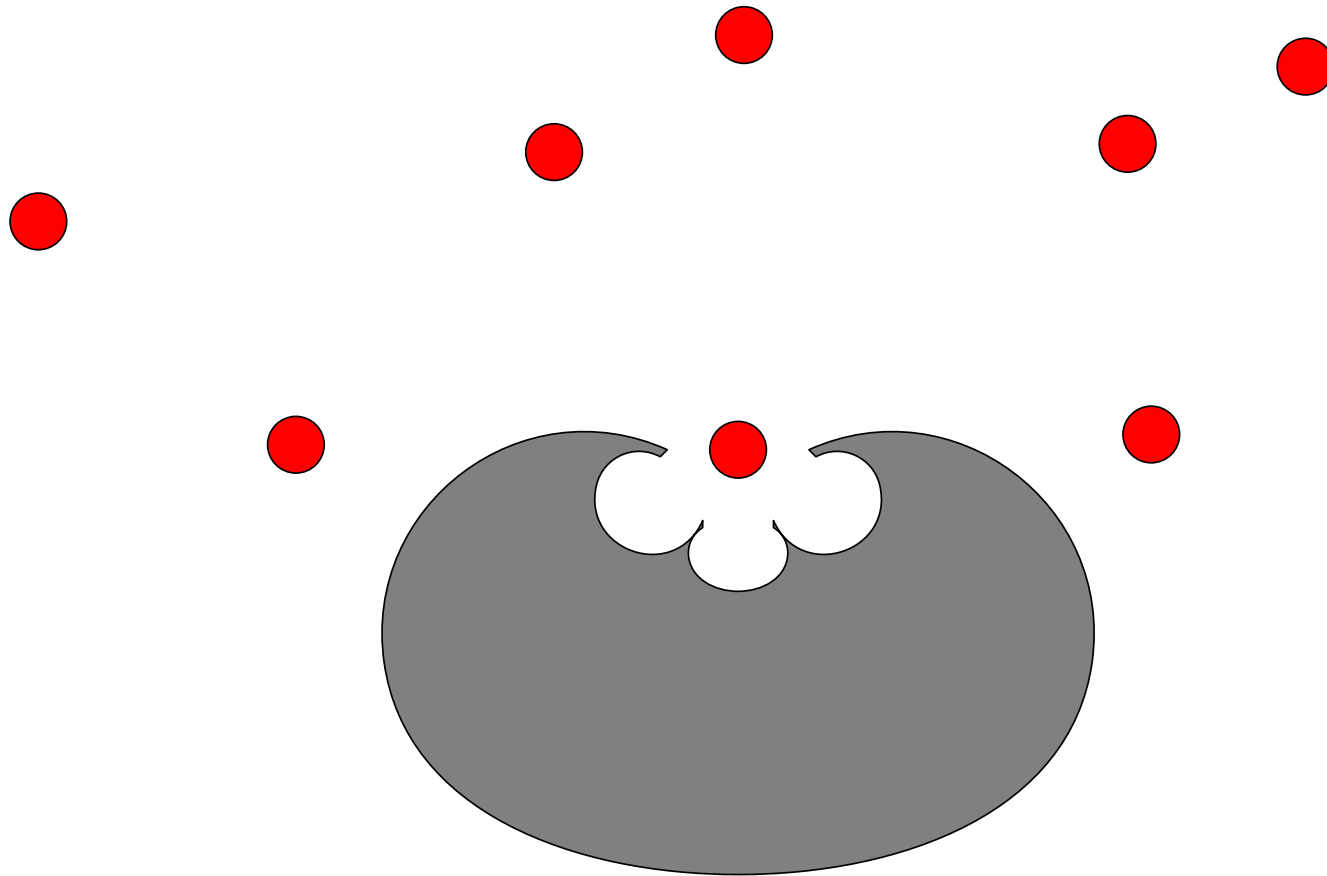
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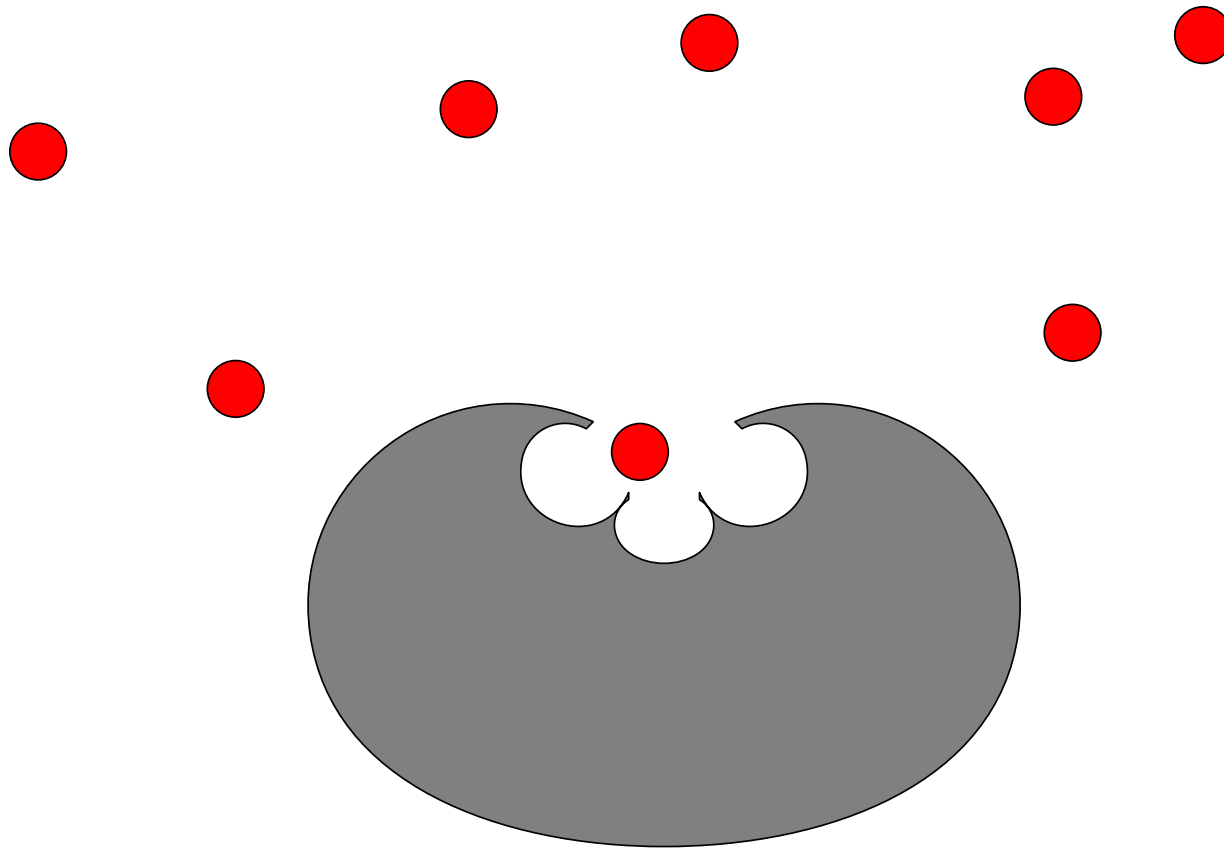
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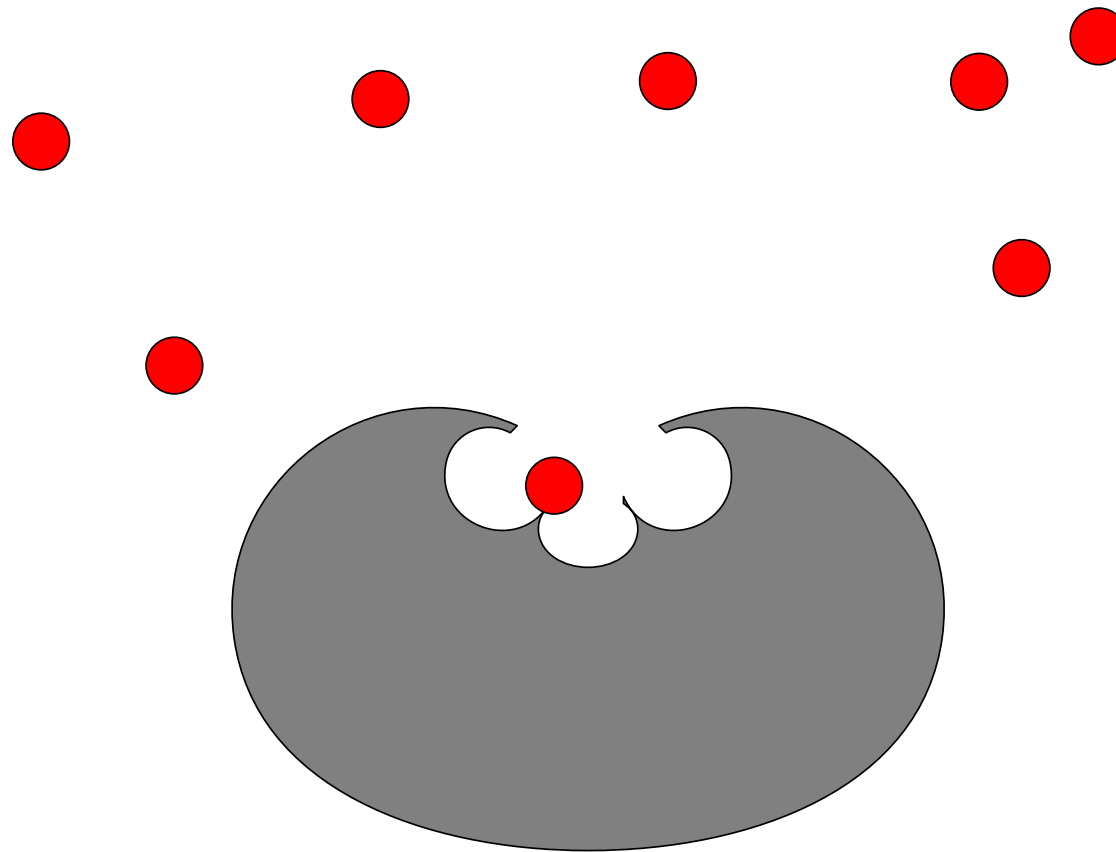
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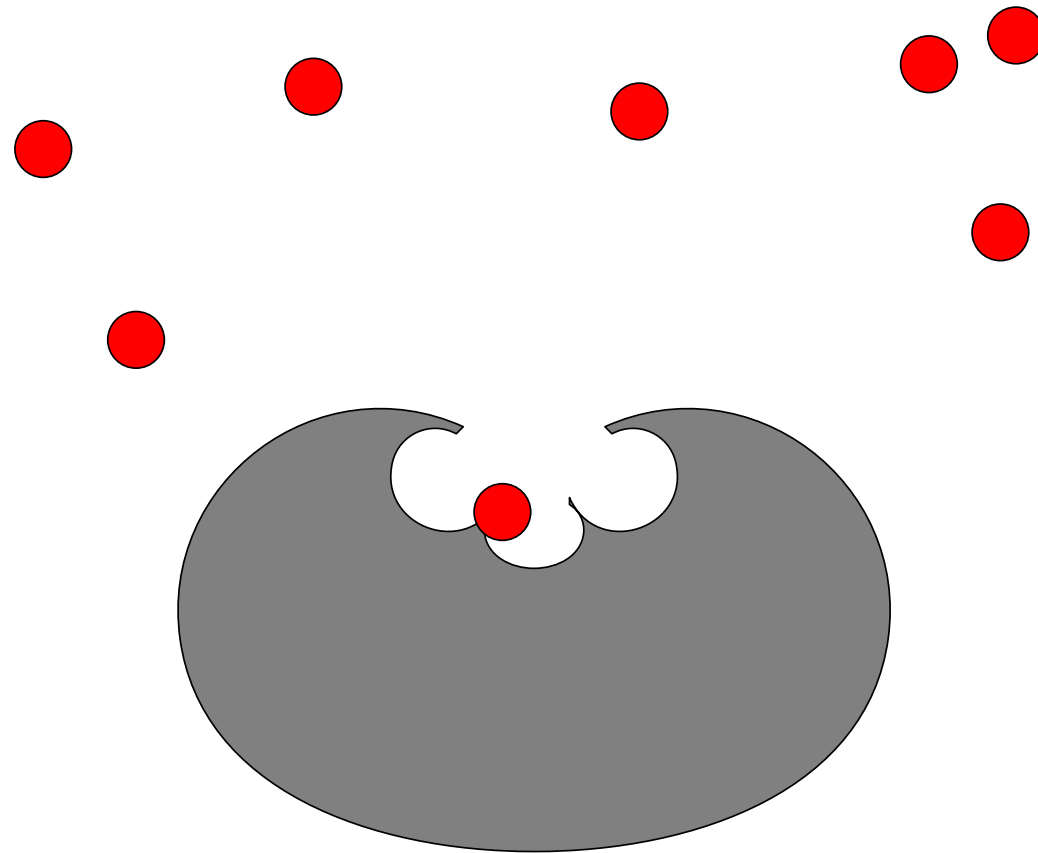
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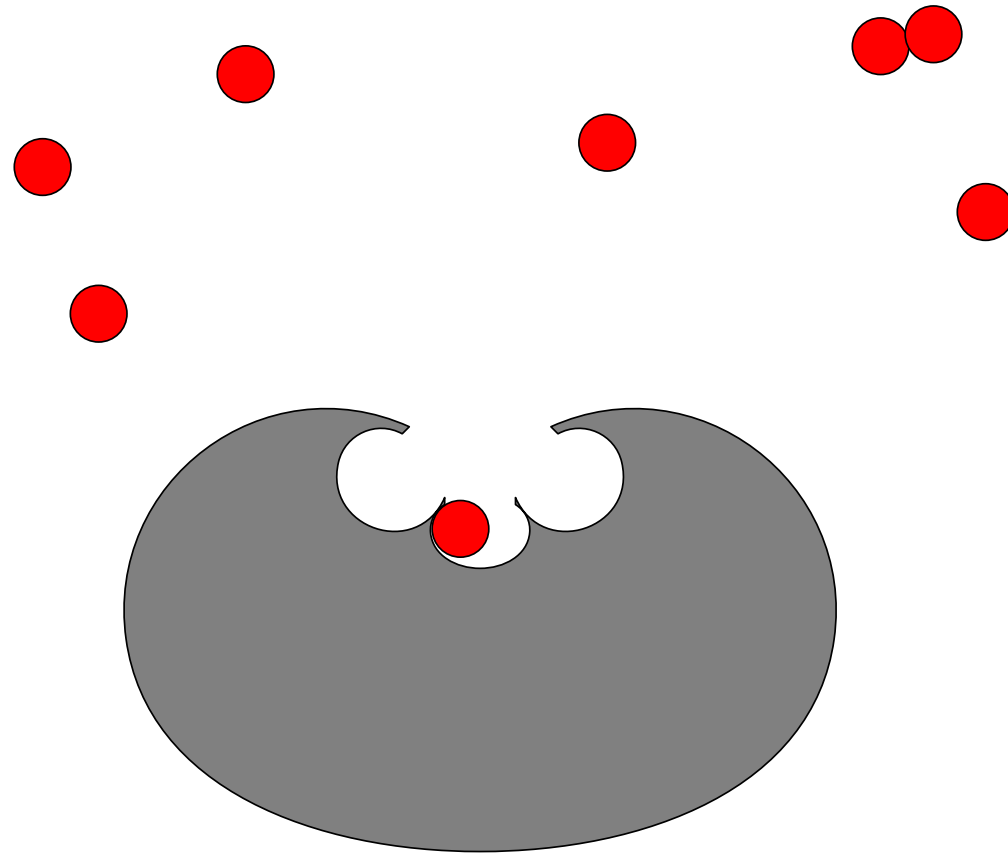
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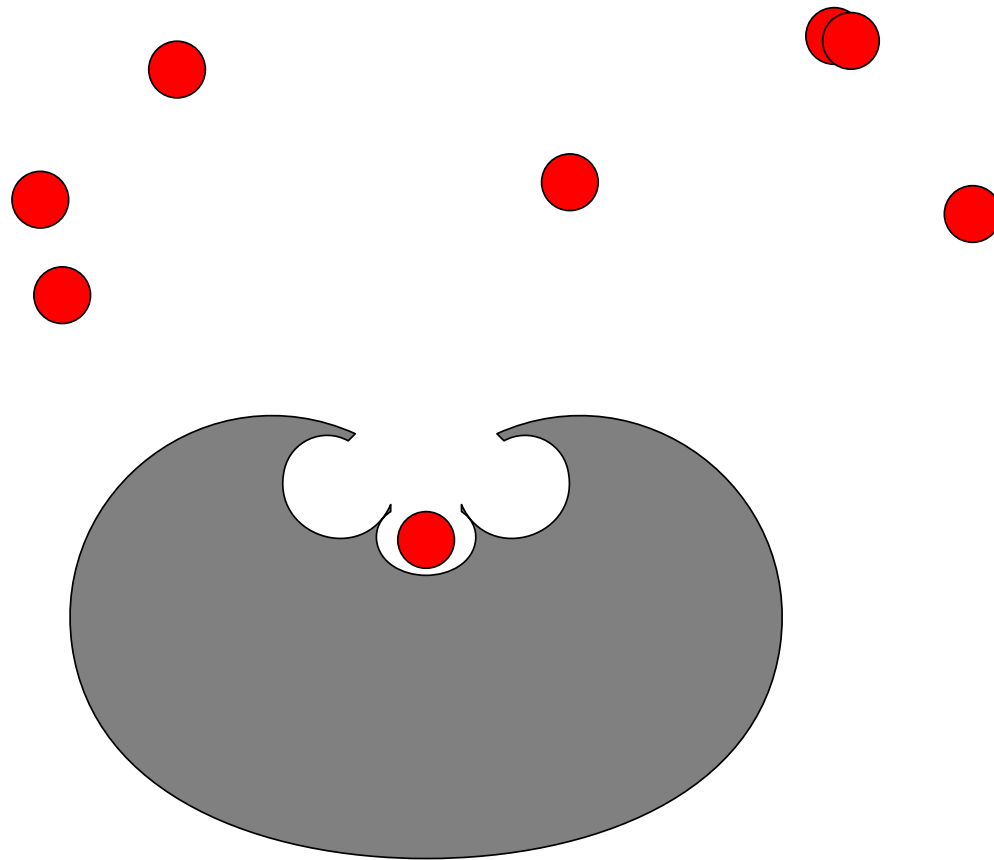
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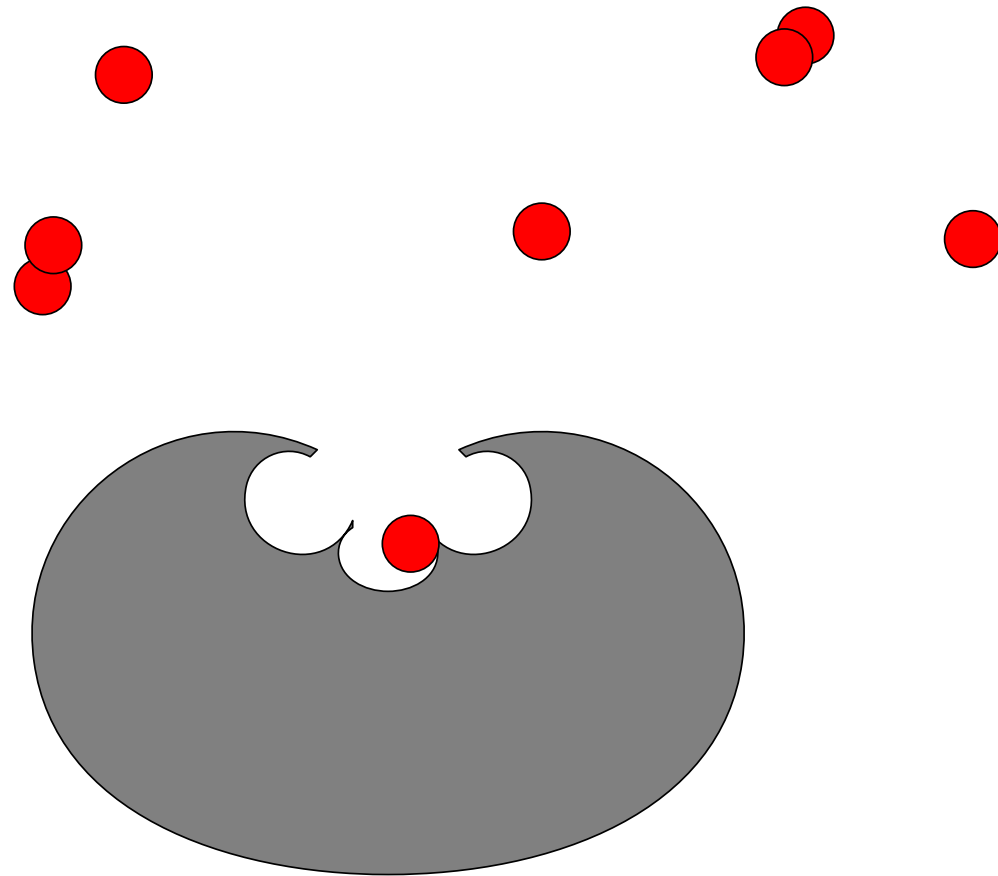
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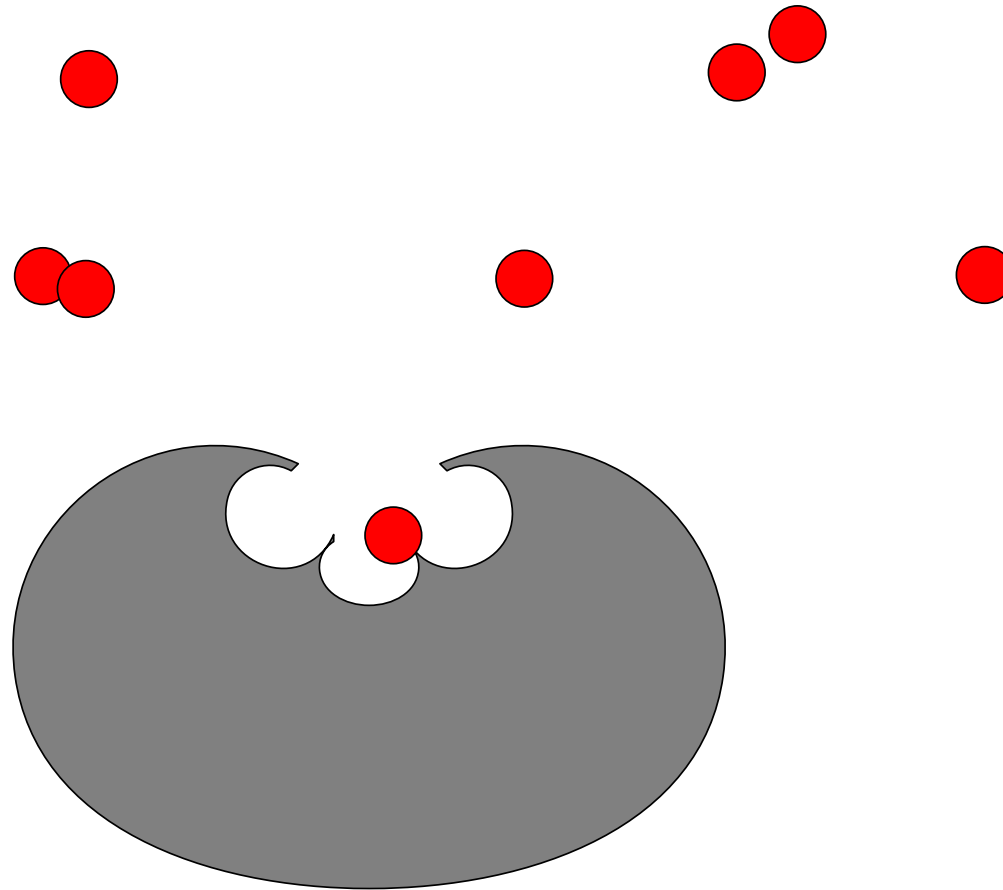
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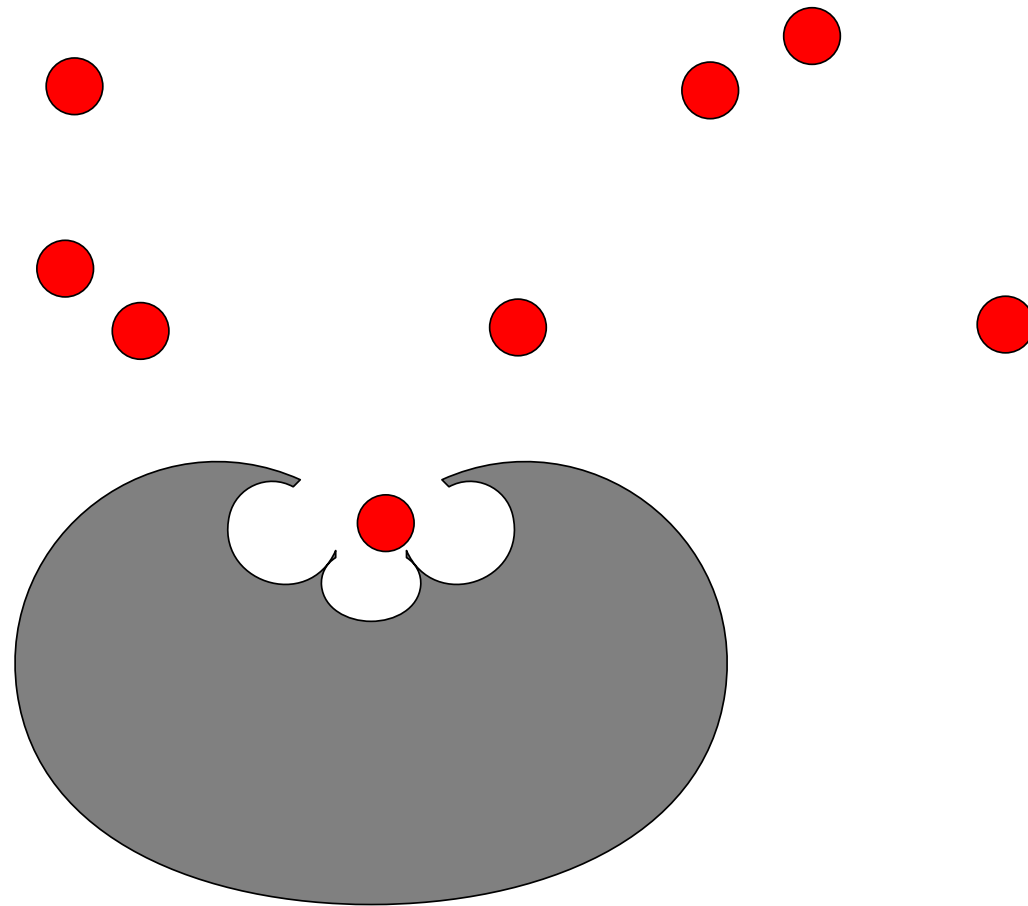
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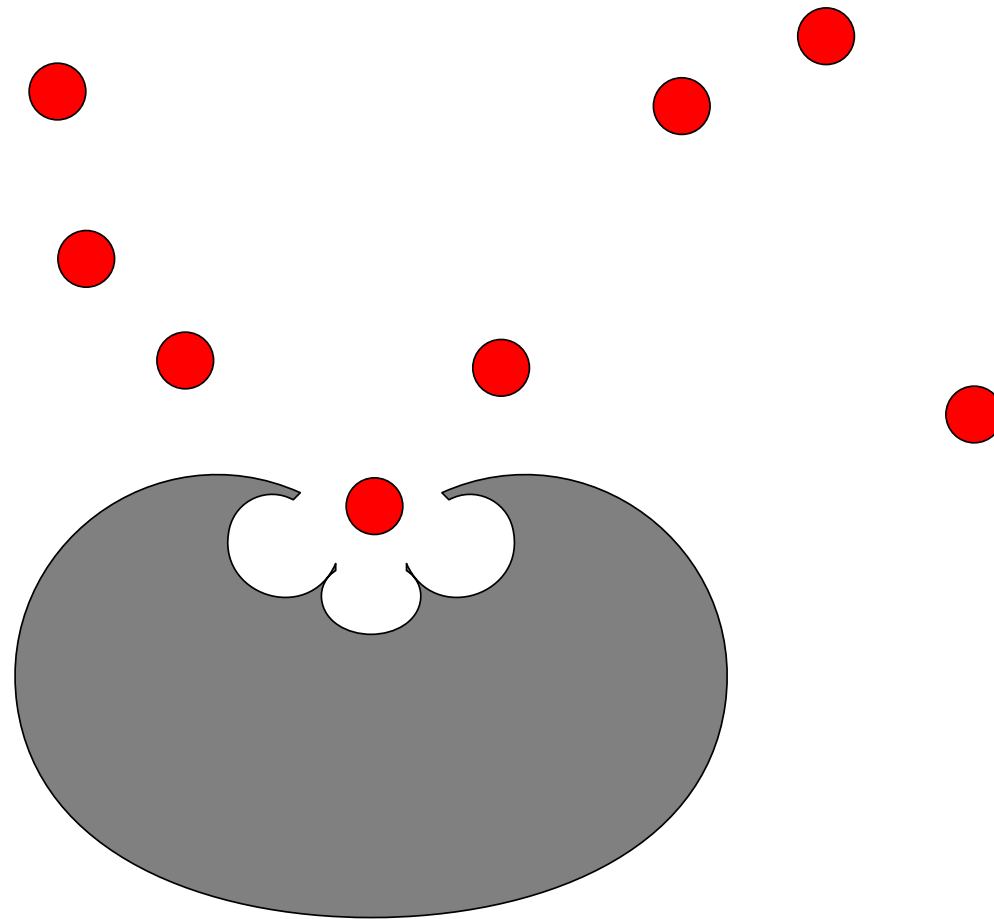
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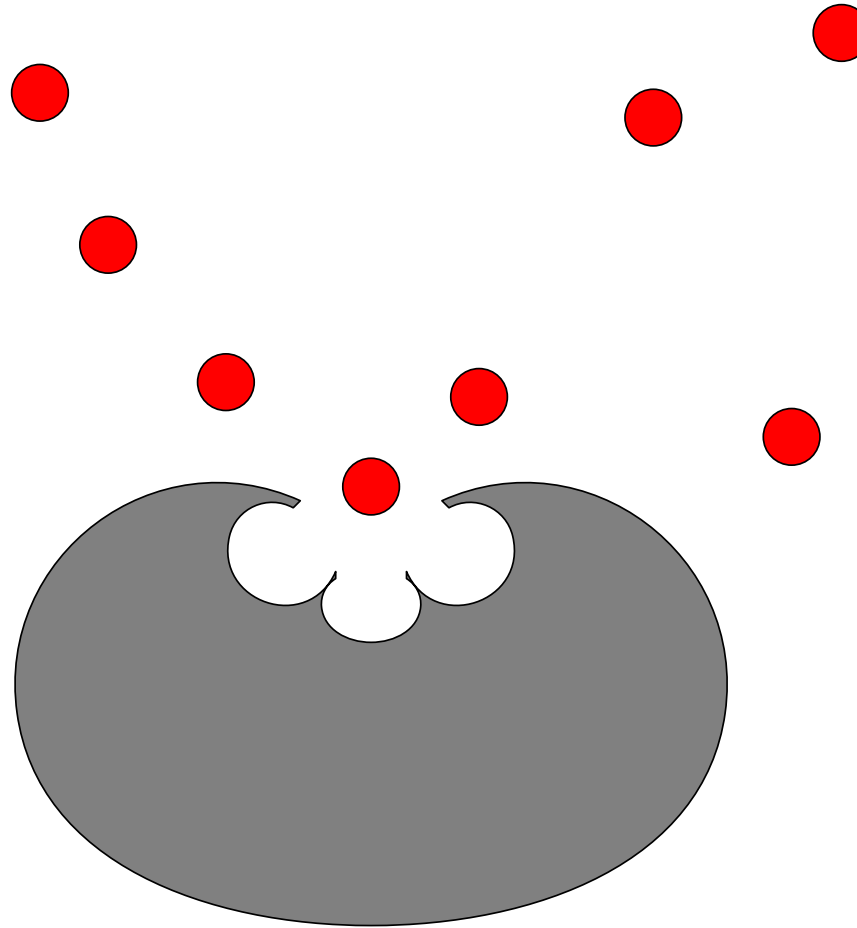
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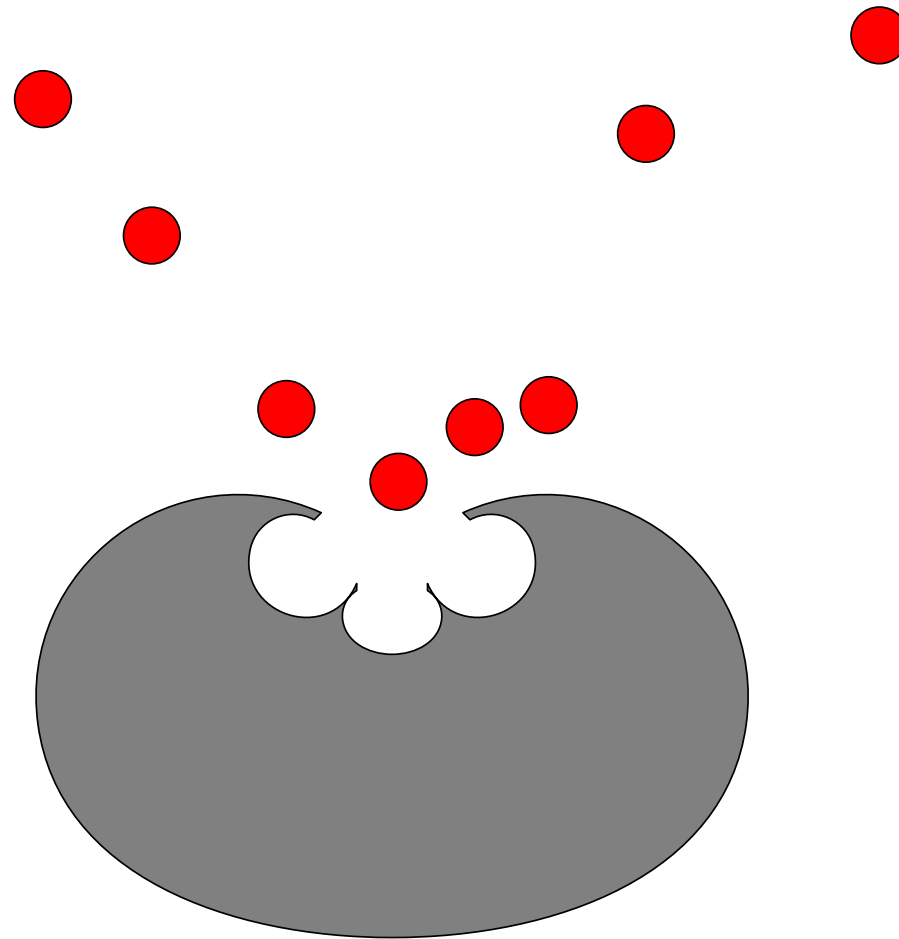
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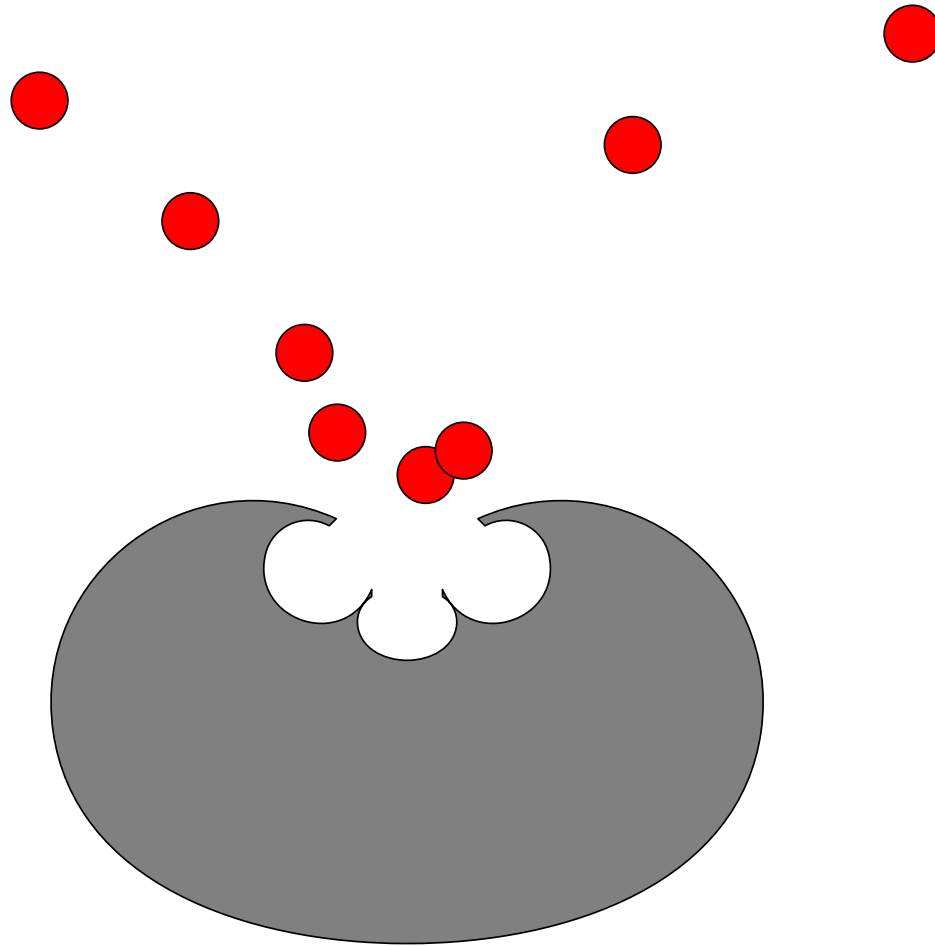
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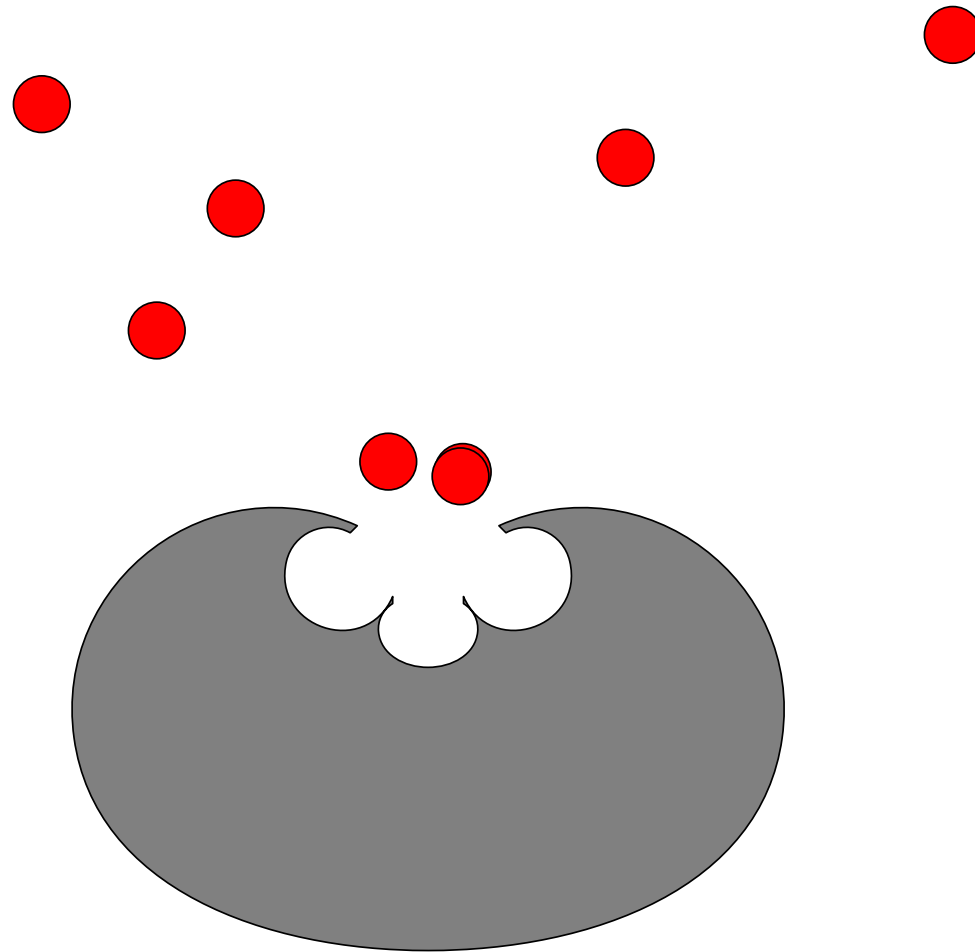
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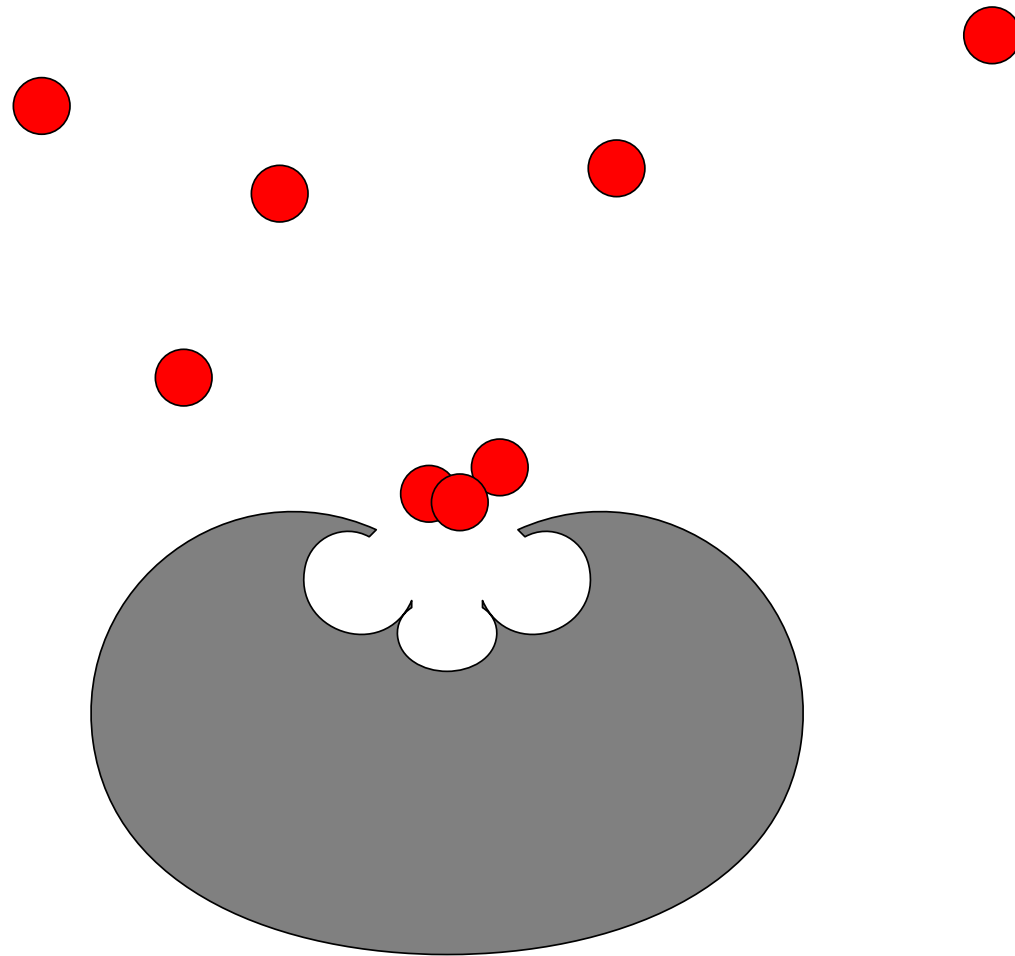
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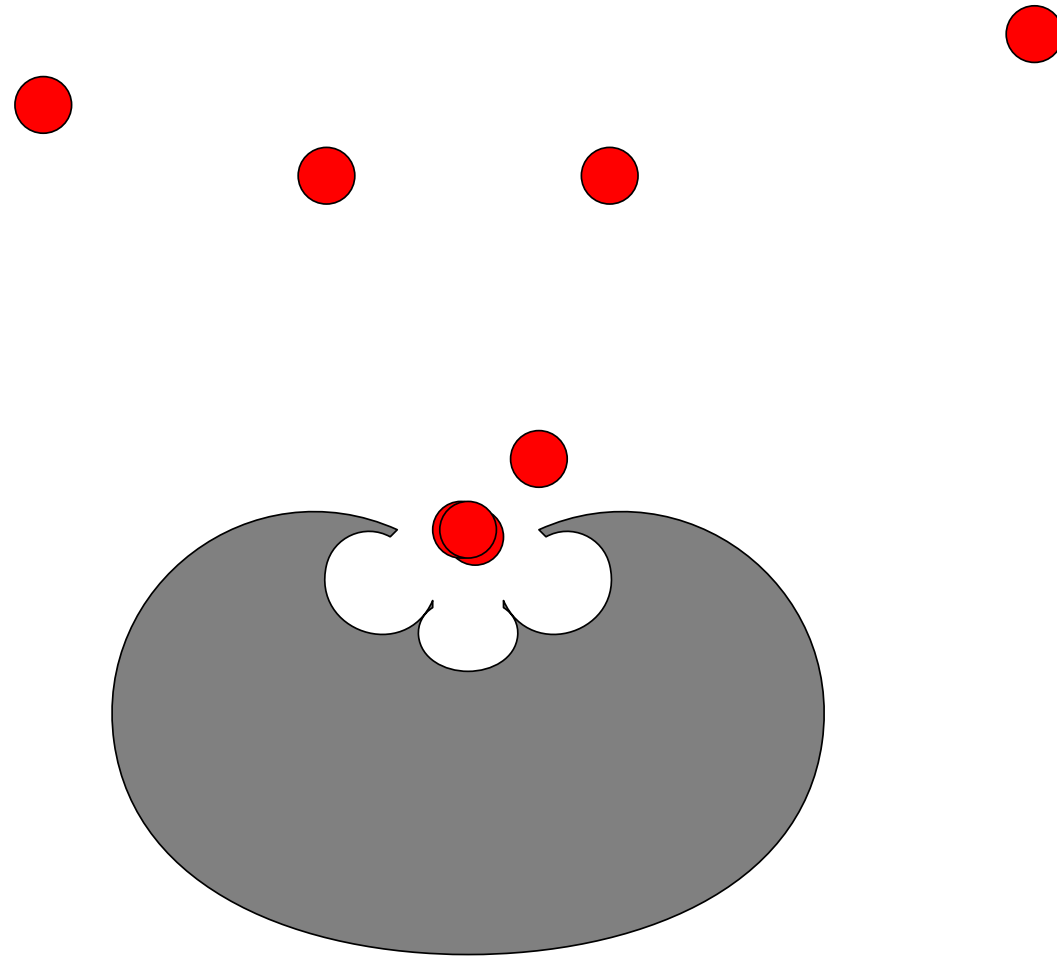
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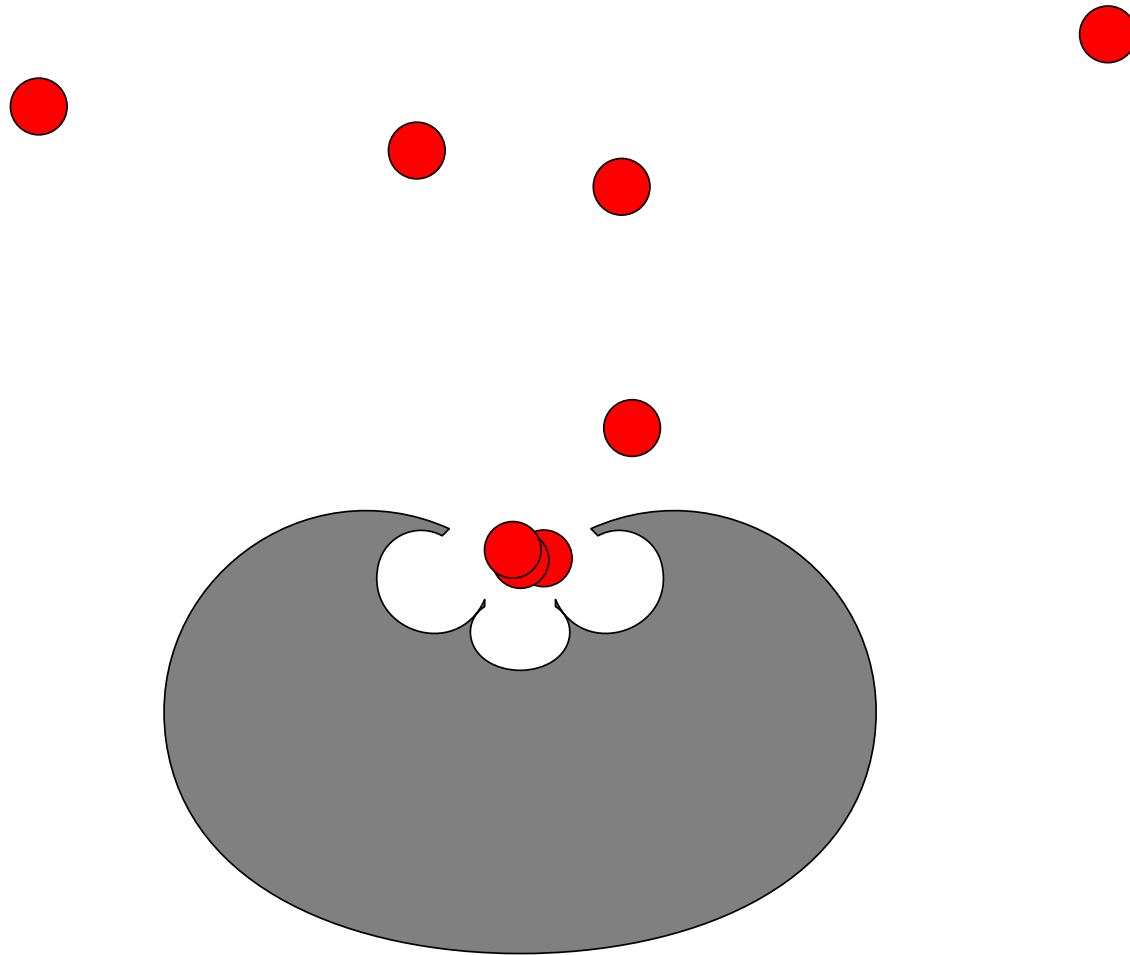
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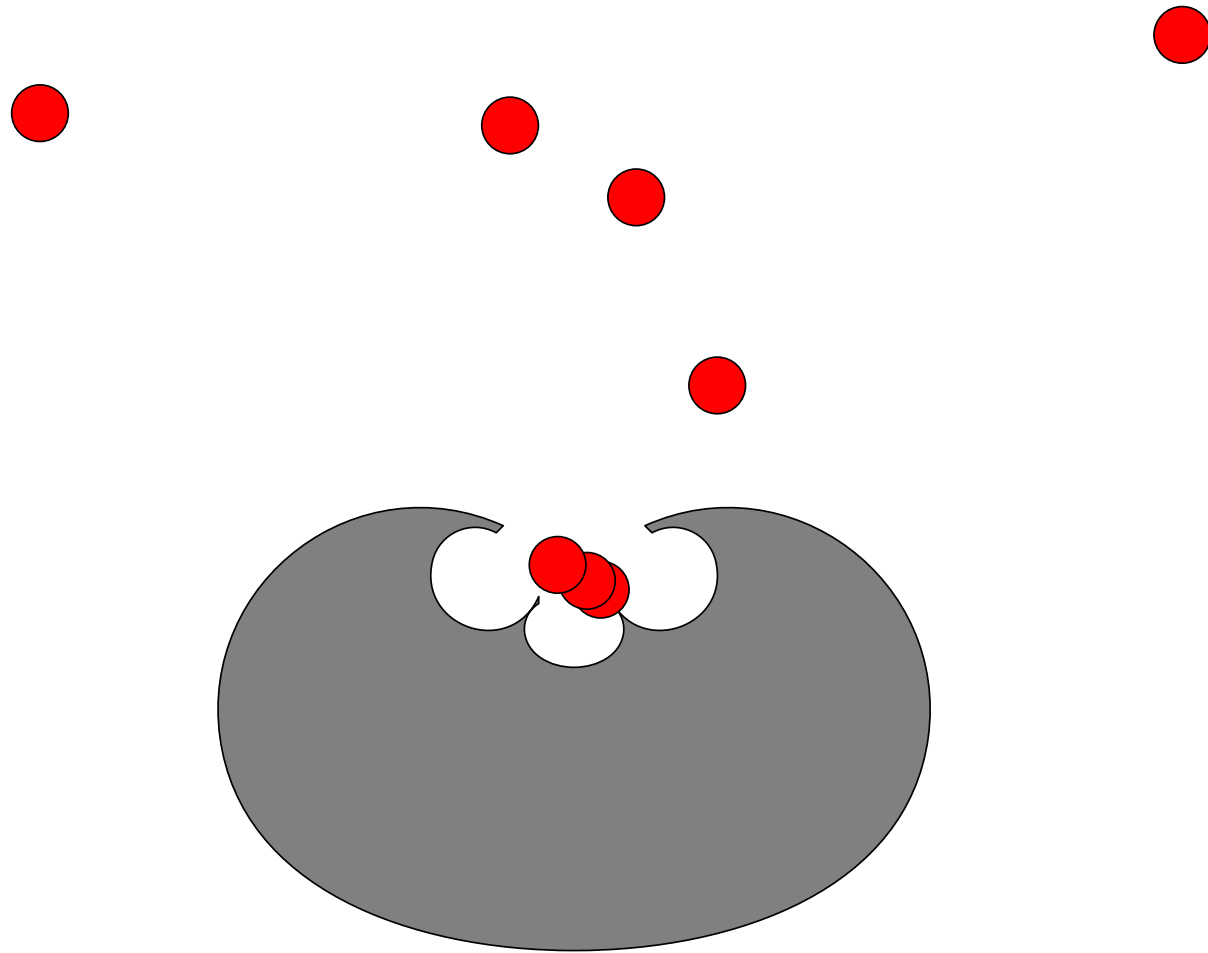
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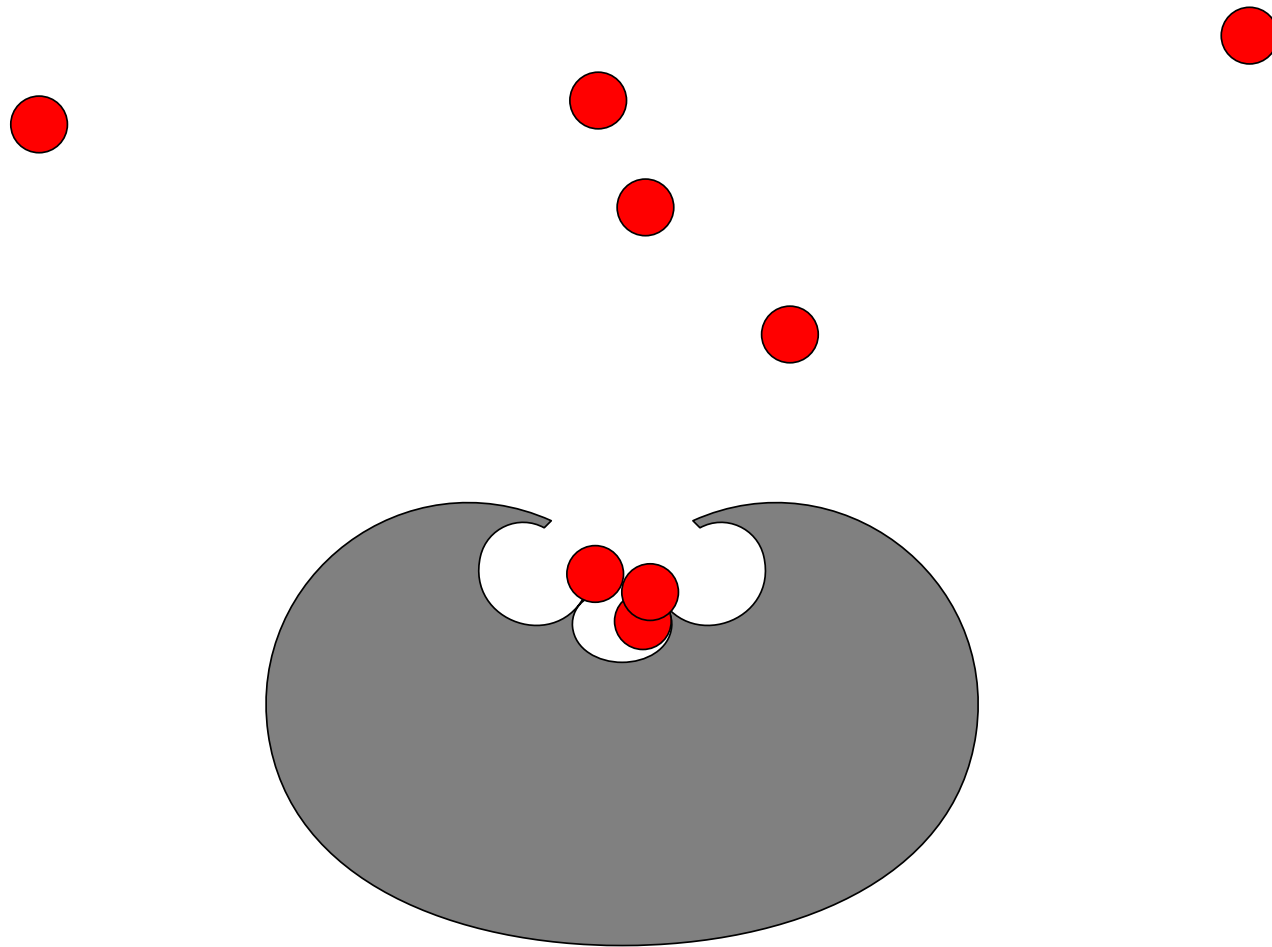
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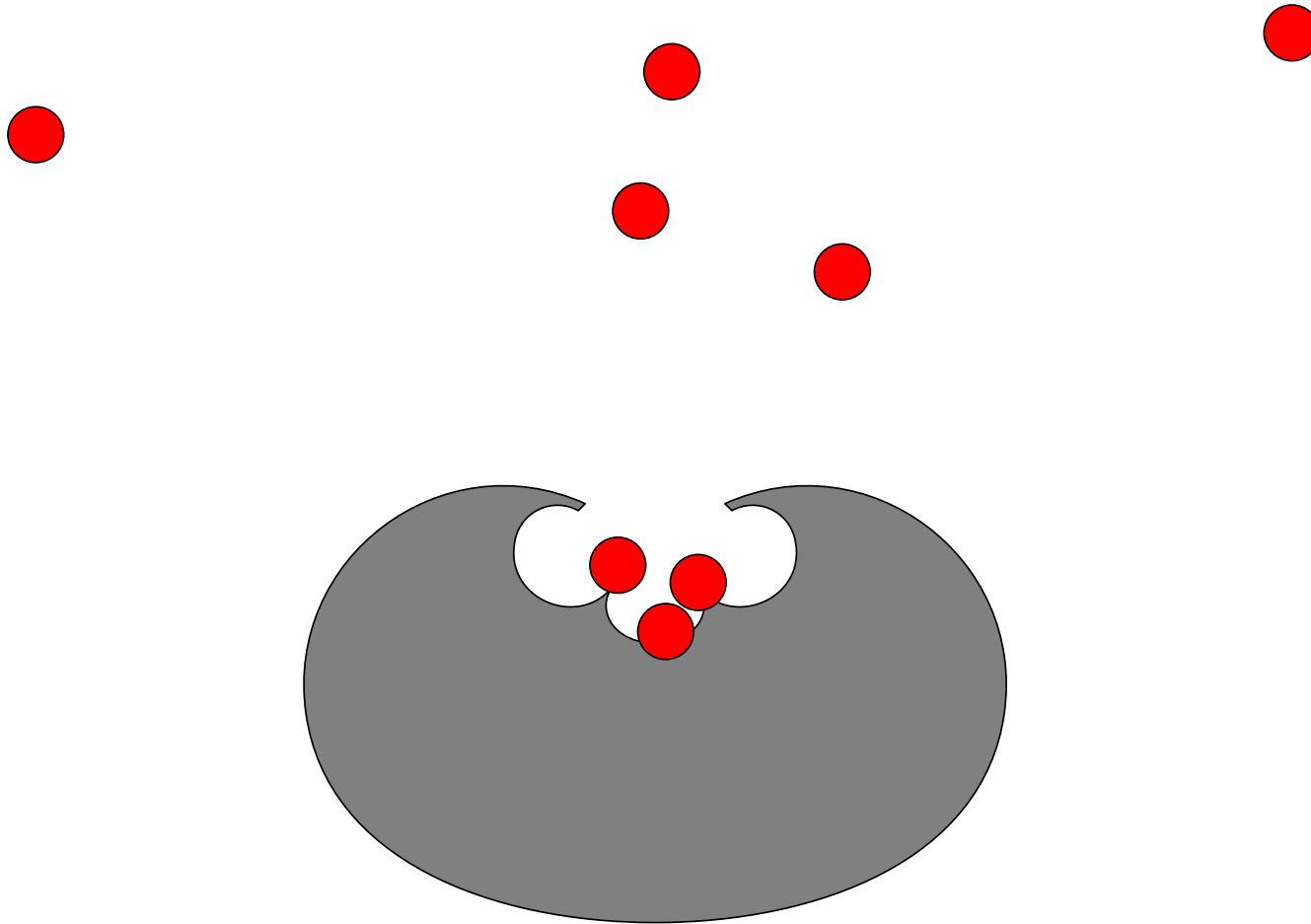
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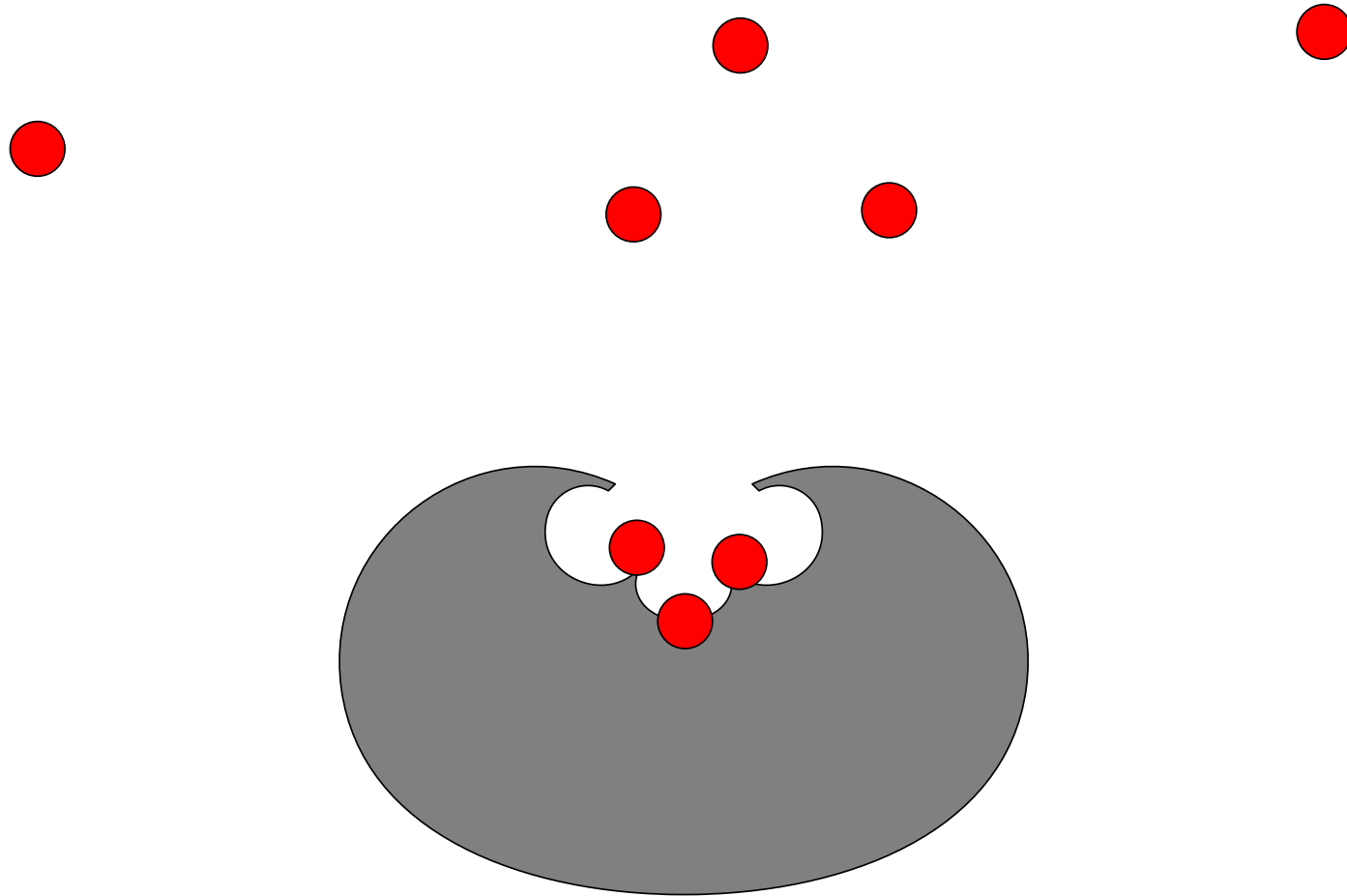
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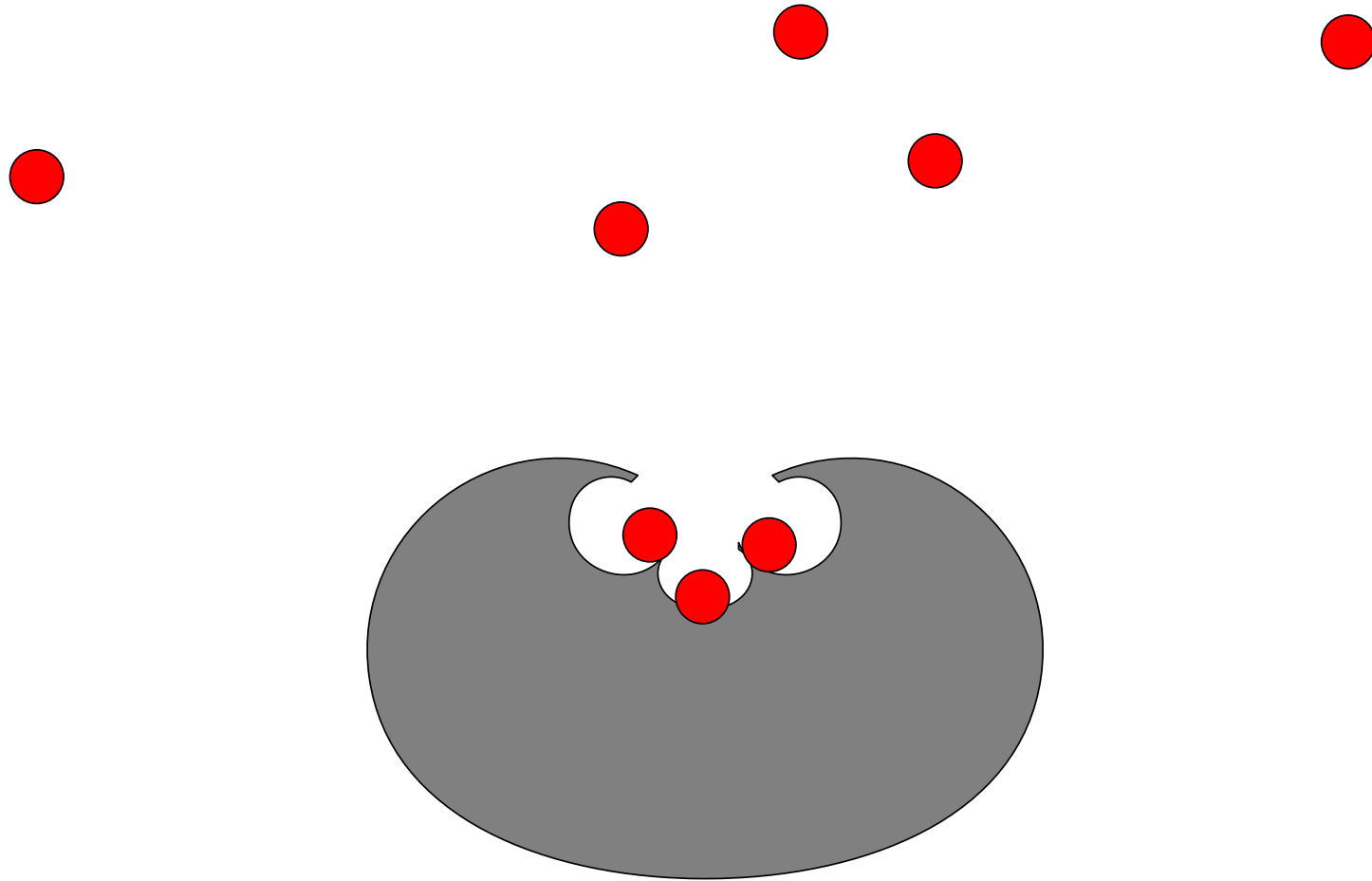
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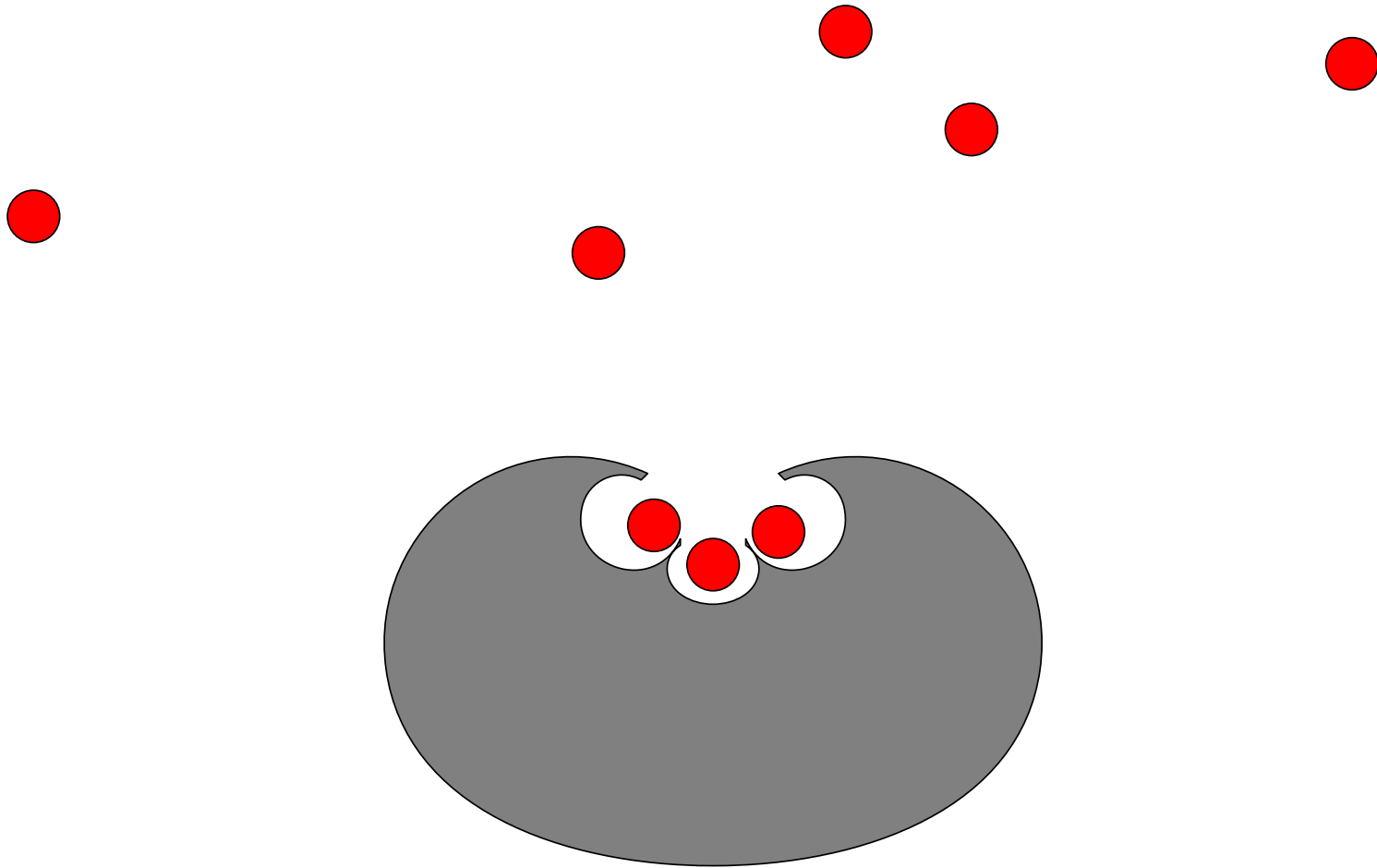
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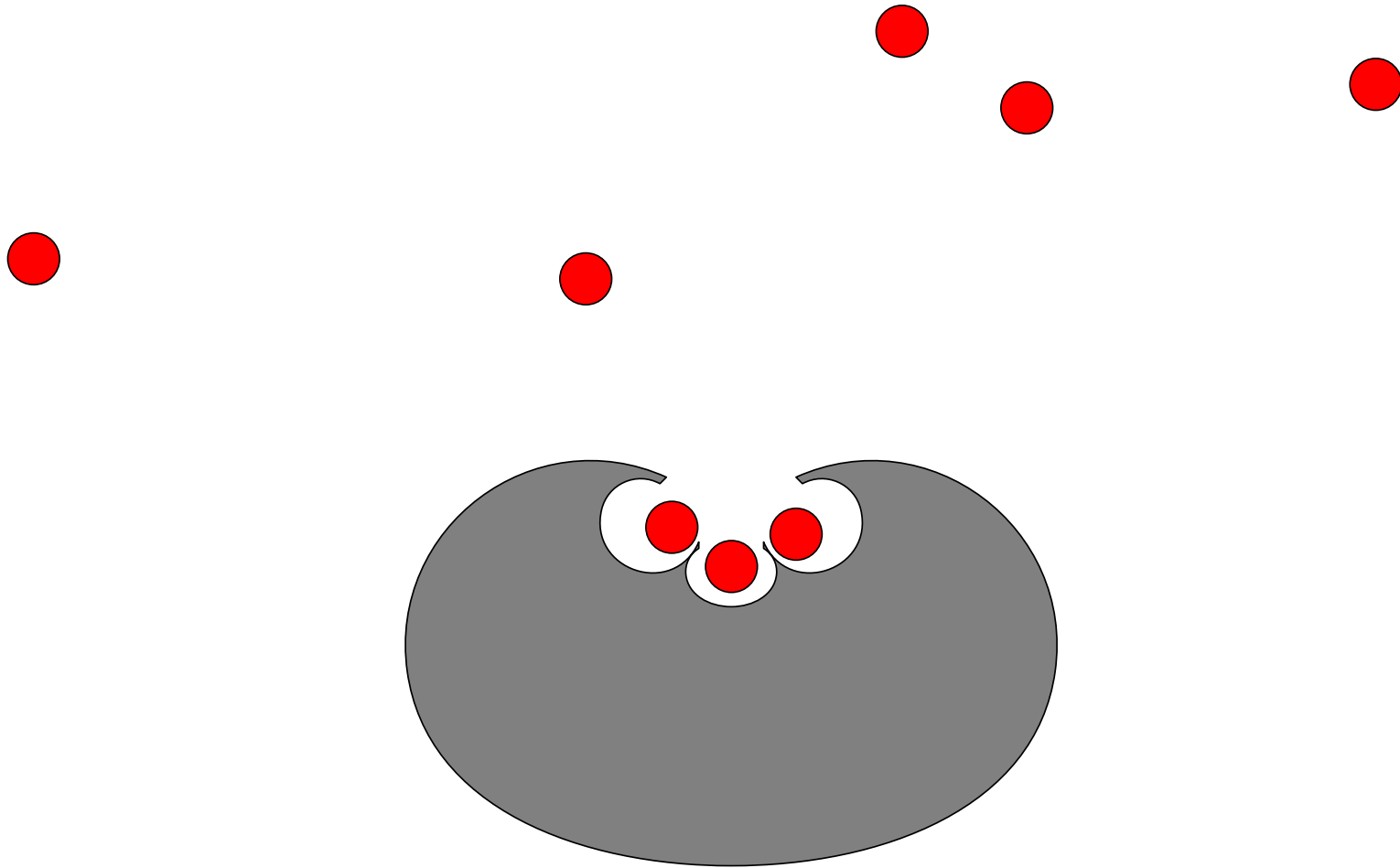
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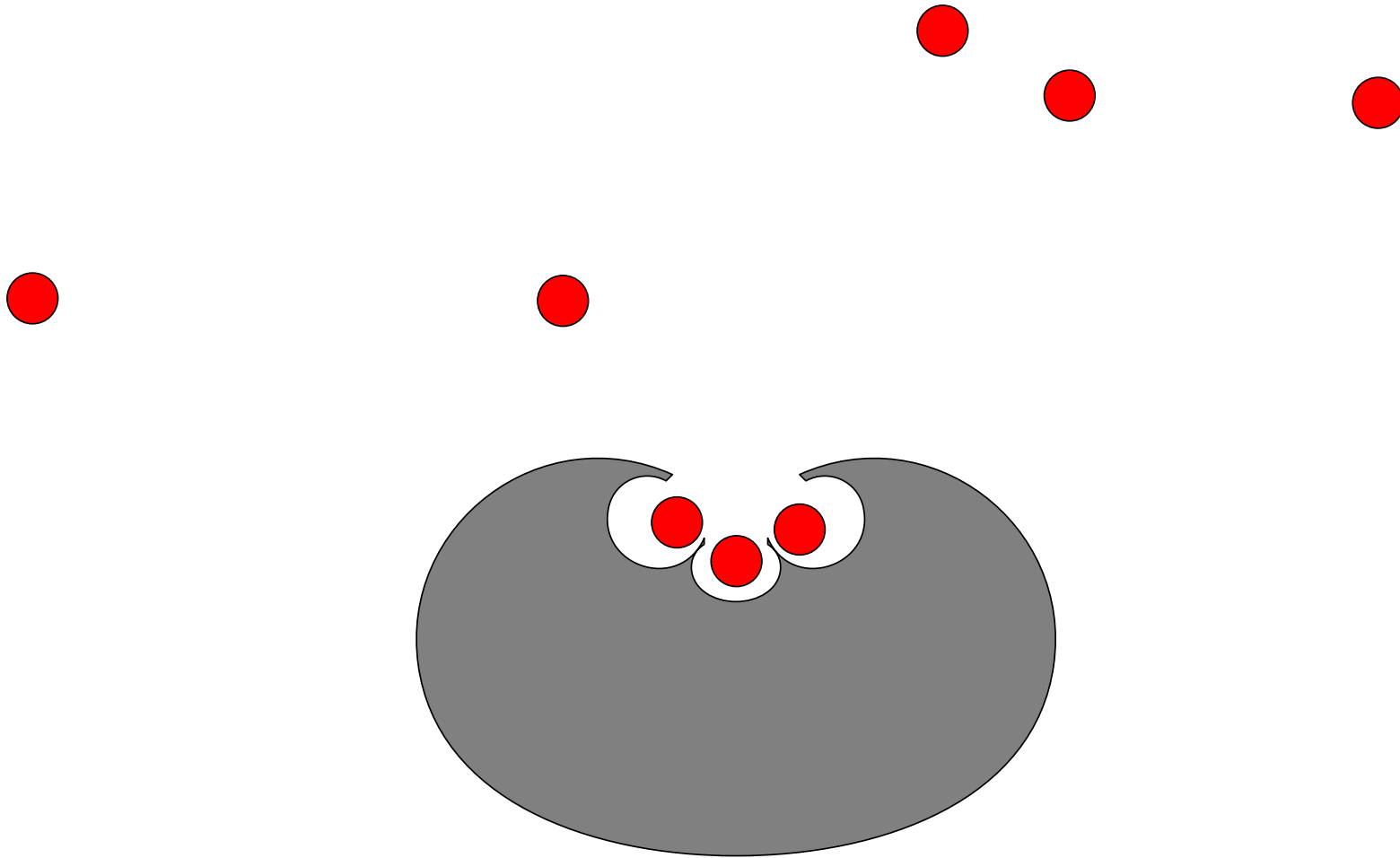
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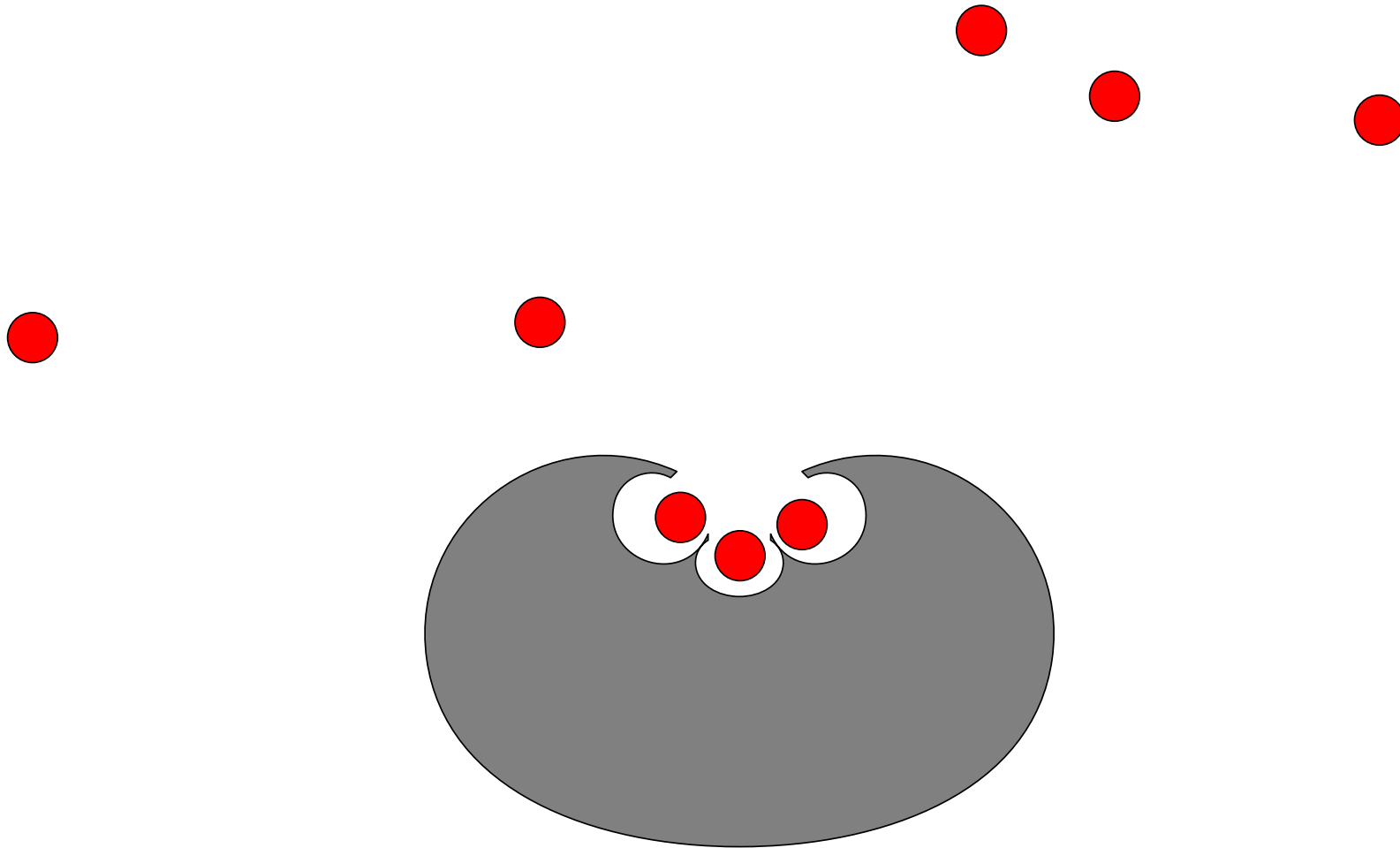
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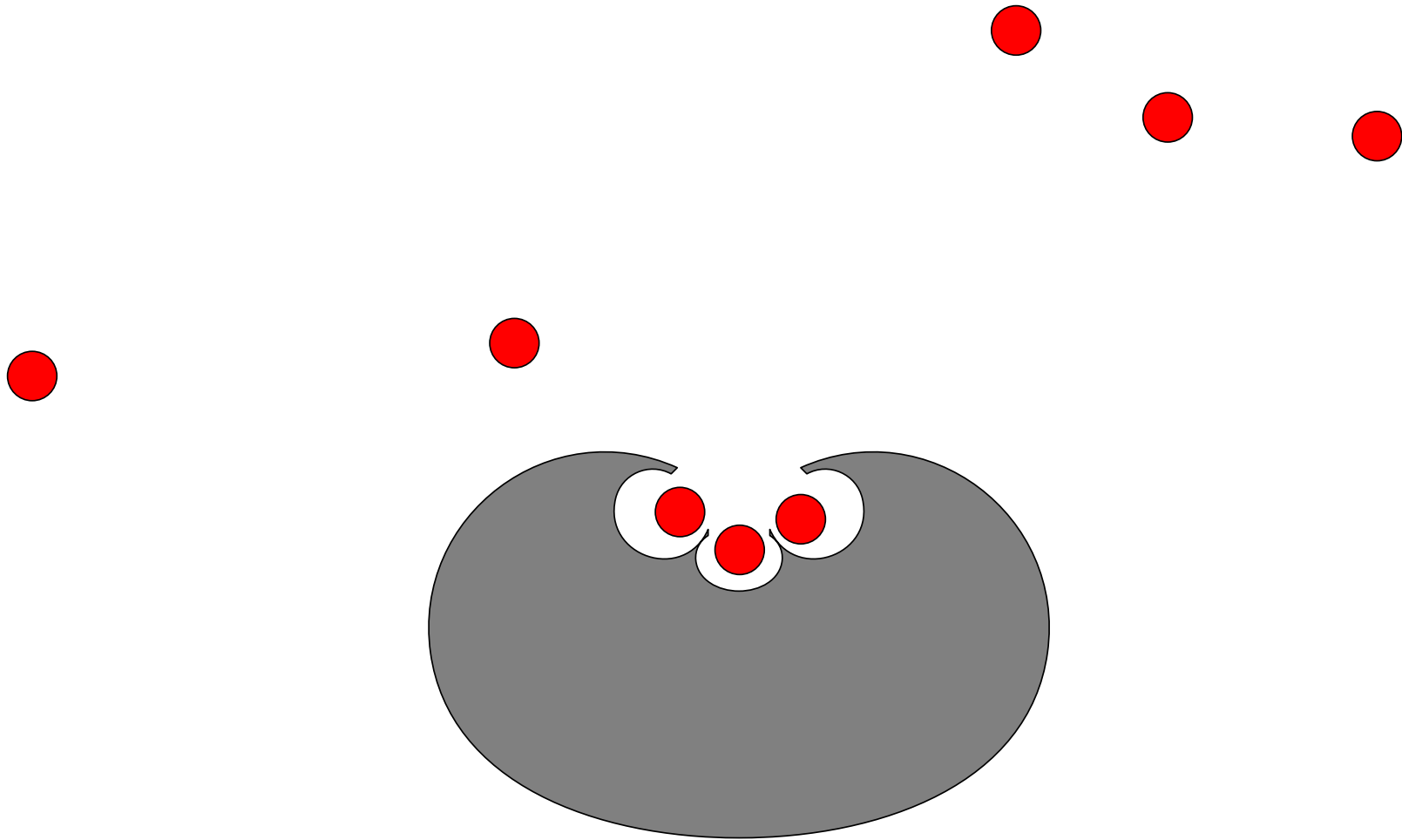
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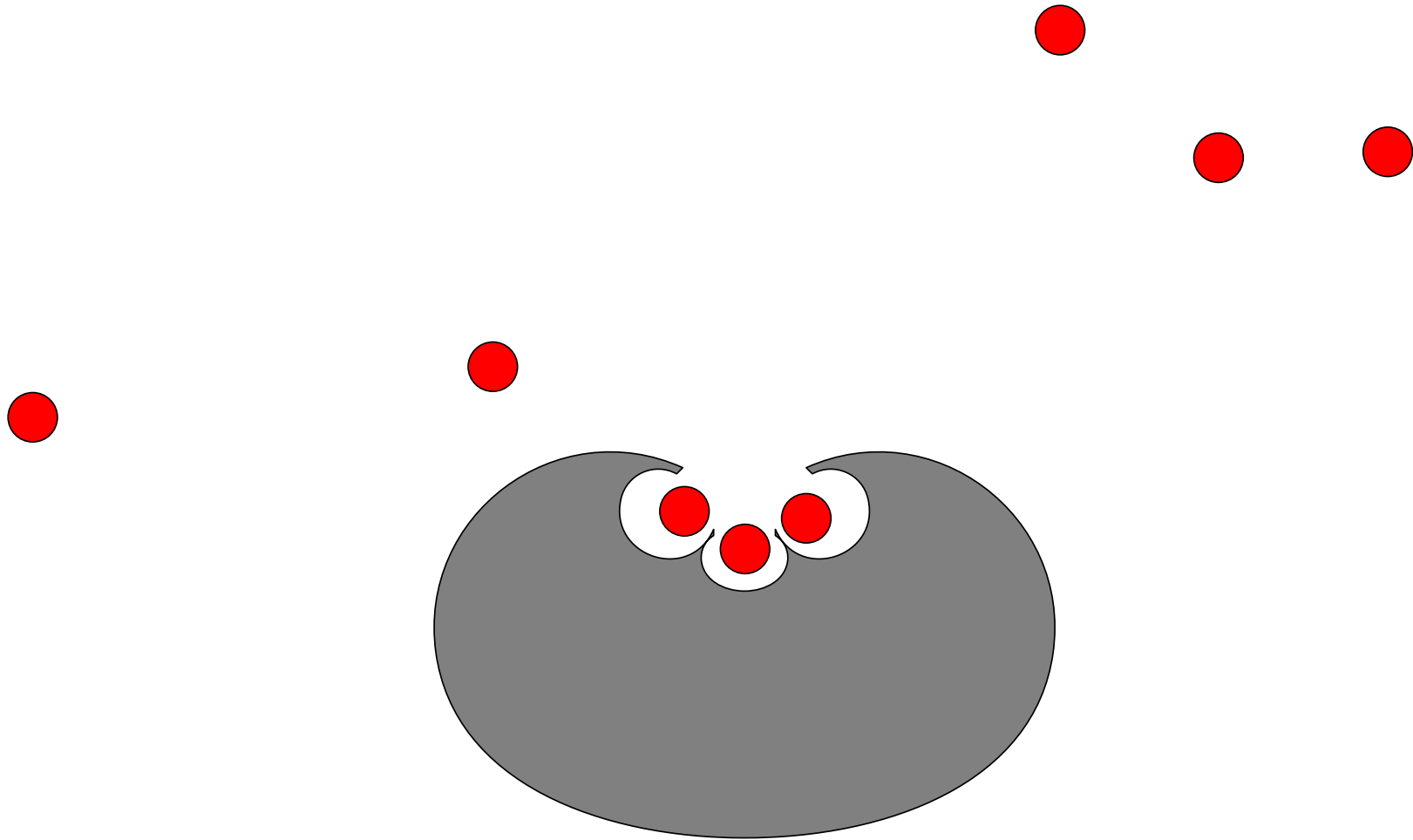
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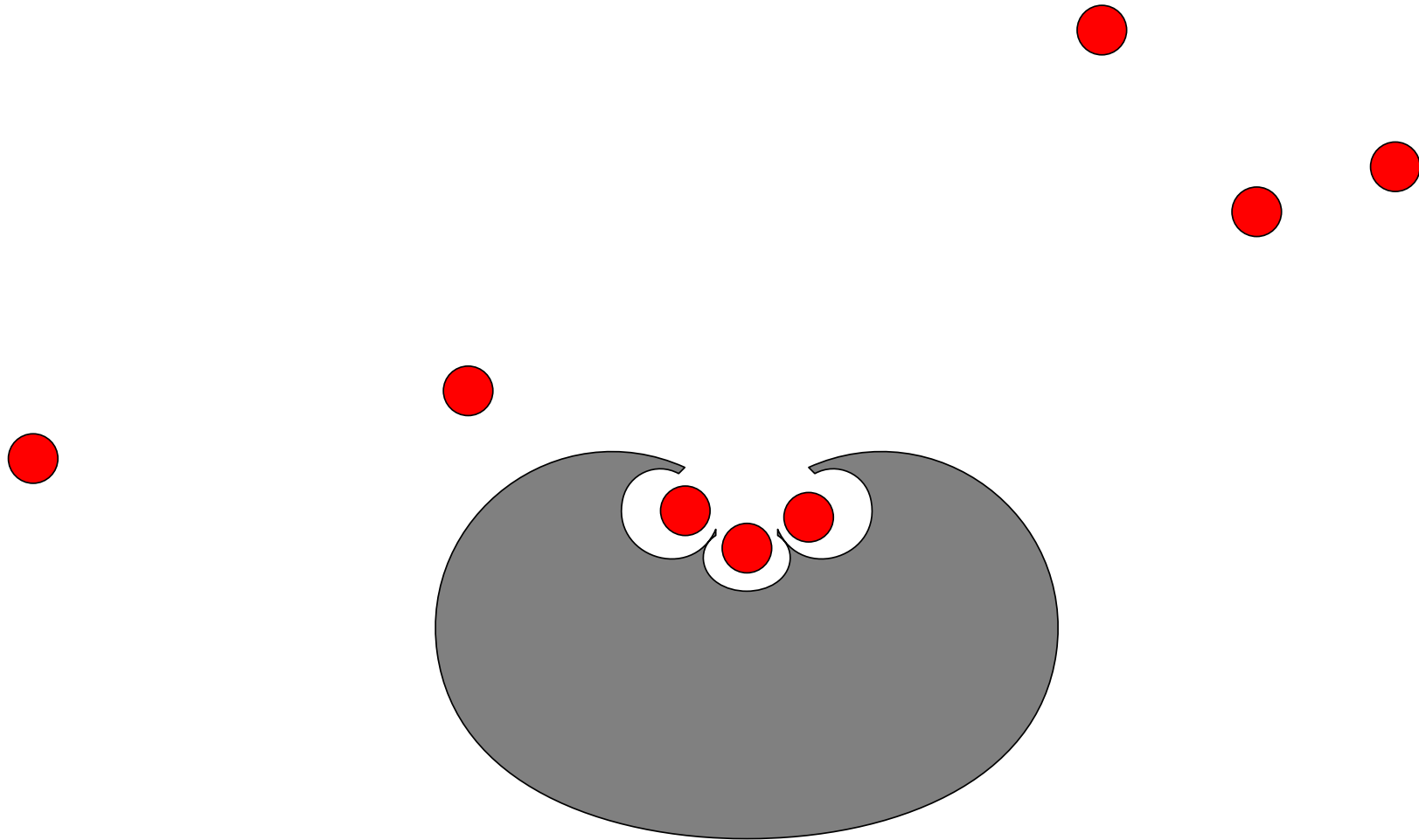
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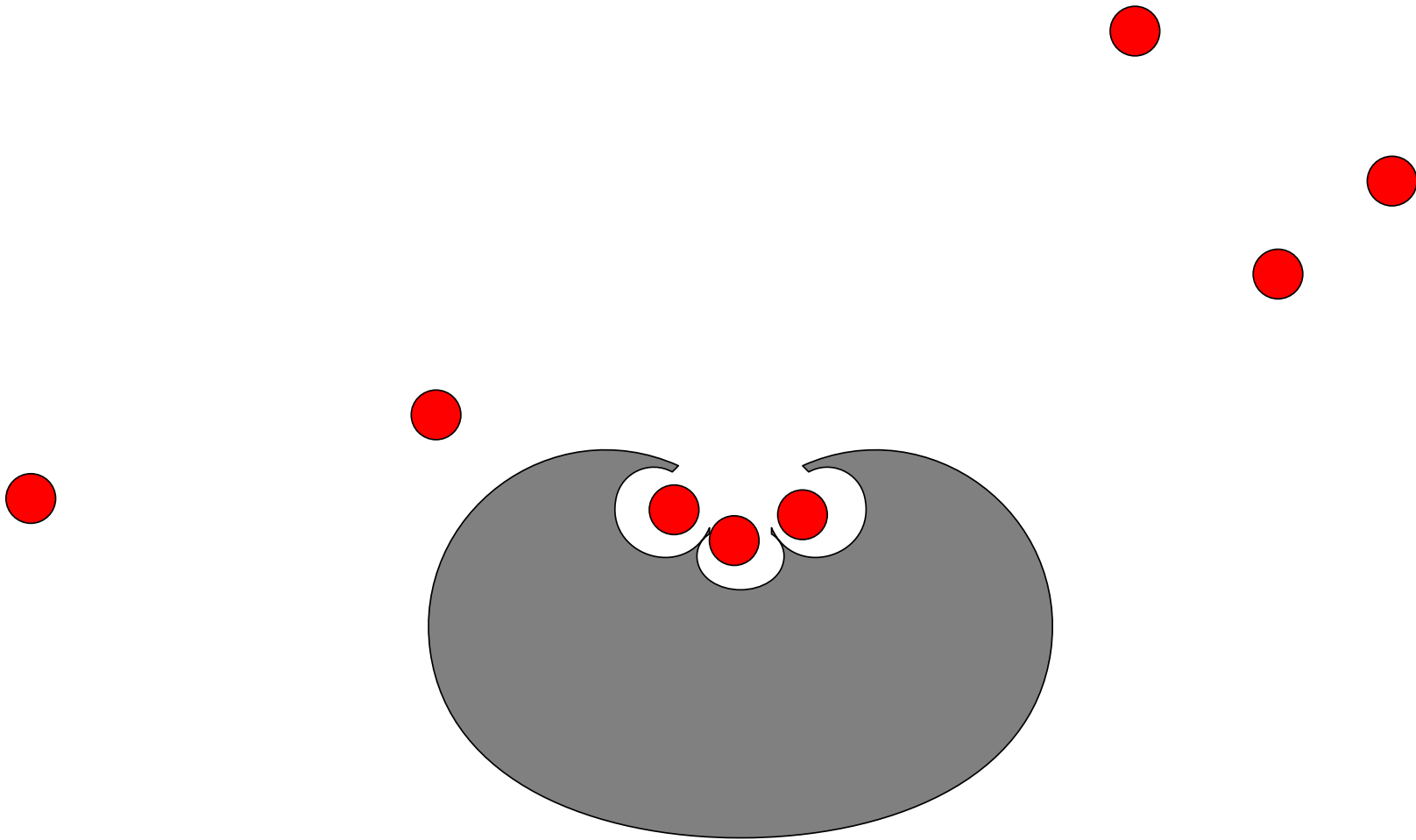
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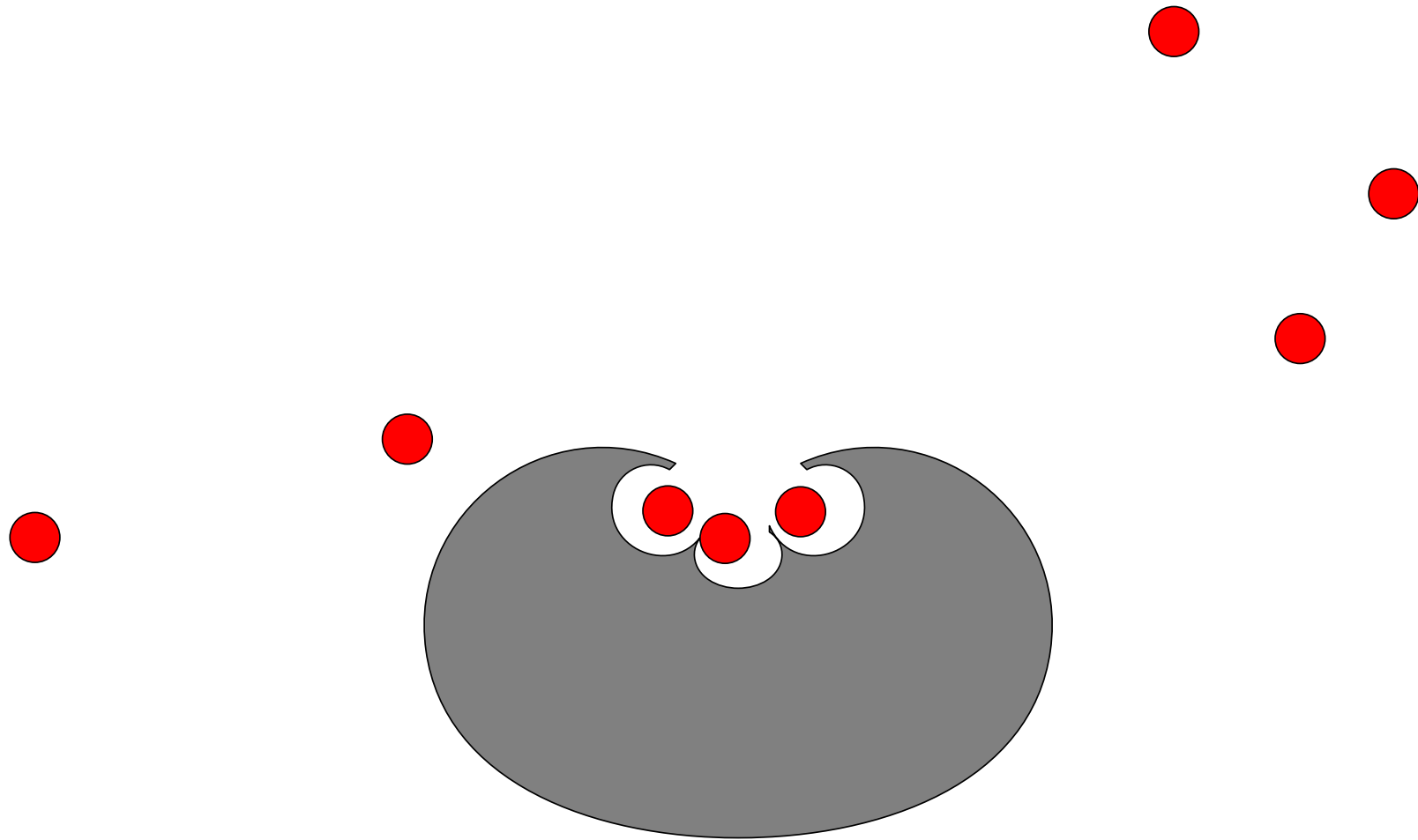
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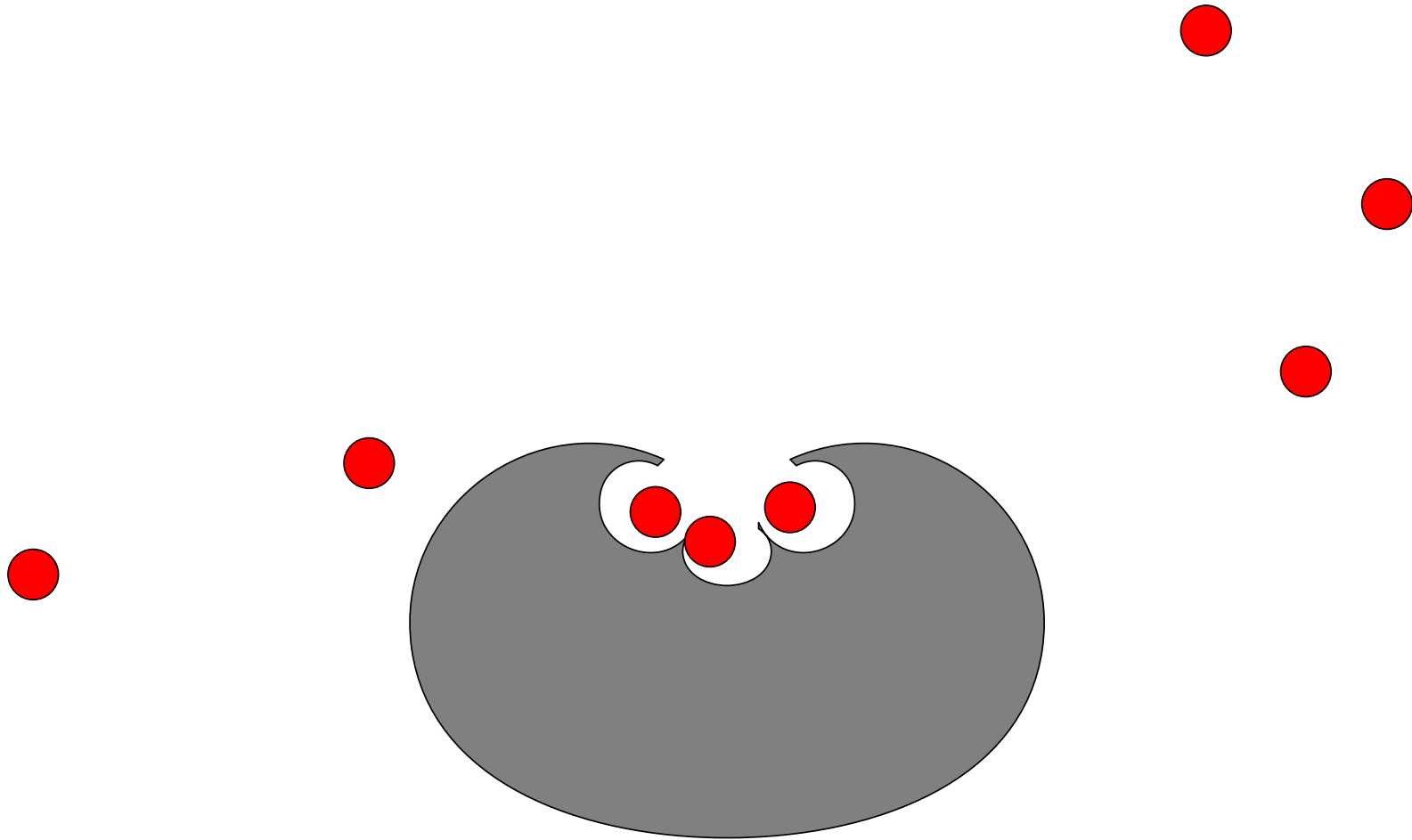
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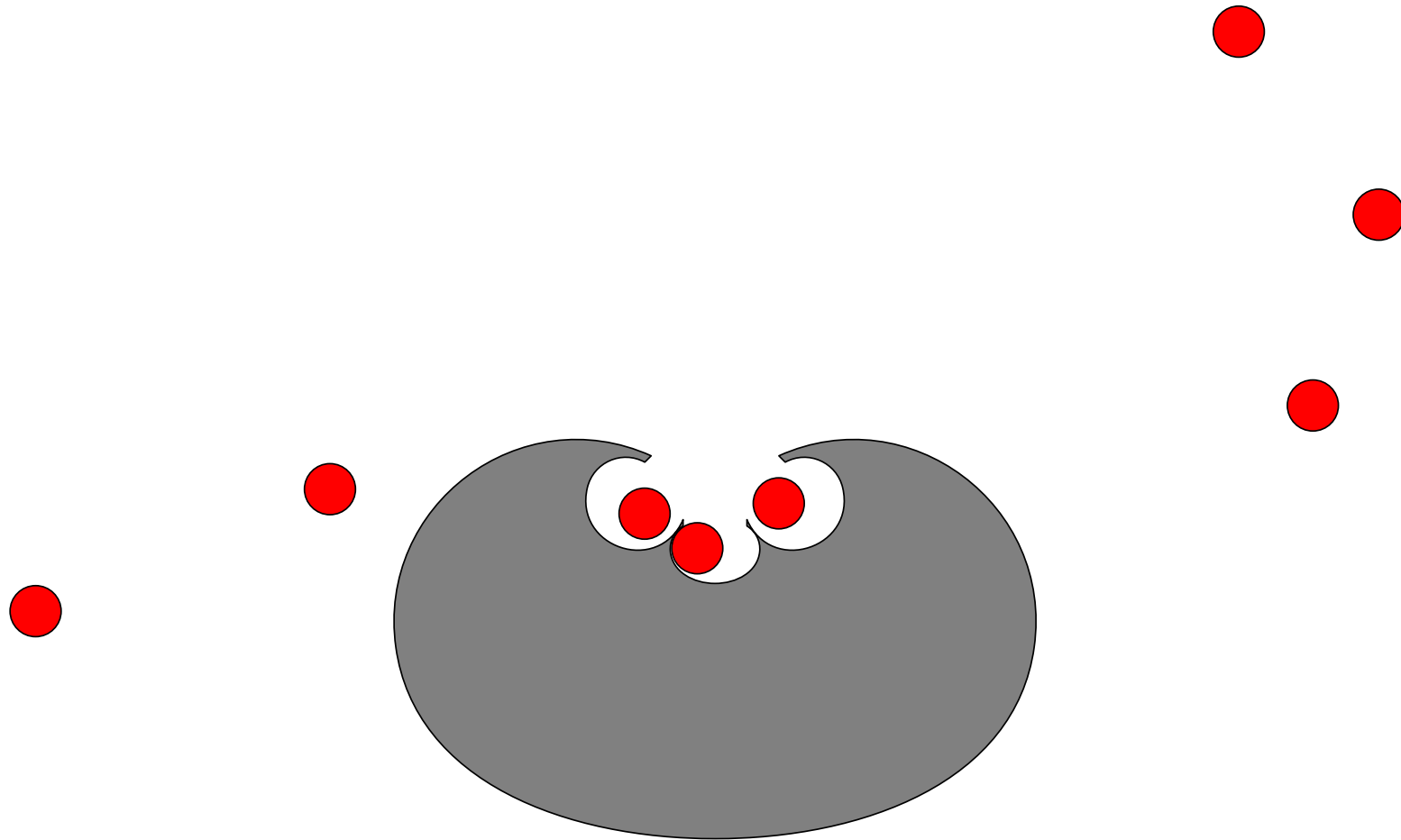
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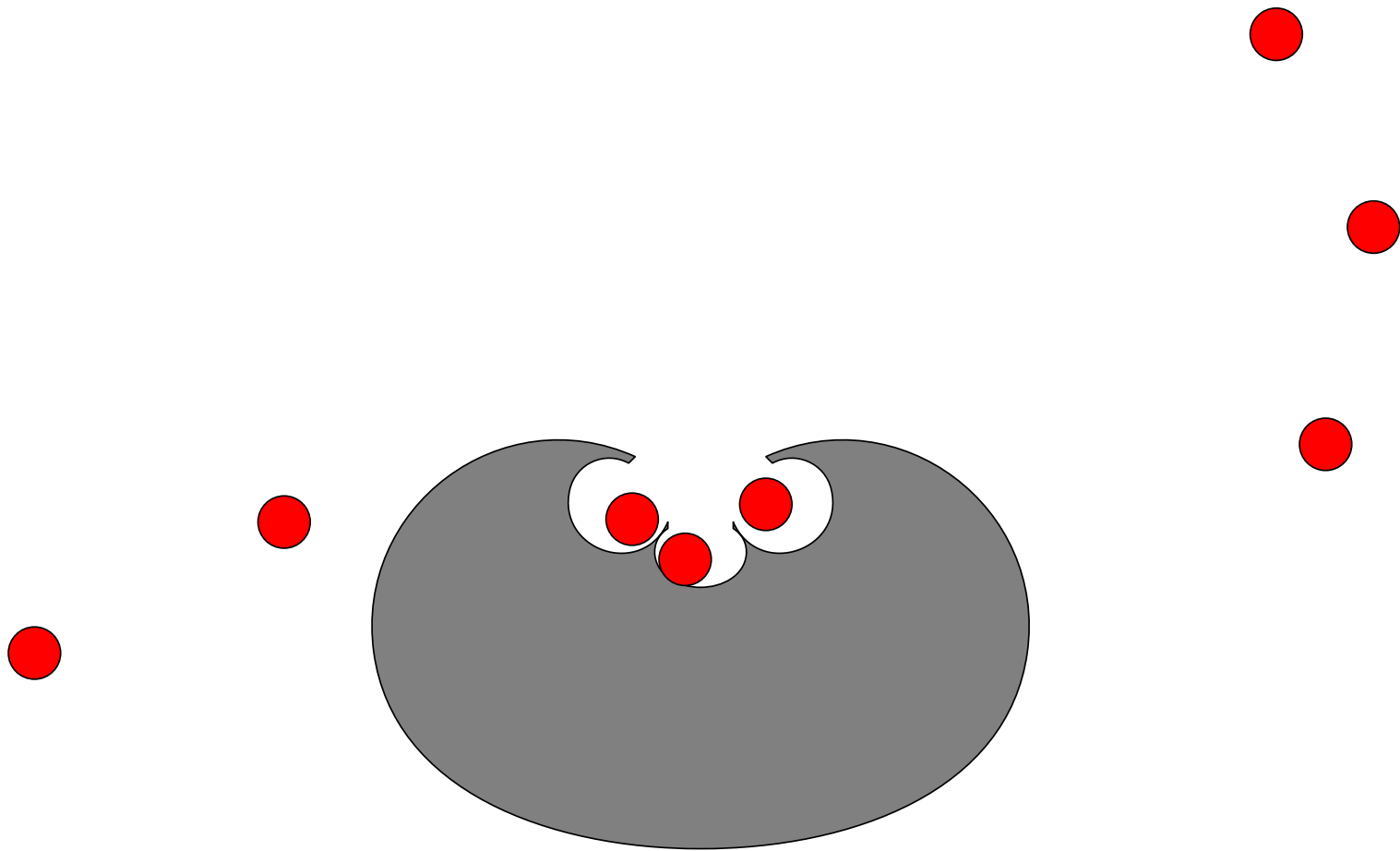
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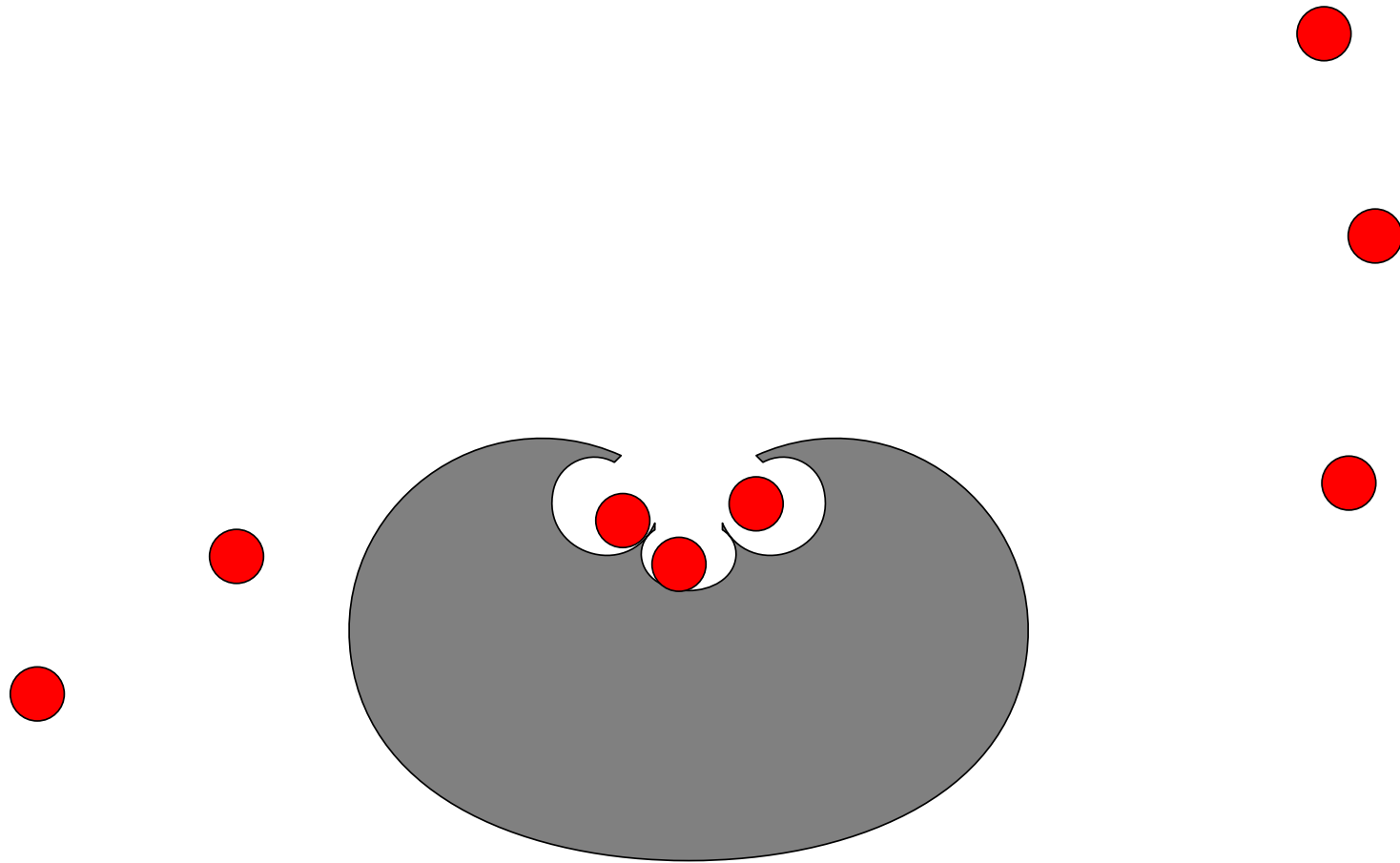
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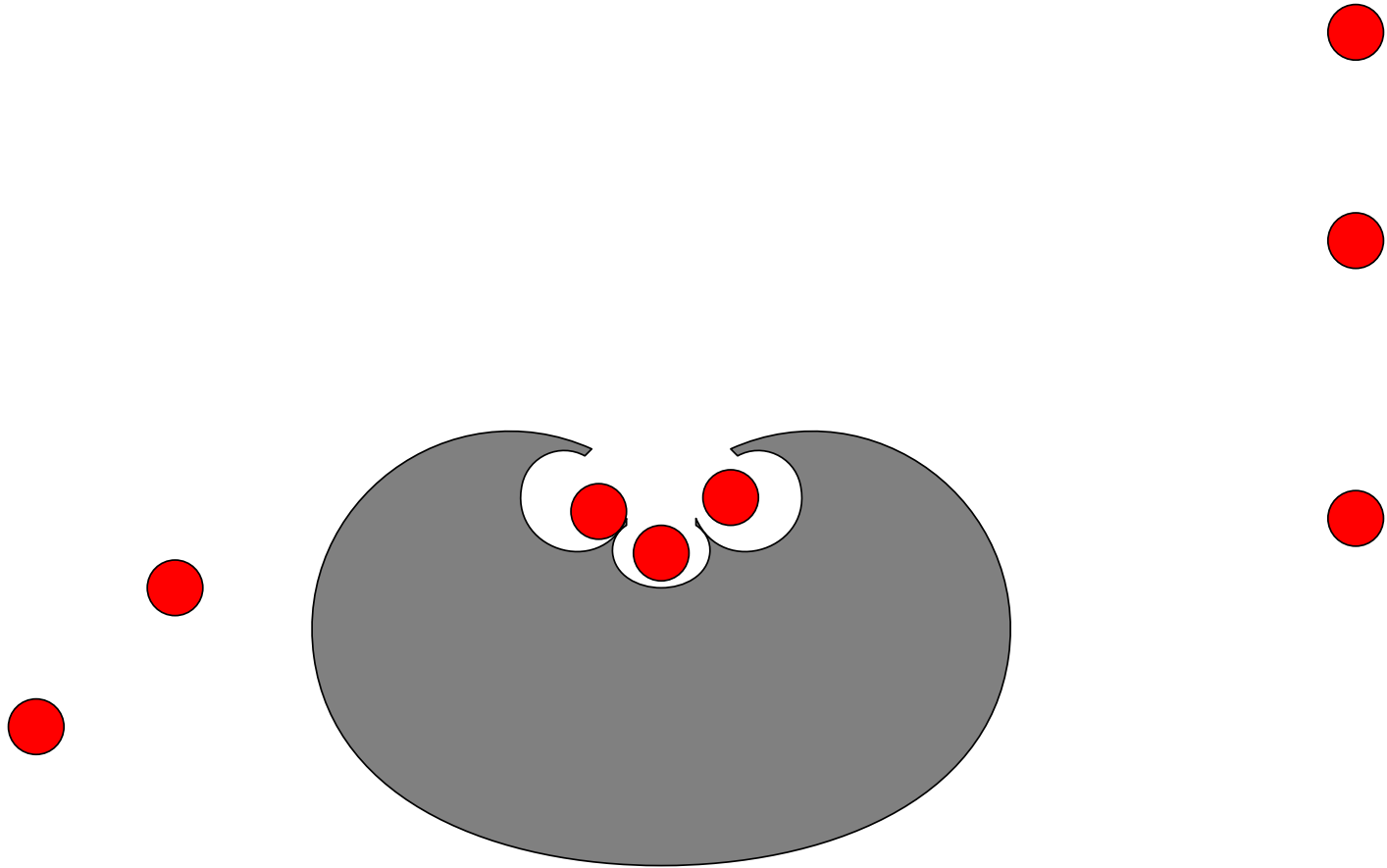
Multi-signalling



Multi-signalling



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Hill Equation

- We assume that we form a complex with n signalling molecules S_X



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- Which we can model using

$$\frac{d[C]}{dt} = k_{on} [X] [S_X]^n - k_{off} [C]$$

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$$[C] = \frac{[X_T] [S_X]^n}{K_X^n + [S_X]^n}$$

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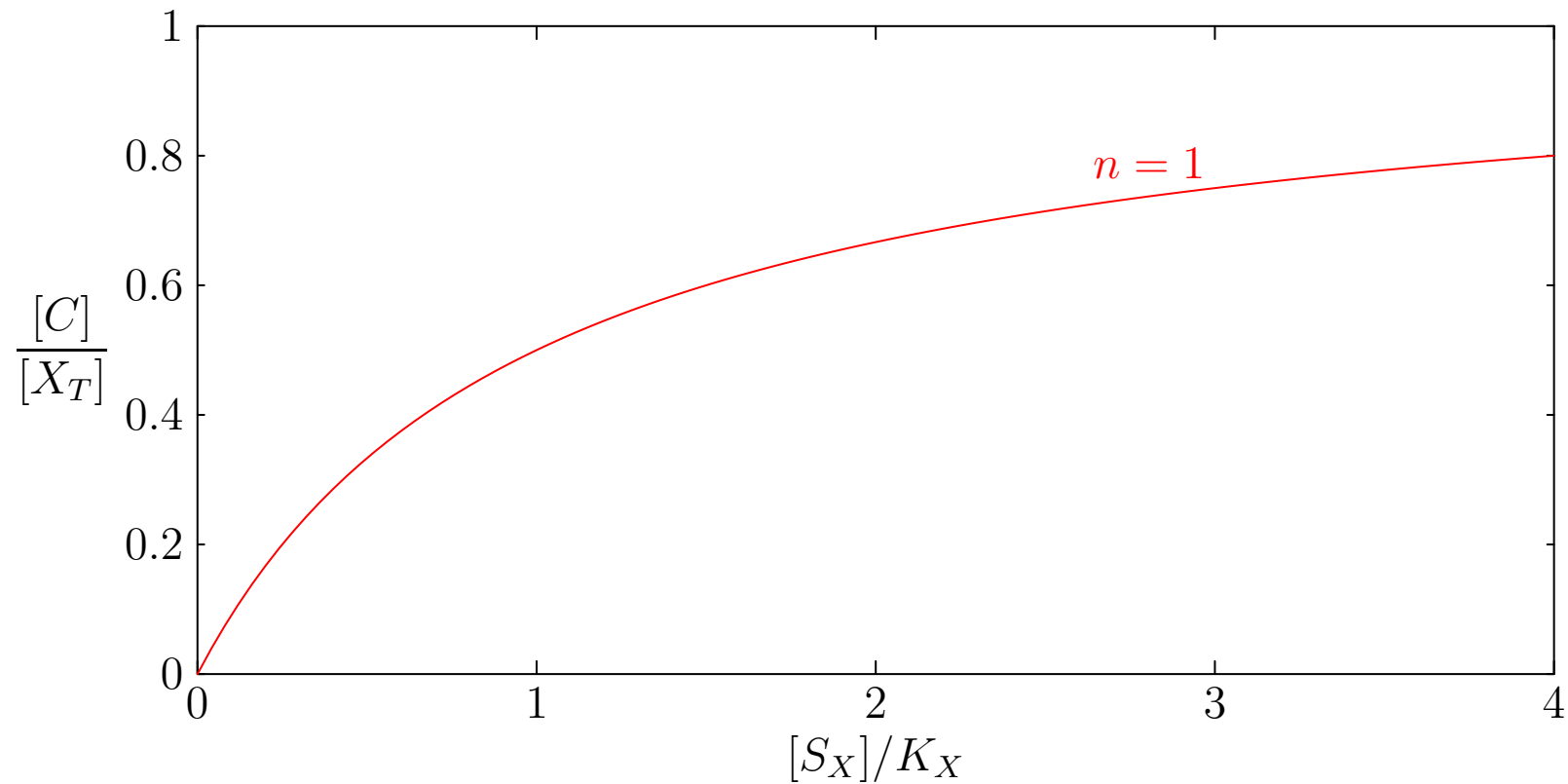


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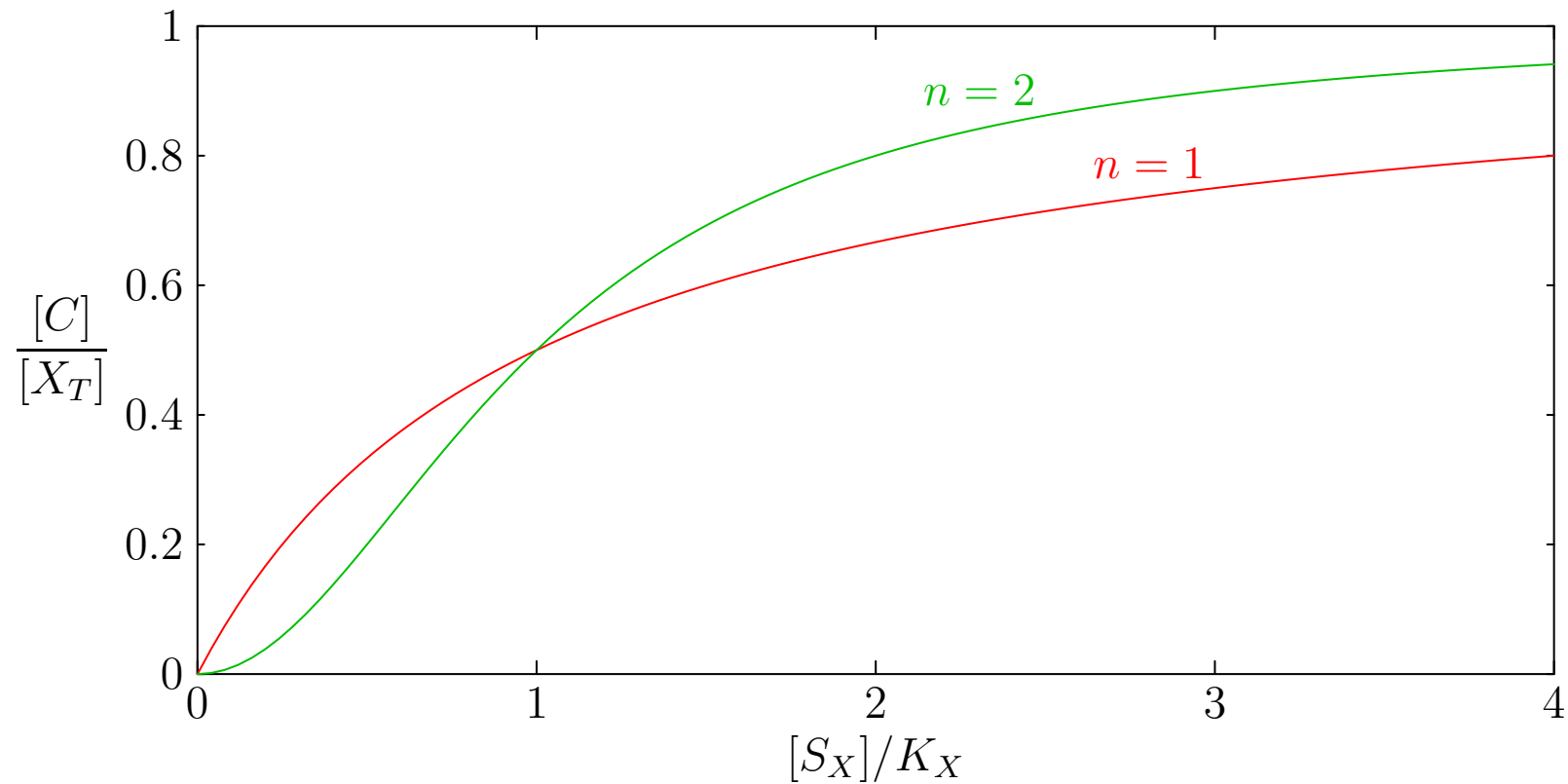
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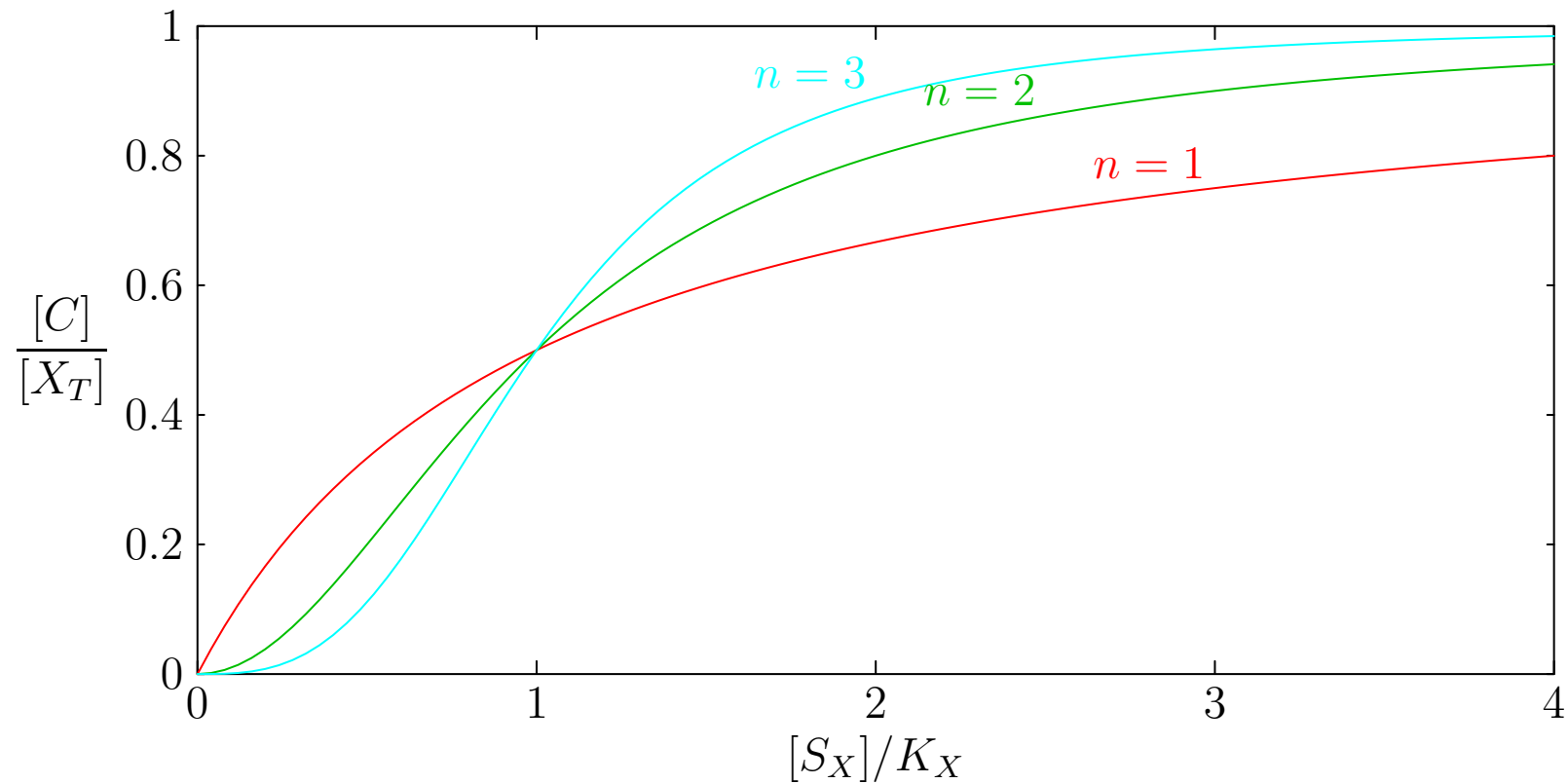
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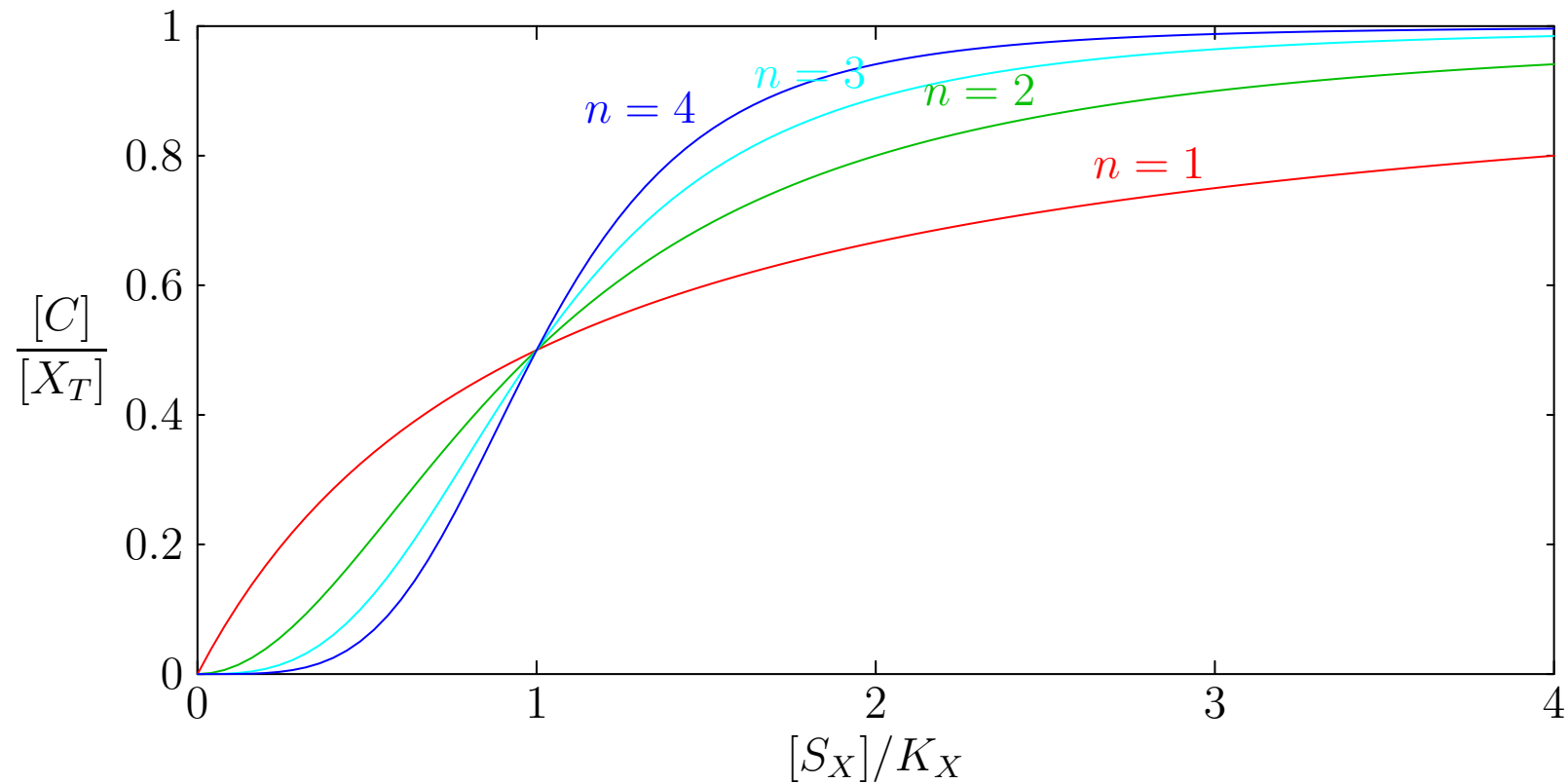
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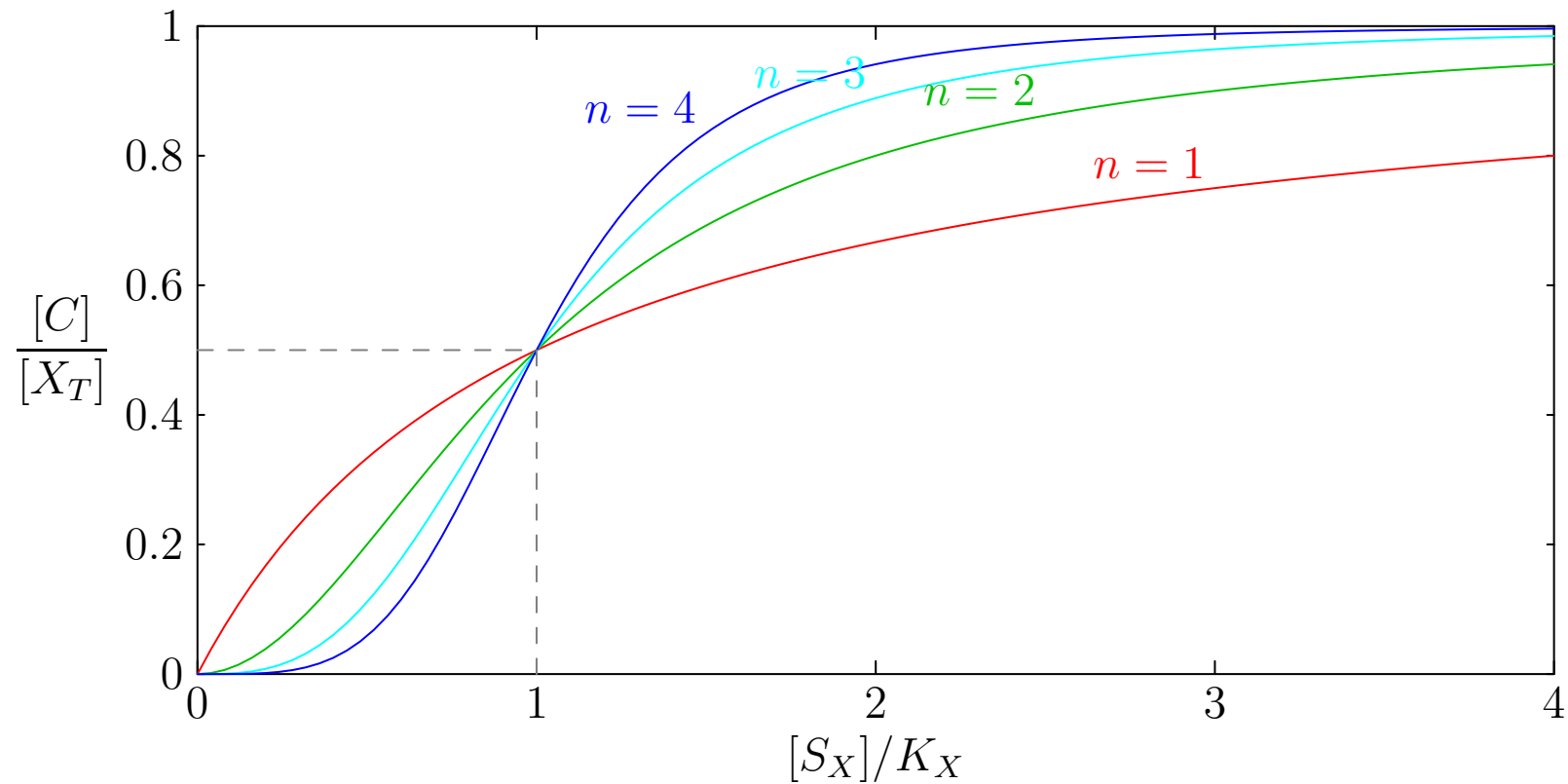
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Activator and Repressor

- A protein that acts to increase the expression of a protein we call an **activator**
- The activation coefficient is often given by the Hill-equation

$$f([X]) = \frac{\beta [X]^n}{K^n + [X]^n}$$

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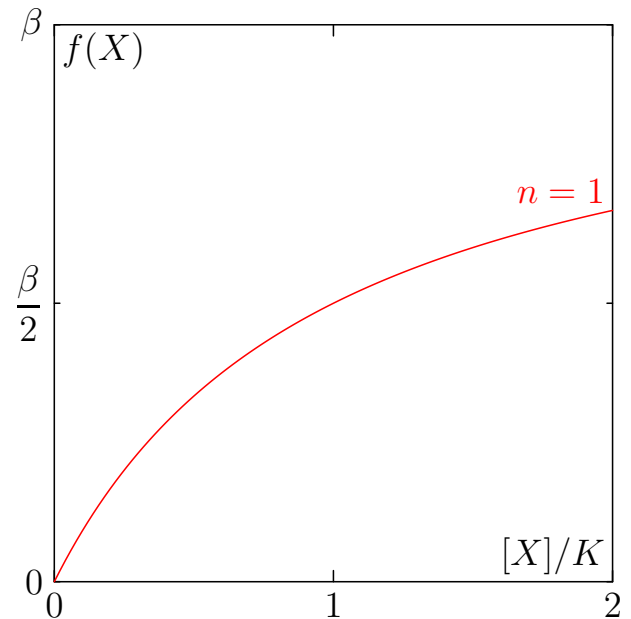
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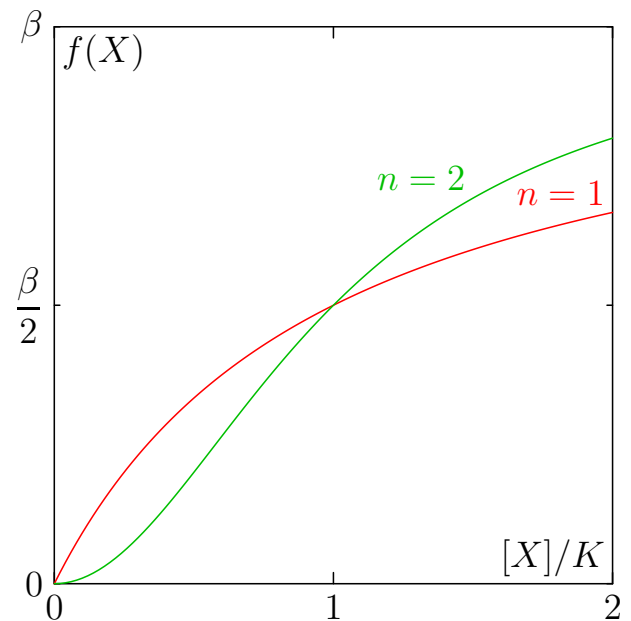
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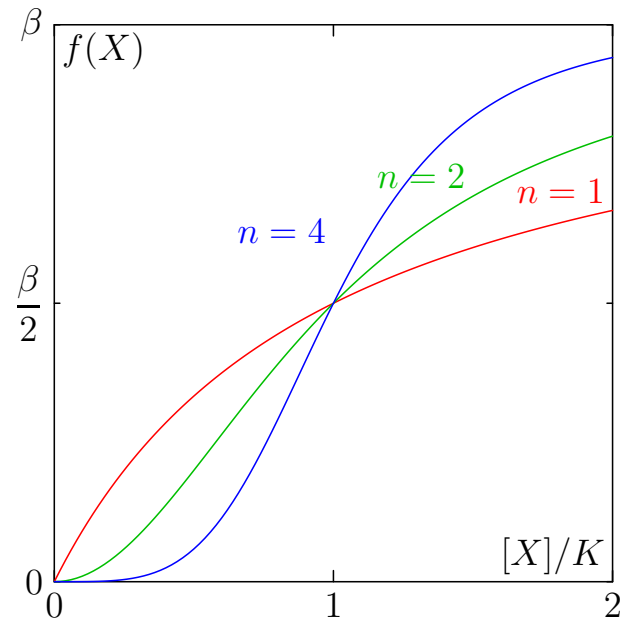
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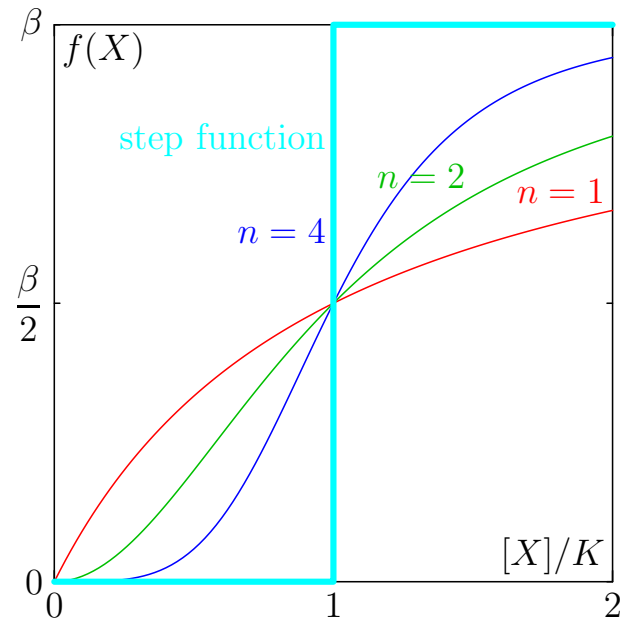
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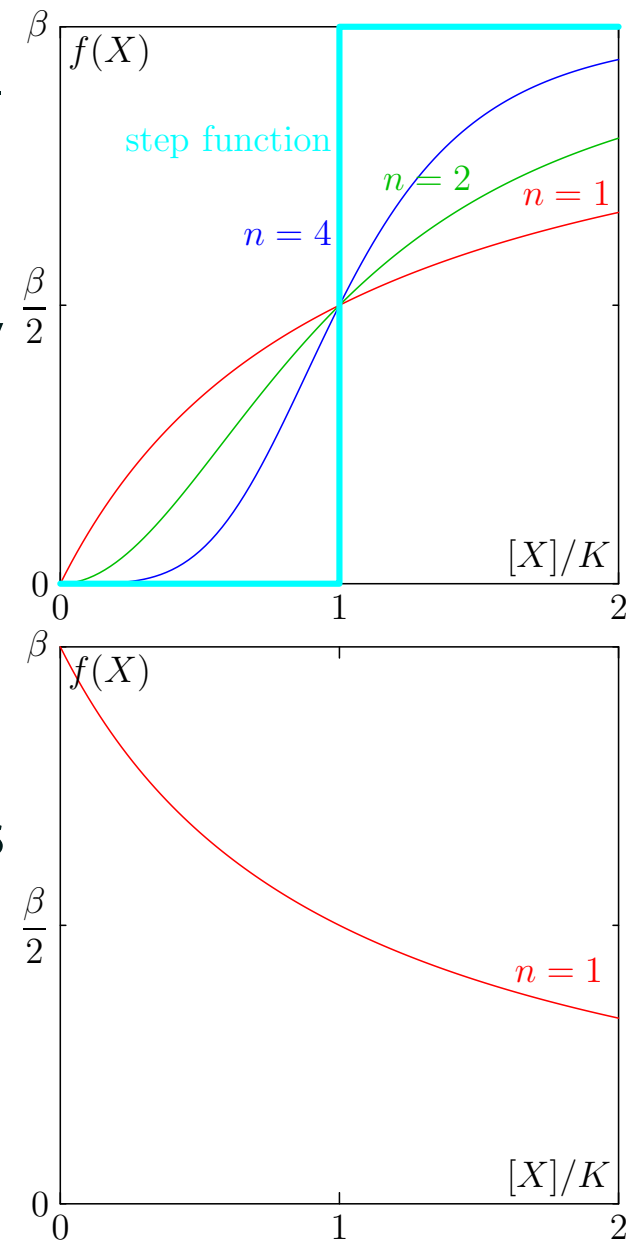
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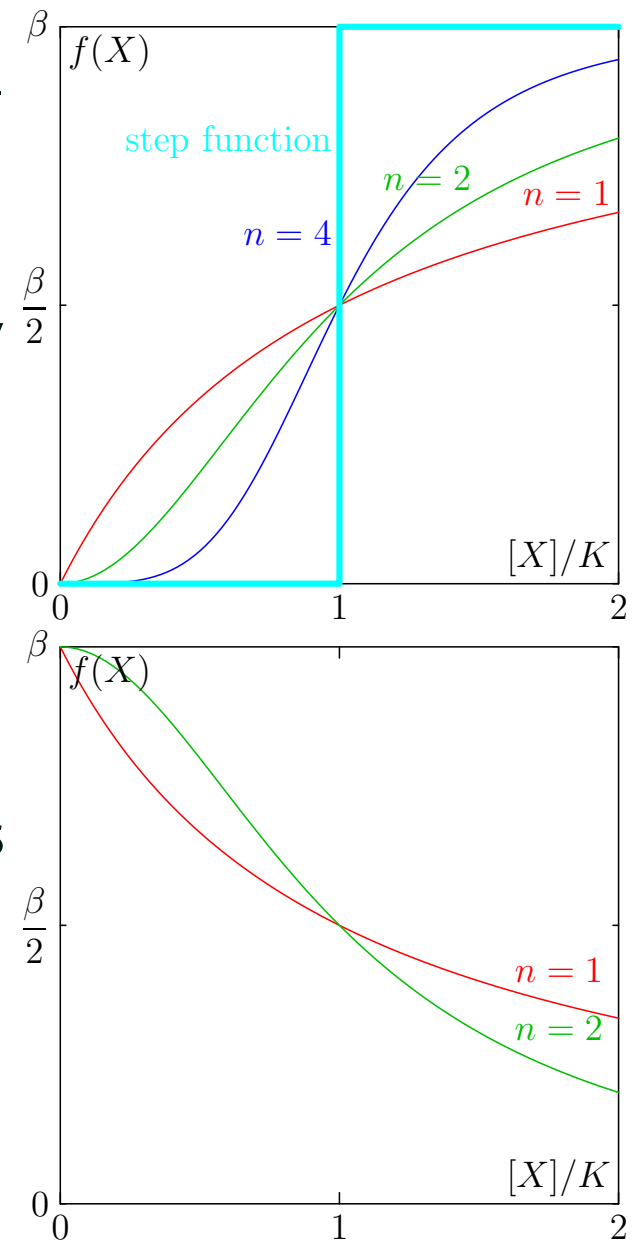
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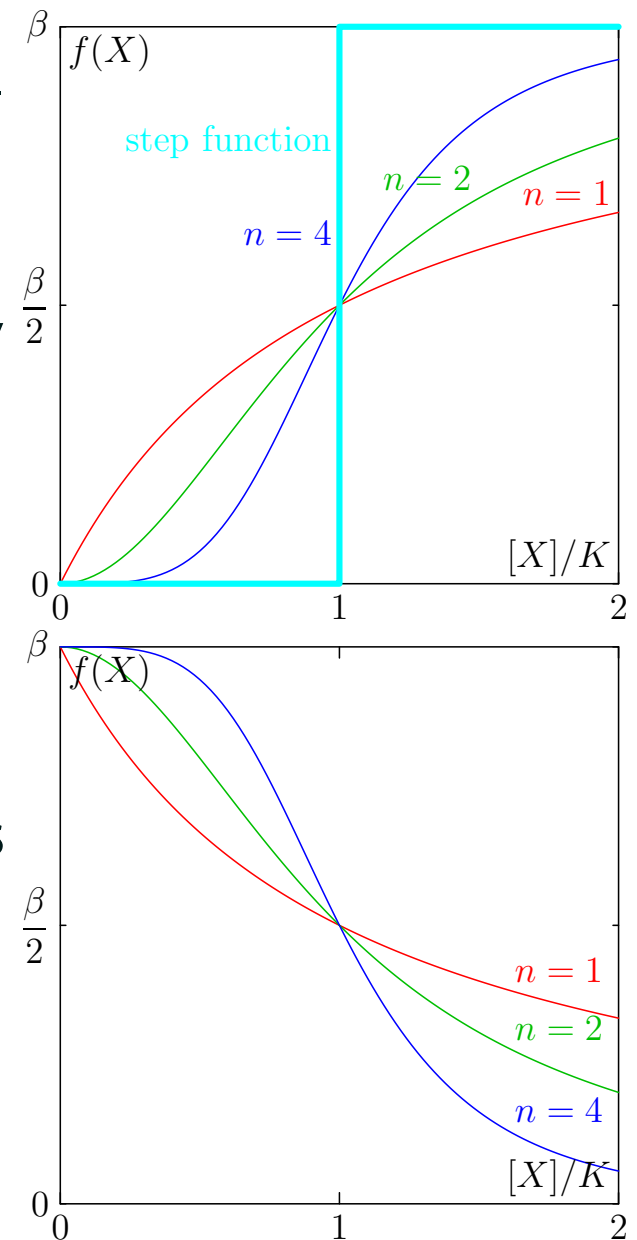
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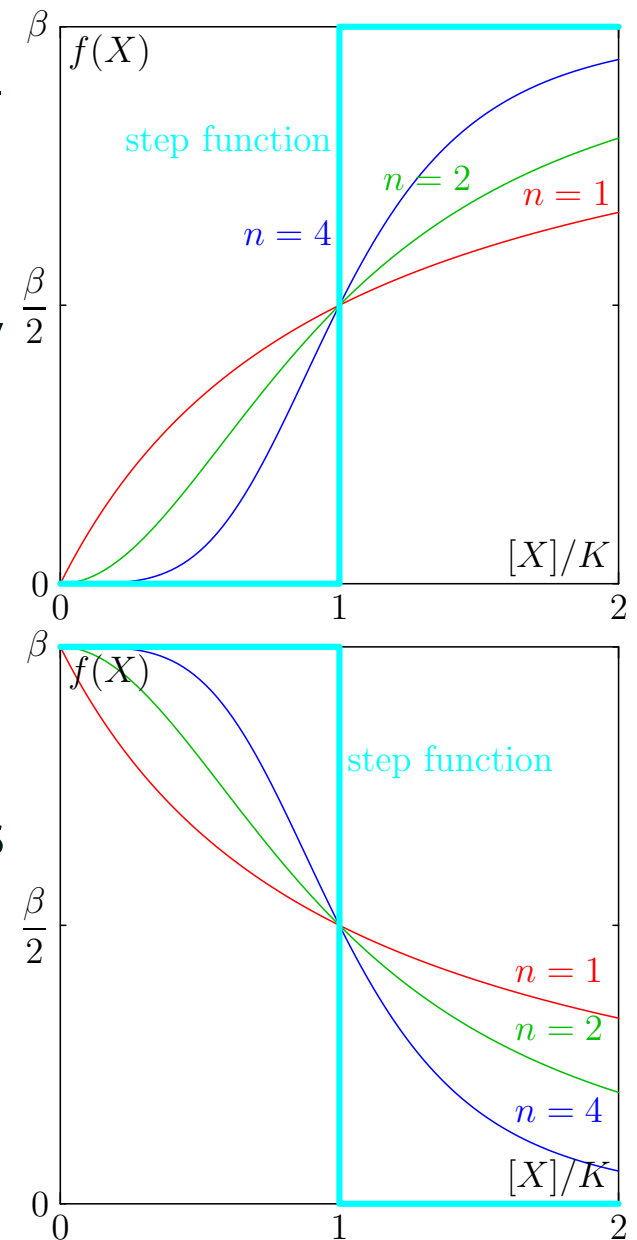
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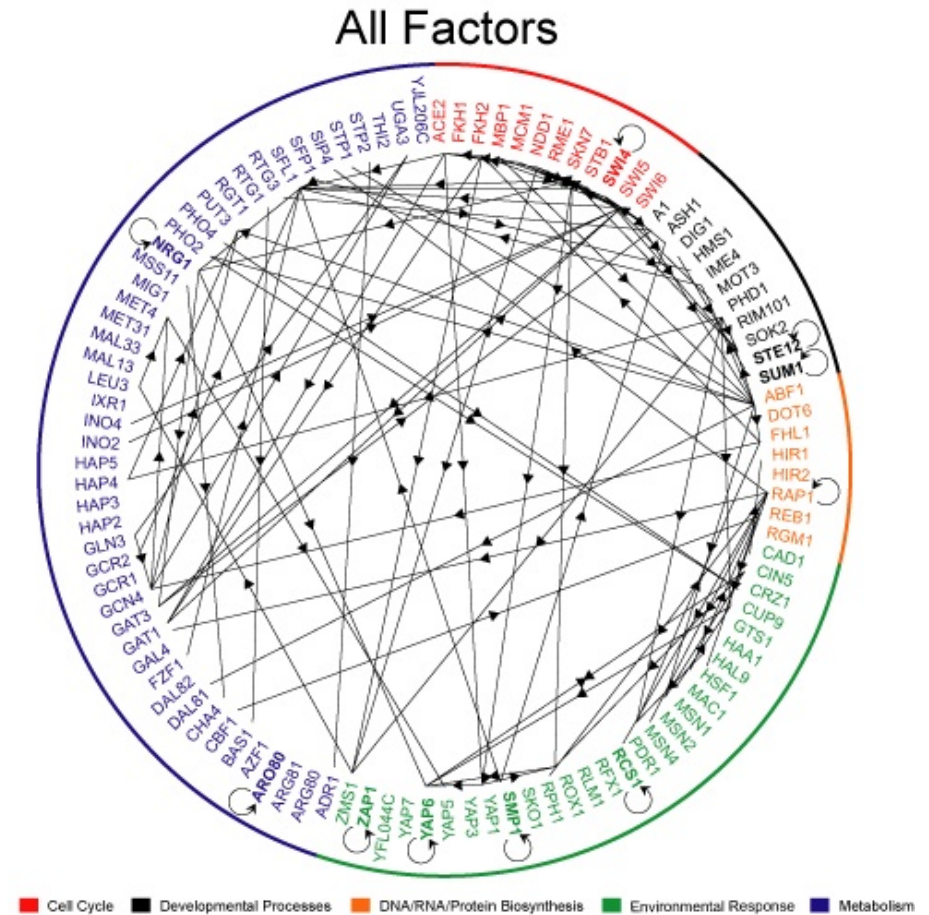
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Outline

1. Gene Regulation
2. Chemical Equations
3. **Motifs**
4. Quiz



Understanding Gene Regulation Networks

- We can infer interactions between proteins as we described in a previous lecture
- Gene regulation networks tend to be rather complicated
- To make sense of them we wish to look for patterns, “**motifs**”, that occur very often
- We look at interactions that occur more often than would be expected to happen in a random network

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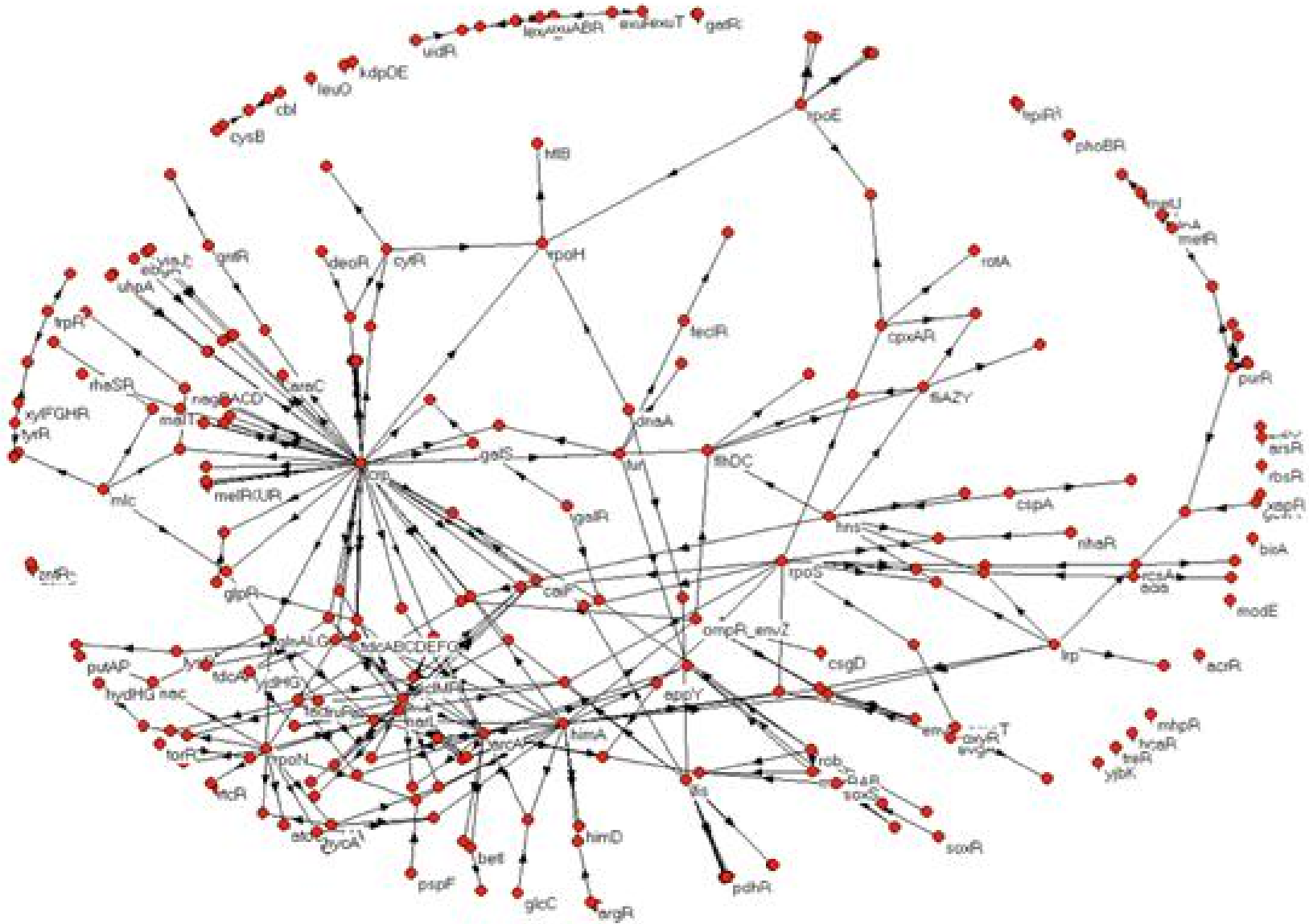
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E-Coli GRN



Auto-regulation

- In E-coli we have $N = 424$ nodes and $E = 519$ edges
- The total number of directed edges (including self-edges) is N^2
- The number of self edges is N so the probability of a randomly chosen edge being a self-edge is $p = 1/N$
- The probability that if we choose E edges k will be self-edges is

$$\mathbb{P}(k|E, p) = \text{Binom}(k|E, p) = \binom{E}{k} p^k (1 - p)^{E-k}$$

- This has mean $\mathbb{E}[N_{\text{self}}] = E p \approx 1.2$ and standard deviation $\sigma = \sqrt{E p (1 - p)} \approx 1.1$
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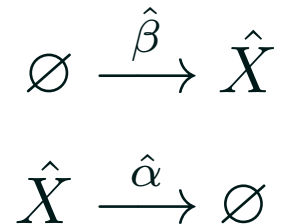
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No Auto-Regulation

- To understand why auto-regulation is interesting let us first consider the case with no negative feedback

- We assume



- So on average

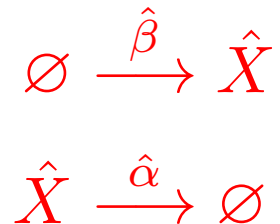
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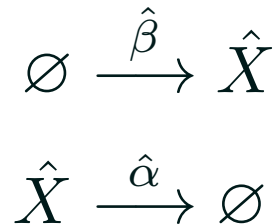
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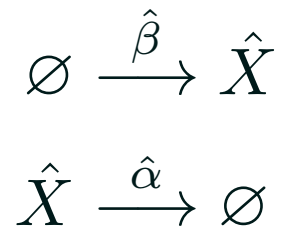
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- The full dynamics is given by the solution to the differential equation and depend on the initial condition, $[\hat{X}(0)]$
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- However, they are usually quite easy to solve numerically (though it is difficult to understand how the solution depend on the parameters)
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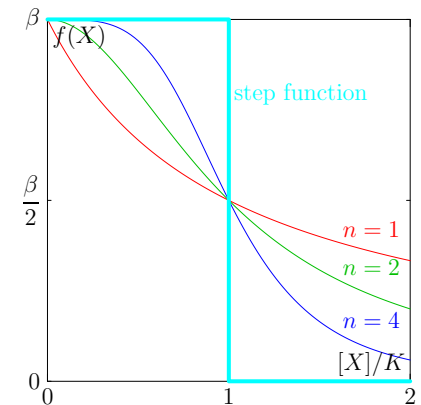
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- We assume a negative feedback so that

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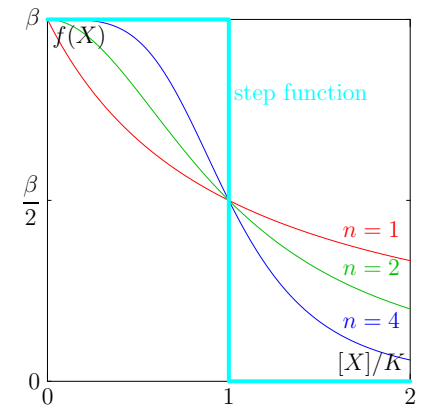
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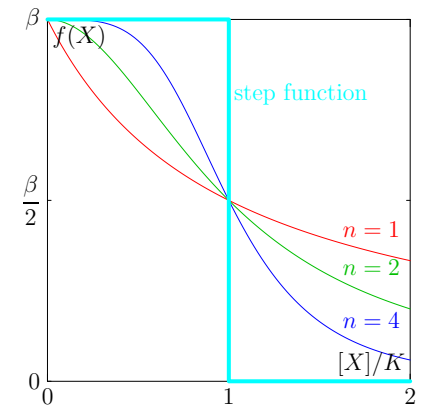
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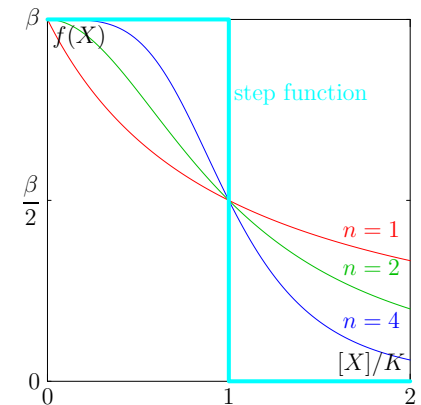
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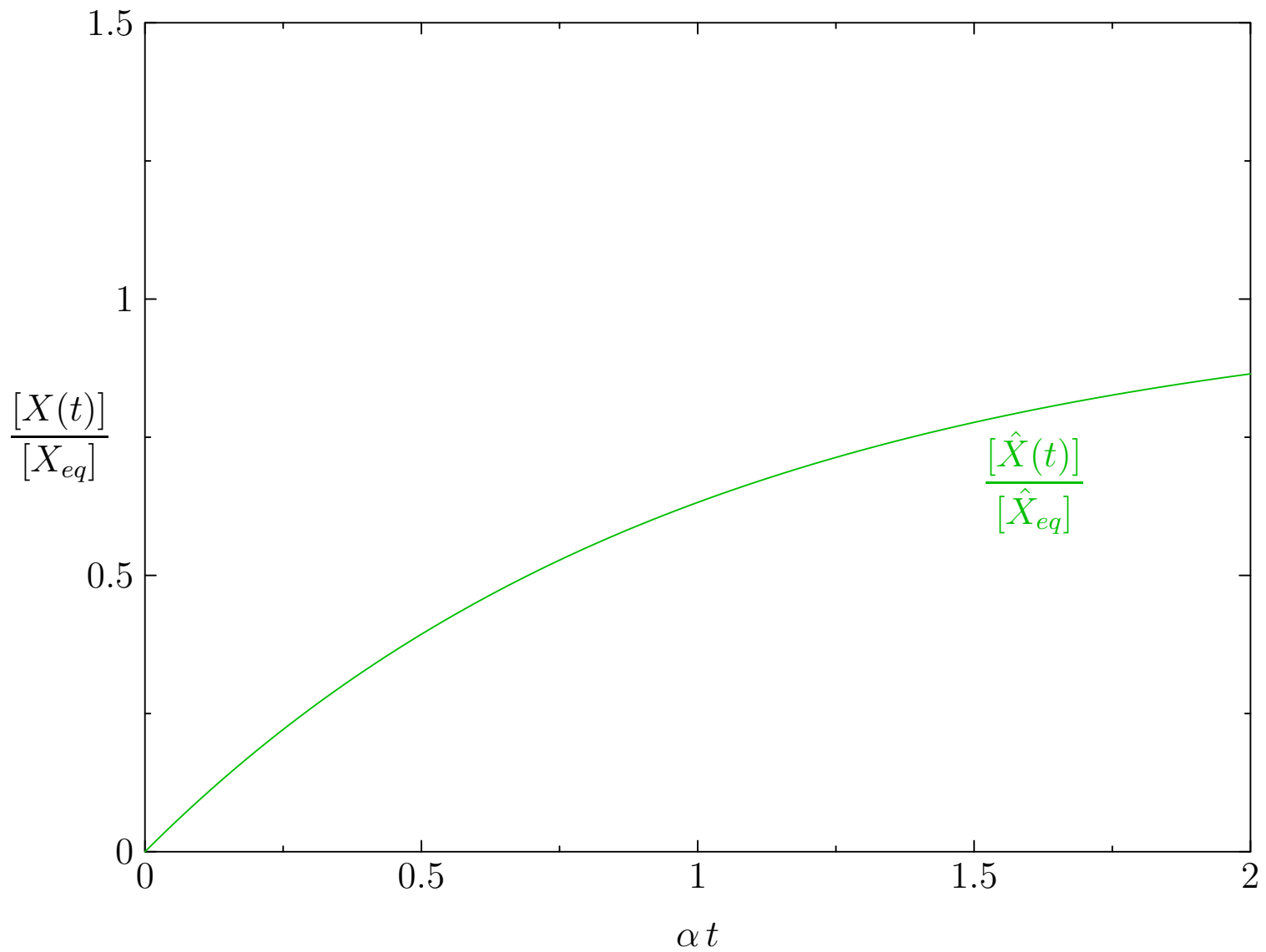
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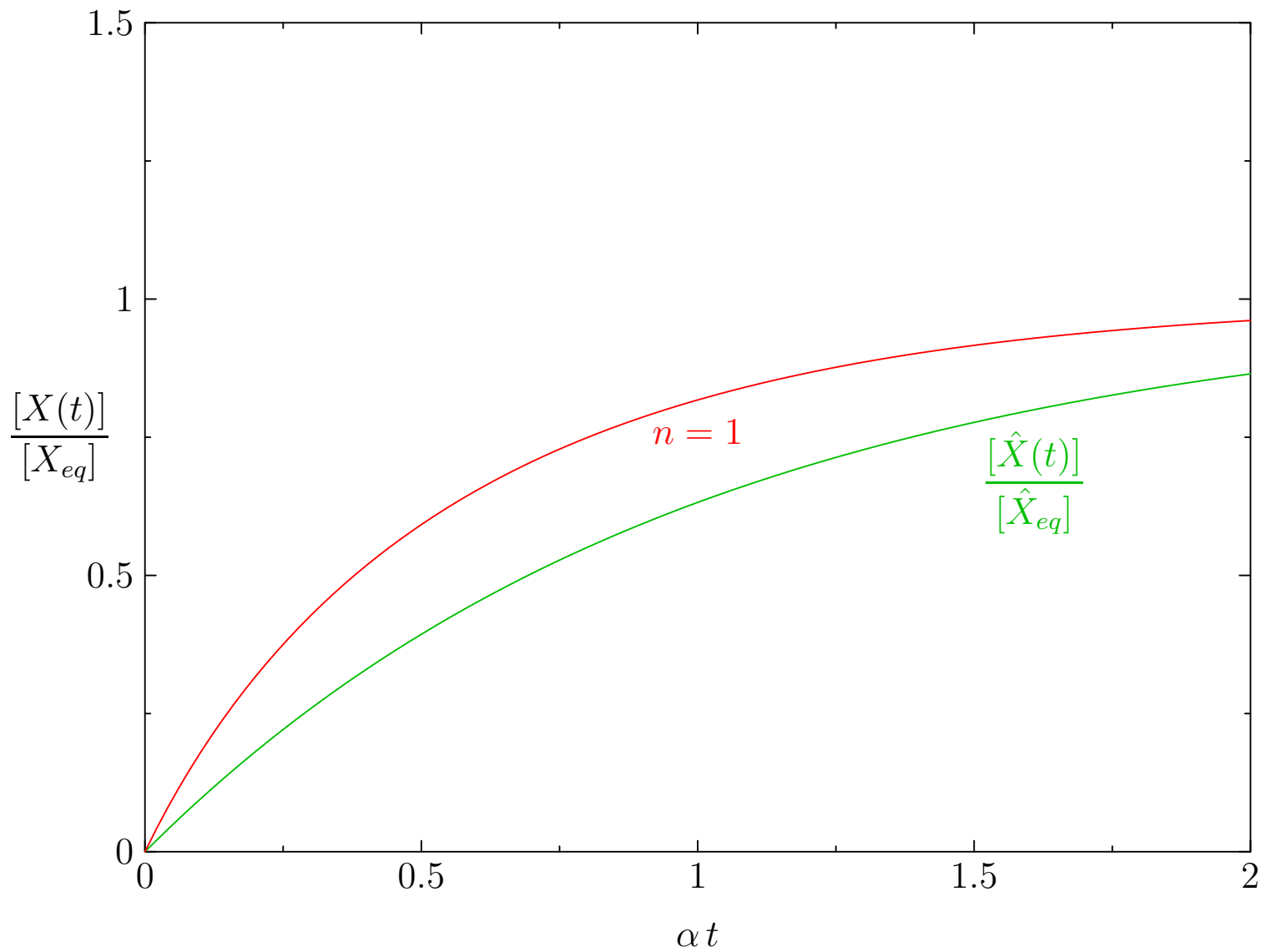
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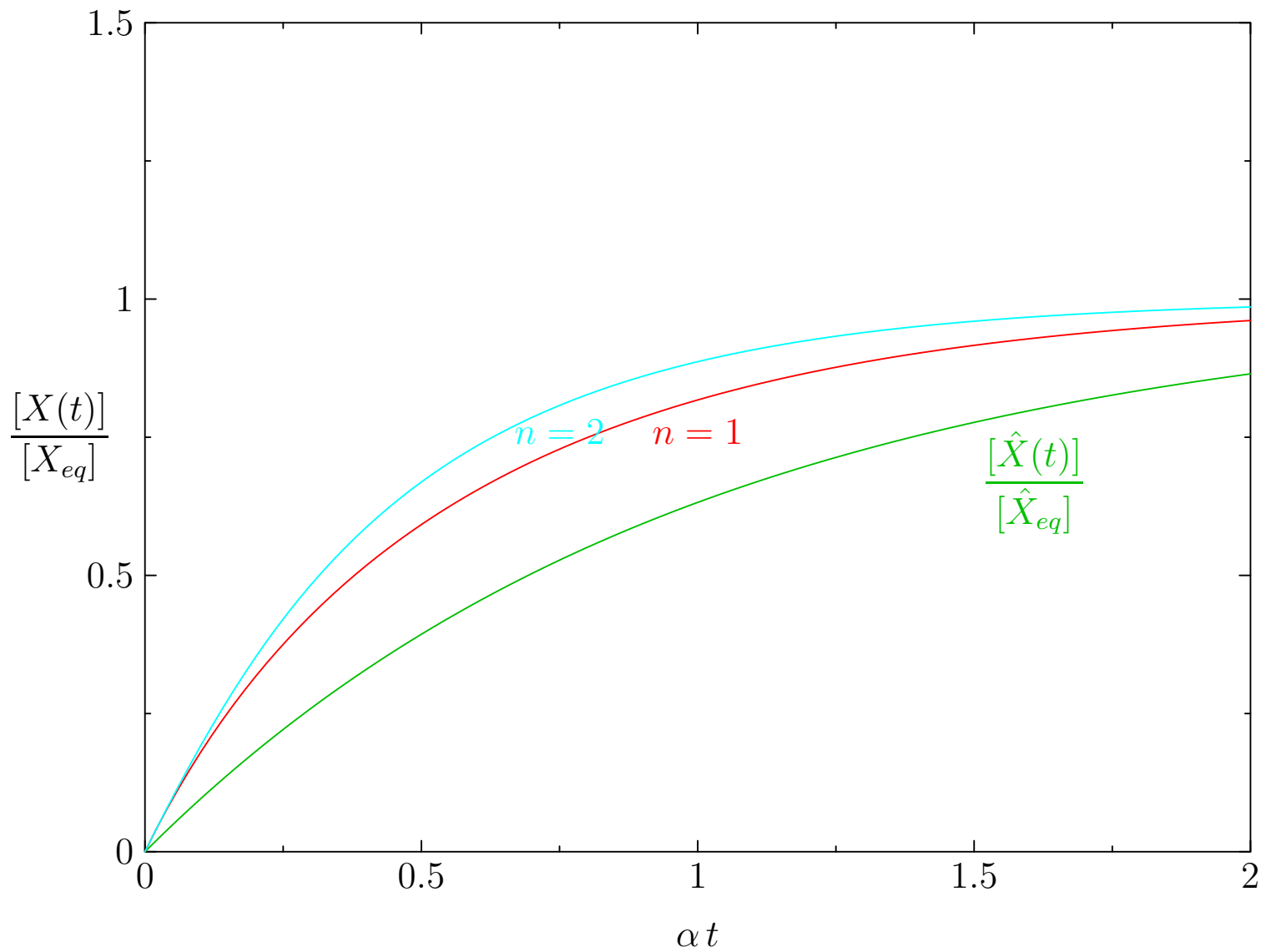
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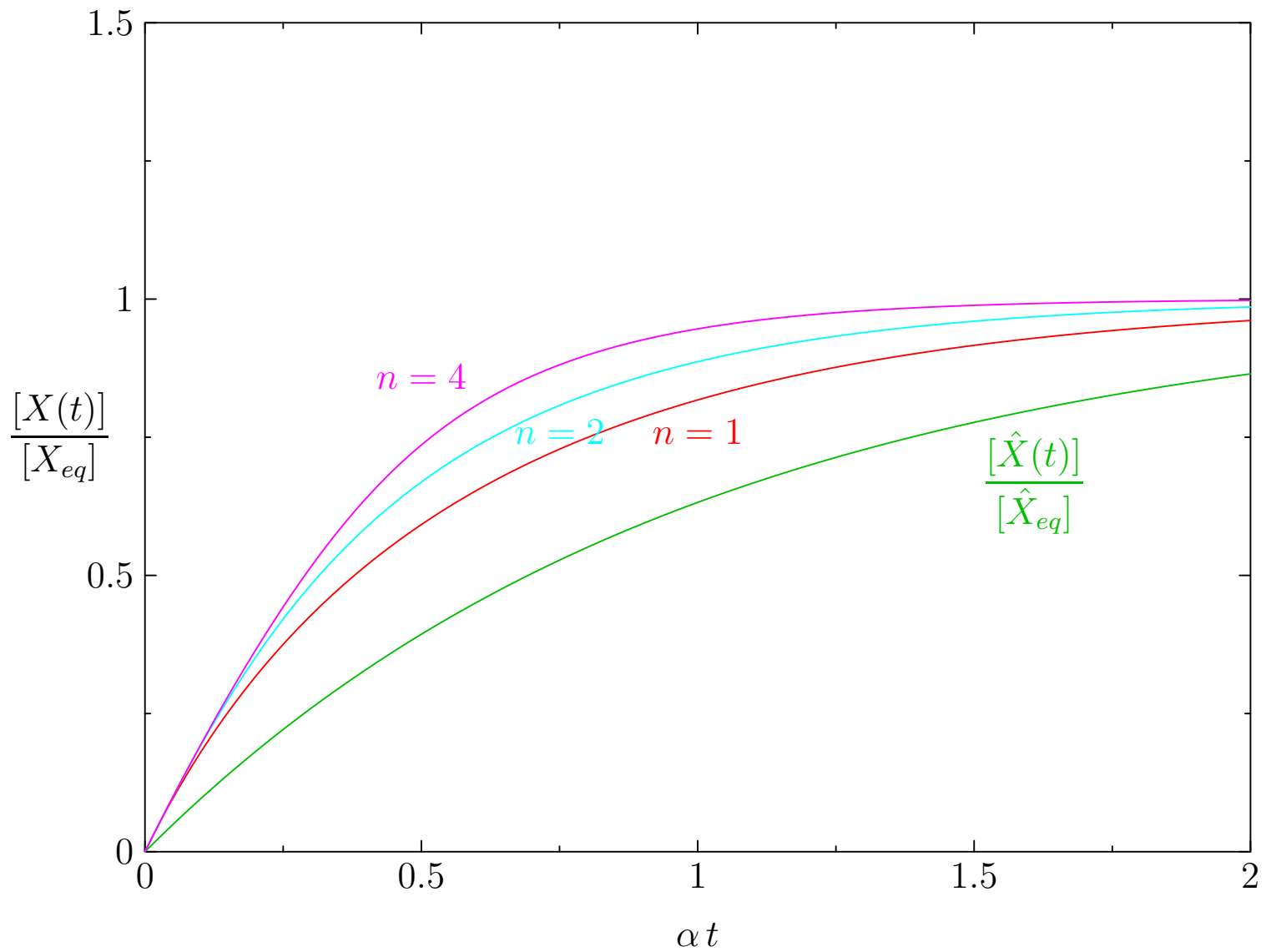
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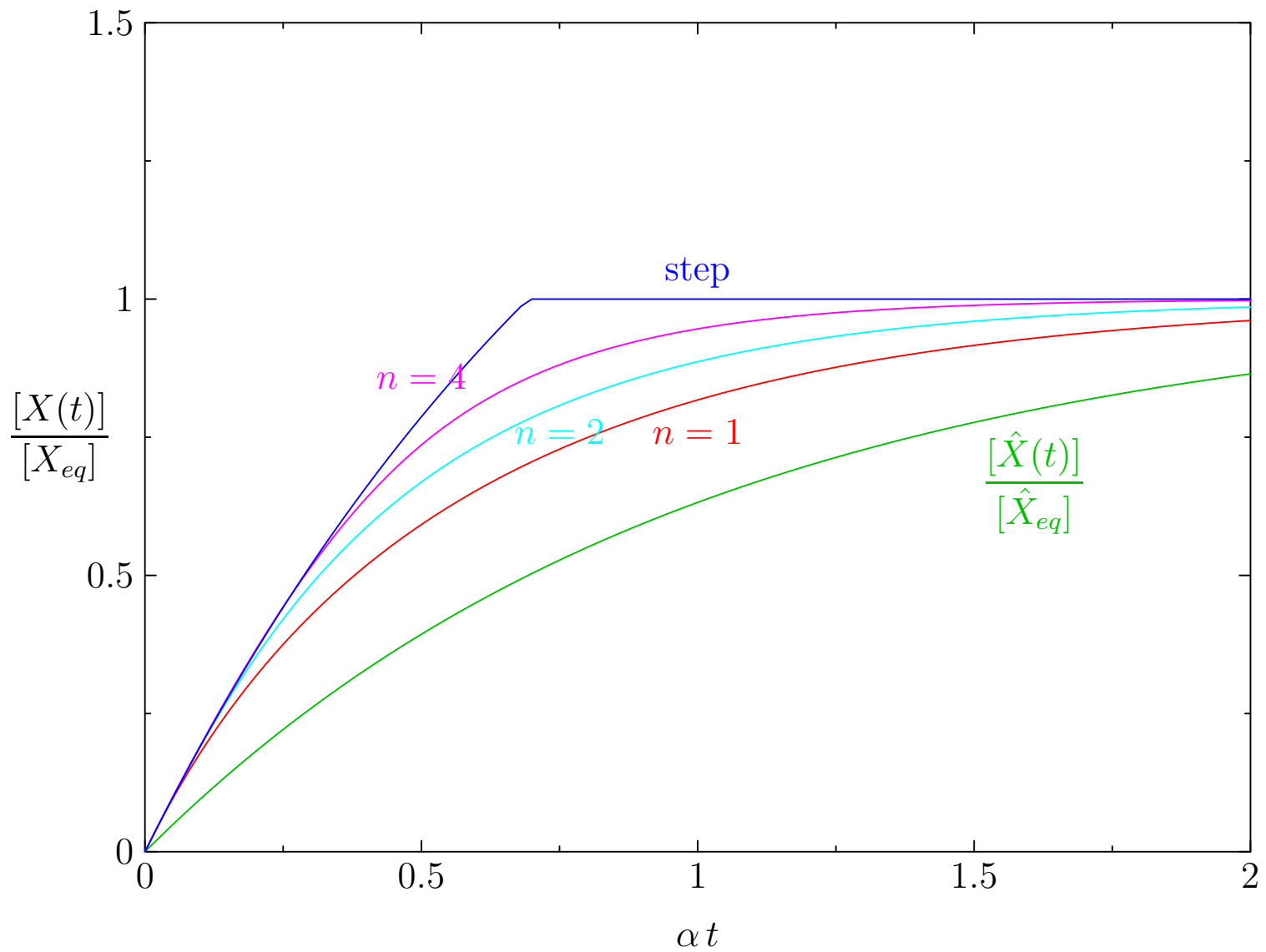
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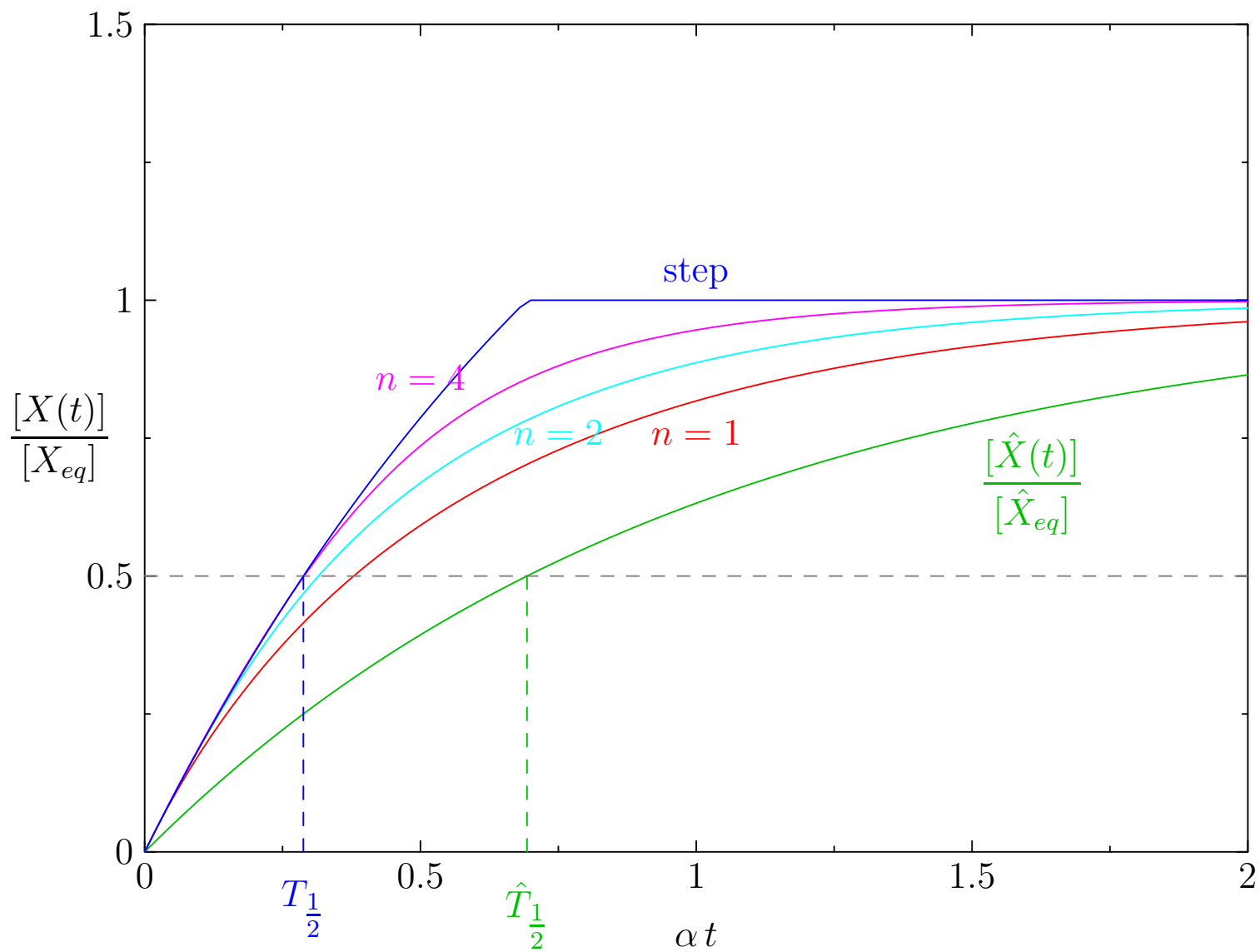
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- Auto-regulation allows us to use a large production rate β for the same equilibrium, speeding up the equilibration time
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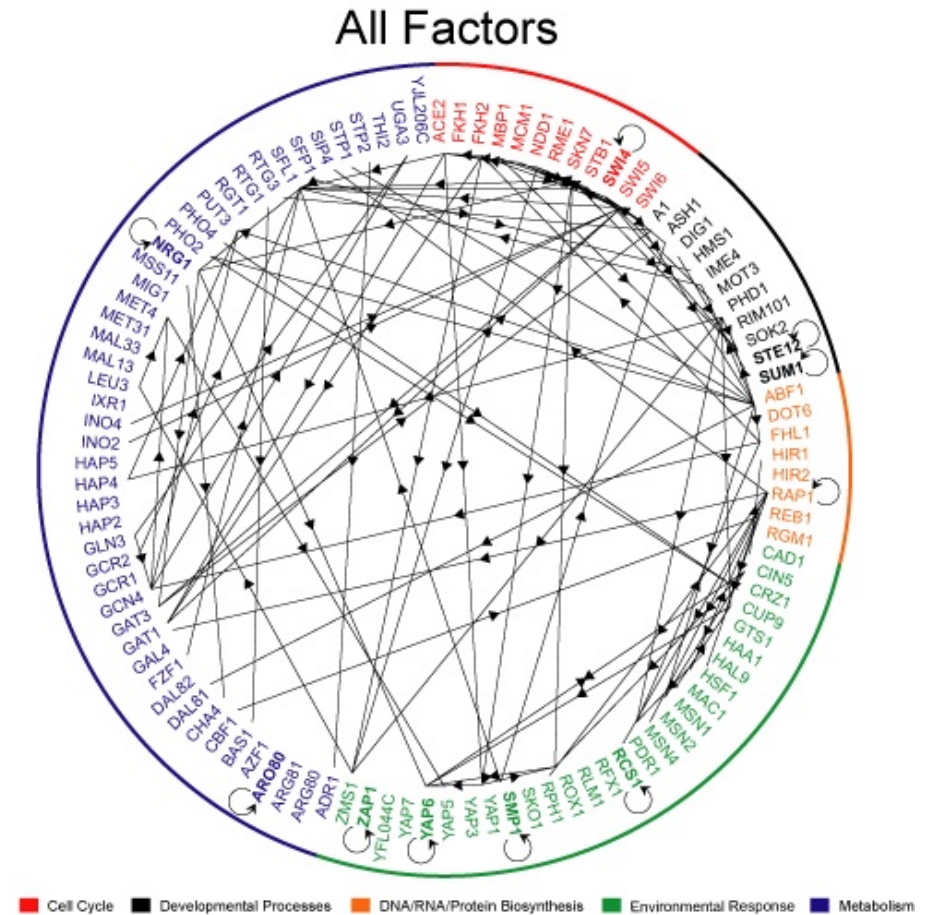
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4. $\frac{d[X]}{dt} = -k [X]$

- Answer: $\frac{d[X]}{dt} = -k[X]$

Question 3

- For the “chemical equation”



what is the correct ODE for Z ?

1. $\frac{d[Z]}{dt} = 2 k [X][Y]$
2. $\frac{d[Z]}{dt} = 2 k [X] + k [Y]$
3. $\frac{d[Z]}{dt} = k [X]^2[Y]$

Question 3

- For the “chemical equation”



what is the correct ODE for Z ?

1. $\frac{d[Z]}{dt} = 2 k [X][Y]$

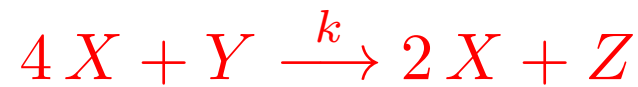
2. $\frac{d[Z]}{dt} = 2 k [X] + k [Y]$

3. $\frac{d[Z]}{dt} = k [X]^2[Y]$

- Answer: $\frac{d[Z]}{dt} = k [X]^2[Y]$

Question 4

- For the “chemical equation”

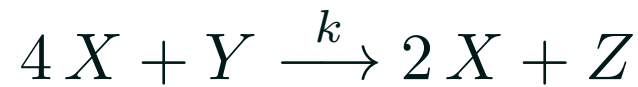


what is the correct ODE for X ?

1. $\frac{d[X]}{dt} = -k [X]^4[Y]$
2. $\frac{d[X]}{dt} = -2 k [X]^4[Y]$
3. $\frac{d[X]}{dt} = 4 k [X]^4[Y]$

Question 4

- For the “chemical equation”



what is the correct ODE for X ?

1. $\frac{d[X]}{dt} = -k [X]^4[Y]$
2. $\frac{d[X]}{dt} = -2 k [X]^4[Y]$
3. $\frac{d[X]}{dt} = 4 k [X]^4[Y]$

- Answer: $\frac{d[X]}{dt} = -2 k [X]^4[Y]$